



**US Army Corps of Engineers
Japan District**

**FY25 REPAIR HVAC
WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN**

**SPECIFICATIONS
Backcheck Submittal**

**CONTRACT NO.: W912HV19D0007
TASK ORDER NO.: W912HV24F0045**

AUGUST 2025

DRAFT

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SECTION 01 11 00.00 10

GENERAL CONTRACT REQUIREMENTS

PART 1 GENERAL

1.1 DEFINITIONS

The term "Government" refers to "United States Government" whenever the term "Government" appears in this Contract, unless otherwise indicated within the Contract.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 189.1 (2017) Standard for the Design of High-Performance Green Buildings Except Low-Rise Residential Buildings

ASHRAE 52.2 (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10 (2022; ERTA 1 2021) Standard for Portable Fire Extinguishers

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. ARMY GARRISON JAPAN (USAG-J)

Regulation 190-13

(2013) Installation Access and Control
Procedures

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Initial Site Conditions Photos

Key Personnel Qualifications; G

Conformed Drawings Submittal

Indoor Air Quality (IAQ) Management Plan; G

SD-07 Certificates

Monthly Progress Photos

SD-11 Closeout Submittals

Construction Completion Photos

1.4 ACCEPTABLE ALTERNATIVE JAPANESE STANDARDS

Where indicated in the technical specifications REFERENCES paragraphs, acceptable Japanese standards are listed which are considered acceptable alternatives to the US Standards within the Contract Specifications. Contract specifications authorize the use of specified acceptable alternative Japanese standards. The use of products not meeting the applicable US standards of the Contract or alternative Japanese standards specifically authorized by the Contract is prohibited unless authorized by the Contracting Officer. Japanese standards not identified in these specifications as being acceptable alternatives to US standards may be submitted to the Contracting Officer for consideration as acceptable alternatives after award using the variation request process set forth in Section 01 33 00 SUBMITTAL PROCEDURES. Prior to the use of such proposed additional alternative Japanese standards, written approval by the Contracting Officer is required. Incorporation of methods, materials, and equipment that will promote cost-effective and timely maintenance, and that are otherwise authorized by the contract, is highly encouraged.

1.5 MANDATORY U.S. TESTED PRODUCTS

The following items shall meet U.S. testing methods (e.g. UL listed, FM approved, ASTM, etc.) and shall be labeled as required. These items shall not be substituted with Japanese testing methods (e.g. JIS, JASS, etc.) for certification:

- a. Fire suppression systems, including valves, alarm valves, sprinklers. Fire suppression systems includes wet chemical, dry chemical, mist, deluge, pre-action, foam, and clean agent, in addition to standard sprinkler systems.

- b. Fire pumps, including motors, controllers, drives, and valves.
- c. Fire alarm and mass notification systems, including panels, initiating devices, notification appliances, smoke alarms.
- d. Engineering technician requirements for development of fire suppression systems and alarm system shop drawings, calculations, and material submittals.
- e. Fire doors, smoke doors, and frames.
- f. Fire dampers and smoke dampers.
- g. Interior finishes with flame spread and smoke development ratings required for installation of rated assemblies.
- h. Insulation with flame spread and smoke development ratings.
- i. Fire proofing and fire stopping materials.
- j. Plenum rated cables.
- k. Domestic water supply piping in the building and plumbing fixtures that directly dispense potable drinking water (NSF stamped and/or labeled). Plumbing fixtures conforming to Japanese standards that do not directly dispense potable drinking water are allowed, such as urinals and water closets, bath and shower faucets, utility/janitor faucets, lavatories and sinks.

1.6 KEY PERSONNEL QUALIFICATIONS

The Contractor's Project Key personnel shall not be assigned duties to any other Contracts (excluding the project manager) without approval from the Contracting Officer. The Project Superintendent, CQC System Manager, and Site Safety Health Officer shall all be separate persons, and shall not have other duties assigned. Within five (5) working days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer an organizational chart, the qualifications and background history of the proposed Key Personnel for approval. The Contracting Officer shall have the explicit right to determine acceptability (or rejection) of the proposed individuals. In addition, the Contractor shall be responsible to replace said individuals upon notification by the Government should performance become inadequate during the Contract period. Key Personnel shall attend the Preconstruction conference.

1.6.1 Project Manager

The Project Manager shall have a minimum of five (5) years experience as a Project Manager on construction projects similar in size and scope to this Contract. Project Manager shall have knowledge of projects with mechanical (HVAC Systems), electrical, structural and fire protection components. The Project Manager shall maintain oversight of Contract proposals prepared by the Contractor staff and be authorized to negotiate Contract terms and sign Contract documents on behalf of the Contractor. The project manager does not need to be present on the site daily, but shall attend weekly progress meetings and be available on site within 24 hours upon request.

1.6.2 Project Superintendent (Supervisor)

The Project Superintendent shall be on the work site when on-site work is being performed and shall be available to the Contracting Officer or his representatives upon request. The Project Superintendent shall have overall responsibility for all operations at the job site and be authorized to make decisions, negotiate Contract terms and sign Contract documents on behalf of the Contractor. The Project Superintendent shall have a minimum of five (5) years experience as a superintendent on construction projects containing mechanical (HVAC systems), electrical, structural and fire protection components similar in size and scope to this Contract.

1.6.3 English Speaking Representative (Interpreter)

At all times during the Contract period the Contractor shall have an employee capable of fluent bilingual speech in the Japanese and English languages at the job site. The bilingual interpreter shall have the capability to receive and issue concise and technical explanation and instructions between the Government representative(s) and Contractor supervisory personnel concerning all aspects of Contract administration and construction. Within fifteen (15) days after receipt of the Notice to Proceed, the Contractor shall submit in writing to the Contracting Officer the qualifications and background history of the proposed interpreter for approval. The Contracting Officer shall have the explicit right to determine acceptability or rejection of the proposed individual. In addition, the Contractor shall be responsible to replace said individual upon notification by the Government should performance become inadequate during the Contract period. The interpreter shall attend the preconstruction conference.

1.6.4 Contractor Quality Control

1.6.4.1 Contractor Quality Control System Manager (CQCSM)

The CQC System Manager is required to be a graduate of an accredited college with an engineering, architecture, or construction management degree, with a minimum of 3 years construction experience on construction of similar size and scope to this Contract. Refer to Section 01 45 00.00 10 QUALITY CONTROL, Paragraph "CQC System Manager" for additional requirements.

1.6.4.2 Contractor Quality Control Personnel

The following Contractor Quality Control (CQC) Personnel are required: Civil, Mechanical, Electrical, Structural, Architectural, Fire Protection, Environmental, Testing, Adjusting, and Balancing. All other specialties in section 01 45 00.00 10 not identified in this paragraph does not apply to this project.

See Section 01 45 00.00 10 QUALITY CONTROL for qualification requirements for CQC personnel. These individuals or specialized technical companies may be employed by the prime contractor or a subcontractor on this project. These individuals can perform other duties but need to be allowed sufficient time to perform the specialized personnel's assigned quality control duties as described in the Quality Control Plan. A single person can cover more than one area provided that the single person is qualified to perform quality control activities in each designated and that workload

allows.

1.6.5 Site Safety and Health Officer (SSHO)

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

1.6.6 Pass Coordinator

A person on the contractor's staff who is familiar with the requirements of USAG-J Regulation 190-13, Individual and Vehicle Access Pass Procedures, and USAG-J vehicle security inspection procedures. The Pass Coordinator must be able to obtain a DBIDS access pass with escort privileges and may be required to escort other contract personnel. Although a TOEIC score is not required, the Pass Coordinator must be able to understand pass requirements and complete pass application forms in the English language.

1.7 TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER

- a. This paragraph specifies the procedure for determination of time extensions for unusually severe weather in accordance with the Contract clause entitled "DEFAULT (FIXED-PRICE CONSTRUCTION)". In order for the Contracting Officer to award a time extension under this clause, the following conditions must be satisfied:
 - (1) The weather experienced at the project site during the Contract period must be found to be unusually severe, that is, more severe than the adverse weather anticipated for the project location during any given month.
 - (2) The unusually severe weather must actually cause a delay to the completion of the project. The delay must be beyond the control and without the fault or negligence of the Contractor.
- b. The listing below defines the monthly anticipated adverse weather days for the Contract period and is based upon 20th Weather Squadron (MAC), U.S. Air Force; Iwakuni U.S. Marine Corps Air Station, Weather Service; Sasebo Naval Pacific Meteorology and Oceanography Detachment; JASDF (Japan Air Self Defense Force), or similar data for the geographic location of the project. (More specific information may be obtained as specified under Special Contract Requirements Clause entitled "Physical Data.")

MONTHLY ANTICIPATED ADVERSE WEATHER DELAYS
WORK DAYS BASED ON (5) DAY WORK WEEK

	Misawa-area	Kanto Plain	Iwakuni	Okinawa	Sasebo	Osaka
JAN	8	1	2	5	7	4
FEB	9	3	4	7	6	4
MAR	6	6	7	8	7	5
APR	3	6	6	6	7	4
MAY	4	4	6	6	6	4
JUN	5	9	8	6	12	5
JUL	6	6	6	6	9	4
AUG	6	6	5	9	9	3
SEP	6	7	6	7	9	4
OCT	4	5	4	4	8	4
NOV	6	4	4	4	5	3

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The above schedule of anticipated adverse weather days shall constitute the base line for monthly (or portion thereof) weather time evaluations.

- c. Upon acknowledgment of the Notice-to-Proceed (NTP) and continuing throughout the Contract, the Contractor shall record on the daily CQC report, the occurrence of adverse weather and resultant impact to normally scheduled work. Actual adverse weather delay days must prevent work on critical activities for 50 percent or more of the Contractor's scheduled work day.
- d. The number of actual adverse weather delay days shall include days impacted by actual adverse weather (even if adverse weather occurred in previous month), be calculated chronologically from the first to the last day of each month, and be recorded as full days. If the number of actual adverse weather delay days exceeds the number of days anticipated in paragraph (b), above, the Contracting Officer shall convert any qualifying delays to calendar days, giving full consideration for equivalent fair weather work days, and issue a modification in accordance with the Contract clause entitled "DEFAULT (FIXED PRICE CONSTRUCTION)".
- e. For all work under this Contract, adverse weather is defined as:
 - (1) Rainfall - Number of occurrences of precipitation greater than or equal to 0.10 inches (2.54mm).
 - (2) Snowfall - Number of occurrences of precipitation greater than or equal to 1.00 inches (25.4mm). (Not applicable to Okinawa Area)
 - (3) Cold Temperature - Number of occurrences when daily maximum temperature does not exceed the monthly mean low temperature or 32 degrees Fahrenheit (0 degrees Celsius), whichever is lower. (Not applicable to Okinawa Area)
 - (4) Concurrence between snowfall and cold temperature is 80 percent, i.e. 80 percent of the time snow falls, the temperature is "cold".
 - (5) Wind - Number of occurrences when the wind is gusting 30 knots (56 kilometers/hour) or greater.
- f. The Contractor's schedule must reflect the above anticipated adverse weather delays on all weather-dependent activities.

1.8 PERMITS AND RESPONSIBILITIES

The Contractor shall, without additional expense to the Government, be responsible for obtaining any necessary licenses and permits, and for complying with any laws, codes, and regulations (including the requirements of material, prefectural, and local Government of Japan, and associated military installation) applicable to the performance of work. The Contractor shall also be responsible for all damages to persons or property that occur as a result of the Contractor's fault or negligence. These damages shall be repaired or replaced by the Contractor at no cost to the Government. The Contractor shall take proper safety and health precautions to protect all work and workers.

Submit a schedule of planned road closures to the Contracting Officer with

the initial project schedule. Notification of specific road closures shall be in writing to the Contracting Officer not less than thirty (30) calendar days in advance of the intended closure. The road closure request shall include the planned traffic control measures as well as the general information about the closure. All road closures shall be coordinated with base officials and are subject to base requirements. No road closure shall be permitted until the Contractor receives written approval from the Contracting Officer. Full road closures is generally not permitted (at least one way traffic shall always be provided).

1.9 SPECIAL CONTRACT REQUIREMENTS

1.9.1 Meetings

- a. Post Award Orientation. A post award orientation (often referred to as a Preconstruction Conference) per FAR 42.503 shall be held. The chairperson shall be the Contracting Officer or his/her designee. The chairperson shall be responsible for providing minutes of the meeting.
- b. Weekly Progress Meetings. A weekly progress meeting shall be conducted, with the day, time, and location to be determined at the Preconstruction Conference. The Contractor's Project Manager, Superintendent, Interpreter, and Quality Control Manager shall attend. Representatives from the Government may include, but are not limited to, Director of Public Works/Facilities Engineering personnel, Project's Resident Office personnel, and Project Manager. The Contractor shall be prepared to discuss work completed during the previous week, work currently in progress, and work forecasted for the following week, as well as the status of any construction issues and open action items. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the progress meeting for review.
- c. Red Zone Meeting. A Red Zone meeting shall be conducted for Contracts with a value in excess of 55,000,000 JPY or those deemed sufficiently complex by the Contracting Officer to warrant one. The purpose of the Red Zone meeting is to discuss closeout requirements for the contract and to establish a timeline to get those items completed (see Section 01 78 00 CLOSEOUT SUBMITTALS for typical closeout requirements). The meeting is a good time for the Contractor to gain approval on format for items such as Operations and Maintenance manuals, the equipment-in-place list, the warranty plan, and any other items. The Initial Red Zone is typically scheduled when project reached 75 percent completion milestone. Provide the meeting agenda for review 24 hours prior to the meeting, take notes during the meeting, and provide electronic copies of the meeting minutes in English within 24 hours of the Red Zone meeting.
- d. Safety and Quality Control Meetings. Safety and quality control meetings shall be held as needed and determined by the Contractor. See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and Section 01 45 00.00 10 USACE QUALITY CONTROL for specific requirements.
- e. Construction Waste Management Meetings. See Section 01 74 19 CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT for requirements.

1.9.2 Conformed Drawings Submittal

Contractor shall submit one (1) hard copy Japanese-style folded and bound set of conformed A3 size drawings. Submittal shall be provided to the Resident Office at the Pre-Construction meeting. Conformed drawings are defined as the original advertised drawings with amended drawing sheets replaced in-kind with the corresponding original sheet number. If applicable, add new sheets and remove deleted sheets in accordance with amendment instructions. If multiple amendments are issued, supersede any previously amended drawing so that only the highest numbered amendment drawing is included.

1.9.3 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 days after Notice to Proceed and not less than 10 days before the preconstruction meeting. The Plan shall include IAQ management practices implemented during construction and preoccupancy phases and describe how each requirement in ANSI/SMACNA 008 and TPC requirements will be met, addressed, and managed on the job site. The plan shall cover the following additional items:

- a. Identify the key players and person(s) responsible for implementing the plan.
- b. Specify procedures for protecting stored and installed absorptive materials from moisture damage.
- c. Prohibit the use of tobacco products inside the building and within 7.5 meters (50 feet) of the building entrance at all times during construction. Consider prohibiting smoking on the entire job site. If smoking will be allowed, locate on a site plan the designated locations and signage. Comply with the tobacco use policy of the local jurisdiction.
- d. Indicate whether air handlers will be operated during construction, and specify compliant filtration procedures for permanent equipment that will be used.
- e. Take photographs of each IAQ measure for documentation. Annotate photographs to indicate each IAQ measure depicted and its general location. Provide photographs of the methods employed to protect stored and installed absorptive materials from moisture damage during construction and preoccupancy.

Revise and resubmit the Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.9.3.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.9.3.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008, Chapter 3, to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.9.3.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

1.9.3.2 Requirements after Construction

After construction ends and prior to occupancy, conduct IAQ testing in accordance with ASHRAE 189.1 Section 10.3.1.4.

In the event that IAQ testing fails, conduct a building flush-out in accordance with ASHRAE 189.1 Section 10.3.1.3. The space shall be ventilated at a minimum rate of 1.5 Liters per second per square meter of outdoor air while maintaining an internal temperature of at least 15 degrees C (60 degrees F) and no higher than 27 degrees C (80 degrees F) and relative humidity no higher than 60%. Total air volume of outdoor air will be determined by ASHRAE 189.1 Section 10.3.1.4 Equation 10-1. Total volume of outdoor air shall be approved by the Contracting Office prior to building flush-out

Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Ambient Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.10 PROGRESS AND COMPLETION PHOTOGRAPHS

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600 by 1200 by 24 bit true color 300 DPI minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 5 digital photographs each week throughout the entire project and 25 digital photographs at completion of project. Submit the Initial Site Conditions Photos, Monthly Progress Photos, and Construction Completion Photos. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Include a date designator in file names. Cross reference submittals in the appropriate daily report. Photographs provided are for unrestricted use by the Government.

1.11 BASE REGULATIONS

The Contractor and Subcontractor(s) shall become familiar with and obey all base regulations, including fire, traffic, safety, and security regulations. All Contractors shall keep within the limits of the work (and avenues of ingress and egress), and shall not enter any restricted areas unless required to do so and are cleared for such entry. The Contractor's equipment shall be conspicuously marked for identification.

1.11.1 Request for Contractor's Employee Passes and Vehicle Passes

No employee or representative of the Contractor shall be admitted to the work site without a Contracting Officer's furnished authorized admittance. Refer to USAG-J Regulation 190-13. Prior to the start of on-site work, submit applications for base passes to the Contracting Officer for key employees (project manager, site supervisor, interpreter, etc.) for long term DBIDS passes. The Contractor (through the employment of a Pass Coordinator), with assistance from the Resident Office, is responsible for securing sufficient passes for other workers, including subcontractor's employees and vehicles, required to access the base for the project duration. Additional personnel data shall also be furnished.

The Contractor shall use the USAG-J Form 1529 "EZ Pass" as the primary means of obtaining passes for all personnel who do not qualify for DBIDS passes. The USAG-J Form 1529 "EZ Pass" One Time/Multiple Access Roster must be complete and submitted to the Provost Marshall Pass and Vehicle Registration Office a minimum of three (3) U.S. workdays prior to access. The use of the USAG-J "one day" pass (AJ Form 47) should be avoided, and used only for urgent situations. Applicants for a "one day" pass may be required to wait for several hours for their pass to be issued. Contractor personnel applying for more than one (1) "one-day" pass within any 15-day period will not be issued multiple one-day passes, and will instead be required to apply for access via the USAG-J Form 1529, and wait three or more days for approval.

Provide all information required for background checks to meet base access requirements to be accomplished by Base Provost Marshal Office, Director of Emergency Services or Security Office. Contractor workforce must

comply with all personal identity verification requirements as directed by DOD, HQDA and/or local policy. The Government reserves the right to make changes to the Contractor security requirements or processes due to a change in the Force Protection Condition (FPCON).

Upon completion of this Contract, return all employee and vehicle passes to the base, and obtain a certification of receipt. Final payment shall be withheld until all passes have been returned.

1.11.2 No Smoking Policy

Smoking is prohibited on installations except in designated smoking areas. This applies to existing buildings, buildings under construction, and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer shall identify designated smoking areas.

1.11.3 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordnance (UXO), Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT. In the event MEC are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer's Representative. Contractor shall provide dispatch and Contracting Officer's Representative with specific location of the item. Contractor personnel shall stop work in the immediate vicinity of the discovery and maintain a minimum distance of 300 meters from the item. Contractor shall maintain flexibility in redirecting personnel and work effort in the event that items possessing an explosive hazard are discovered and construction personnel are excluded from areas during the destruction/removal process.

1.12 ORDER OF WORK

1.12.1 Schedule

Schedule all work to cause the least amount of interference with activity operations. Permission to interrupt any activity or roads shall be requested in writing a minimum of 30 calendar days prior to the desired date of interruption. Interruptions of activities, roads, and utility services shall be allowed only when they will not cause interference with the operations of the activity. The Contractor shall remove and dispose off Government property all Contractor generated debris at the end of each shift, or more frequently if required, to keep the space usable. Upon award of this Contract, the Contractor shall begin and complete all required work; ready for use and including cleanup, within the time period specified on this Contract. All work scheduled in occupied areas shall be accomplished in such a manner as to cause the least possible inconvenience to the occupants.

1.12.2 Working Hours

Normal Working Hours and Days are Monday through Friday, 0800 hours through 1700 hours. Saturday work is allowed on this contract and does not require additional approvals. Notify the Contracting Officer at least three days in advance when scheduling Saturday work.

1.12.3 Noise and Vibration

Noise greater than 70 decibels and vibration producing work in occupied buildings shall be conducted outside core school/business hours (0700 - 1430) unless written permission is given by the Contracting Officer.

1.12.4 Work Outside of Regular Hours

If the Contractor desires to carry on work outside the regular hours, on Sundays, or holidays, a written application shall be submitted to the Contracting Officer or his representative for approval. The Contractor shall allow three working days notice to enable satisfactory arrangements to be made by the Government for inspecting the work in progress. If work is to be accomplished after daylight hours, the Contractor shall illuminate the area in a manner approved by the Contracting Officer or their representative and in accordance with EM 385-1-1. Unless directed by a Contracting Officer, work accomplished outside regular working hours shall be at no additional cost to the Government.

1.13 FIRE PREVENTION DURING CONSTRUCTION

1.13.1 General

Comply with all pertinent fire prevention provisions of the US Army Corps of Engineers Manual EM 385-1-1, NFPA 241 and shall follow the Installation fire regulations. Prior to commencement of welding or other hot work operations, obtain approval from the Installation Fire Chief.

1.13.2 Supply

No more than one day's supply of paint, paint materials, or compounds shall be allowed within the area of the building, and shall be removed from the job site after each working day. No gasoline or similar low flash point flammable liquid shall be allowed within the building area. After proper coordination with the Facility's Emergency Services (to confirm the maximum allowable quantities and location of storage), storage of additional product may be authorized.

1.13.3 Fire Extinguishers

Provide, as a minimum, the number, size, and type of fire extinguishers in accordance with the latest NFPA 10. The Contractor shall comply with the Installation Fire Chief's policies if they are more stringent than NFPA 10. Fire extinguishers shall remain the property of the Contractor and shall be removed upon completion of the project.

1.13.4 Housekeeping

Accumulations of combustible material shall be removed from the building area on a daily basis.

1.13.5 Handling of Gasoline

Gasoline shall be stored in industry standard approved safety containers. Adequate ventilation shall be provided to safely dispose of flammable vapors where flammable liquids are utilized. Gasoline powered equipment shall be refueled a minimum 6 meters away from the building area.

1.13.6 Notification of Fire

Be familiar with methods for notifying the Installation Fire Department. The Installation's fire poster shall be posted in conspicuous locations and at telephones in construction shacks.

1.14 CONTRACTOR FURNISHED MATERIAL AND WORKMANSHIP

The Contractor shall furnish all materials necessary for performance of the work of this contract unless otherwise specified. Materials procured shall be new and shall meet any specifications and standards listed in these specifications. If no specification for a material needed to perform this contract are stated, the material shall be new, of acceptable industrial grade and quality, equal to or better than the manufacturer's original equipment for equipment being repaired or replaced, and will be compatible with existing materials and systems. All materials provided under this contract shall be free of asbestos, lead in paint, and PCB. The Contracting Officer reserves the right to request submittal of any material being provided under this contract. The Contracting Officer shall make the final determination of the acceptability of any material used on this contract.

1.15 CORRESPONDENCE

All correspondence addressed to the Government shall be made through serialized letters furnished with one original and two copies. Serialized letters shall begin with the number S-0001 and shall be continuous without a break in numbering. Serialized letters shall include the contract title and number, date, subject, and shall be signed by an authorized representative of the Contractor.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

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SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

PART 1 GENERAL

1.1 GENERAL

Payment items for the work of this Contract for which payments shall be made are listed in the CONTRACT LINE ITEM NUMBER (CLIN) SCHEDULE and described below. All costs for items of work, which are not specifically mentioned to be included in a particular job or unit price payment item, are included in the listed job item most closely associated with the work involved. The price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, topographic surveys, cost of bond premiums, and for performing all work required by drawings and specifications for which separate payment is not otherwise provided. Work paid for under one item shall not be paid for under any other item. No separate payment shall be made for the work, services, or operations required by the Contractor, as specified in DIVISION 01, GENERAL REQUIREMENTS, to complete the project in accordance with these specifications; all costs thereof shall be considered as incidental to the work.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; DESIGN-BID-BUILD: submittals not having a "G" designation are for information only; DESIGN-BUILD: submittals not having a "G" designation are for Contractor Quality Control or Designer of Record approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-07 Certificates

Bill of Lading; G

1.3 LINE ITEMS

1.3.1 CLIN 0001, Construction of Primary Facility

1.3.1.1 Payment

Payment shall be made at the Contract job price for CLIN 0001 "Construction of Primary Facility" for work completed including, but not limited to []. Payment for work completed shall constitute full compensation of CLIN 0001, complete.

1.3.1.2 Unit of Measure

Unit of measure: job.

1.3.2 CLIN 0002, Construction of Supporting Facilities, 1.5 meters outside of Buildings

1.3.2.1 Payment

Payment shall be made at the Contract job price for CLIN 0002 "Construction of Supporting Facilities, 1.5 meters outside of buildings" for work completed including, but not limited to, construction of Supporting Facilities including special construction features, canopies, electrical utilities, communication utilities, water/sewer/utilities, site preparation, roads, sidewalks and parking, site improvements, AT/FP, demolition, low impact development, and environmental mitigation as shown on the drawings and specifications 1.5 meter outside of buildings, complete. Payment for work completed shall constitute full compensation of CLIN 0002, complete.

1.3.2.2 Unit of Measure

Unit of measure: job.

1.3.3 CLIN 0003, Excavation

1.3.3.1 Payment

Payment will be made for costs associated with excavation for the channel and for the structure, which includes performing required excavation and other operations incidental thereto, Contractor-furnished disposal area(s) and disposition of excess excavated material and unsuitable and frozen materials.

1.3.3.2 Measurement

The total quantity of excavated material for which payment will be made will be the theoretical quantity between the ground surface as determined by a survey and the grade and slope of the theoretical cross sections indicated. No allowance will be made for overdepth excavation or for the removal of any material outside the required slope lines unless authorized.

1.3.3.3 Unit of Measure

Unit of measure: cubic yard (CY).

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PROGRESS PAYMENT INVOICE

Requests for payment shall be submitted in accordance with Federal Acquisition Regulations (FAR) Subpart 32.9, entitled "PROMPT PAYMENT", and Paragraphs 52.232-5 and 52.232-27, entitled "Payments Under Fixed-Price Construction Contracts", and "Prompt Payment for Construction Contracts", respectively. In addition each request shall be submitted in the number of copies and to the designated billing office as shown in the Contract.

When submitting payment requests, the Contractor shall complete Blocks 1 through 12 of the "PROGRESS PAYMENT INVOICE" Form as directed by the Contracting Officer (provided in RMS CM). The completed form shall then become the cover document to which all other support data shall be

attached.

One additional copy of the entire request for payment, to include the "PROGRESS PAYMENT INVOICE" cover document, shall be forwarded to a separate address as designated by the Contracting Officer.

The Contractor shall submit with each pay request, a list of subcontractors that have worked during that pay period. The listing shall be broken down into weeks, identifying each subcontractor that has worked during a particular week, and indicate the total number of employees that have worked on site for each subcontractor for each week. The prime Contractor shall also indicate the total number of employees for its on site staff for each week.

Formal invoices shall be submitted every month between the first and the tenth of the month. Any invoice received after the tenth will be considered as not meeting the contract requirements and rejected. It may be resubmitted on the first of the following month.

3.2 TRANSPORTATION OF SUPPLIES BY SEA, BILL OF LADING

The Contractor shall, within 30 days after each shipment covered by Defense Federal Acquisition Regulation (DFAR) Subpart 252.247-7023, provide the Contracting Officer and the Maritime Administration (MARAD) one copy of the rated on board vessel operating carrier's ocean Bill of Lading. See DFAR 252.247-7023 for information required in the bill of lading. MARAD contact information: usarmy.scott.sddc.mbx.g3-ffw-team@mail.mil and msc.n101.ffw@navy.mil.

The Contractor shall use U.S.-flag vessels when transporting any supplies by sea under this contract. If U.S.-flag vessels are not available, the Contractor may submit a request for use of foreign-flag vessels in writing to the Contracting Officer at least 45 days prior to the sailing date necessary to meet its delivery schedules. Refer to DFAR 252.247-7023 for required documentation when making this request. The Contracting Officer will process requests submitted as expeditiously as possible, but the Contracting Officer's failure to grant approvals to meet the shipper's sailing date will not of itself constitute a compensable delay. Failure to get approval prior to the use of foreign-flag vessels may result in monetary penalty.

3.3 CONTRACT COST BREAKDOWN

The Contractor must furnish within 30 days after the date of Notice to Proceed, and prior to the submission of its first partial payment estimate, a breakdown of its single job pay item or items which will be reviewed by the Contracting Officer as to propriety of distribution of the total cost to the various accounts. Any unbalanced items as between early and late payment items or other discrepancies will be revised by the Contracting Officer to agree with a reasonable cost of the work included in the various items. This contract cost breakdown will then be utilized as the basis for progress payments to the Contractor.

3.4 CONTRACT ADMINISTRATION

3.4.1 Contracting Officer (K0)

This Contract shall be administered by a Contracting Officer (K0) assigned to the U.S. Army Corps of Engineers, Japan Engineer District, APO AP

96338-5010.

3.4.2 Administrative Contracting Officer (ACO)

In accordance with the USACE Acquisition Instruction (UAI) 5101.603-3-100, Appointment, Administrative Contracting Officer(s) (ACO's) may be delegated a part of the Contracting Officer's authority to modify this Contract and perform Contract administration functions under Federal Acquisition Regulation (FAR) Section 42.302, subject to the authority and limitations set forth in the appointment and delegation letter.

3.4.3 Contracting Officer's Representative (COR)

The Contracting Officer may designate one or more individuals as his/her authorized representatives in administering this Contract. Refer to the clause in Section 00 70 00 CONDITIONS OF THE CONTRACT entitled CONTRACTING OFFICER'S REPRESENTATIVE (DFARS 252.201-7000) for the definition and limited authority of CORs.

3.5 DESIGNATED BILLING OFFICE

The designated billing office for this Contract is:

Resident Engineer, Zukeran Resident Office
U.S. Army Corps of Engineers, Japan Engineer District
Unit 35132
APO AP 96376-5132

-- End of Section --

SECTION 01 32 16.00 20

SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES
08/18

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Baseline Construction Schedule; G

Monthly Updates; G

1.2 PRE-CONSTRUCTION SCHEDULE REQUIREMENT

Prior to the start of work, prepare and submit to the Contracting Officer a Baseline Construction Schedule in the form of a Network Analysis Schedule (NAS) or Bar Chart Schedule in accordance with the terms in Contract Clause FAR 52.236-15 Schedules for Construction Contracts, except as modified in this contract.

The approval of a Baseline Construction Schedule is a condition precedent to:

- a. The Contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoice(s) for construction activities/items of work.
- c. Review of any schedule updates.

Submittal of the Baseline Construction Schedule, and subsequent schedule updates, is understood to be the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents, represents the Contractor's plan on how the work will be accomplished, and accurately reflects the work that has been accomplished and how it was sequenced (as-built logic).

1.3 SCHEDULE FORMAT

Provide schedule in either NAS or Bar Chart format as specified in the task order.

1.3.1 Network Analysis Schedule (NAS)

Use the critical path method (CPM) to schedule and control project activities. Prepare and maintain project schedules using Primavera P6 or Microsoft Project 2010.

Within 15 calendar days after approval of the Initial Schedule, submit to the Contracting Officer a final NAS schedule.

1.3.1.1 Activity Requirements

- a. At a minimum, identify the following in the schedule:
 - (1) Construction time for major systems and components
 - (2) Each activity assigned with its appropriate Responsibility Code
 - (3) Each activity assigned with its appropriate Phase and Area Codes
 - (4) Major submittals and submittal processing time
 - (5) Major equipment lead time
- b. Build the Schedule as follows:
 - (1) Show submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities.
 - (2) With the exception of the Contract Award and End Contract milestone activities, use of open-ended activities is not allowed; each activity must have predecessor and successor ties. No activity must have open start or open finish (dangling) logic. Minimize redundant logic ties. Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the Contracting Officer. While an activity cannot be deleted, where said activity is no longer applicable to the schedule but must remain within the logic stream for historical record, it can be changed to a milestone. Document any such change in the milestone's "Notebook," including a date and explanation for the change. The ID number for a deleted activity must not be re-used for another activity.
 - (3) Assign each activity its appropriate Responsibility Code and Area Code, indicating location and responsibility to accomplish the work indicated by the activity, Phase Code, and Work Location Code. Include anticipated tasks to be assigned Government responsibility.
 - (4) Date/time constraints or lags, other than those required by the contract, are not allowed unless approved by the Contracting Officer. Include as the last activity in the contract schedule, a milestone activity named "Contract Completion Date".
 - (5) Include the following Contract Milestones:
 - (a) Include as the first activity on the schedule a start milestone titled "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date;
 - (b) Include Interim or Phased Completion Milestones required by the Contract or as approved by the Contracting Officer;
 - (c) Include Facility Turnover Planning Meeting Milestones;
 - (d) Include an unconstrained finish milestone on the schedule titled "Substantial Completion". Substantial Completion is

defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor. Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g. punch list and other close out submittals).

(e) Include an unconstrained finish milestone on the schedule titled "Projected Completion". Projected Completion is defined as the point in time the Government would consider the project complete. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

(f) Include as the last activity on the schedule a finish milestone titled "Contract Completion (CCD)" with constraint type "Must Finish No Later Than". Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float will be calculated on the longest path and if the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path. This milestone must be set to 5:00 pm.

(6) Provide lead time for major equipment.

1.3.1.2 Anticipated Weather Lost Work Days

Refer to Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, paragraph TIME EXTENSIONS FOR UNUSUALLY SEVERE WEATHER.

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay occurs and the adverse weather delayed activities are critical to contract completion. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity.

1.3.1.3 Activity Identification

- a. Identify Government, Construction Quality Management (CQM), Construction activities planned for the project and other activities that could impact project completion if delayed.
- b. Identify administrative type activity/milestones including pre-construction submittal and permit requirements prior to demolition or construction stage.
- c. Create separate activities for each Phase, Area, Floor Level, and Location the activity is occurring.
- d. Do not use construction category activity to represent non-work type reference (Such as, Serial Letter or Request for Information) in NAS.
- e. Place non-work reference within P6 activity details notebook. Activity categories included in the schedule are specified below.

1.3.1.4 Responsibility Code

Assign each activity its appropriate Responsibility Code indicating responsibility to accomplish the work indicated by the activity, Phase Code and Work Location Code.

1.3.1.5 Primavera P6 Settings and Parameters

Use the following Primavera P6 settings and parameters in preparing the Baseline Schedule. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of schedule submission.

- a. General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.
- b. Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:
 - (1) Set time periods for P6 for Hours/Day, Hours/Week, Hours/Month and Hours/Year based on Section 01 11 00.00 10 Paragraph "WORKING HOURS".
 - (2) Use assigned calendar to specify the number of work hours for each time period: Must be checked.
- c. Admin Drop-Down Menu, Admin Preferences, Earned Value Tab: Earned Value Calculation: Use "Budgeted values with current dates".
- d. Project Level, Dates Tab: Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.
- e. Project Level, Defaults Tab:
 - (1) Duration Type: Set to "Fixed Duration & Units".
 - (2) Percent Complete Type: Set to "Physical".
 - (3) Activity Type: Set to "Task Dependent".
 - (4) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.
- f. Project Level, Calculations Tab:
 - (1) Activity percent complete based on activity steps: Must be Checked.
 - (2) Reset Remaining Duration and Units to Original: Must be Checked.
 - (3) Subtract Actual from At Completion: Must be Checked.
 - (4) Recalculate Actual units and Cost when duration percent (%) complete changes: Must be Checked.
 - (5) Link Actual to Date and Actual This Period Units and Cost: Must be Checked.

(6) Price/Unit: Set to "\$1/h".

(7) Update units when costs change on resource assignments: Must be Unchecked.

g. Project Level, Settings Tab:

(1) Define Critical Activities: Check "Longest Path".

h. The NAS must have a minimum of 30 construction activities. No on-site construction activity may have durations in excess of 20 working days.

1.3.1.6 Microsoft Project 2010 Settings and Parameters

Use the following MS Project 2010 settings and parameters in preparing the Baseline Schedule:

- a. The Network must have a minimum of 30 construction activities. In general, all definable features of work identified in the Quality Control Plan should be represented by construction activities where practicable or sensible.
- b. No on-site construction activity may have durations in excess of 20 working days.
- c. Critical is defined as having zero days of Total Slack. Within the Baseline Schedule no more than 20 percent of the activities shall be critical.
- d. Logic: include the following setting: File, Options, Schedule tab - Split in-progress tasks - must be selected.
- e. Status Date gridline is displayed in the Gantt Chart view.
- f. Task Type is set to Fixed Work for "boots-on-the-ground" construction activities.
- g. Task Type is set to Fixed Duration for design activities, submittals, Government reviews, procurement, material/equipment delivery, and utility outages.
- h. "Effort Driven" is turned ON for Fixed Duration tasks.
- i. Set time periods for the project for Hrs/Day, Hrs/Week, and days/month based on Section 01 11 00.00 10 Paragraph "WORKING HOURS."

1.3.1.7 Cost Loading Microsoft Project 2010 Schedules

Assign material, labor and equipment costs to their respective Construction Activities. Assign material and equipment costs, for which payment will be requested in advance of installation, to their respective procurement activity (i.e. the material/equipment on-site activity). Evenly disperse overhead and profit to each activity over the duration of the project. Cost loading must total to 100 percent of the value of the contract.

1.3.1.7.1 Software Settings

- a. Resource Sheet

- (1) Resource Name: Enter each code and resource for the project
- (2) Type: Set to "Material"
- (3) Material Label: Enter units of measurement for each resource
- (4) Std. Rate: Enter unit cost for each resource
- (5) Accrue at: Set to "Prorated"

b. Assigning Resources to Each Activity

- (1) Select each activity in Gantt Chart
- (2) Assign resources, Resource Tab
- (3) Select each resource and enter the quantity of the units; then, assign the resource(s) to the activity

c. Baseline for Earned Value Calculation, File Tab, Options, Advanced, Default task Earned Value method: Set to "Physical % Complete" or as directed by the Contracting Officer

1.3.1.7.2 Tabular Reports

1.3.1.7.2.1 Tracking Gantt Schedule with Cost Table

Submit a Tracking Gantt Schedule with each schedule update showing activity baseline cost, cost percent complete, and Budgeted Cost of Work Performed (BCWP), as directed by the Contracting Officer.

1.3.1.7.2.2 Earned Value Over Time Report

- a. With each schedule submission, submit Earned Value Over Time Report S-Curves indicating Planned Value to the contract completion date based on projected early and late activity finish dates and Earned Value.
- b. Revise Earned Value Over Time Report S-Curves when the contract is modified, or as directed by the Contracting Officer.

1.3.1.7.3 Pay Activity Data

Manually enter pay activity data in the RMS 3 database according to the requirements of Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The pay activity schedule shall be identical to the activities in the NAS.

1.3.2 Bar Chart Schedule

The Bar Chart must, as a minimum, show work activities, submittals, Government review periods, material/equipment delivery, utility outages, on-site construction, inspection, testing, and closeout activities. The Bar Chart must be time scaled and generated using an electronic spreadsheet program.

1.3.2.1 Schedule of Prices

Within 15 calendar days after the NTP is acknowledged, prepare and deliver to the Contracting Officer a Schedule of Prices. Provide a detailed breakdown of the task order price that assigns costs to activities shown on the Bar Chart Schedule.

Costs may be considered for Bonding, Design Submittals, and on-site

construction activities only. Cost may not be considered for activities that do not result in tangible work in place including mobilization, temporary construction, Environmental Protection, Quality Control, Safety, submittals, pre-construction planning, or similar.

Overhead and profit costs shall be evenly dispersed over each pay activity and over the entire duration of the project. Cost loadings must total 100 percent of the task order value. The Schedule of Prices shall be manually entered in the RMS 3 database as pay activities according to the requirements of Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM).

1.3.3 Schedule Submittals and Procedures

Submit Schedules and updates in hard copy and on electronic media that is acceptable to the Contracting Officer. Submit an electronic back-up of the project schedule in an import format compatible with the Government's scheduling program.

1.4 PAYMENTS FOR STORED MATERIALS

Per FAR 52.232-5(B)(2), Payments Under Fix-Price Construction: "In preparation of estimates, the Contracting Office may authorize material delivered on the site and preparatory work done to be taken into consideration." Material costs for which payment will be requested in advance of installation must be assigned to their respective on-site construction activity. These Stored Materials payments require evidence of payment, proper storage on site, and inspection for compliance.

1.5 COST LOADING OF CLOSEOUT ACTIVITIES

Cost load the "Correction of punch list from Government pre-final inspection" activity(ies) not less than 5 percent of the present task order value. Activity(ies) may be declared 100 percent complete upon the Government's verification of completion and correction of all punch list work identified during Government pre-final inspection(s).

1.6 SCHEDULE MONTHLY UPDATES

Update the Construction Schedule at monthly intervals. The updated schedule must be kept current, reflecting actual activity progress and plan for completing the remaining work. Submit copies of purchase orders and confirmation of delivery dates as directed by the Contracting Officer.

a. Narrative Report: Identify and justify the following:

- (1) Progress made in each area of the project;
- (2) Critical Path
- (3) Date/time constraint(s), other than those required by the contract;
- (4) Listing of changes made between the previous schedule and current updated schedule including: added or removed activities, original and remaining durations for activities that have not started, logic (sequence, constraint, lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading.

- (5) Status of any long lead materials/equipment
- (6) Status of critical testing requirements, (i.e. TAB (testing, adjusting and balancing), sewer line testing, commissioning).
- (7) Status of Contract Completion Date and interim milestones;
- (8) Current and anticipated delays (describe cause of delay and corrective actions(s) and mitigation measures to minimize);
- (9) Description of current and future schedule problem areas.

For each entry in the narrative report, cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

1.7 CONTRACT MODIFICATION (applicable when use of NAS is specified)

Submit a Time Impact Analysis (TIA) with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, and extends the Projected Finish beyond the Contract Completion Date.

- a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact, successor and predecessor activity to impact period, responsible party, describe how it originated, and how it impacts the schedule. The schedule submission must consist of three native files:
 - (1) Schedule Sub-network - Model the scope of the change condition by showing only the schedule activities to be added as a result of the change, their immediate predecessor and successor activities in the existing schedule, and how the change activities are to be logically tied together.
 - (2) Update Schedule - Provide the most recently accepted update schedule in effect at the time of the change. This update schedule shall not contain any revisions or additions to the existing logic or activities.
 - (3) Revised Update Schedule - Provide a schedule that models the effects of the disruption on the update schedule in effect at the time of the change. The revised update shall have the Schedule Sub-network inserted into the Update Schedule to show how the project and scheduled completion date was impacted as a result of the disruption.
- b. TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.
 - (1) Identify and classify types of delays as follows:

- (a) Force majeure delay (e.g. weather delay): Any delay event caused by something or someone other than the Government (including its agents) or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the critical path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.
 - (b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the critical path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, provide a Corrective Action Plan, identifying plan to mitigate delay, to the Contracting Officer.
 - (c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.
- (2) Use functional theory to analyze concurrent delays, where: Separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined in the following combinations:
- (a) Government-delay concurrent with Contractor-delay: Excusable time extension, classified non-compensable event.
 - (b) Government-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.
 - (c) Contractor-delay concurrent with force majeure delay: Excusable time extension, classified non-compensable event.
- (3) A pacing delay, reacting to another delay (parent delay) equally or more critical than paced activity, must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:
- (a) Government-delay concurrent with Contractor-pacing: Excusable time extension, classified compensable event.
 - (b) Contractor-delay concurrent with Government-pacing: Inexcusable time extension, classified non-compensable event.
- c. Submit data containing the narrative and native schedule files.
- d. Unless the Contracting Office request otherwise, only confirmed task

order modifications must be added into the Construction Schedule.

1.8 3-WEEK LOOK AHEAD SCHEDULE

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Construction Schedule. Key the work plans to activity numbers when a NAS is required and update each week to show the planned work for the current and following two-week period. Additionally, include upcoming outages, closures, preparatory meetings, and initial meetings. Identify critical path activities on the Three-Week Look Ahead Schedule. The detail work plans are to be bar chart type schedules, maintained separately from the Construction Schedule. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.

1.9 CORRESPONDENCE AND TEST REPORTS:

All correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) must reference Schedule Activities that are being addressed. All test reports (e.g., concrete, soil compaction, weld, pressure) must reference Schedule Activities that are being addressed.

1.10 ADDITIONAL SCHEDULING REQUIREMENTS

Any references to additional scheduling requirements, including systems to be inspected, tested and commissioned, that are located throughout the remainder of the Contract Documents, are subject to all requirements of this section.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 SUMMARY

1.1.1 Submittal Information

The Contracting Officer may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections. Each submittal is to be complete and in sufficient detail to allow ready determination of compliance with contract requirements.

Units of weights and measures used on all submittals are to be the same as those used in the contract drawings.

1.1.2 Project Type

The Contractor's Quality Control (CQC) System Manager are to check and approve all items before submittal and stamp, sign, and date indicating action taken. Proposed deviations from the contract requirements are to be clearly identified. Include within submittals items such as: Contractor's, manufacturer's, or fabricator's drawings; descriptive literature including (but not limited to) catalog cuts, diagrams, operating charts or curves; test reports; test cylinders; samples; O&M manuals (including parts list); certifications; warranties; and other such required submittals.

1.1.3 Submission of Submittals

Schedule and provide submittals requiring Government approval before acquiring the material or equipment covered thereby. Pick up and dispose of samples not incorporated into the work in accordance with manufacturer's Safety Data Sheets (SDS) and in compliance with existing laws and regulations.

1.2 DEFINITIONS

1.2.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan
SD-02

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project

and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS) concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by

manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.2.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.2.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.4 SUBMITTAL CLASSIFICATION

1.4.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.4.2 Design-Build Submittal Classifications

1.4.2.1 Designer of Record Approved (DA)

Designer of Record (DOR) approval is required for extensions of design; critical materials; any variations from the Solicitation, the Accepted Proposal, or the completed design; equipment whose compatibility with the entire system must be checked; and other items as designated by the Contracting Officer. Provide the Government with the number of copies designated hereinafter of all DOR approved submittals. The Government may review any or all Designer of Record approved submittals for conformance with the Solicitation, the Accepted Proposal, and the completed design. The Government will review all submittals designated as varying from the Solicitation or Accepted Proposal, as described below.

1.4.2.2 Government Conformance Review of Design (CR)

The Government will review all intermediate and final design submittals for conformance with the technical requirements of the Solicitation. Review will be only for conformance with the applicable codes, standards, and contract requirements.

1.4.2.3 Designer of Record Approved/Government Conformance Review (DA/CR)

1.4.2.3.1 Variations from the Accepted Design

DOR approval and the Government's concurrence are required for any proposed variation from the accepted design that still complies with the contract before the Contractor is authorized to proceed with material acquisition or installation. If necessary to facilitate the project schedule, before official submission to the Government, the Contractor and the DOR may discuss with the Contracting Officer's Representative a submittal proposing a variation. However, the Government reserves the right to review the submittal before providing an opinion. In any case, the Government will not formally agree to or provide a preliminary opinion on any variation without the DOR's approval or recommended approval. The Government reserves the right to reject any design, variation that may affect furniture, furnishings, equipment selections, or operational decisions that were made, based on the reviewed and concurred design.

1.4.2.3.2 Substitutions

Unless prohibited or otherwise provided for elsewhere in the contract,

where the Accepted Proposal named products, systems, materials or equipment by manufacturer, brand name, model number, or other specific identification, and the Contractor desires to substitute a manufacturer or model after award, submit a requested substitution for Government concurrence. Include substantiation, through identifying information and the DOR's approval, that the substitute meets the contract requirements and that it is equal in function, performance, quality, and salient features to that in the accepted contract proposal. If the contract otherwise prohibits substitutions of equal named products, systems, materials or equipment by manufacturer, brand name, model number or other specific identification, the request is considered a "variation" to the contract. Variations are discussed below in paragraphs: "DESIGNER OF RECORD APPROVED/GOVERNMENT APPROVED" and VARIATIONS.

1.4.2.4 Designer of Record Approved/Government Approved (DA/GA)

In addition to the above-stated requirements for proposed variations to the accepted design, both DOR and Government Approval and, where applicable, a contract modification are required before the Contractor is authorized to proceed with material acquisition or installation for any proposed variation to the contract (the Solicitation or the Accepted Proposal), that constitutes a change to the contract terms. The Government reserves the right to accept or reject any such proposed variation.

1.4.3 For Information Only

Submittals not requiring Government approval will be for information only. For Design-build construction all submittals not requiring DOR or Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.5 PREPARATION

1.5.1 Transmittal Form

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms are included in the RMS CM software that the Contractor is required to use for this contract. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Submittal Format

1.5.2.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.2.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 210 by 297 mm nor more than 1189 by

841 mm, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Present 210 by 297 mm shop drawings sized as part of the bound volume for submittals. Present larger drawings in sets. Submit an electronic copy of drawings in PDF format.

1.5.2.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 50 millimeter on the right-hand side of each sheet for the Government disposition stamp.

1.5.2.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.2.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in metric dimensions. Where product data are included in preprinted catalogs with English units only, submit metric dimensions on separate sheet.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.2.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.2.4 Format of SD-04 Samples

1.5.2.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 50 by 75 mm: Built up to 210 by 297 mm.
- c. Sample of Materials Exceeding 210 by 297 mm: Cut down to 210 by 297 mm and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 250 mm length or length to be supplied, if less than 250 mm. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: 750 ml. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 50 by 100 mm. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 1200 by 1200 mm.
- h. Sample Installation: 10 square meters.

1.5.2.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.2.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.2.5 Format of SD-05 Design Data

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.2.6 Format of SD-06 Test Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.7 Format of SD-07 Certificates

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.2.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.2.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified

organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.2.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.2.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.2.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

Provide all dimensions in administrative submittals in metric. Where data are included in preprinted material with English units only, submit metric dimensions on separate sheet.

1.5.3 Source Drawings for Shop Drawings

1.5.3.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.3.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents.

Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.4 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. In addition to the electronic submittal, refer to paragraph QUANTITY OF SUBMITTALS of this section. Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the AMRDEC SAFE Web Application located at the following website: <https://safe.amrdec.army.mil/safe/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.6.2 Number of SD-02 Shop Drawing Copies

Submit two copies of submittals of shop drawings requiring review and approval by a QC organization. Submit two copies of shop drawings requiring review and approval by the Contracting Officer.

1.6.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.4 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

1.6.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.6.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC reports.

1.6.10 Number of SD-10 Operation and Maintenance Data Copies

Submit five copies of O&M data to the Contracting Officer for review and approval.

1.6.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit two sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will

not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER

A Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as Submittal Register at the end of this section.

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with Section 01 45 00.15 10 RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE(RMS CM). The Government will provide the initial submittal register in electronic format with the following fields completed, to the extent that will be required by the Government during subsequent usage.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

1.8.5.1 Contractor Action Codes

DESIGN BID BUILD SUBMITTALS			
Submittal Classifications shown in UFGS Sections	Submittal Classification	Corresponding SpecsIntact Submittal Register Code which is populated in the SI Submittal Register. Software Limitations: (The software shows one character delineation in the SpecsIntact Submittal Register)	RMS - The following Submittal Classifications are populated in RMS when the SpecsIntact Submittal Data File is pulled into RMS)
G	Submittal requires Government Approval	G	GA
BLANK	Submittal is For Information Only (FIO)	BLANK	FIO
S	Submittal is for documentation of Sustainable requirements	S	S/FIO

1.8.6 Delivery of Copies

Submit an updated electronic copy of the submittal register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission will help ensure that functional and quality requirements are met and minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

1.9.2 Proposing Variations

When proposing variation, deliver a submittal, clearly marked as a "VARIATION" to the Contracting Officer, with documentation illustrating the nature and features of the variation including any necessary technical submittals and why the variation is desirable and beneficial to Government. If lower cost is a benefit, also include an estimate of the cost savings. In addition to documentation required for variation, include the submittals required for the item. Clearly mark the proposed variation in all documentation.

The Contracting Officer will indicate an approval or disapproval of the variation request; and if not approved as submitted, will indicate the Government's reasons therefore. Any work done before such approval is received is performed at the Contractor's risk."

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

Check the column "variation" of ENG Form 4025 for submittals that include variations proposed by the Contractor. Set forth in writing the reason for any variations and note such variations on the submittal. The Government reserves the right to rescind inadvertent approval of submittals containing unnoted variations.

1.9.3 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.4 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 calendar days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.
- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not

relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 30 calendar days after the date of submission.

1.10.1 Government Reviewed Design

The Government will review design submittals for conformance with the technical requirements of the Solicitation. Government review is required for variations from the completed design. Review will be only for conformance with the contract requirements. Included are only those construction submittals for which the DOR's design documents do not include enough detail to ascertain contract compliance. The Government may, but is not required to, review extensions of design such as structural steel or reinforcement shop drawings.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Two copies of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal

is approved.

- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory. the design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____NAME OF CONTRACTOR____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

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U.S. Army Corps of Engineers (USACE) TRANSMITTAL OF SHOP DRAWINGS, EQUIPMENT DATA, MATERIAL SAMPLES, OR MANUFACTURER'S CERTIFICATES OF COMPLIANCE For use of this form, see ER 415-1-10; the proponent agency is CECW-CE.					DATE		TRANSMITTAL NO.	
SECTION I - REQUEST FOR APPROVAL OF THE FOLLOWING ITEMS <i>(This section will be initiated by the contractor)</i>								
TO:		FROM:		CONTRACT NO.			CHECK ONE: ☐ THIS IS A NEW TRANSMITTAL ☐ THIS IS A RESUBMITTAL OF TRANSMITTAL _____	
SPECIFICATION SEC. NO. <i>(Cover only one section with each transmittal)</i>			PROJECT TITLE AND LOCATION			THIS TRANSMITTAL IS FOR: <i>(Check one)</i> ☐ FIO ☐ GA ☐ DA ☐ CR ☐ DA/CR ☐ DA/GA		
ITEM NO. <i>(See Note 3)</i> a.	DESCRIPTION OF SUBMITTAL ITEM <i>(Type size, model number/etc.)</i> b.	SUBMITTAL TYPE CODE <i>(See Note 8)</i> c.	NO. OF COPIES d.	CONTRACT DOCUMENT REFERENCE		CONTRACTOR REVIEW CODE g.	VARIATION <i>Enter "Y" if requesting a variation (See Note 6)</i> h.	USACE ACTION CODE <i>(Note 9)</i> i.
				SPEC. PARA. NO. e.	DRAWING SHEET NO. f.			
REMARKS				I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.				
				NAME OF CONTRACTOR			SIGNATURE OF CONTRACTOR	
SECTION II - APPROVAL ACTION								
ENCLOSURES RETURNED <i>(List by item No.)</i>		NAME AND TITLE OF APPROVING AUTHORITY			SIGNATURE OF APPROVING AUTHORITY		DATE	

INSTRUCTIONS

1. Section I will be initiated by the Contractor in the required number of copies.
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.
8. When submittal items are transmitted, indicate the "Submittal Type" (*SD-01 through SD-11*) in column c of Section I.
Submittal types are the following:

SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data	SD-04 - Samples	SD-05 - Design Data	SD-06 - Test Reports
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports	SD-10 - O&M Data	SD-11 - Closeout	
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:

A -- Approved as submitted.	F -- Receipt acknowledged.
B -- Approved, except as noted on drawings. Resubmission not required.	X -- Receipt acknowledged, does not comply with contract requirements, as noted.
C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required.	G -- Other action required (<i>Specify</i>)
D -- Will be returned by separate correspondence.	K -- Government concurs with intermediate design. (<i>For D-B contracts</i>)
E -- Disapproved. Refer to attached comments.	R -- Design submittal is acceptable for release for construction. (<i>For D-B contracts</i>)
10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

REPAIR HVAC WIDEBAND SATCOM OPS B100

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	CLASSIFICATION GOVT OR A/E REVIEWER	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 11 00.00 10	SD-01 Preconstruction Submittals														
			Initial Site Conditions Photos	1.10													
			Key Personnel Qualifications	1.6	G												
			Conformed Drawings Submittal	1.9.2													
			Indoor Air Quality (IAQ)	1.9.3	G												
			Management Plan														
			SD-07 Certificates														
			Monthly Progress Photos	1.10													
			SD-11 Closeout Submittals														
			Construction Completion Photos	1.10													
		01 20 00	SD-07 Certificates														
			Bill Of Lading	3.2	G												
		01 32 16.00 20	SD-01 Preconstruction Submittals														
			Baseline Construction Schedule	1.2	G												
			Monthly Updates	1.6	G												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G												
			Indoor Air Quality (IAQ)	1.15	G												
			Management Plan														
			SD-06 Test Reports														
			Accident Reports	1.12.2	G												
			LHE Inspection Reports	1.12.3													
			Monthly Exposure Reports	1.5	G												
			SD-07 Certificates														

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Crane Operators/Riggers	1.7.4													
			Activity Hazard Analysis (AHA)	1.9	G												
			Certificate of Compliance	1.12.4													
			Hot Work Permit	1.13													
			Standard Lift Plan	1.12.4	G												
			Certifications	3.10.2													
			Licenses	3.10.2													
		01 45 00.00 10	SD-01 Preconstruction Submittals														
			Contractor Quality Control (CQC)	3.2	G												
			Plan														
			SD-06 Test Reports														
			Verification Statement	3.9.2													
		01 50 00	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												
			Traffic Control Plan	3.4.1	G												
			Temporary Utility Connections	3.3.2	G												
			Plan														
		01 57 19	SD-01 Preconstruction Submittals														
			Environmental Protection Plan	1.9													
			Environmental Protection Plan	3.5.2.2													
			Environmental Manager	1.8.3													
			Qualifications														
			Written Assessment Of Friable	3.8													
			Asbestos Disturbance														
			SD-06 Test Reports														
			Solid Waste Management Report	3.5.2.1													

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19	Nonhazardous Solid Waste Diversion Report	3.4.2													
			SD-07 Certificates														
			Asbestos Certification	1.7.1	G												
			Lead Certification	1.7.2	G												
			ECATTS Certificate Of Completion	1.6.1.2	G												
			PCB Certifications	1.7.4	G												
			SD-11 Closeout Submittals														
			Disposal Documentation for Hazardous and Regulated Waste	3.5.3.4													
			Assembled Employee Training Records	1.8.4													
			Solid Waste Management Report	3.5.2.1													
			Hazardous Waste/Debris Management	3.5.3.1													
			Environmental Records Binder	1.11													
			Environmental Records Binder	3.5.1													
			Environmental Records Binder	3.5.1.1.2													
			Environmental Records Binder	3.5.3.4													
		01 74 19	SD-01 Preconstruction Submittals														
			Waste Management Plan	1.6	G												
			SD-11 Closeout Submittals Records	1.7	G												
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.7.1	G												

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS OR FIC ATION REV WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 00	Warranty Tags	1.7.5													
			Final Cleaning	3.8													
			Spare Parts Data	1.5													
			Spare Parts And Spare Parts Data	1.5													
			SD-08 Manufacturer's Instructions														
			Instructions	1.7.1													
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.7	G												
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												
			Record Drawings	3.3	G												
			As-Built Record of Equipment and Materials	1.7.1	G												
			As-Built Record of Equipment and Materials	3.6	G												
			Final Approved Shop Drawings	3.4	G												
			Construction Contract Specifications	3.5	G												
			Interim DD FORM 1354	3.9	G												
			Checklist for DD FORM 1354	3.9	G												
		01 78 23	SD-10 Operation and Maintenance Data														
			O&M Database	1.4	G												

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS S I F I C A T I O N R E V I E W E R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 23	Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												
			Training Content	3.1.2	G												
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
		01 91 00.15 10	SD-01 Preconstruction Submittals														
			Commissioning Firm	1.7	G												
			Lead Commissioning Specialist	1.7.1													
			Technical Commissioning Specialists	1.7.2													
			Commissioning Firm's Contract	1.7													
			SD-06 Test Reports														
			Design Review Report	3.1.3	G												
			Interim Construction Phase	3.1.2.1	G												
			Commissioning Plan														
			Final Construction Phase	3.1.2.2	G												
			Commissioning Plan														
			Pre-Functional Checklists	3.1.5.2	G												
			Issues Log	1.8													
			Commissioning Report	3.2	G												
			Post-Construction Trend Log Report	3.3.1	G												
			SD-07 Certificates														
			Certificate of Readiness	1.9	G												

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ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 91 00.15 10	SD-10 Operation and Maintenance Data														
			Training Plan	3.1.6	G												
			Training Attendance Rosters	3.1.6													
			Systems Manual	3.1.7	G												
			Maintenance and Service Life Plans	3.1.8													
			SD-11 Closeout Submittals														
			Construction Phase	3.1.2.1	G												
			Commissioning Plan														
			Final Commissioning Report	3.2	G												
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G												
			Existing Conditions	1.10													
			SD-07 Certificates														
			Notification	1.7	G												
			SD-11 Closeout Submittals														
			Receipts	3.2.4													
		02 82 00	SD-01 Preconstruction Submittals														
			Asbestos Hazard Abatement Plan	1.3.10	G												
			SD-03 Product Data														
			Amended Water	1.2.1													
			Safety Data Sheets (SDS) for All Materials	1.3.9													
			Encapsulants	2.1													
			Respirators	3.1.2.1													

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						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		02 82 00	Local Exhaust Equipment	3.1.7													
			Pressure Differential Automatic	3.1.7													
			Recording Instrument														
			Vacuums	3.1.8													
			SD-06 Test Reports														
			Air Sampling Results	1.5.6	G												
			Pressure Differential Recordings	1.5.7													
			for Local Exhaust System														
			Clearance Sampling	3.2.13.5	G												
			Asbestos Disposal Quantity	3.3.3.2													
			Report														
			SD-07 Certificates														
			Employee Training	1.3.4	G												
			Notifications	1.3.5													
			Respiratory Protection Program	1.3.7													
			Testing Laboratory	1.3.11	G												
			Landfill Approval	1.3.12	G												
			Delivery Tickets	1.3.12													
			Waste Shipment Records	1.3.12													
			Transporter Certification	1.3.13	G												
			Medical Certification	1.3.14													
			Private Qualified Person	1.5.2	G												
			Documentation														
			Competent Person	1.5.3	G												
			Worker's License	1.5.4	G												
			Contractor's License	1.5.5	G												

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		02 82 00	Encapsulants	2.1													
			Equipment Used to Contain Airborne Asbestos Fibers	3.1													
			Water Filtration Equipment	3.1.3.3													
			Vacuums	3.1.8													
			Ventilation Systems	3.1.8													
			SD-11 Closeout Submittals														
			Permits	1.3.5													
			Notifications	1.3.5													
			Respirator Program Records	1.3.7.1													
			Rental Equipment	1.7.1													
		02 83 00	SD-01 Preconstruction Submittals														
			Occupational and Environmental Assessment Data Report	1.5.2.3													
			Medical Examinations	1.5.2.4													
			Licenses, Permits and Notifications	1.5.4	G												
			Written Evidence of TSD Approval	3.5.2.1	G												
			Lead, Cadmium, Chromium Compliance Plan	1.5.2.2	G												
			Lead, Cadmium, Chromium Compliance Plan	3.1.1.5	G												
			SD-03 Product Data														
			Respirators	1.6.1													
			Vacuum Filters	1.6.4													

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		02 83 00	Negative Air Pressure System	1.6.7													
			Materials and Equipment	2.1													
			Expendable Supplies	2.1.1													
			Local Exhaust Equipment	3.1.1.4													
			Pressure Differential Automatic	3.1.1.4													
			Recording Instrument														
			Pressure Differential Log	3.1.1.5													
			SD-06 Test Reports														
			Occupational and Environmental	1.5.2.3	G												
			Assessment Data Report														
			Sampling Results	1.5.2.3	G												
			SD-11 Closeout Submittals														
			Hazardous Waste Manifest	3.5.2.1													
			Turn-In Documents or Weight	3.5.2.1													
			Tickets														
		02 84 16	SD-01 Preconstruction Submittals														
			Notification	1.4.7													
			PCB and Lamp Management and	1.4.1.1	G												
			Disposal Plan														
			SD-11 Closeout Submittals														
			Disposal Request Form	3.5.5.2													
			Transporter Certification	3.5.4	G												
			Certification Of Decontamination	3.2.4													
		03 01 00	SD-01 Preconstruction Submittals														
			Qualifications	1.5.3	G												
			Work Plan	1.5.5	G												

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		03 01 00	Work Plan	1.6.1	G												
			Quality Control Plan	1.5.2	G												
			SD-03 Product Data														
			Conventional Concrete	2.5													
			Polymers	2.6													
			Miscellaneous Materials And Equipment	2.7													
			SD-04 Samples														
			Reinforcement And	2.4													
			Reinforcement Supports														
			Polymers	2.6													
			Miscellaneous Materials And Equipment	2.7													
			SD-05 Design Data														
			Formwork And Shoring	3.1.3	G												
			Repair Procedures	1.5.1	G												
			Mixture Proportioning	2.8	G												
			SD-06 Test Reports														
			Mixture Proportioning	2.8													
			Quality Control	3.1.5													
			Quality Control	3.2.3													
			Quality Control	3.3.3													
			Tolerance Report	1.6.2													
			Reinforcement And	2.4													
			Reinforcement Supports														
			Conventional Concrete	2.5													

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		03 01 00	Polymers	2.6													
			Miscellaneous Materials And Equipment	2.7													
			SD-07 Certificates														
			Qualifications	1.5.3													
			Reinforcement And Reinforcement Supports	2.4													
			Conventional Concrete	2.5													
			Polymers	2.6													
			SD-08 Manufacturer's Instructions														
			Equipment For Concrete Preparation	2.2													
			Conventional Concrete	2.5													
			Polymers	2.6													
			Miscellaneous Materials And Equipment	2.7													
		03 30 53	SD-02 Shop Drawings														
			Installation Drawings	1.4	G												
			SD-03 Product Data														
			Air-Entraining Admixture	2.2.3.1													
			Water-Reducing or Retarding Admixture	2.2.3.2													
			Curing Materials	2.2.8													
			Batching and Mixing Equipment	3.1.3.3													
			Conveying and Placing Concrete	3.2													
			Formwork	2.2.6													

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		03 30 53	Ready-Mix Concrete	2.3													
			Curing Compound	2.4.1													
			Mechanical Reinforcing Bar	2.2.5													
			Connectors														
			SD-06 Test Reports														
			Aggregates	2.2.2													
			Concrete Mixture Proportions	2.1.3	G												
			Compressive Strength Testing	3.8.3	G												
			Slump	3.8.3	G												
			Air Content	3.8.3													
			Water	2.2.4													
			SD-07 Certificates														
			Cementitious Materials	2.2.1													
			Aggregates	2.2.2													
			Delivery Tickets	2.3													
			SD-08 Manufacturer's Instructions														
			Curing Compound	2.4.1													
		06 10 00	SD-03 Product Data														
			Fire-retardant Treatment	1.7													
		06 41 16.00 10	SD-02 Shop Drawings														
			Shop Drawings	2.6													
			Installation	3.1													
			SD-04 Samples														
			Plastic Laminates	2.2													
		06 61 16	SD-02 Shop Drawings														
			Installation	3.1	G												

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		06 61 16	SD-03 Product Data														
			Solid Polymer	2.1.1	G												
			SD-04 Samples														
			Material	2.1	G												
			Counter Tops	2.3.4	G												
			SD-06 Test Reports														
			Test Report Results	2.1.1													
			SD-07 Certificates														
			Qualifications	1.4.1													
			SD-10 Operation and Maintenance Data														
			Solid Polymer	2.1.1	G												
		07 21 13	SD-02 Shop Drawings														
			Manufacturer's Standard Details	1.2	G												
			SD-03 Product Data														
			Board Insulation	2.1	G												
			Protection Board or Coating	2.2	G												
			Accessories	2.3	G												
			SD-08 Manufacturer's Instructions														
			Insulation Installation and Handling	3.2.1													
			Protection Board or Coating Installation	3.3.2													
			Adhesive	2.3.1													
			SD-11 Closeout Submittals														
			Draft Warranty	1.5.1	G												

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		07 21 13	Final Warranty	1.5.1	G												
		07 21 16	SD-03 Product Data														
			Blanket Insulation	2.1													
			SD-08 Manufacturer's Instructions														
			Insulation	3.2.1													
		07 24 00	SD-02 Shop Drawings														
			Shop Drawings	1.4.6	G												
			SD-03 Product Data														
			Thermal Insulation	2.5													
			Adhesive	2.2													
			Mechanical Fasteners	2.4													
			Accessories	2.11													
			Base Coat	2.6													
			Portland Cement	2.7													
			Reinforcing Fabric	2.8													
			Finish Coat	2.9													
			Joint Sealant	2.12													
			Sealant Primer	2.10													
			Bond Breaker	2.13													
			Backer Rod	2.14													
			Insulation Board	1.4.5													
			Warranty	1.7													
			SD-04 Samples														
			Sample Boards	1.2.3.7	G												
			SD-05 Design Data														
			Moisture Analysis	1.2.4													

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		07 24 00	SD-06 Test Reports														
			Abrasion Resistance	1.2.3.1													
			Accelerated Weathering	1.2.3.2													
			Impact Resistance	1.2.2.3													
			Mildew Resistance	1.2.3.3													
			Salt Spray Resistance	1.2.3.4													
			Vapor Transmission	1.2.4													
			Absorption-Freeze-Thaw	1.2.3.6													
			Wall Fire Test	1.2.1.3													
			Water Penetration	1.2.1.1													
			Water Resistance	1.2.3.5													
			Full Scale or Intermediate Scale	1.2.1.3													
			Fire Test														
			Surface Burning Characteristics	1.2.2.1													
			Radiant Heat	1.2.2.2													
			Substrate	3.1													
			Wind Load Test	1.2.1.2													
			SD-07 Certificates														
			Qualifications of EIFS	1.4.1													
			Manufacturer														
			Qualification of EIFS Installer	1.4.2													
			Qualification of Sealant Applicator	1.4.3													
			Qualifications of Third-Party	1.4.4													
			Inspector														
			Inspection Checklist	3.5.2	G												
			SD-08 Manufacturer's Instructions														

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		07 24 00	Manufacturer's Instructions	3.3													
		07 57 12.42	SD-03 Product Data														
			Protective Water-based	2.2													
			Elastomeric Silicone Coating														
			Primer	2.3													
			Sealants	2.5													
			SD-04 Samples														
			Elastomeric Silicone Wall System	1.4.4													
			Mock-up														
			Sample Warranty	1.4.5													
			SD-07 Certificates														
			Qualification of Manufacturer	1.4.1													
			Qualification of Applicator	1.4.2													
			SD-08 Manufacturer's Instructions														
			Protective Coating	2.2													
			Manufacturer's Application	2.1.3													
			Instructions														
			SD-11 Closeout Submittals														
			Warranty	1.11													
		08 11 13	SD-02 Shop Drawings														
			Doors	2.1	G												
			Doors	2.1	G												
			Frames	2.5	G												
			Frames	2.5	G												
		08 33 23	SD-02 Shop Drawings														
			Overhead Coiling Doors	2.2.1	G												

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		08 33 23	Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												
			Bottom Bars	2.2.1.3	G												
			Guides	2.1.1.1	G												
			Mounting Brackets	2.2.3.1	G												
			Hood	2.2.2.2	G												
			Installation Drawings	2.1.1.1	G												
			SD-03 Product Data														
			Overhead Coiling Doors	2.2.1	G												
			Hardware	2.2.2	G												
			Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												
			SD-05 Design Data														
			Overhead Coiling Doors	2.2.1	G												
			Hardware	2.2.2	G												
			Counterbalancing Mechanism	2.2.3	G												
			Manual Door Operators	2.2.4	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	1.3.2	G												
			Manuals														
			SD-11 Closeout Submittals														
			Warranty	1.3.1	G												
		08 34 73	SD-02 Shop Drawings														
			Fabrication Drawings	2.1													
			SD-03 Product Data														

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		08 34 73	Hollow Metal Sound Retardant Doors	2.1	G												
			Door Frames	2.1	G												
			Door Hardware	2.1	G												
			Door Frame Sound Infill	2.3.2	G												
			Gasketing	2.1	G												
			Thresholds	2.1	G												
			SD-06 Test Reports														
			Acoustical Tests	2.4.4	G												
			Positive Pressure Tests	2.4.4	G												
			SD-07 Certificates														
			Hollow Metal Sound Retardant Doors	2.1	G												
			Door Frames	2.1	G												
			Door Hardware	2.1	G												
			Gasketing and Door Bottoms	1.3.1.2	G												
			Thresholds	2.1	G												
			Assembly Test Reports	3.3.1													
		08 71 00	SD-02 Shop Drawings														
			Manufacturer's Detail Drawings	1.3	G												
			Verification of Existing Conditions	1.3	G												
			Hardware Schedule	1.5	G												
			Keying System	2.3.6	G												
			SD-08 Manufacturer's Instructions														
			Installation	3.1													

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		08 71 00	SD-10 Operation and Maintenance Data														
			Hardware Schedule	1.5	G												
			SD-11 Closeout Submittals														
			Key Bitting	1.6.1													
		08 91 00	SD-02 Shop Drawings														
			Wall Louvers	1.4													
			SD-03 Product Data														
			Metal Wall Louvers	2.2													
			SD-04 Samples														
			Wall Louver Samples	1.5	G												
		09 22 00	SD-02 Shop Drawings														
			Metal Support Systems	2.1	G												
			SD-03 Product Data														
			Metal Support Systems	2.1													
		09 29 00	SD-08 Manufacturer's Instructions														
			Safety Data Sheets	2.1													
			SD-10 Operation and Maintenance Data														
			Manufacturer Maintenance Instructions	2.1													
		09 30 10	SD-02 Shop Drawings														
			Detail Drawings	3.2	G												
			SD-10 Operation and Maintenance Data														
			Installation	3.2	G												

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		09 51 00	SD-02 Shop Drawings														
			Approved Detail Drawings	2.1	G												
		09 65 00	SD-02 Shop Drawings														
			Resilient Flooring and	2.8	G												
			Accessories														
			SD-10 Operation and Maintenance														
			Data														
			Resilient Flooring and	2.8	G												
			Accessories														
		09 69 13	SD-02 Shop Drawings														
			Detailed Installation Drawings	2.2.1	G												
			SD-04 Samples														
			Floor Panels	2.2													
			Floor Covering	2.2.3													
			Panel Support System	2.3													
			Accessories	2.2.4													
			SD-07 Certificates														
			Qualification of Manufacturer	1.3.1													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.4	G												
			Manuals														
			SD-11 Closeout Submittals														
			Lifting Device	2.2.5													
		09 90 00	SD-08 Manufacturer's Instructions														
			Application Instructions	3.3.1													

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		09 90 00	Mixing	3.7.2													
		10 28 13	SD-04 Samples														
			Finishes	2.1.2	G												
			Accessory Items	2.2													
		10 44 16	SD-02 Shop Drawings														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Schedule	1.5	G												
			SD-03 Product Data														
			Fire Extinguishers	2.1.1	G												
			Accessories	Part 2	G												
			Cabinets	Part 2	G												
			Wall Brackets	2.2.2	G												
			Replacement Parts List	3.2.1	G												
			SD-07 Certificates														
			Fire Extinguishers Certifications	2.1.1	G												
			Manufacturer's Warranty with	1.4	G												
			Inspection Tag														
		13 48 73	SD-02 Shop Drawings														
			Resilient Vibration Isolation	3.4	G												
			Devices														
			SD-03 Product Data														
			Material Supports	2.2.1	G												
			Common Bolts	2.2.2	G												

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		13 48 73	High-Strength Bolts	2.2.2	G												
			Anchors Certified Manufacturer's	2.2.3	G												
			Mill Reports														
			Vibration Isolators	2.1.3	G												
			Snubbers	2.1.3	G												
			Welding Electrodes And Rods	2.2.4													
			SD-05 Design Data														
			Design Calculations and	2.1	G												
			Drawings														
			Details of Equipment Supports	3.5.1	G												
			and Attachments														
			SD-06 Test Reports														
			Anchor Test Results	3.3.1	G												
			Shake Table Test	3.6	G												
			SD-07 Certificates														
			Certificate Of Completion	3.7	G												
			Welder Qualifications	3.1	G												
		22 00 00	SD-02 Shop Drawings														
			Plumbing System	3.5.1	G												
			SD-06 Test Reports														
			Tests, Flushing and Disinfection	3.5													
			SD-10 Operation and Maintenance														
			Data														
			Plumbing System	3.5.1													
		23 01 30.41	SD-01 Preconstruction Submittals														
			Record of Existing Conditions	3.1.1.1													

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		23 01 30.41	Coordination Plan	3.1.1.2													
			NADCA Firm	1.4.1													
			NADCA Team Assistants	1.4.1													
			NADCA Air System Cleaning Specialist (ASCS)	1.4.1													
			NADCA Supervisor Qualifications	1.4.1													
			Records of Experience in the Field of HVAC System Cleaning	1.4.2													
			NADCA Work Execution Schedule	1.5.1													
			SD-03 Product Data														
			Safety Data Sheets (SDS)	1.4.3													
			SD-06 Test Reports														
			Testing Procedures Summary	3.3.2													
			Gravimetric Analysis	3.3.1.3													
			Post-Project Report	3.3.2													
		23 05 15	SD-01 Preconstruction Submittals														
			Material, Equipment, and Fixture Lists	1.2	G												
			SD-02 Shop Drawings														
			Record Drawings	1.2	G												
			Coordination Drawings	1.2	G												
			SD-10 Operation and Maintenance Data														
			Operation and Maintenance Manuals	3.9													

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		23 05 48.19	SD-02 Shop Drawings														
			Coupling and Bracing	3.1													
			Flexible Couplings or Joints	3.2													
			Equipment Restraint	2.2													
			Contractor Designed Bracing	1.2.4	G												
			SD-03 Product Data														
			Coupling and Bracing	3.1	G												
			Flexible Couplings Or Joints	3.2	G												
			Equipment Restraint	2.2	G												
			Contractor Designed Bracing	1.2.4	G												
			Anchor Bolts	3.8													
			Vibration Isolators	2.2.2													
			SD-05 Design Data														
			Design Calculations	1.2.4													
			SD-06 Test Reports														
			Anchor Bolts	3.8	G												
		23 05 93	SD-01 Preconstruction Submittals														
			Records of Existing Conditions	1.3.3	G												
			TAB Firm	1.5.3.1	G												
			TAB Team Assistants	1.2	G												
			TAB Team Engineer	1.2	G												
			TAB Specialist	1.5.3.2	G												
			TAB Team Field Leader	1.2	G												
			SD-02 Shop Drawings														
			TAB Schematic Drawings and	1.3.3	G												
			Report Forms														

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		23 05 93	SD-03 Product Data														
			Equipment and Performance Data	1.3	G												
			TAB Related HVAC Submittals	1.5.3.4	G												
			TAB Procedures	1.5.2	G												
			Calibration	1.5.2	G												
			Systems Readiness Check	1.3.3	G												
			TAB Execution	1.5.4	G												
			SD-06 Test Reports														
			TAB Design Review Report	1.7.2.1	G												
			TAB Report for Season 1	1.5.5.1	G												
			SD-07 Certificates														
			Independent TAB Agency and Personnel Qualifications	1.5.1	G												
			TAB Submittal and Work Schedule	1.7.2	G												
			TAB Pre-Field Engineering Report	1.7.2.2	G												
			TAB Firm	1.5.3.1	G												
			Design Review Report	1.3.3	G												
			Advanced Notice for Season 1	1.7.2	G												
			TAB Field Work														
			Prerequisite HVAC Work Check Out List For Season 1	1.7.2	G												
			Advanced Notice for Season 2	1.7.2	G												
			TAB Field Work														

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		23 05 93	Prerequisite HVAC Work Check Out List For Season 2	1.7.2	G												
		23 07 00	SD-02 Shop Drawings														
			Pipe Insulation Systems	2.3													
			Duct Insulation Systems	3.3													
			Equipment Insulation Systems	3.4													
		23 08 00	SD-03 Product Data														
			Test Equipment	2.1	G												
			SD-06 Test Reports														
			Pipe Flushing, Testing, And Water Treatment Reports	1.5	G												
			Completed Pre-Functional Checklists	3.3	G												
			Seasonal Test Report	3.11	G												
			Full-Load Test Report	3.12	G												
			Post-Construction Trend Log Report	3.15	G												
			SD-07 Certificates														
			Certificate Of Readiness	1.6	G												
		23 09 00	SD-02 Shop Drawings														
			DDC Contractor Design Drawings	3.3	G												
			Draft As-Built Drawings	3.3	G												
			Final As-Built Drawings	3.3	G												
			SD-03 Product Data														
			Programming Software	1.8.3	G												
			Controller Application Programs	1.8.4	G												

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		23 09 00	Configuration Software	1.8.1	G												
			Controller Configuration Settings	1.8.2	G												
			Proprietary Multi-Split	1.1.1.3	G												
			Engineering Tool Software														
			Manufacturer's Product Data	2.2	G												
			Niagara Framework Supervisory	1.8.5	G												
			Gateway Backups														
			Niagara Framework Engineering	1.8.6	G												
			Tool														
			SD-05 Design Data														
			Boiler Or Chiller Plant Gateway	1.9													
			Request														
			SD-06 Test Reports														
			Existing Conditions Report	3.1.1													
			Pre-Construction Quality Control	1.10.1	G												
			(QC) Checklist														
			Post-Construction Quality Control	1.10.2	G												
			(QC) Checklist														
			Start-Up Testing Report	3.5.2	G												
			PVT Procedures	3.6.1	G												
			PVT Report	3.6.3	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.7	G												
			(O&M) Instructions														
			Training Documentation	3.9.1	G												

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		23 09 00	SD-11 Closeout Submittals														
			Enclosure Keys	2.5	G												
			Password Summary Report	3.2.6.1	G												
			Closeout Quality Control (QC)	1.10.3	G												
			Checklist														
		23 21 23	SD-02 Shop Drawings														
			System Coordination	2.1.2	G												
			SD-03 Product Data														
			Instructions	2.2.2	G												
			Equipment Data	2.2.5	G												
			Training Period	3.5.2	G												
			SD-06 Test Reports														
			Field Quality Control	3.3													
			SD-07 Certificates														
			Manufacturer's Representative	1.3.1													
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.5.1	G												
			Manuals														
			Training	3.5.2	G												
		23 23 00	SD-02 Shop Drawings														
			Refrigerant Piping System	2.3	G												
			SD-03 Product Data														
			Refrigerant Piping System	2.3													
			Spare Parts	1.5.2													
			Qualifications	1.3.1													

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		23 23 00	Refrigerant Piping Tests	3.5													
			Verification of Dimensions	3.1													
			SD-06 Test Reports														
			Refrigerant Piping Tests	3.5													
			SD-07 Certificates														
			Service Organization	2.1													
			SD-10 Operation and Maintenance														
			Data														
			Maintenance	1.5	G												
			Operation and Maintenance	3.4	G												
			Manuals														
			Demonstrations	3.4	G												
		23 25 00	SD-03 Product Data														
			Water Analysis	2.1.3	G												
			Spare Parts	1.3													
			Field Instructions	3.4.1													
			Tests	3.3.1	G												
			Training Course	3.4.1	G												
		23 30 00	SD-02 Shop Drawings														
			Detail Drawings	1.4.4	G												
			SD-03 Product Data														
			Fire Dampers	2.8.3													
			Air Handling Units	2.10	G												
			Variable Volume, Single Duct	2.11.1.1	G												
			Terminal Units														
			Test Procedures	1.4.5													

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		23 30 00	Diagrams	1.2.1.2	G												
			SD-06 Test Reports														
			Performance Tests	3.12													
			Damper Acceptance Test	3.10	G												
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.14.1													
			Manuals														
			Fire Dampers	2.8.3													
			Manual Balancing Dampers	2.8.4													
			Centrifugal Fans	2.9.1.1													
			In-Line Centrifugal Fans	2.9.1.2													
			Air Handling Units	2.10													
			Variable Volume, Single Duct	2.11.1.1													
			Terminal Units														
		23 64 10	SD-03 Product Data														
			Water Chiller	2.4	G												
			Posted Instructions	3.1.3													
			Verification of Dimensions	1.6.1													
			Factory Tests	2.8													
			System Performance Tests	3.6													
			Demonstrations	3.7													
			Refrigerant	2.5.1													
			Water Chiller - Field Acceptance	3.5.1													
			Test Plan														
			SD-06 Test Reports														

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		23 64 10	Field Acceptance Testing	3.5													
			Water Chiller - Field Acceptance	3.5.2													
			Test Report														
			Factory Tests	2.8													
			System Performance Tests	3.6													
			SD-07 Certificates														
			Refrigeration System	3.1.9	G												
			Ozone Depleting Substances	1.3.1													
			Technician Certification														
			SD-08 Manufacturer's Instructions														
			Water Chiller - Installation	3.1	G												
			Instructions														
			SD-10 Operation and Maintenance														
			Data														
			Operation and Maintenance	3.7	G												
			Manuals														
			SD-11 Closeout Submittals														
			Indoor Air Quality During	3.4	S												
			Construction														
		23 64 26	SD-03 Product Data														
			Pumps	2.6	G												
			SD-06 Test Reports														
			Piping Welds NDE Report	3.1.1.3													
			Pressure Tests Reports	3.4.2	G												
			SD-10 Operation and Maintenance														
			Data														

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		23 64 26	Water Treatment Systems	2.9													
			Calibrated Balancing Valves	2.4.8													
			Automatic Flow Control Valves	2.4.9													
			Pump Discharge Valve	2.4.10													
			Water Temperature Mixing Valve	2.4.11													
			Water Temperature Regulating Valves	2.4.12													
			Water Pressure Reducing Valve	2.4.13													
			Pressure Relief Valve	2.4.14													
			Combination Pressure and Temperature Relief Valves	2.4.15													
			Expansion Joints	2.5.8													
			Pumps	2.6	G												
			Combination Strainer and Pump Suction Diffuser	2.5.2													
			Expansion Tanks	2.7	G												
			Air Separator Tanks	2.8	G												
		23 81 00	SD-02 Shop Drawings														
			Field-Assembled Refrigerant Piping	2.7.2													
			Control System Wiring Diagrams	1.3.2													
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			Packaged Terminal Units	2.2													
			Air Conditioners	2.3													
			Training	3.8	G												

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		23 81 00	Posted Instructions	3.8													
			Spare Parts	3.9.1													
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			Start-Up and Initial Operational Tests	3.7.3													
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		23 81 23	SD-03 Product Data														
			Computer Room Air Conditioner	2.1	G												
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		23 81 23	Factory Test Reports	2.7.4	G												
			Field Test Schedule	3.3.2	G												
			Manufacturer's Field Test Plans	3.3.1	G												
			Field Test Reports	3.3.6	G												
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			Credentials of the Manufacturer's Field Test Representative	3.3.3	G												
			Ozone Depleting Substances	1.5.1													
			Technician Certification														
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			Seismic Certification	2.1.6	G												
			SD-08 Manufacturer's Instructions														
			Installation Manual for Each Type of CRAC	3.1.2													
			SD-10 Operation and Maintenance Data														
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			SD-11 Closeout Submittals														
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		25 05 11.21	SD-01 Preconstruction Submittals														
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			Request														

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		25 05 11.21	Contractor Computer Cybersecurity Compliance Statements	1.5.1.4	G												
			Contractor Temporary Network Cybersecurity Compliance Statements	1.5.6	G												
			SD-02 Shop Drawings														
			User Interface Banner Schedule	3.1.3.1	G												
			Network Communication Report	1.3.2	G												
			Control System Inventory Report	1.3.3	G												
			Cybersecurity Interconnection Schedule	1.3.1	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.5	G												
			SD-11 Closeout Submittals														
			Password Summary Report	3.5.2.2.2	G												
			Device Audit Record Upload Software	3.2.2.1	G												
		25 05 11.23	SD-01 Preconstruction Submittals														
			Device Account Lock Exception Request	3.1.2.2	G												
			Multiple IP Connection Device Request	3.8	G												
			Contractor Computer Cybersecurity Compliance Statements	1.5.1.4	G												

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		25 05 11.23	Contractor Temporary Network Cybersecurity Compliance Statements	1.5.6	G												
			SD-02 Shop Drawings														
			User Interface Banner Schedule	3.1.3.1	G												
			Network Communication Report	1.3.2	G												
			Cybersecurity Riser Diagram	1.3.4	G												
			Control System Inventory Report	1.3.3	G												
			Cybersecurity Interconnection Schedule	1.3.1	G												
			SD-03 Product Data														
			Control System Cybersecurity	1.5	G												
			SD-11 Closeout Submittals														
			Password Summary Report	3.5.2.2.5	G												
			Device Audit Record Upload Software	3.2.2.1	G												
		26 05 48.00 10	SD-02 Shop Drawings														
			Lighting Fixtures in Buildings	3.1													
			Equipment Requirements	1.3													
			SD-03 Product Data														
			Lighting Fixtures in Buildings	3.1	G												
			Equipment Requirements	1.3	G												
			Contractor Designed Bracing	1.2.4	G												
		26 05 73	SD-01 Preconstruction Submittals														
			Field Examination Plan	3.1	G												
			Arc Flash Label Formats	3.2.9	G												

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		26 05 73	SD-06 Test Reports														
			Field Examination	3.1.1													
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			System Analyzer	1.4.1	G												
			SD-11 Closeout Submittals														
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			Load Flow Study	3.2.4	G												
			Fault Current Study	3.2.5	G												
			System Coordination Study	3.2.6	G												
			Arc Flash Hazard Study	3.2.7	G												
		26 20 00	SD-02 Shop Drawings														
			Wireways	2.16	G												
			Marking strips	3.1.7.1	G												
			SD-03 Product Data														
			Receptacles	2.9													
			Switches	2.8	G												
			Motor controllers	2.10	G												
			Manual motor starters	2.11	G												
			Surge protective devices	2.17	G												
			SD-06 Test Reports														
			600-volt wiring test	3.5.2													
			Ground-fault receptacle test	3.5.3													
		26 29 23	SD-02 Shop Drawings														
			Schematic Diagrams	1.5.1	G												
			Interconnecting Diagrams	1.5.2	G												
			Installation Drawings	1.5.3	G												

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		26 29 23	As-Built Drawings	1.5.3	G												
			SD-03 Product Data														
			Adjustable Speed Drives	2.1	G												
			Wires and Cables	2.3													
			Equipment Schedule	1.5.4													
			SD-06 Test Reports														
			ASD Test	3.3.1													
			Performance Verification Tests	3.3.2													
			Endurance Test	3.3.3													
			SD-08 Manufacturer's Instructions														
			Installation instructions	1.5.5													
			SD-09 Manufacturer's Field														
			Reports														
			ASD Test Plan	2.5.1													
			Standard Products	1.5.6													
			SD-10 Operation and Maintenance														
			Data														
			Adjustable Speed Drives	2.1													
		26 51 00	SD-02 Shop Drawings														
			Occupancy/Vacancy Sensor	1.5.2													
			Coverage Layout														
			SD-03 Product Data														
			Luminaires	2.2	G												
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			Drivers, Ballasts and Generators	2.3	G												
			LED Luminaire Warranty	1.6.1													

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		26 51 00	Vacancy Sensors	2.5.3.2	G												
			Dimming Controllers (Dimmers)	2.5.2	G												
			Power Hook Luminaire Hangers	2.8	G												
			Exit Signs	2.6.1	G												
			Emergency Lighting Unit (EBU)	2.6.2	G												
			LED Emergency Drivers	2.6.3	G												
			Occupancy Sensors	2.5.3.1	G												
			Ambient Light Level Sensor	3.1.7	G												
			SD-06 Test Reports														
			Occupancy/Vacancy Sensor	1.5.7													
			Verification Tests														
			Energy Efficiency	1.5.10.3													
		27 10 00	SD-02 Shop Drawings														
			Telecommunications drawings	1.5.1.1	G AO												
			Telecommunications Space	1.5.1.2	G AO												
			Drawings														
			SD-03 Product Data														
			Telecommunications cabling	2.3	G AO												
			Patch panels	2.4.3	G AO												
			Telecommunications	2.5	G AO												
			outlet/connector assemblies														
			Equipment support frame	2.4.1	G AO												
			SD-06 Test Reports														
			Telecommunications cabling	3.5.1													
			testing														

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		27 10 00	SD-10 Operation and Maintenance Data														
			Telecommunications cabling and pathway system	1.9.1													
			SD-11 Closeout Submittals														
			Record Documentation	1.9.2													
		28 31 76	SD-01 Preconstruction Submittals														
			Qualified Fire Protection Engineer (QFPE)	1.3.2	G												
			Fire alarm system designer	1.9.2.1	G												
			Supervisor	1.9.2.2	G												
			Technician	1.9.2.3	G												
			Installer	1.9.2.4	G												
			Test Technician	1.9.2.5	G												
			Fire Alarm System Site-Specific Software Acknowledgement	1.7	G												
			SD-02 Shop Drawings														
			Nameplates	1.9.1.3	G												
			Instructions	2.2.4	G												
			Wiring Diagrams	1.9.1.4	G												
			System Layout	1.9.1.5	G												
			Notification Appliances	1.9.1.6	G												
			Initiating devices	1.9.1.7	G												
			Amplifiers	1.9.1.8	G												
			Battery Power	1.9.1.9	G												
			Voltage Drop Calculations	1.9.1.10	G												

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		28 31 76	SD-03 Product Data															
			Amplifiers	1.9.1.8	G													
			Tone Generators	2.4	G													
			Digitalized voice generators	2.4	G													
			Manual Stations	2.5	G													
			Smoke Detectors	2.6	G													
			Duct Smoke Detectors	2.6.2	G													
			Notification Appliances	1.9.1.6	G													
			Textual Display Sign Control Panel	2.7.3	G													
			Textual Display Signs	2.7.3	G													
			Batteries	2.9.1	G													
			Battery Chargers	2.9.2	G													
			Supplemental Notification Appliance Circuit Panels	2.9.1.1	G													
			Auxiliary Power Supply Panels	2.9.1.1	G													
			Surge Protective Devices	2.10	G													
			Alarm Wiring	2.10	G													
			Back Boxes and Conduit	3.3.4	G													
			Terminal Cabinets	3.3.2	G													
			Environmental Enclosures or Guards	2.13	G													
			SD-06 Test Reports															
			Test Procedures	3.7.1	G													
			SD-07 Certificates															

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GOVERNMENTAL SAFETY REQUIREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34 (2021) Protection of the Public on or Adjacent to Construction Sites

ANSI/ASSP A10.44 (2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations

ANSI/ASSP A10.8 (2019) Scaffolding Safety Requirements

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air

ASTM F855 (2020) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048 (2016) Guide for Protective Grounding of Power Lines

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard for Industrial Head Protection

Japanese Standards Association (JSA)

JIS A 8951 (1995) Tubular Steel Scaffolds

MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)

MHLW (1972) Amendment No. 57 - 2006 Industrial Safety and Health Law

ORDINANCE NO. 34 (1972) The Safety Ordinance for Cranes

ORDINANCE NO. 47 (2007) Ordinance on Industrial Safety and Health

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 51B (2024) Standard for Fire Prevention During Welding, Cutting, and Other Hot Work

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in the Workplace

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.1000 Air Contaminants

29 CFR 1915 Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment

29 CFR 1926 Safety and Health Regulations for Construction

1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, Chapter 2 applicable federal, state, and local requirements.

1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1, Chapter 2-3. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractors for their actions or liability.

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in

accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

Indoor Air Quality (IAQ) Management Plan; G

SD-06 Test Reports

Accident Reports; G

LHE Inspection Reports

Monthly Exposure Reports; G

SD-07 Certificates

Crane Operators/Riggers

Activity Hazard Analysis (AHA); G

Certificate of Compliance

Hot Work Permit

Standard Lift Plan; G

Certifications

Licenses

1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by installation emergency management requirement.

1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher. Monthly exposure reports are to be tracked and generated through RMS.

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, MHLW Laws and follow host nation laws, ordinances, criteria, rules and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Site Safety and Health Officer (SSHO)

1.7.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1, Chapter 2 and this Contract. In conjunction with the local Safety and Occupational Health Office (SOHO), the Contracting Officer will evaluate proposed Japanese equivalent training for applicability to the contract scope of work being performed.

1.7.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1, Chapter 2. All SSHOs will be designated on an ENG Form 6282, as approved by the Contracting Officer. Superintendent, QC Manager, and SSHO are subject to dismissal if their required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.1.3 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.7.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1, Chapter 2 are met for the project. The Prime Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed. For features of work with work with multiple high or extremely high RAC codes, a Level One SSHO or Alternate Level One SSHO presence is required on site at all times the work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs will:

(1) Are designated on an ENG Form 6282.

(2) Meet minimum training and experience requirements identified in EM 385-1-1, Chapter 2.

(3) Execute roles and responsibilities identified in EM 385-1-1, Chapter 2.

1.7.3 Contract Site Safety And Health Officer(s)(SSHOs) Minimum Requirements

Provide a minimum of one Level One SSHO that meets the requirements of EM 385-1-1, Chapter 2 for this project.

1.7.4 Crane Operators/Riggers

Provide Operators meeting the requirements in EM 385-1-1, Chapter 15 for Riggers and Chapter 16 for Crane Operators. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.4.1 Load Handling Equipment Operator Accepted Alternative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance to achieve the required protections for crane operator licensing and practical training requirements employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Load Handling Equipment, Qualifications, dated 8 January 2020.

Provisions within the JISHA (ORDINANCE NO. 34) Articles 223-228 provide sufficient confirmation of an operator's certification/qualification and practical training to achieve the intent of EM 385-1-1, Chapter 16-3.

JISHA (ORDINANCE NO. 34) Articles 223 establishes the Director of the Prefectural Labour Bureau as the licensing authority. Based upon this Article the Ministry of Labour is the recognized auditor, established for local accreditation of certification and training programs.

1.7.5 Preconstruction Conference

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction conference. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. The creation of the APP

and Schedule will be created after being given Notice to Proceed.

- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 Construction Accident Prevention Plan (APP)

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government at least 15 calendar days prior to the start, after being given Notice to Proceed. A competent person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, ENG Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and occupational health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer as attachment 01 35 26-A "ACCIDENT PREVENTION PLAN (APP) WORKSHEET; ENG Form 6293" 15 calendar days prior to the date of the preconstruction conference for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e., imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

Copies of the accepted APP shall be maintained at the resident engineer's office and at the job site. Continuously review and amend the APP, as necessary, throughout the life of the contract. Incorporate unusual or high-hazard activities not identified in the original APP as discovered.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

1.8.3 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including the following:

1.8.3.1 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium and chromium, and prepare in accordance with Section 02 83 00 LEAD REMEDIATION.

1.8.3.2 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00 ASBESTOS REMEDIATION.

1.8.3.3 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION and referenced sources.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Chapter 2 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the

start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.11 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. The Government has no responsibility to provide emergency medical treatment.

1.12 NOTIFICATIONS AND REPORTS

1.12.1 Accident Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

Table Accident Reporting Required Timeline		
Accident Type	Notify KO or COR	Complete Final Accident Report on ENG 3394 and provide to KO or COR

Table Accident Reporting Required Timeline		
Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000.	Immediately, no later than (NLT) 8 Hours	Within 7 Days
All other accidents and near misses	Immediately, no later than (NLT) 24 Hours	Within 7 Days

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any accident or near miss.

1.12.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1 to establish the root cause(s) of the accident. All accidents are reportable, regardless of whether or not it is recordable. Complete the applicable USACE Accident Report, ENG Form 3394, and provide the report to the Contracting Officer within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms. All accidents are reportable, regardless of whether or not it is recordable.
- b. Near Misses: Report all "Near Misses" to the Contracting Officer, using local accident reporting procedures, within 24 hours. The Contracting Officer will provide the Contractor the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

1.12.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.12.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1, Chapter 16-10, paragraph b (16-10.b). Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Chapter 16-10, paragraph a (16-10.a) and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.13 HOT WORK PERMIT

1.13.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Installation Fire Department. Contractors are required to meet all criteria before a permit is issued. Provide at least two (2) 9kg 4A:20 BC rated extinguishers for normal "Hot Work". Extinguishers shall be current inspection tagged, and have approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch shall be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Base Fire Department phone number. Report any fire, no matter how small, to the responsible Base Fire Department immediately.

1.13.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Chapter 6.

1.14 RADIATION SAFETY REQUIREMENTS

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

1.15 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days after notice to proceed and not less than 10 calendar days before the preconstruction conference. Revise and resubmit Plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the Plan to meet the requirements specified below and to ensure safe, healthy air for construction workers and building occupants.

1.15.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

1.15.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008 to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.15.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.

- e. Remove and replace items which remain damp for more than a few hours.

1.15.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

1.16 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Head Protection that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable

means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. The Contractor shall request approval in writing to the Contracting Officer not less than thirty (30) calendar days in advance of any request for exception. Following Contracting Officer approval of any radiation related exception, the Contractor shall coordinate with the Contracting Officer's Representative at least fifteen (15) calendar days in advance to notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.3 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e., 29 CFR 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during demolition, repair, renovation, or construction operations. Stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 45 days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Chapter 12. In accordance with EM 385-1-1, Chapter 12 where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.2.1 Okinawa Utility Outages

Outages (power, water, sewer, and communications) for the following listed dates, times, and locations shall not be permitted during the performance of this Contract, unless approved by the authorized agent of the Installation Commander and Contracting Officer's Representative:

- a. DoDEA Schools: No outages from 01 September through 31 July, Monday through Friday between the hours of 0700 and 1800.
- b. Family Housing Areas: No outages will be allowed on Saturdays when DoDEA schools are not in session. No more than two (2) outages for the same utility service shall be allowed in a one-month period (30 calendar days), and no more than one (1) outage in a two-week period (14 calendar days).
- c. U.S. Government Holidays: No outages during the following periods:
 - (1) New Year's Day - to include seven (7) calendar days before and one (1) calendar day after this date (contiguous with the Christmas Day outage requirement).
 - (2) Martin Luther King's Birthday - to include the following Tuesday.
 - (3) President's Day - to include the following Tuesday.
 - (4) Memorial Day - to include the following Tuesday.
 - (5) Juneteenth - to include one (1) calendar day before and after this date.
 - (6) Independence Day - to include one (1) calendar day before and after this date.
 - (7) Labor Day - to include the following Tuesday.
 - (8) Columbus Day - to include the preceding Friday.
 - (9) Veteran's Day - to include two (2) calendar days before and after this date.
 - (10) Thanksgiving Day - to include one (1) calendar day before and after this date.
 - (11) Christmas Day - to include seven (7) calendar days before and after this date (contiguous with the New Year's Day outage requirement).
- d. Child Development Centers (CDC): Outages allowed only when the CDC is closed or on Sundays.
- e. Marine Corps Bases: No outages during Marine Corps Balls or Change of Commands which are normally scheduled between 20 October and 21 November. Specific dates and locations shall be provided during or subsequent to the Preconstruction Conference.
- f. Kadena Air Base: No outages during Change of Commands. Specific dates and locations shall be provided during or subsequent to the

Preconstruction Conference.

- g. Torii Station: No outages during Change of Commands. Specific dates and locations shall be provided during or subsequent to the Preconstruction Conference.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Chapter 12. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Chapter 12. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, Chapter 12 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Chapter 12. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Chapter 12. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Chapter 12.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Chapter 12, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide

formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with EM 385-1-1, Chapter 21.

3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Chapter 21-4.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal floatation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1, Chapter 21-8.t. Personal fall protection systems and equipment are required when working from an articulating or extendible boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with

EM 385-1-1, Chapter 21-8.t.

3.6 SCAFFOLDING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for supported metal scaffolding employed as part of contracted activities provided for U.S. Army Corps of Engineers (USACE) within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Scaffolding Standards Review, dated 24 September 2019.

This determination only includes supported metal scaffolding structures and does not include other forms of scaffolding methods included in EM 385-1-1, Chapter 22.

JISHA requirements for steel scaffolding structures to conform to the Japanese Industrial Standard JIS A 8951 are comparable to those of ANSI/ASSP A10.8, providing sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 22.

3.7 RIGGING ACCEPTED ALTERNATIVE

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan (GOJ) Ordinance, which warrant a partial acceptance for rigging procedures and rigging equipment, employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Rigging Requirements and Procedures, dated 15 January 2020.

This acceptance excludes Multiple Lift Rigging (MLR) procedures as well as standards for all hooks used for lifting or load handling purposes. MLR and hooks used for load handling purposes will adhere to the respective sections of the EM 385-1-1, Chapter 15 without exception.

Accepted standards include, JISHA ORDINANCE NO. 34, "The Safety Ordinance for Cranes", Chapter VIII Sling Work, Section 1 Slings Equipment Articles 213-220. Also referenced, is the JISHA's "Guideline for sling work safety (2000)".

3.8 EXCAVATIONS

Soil classification shall be performed by a competent person in accordance with EM 385-1-1, Chapter 25.

3.8.1 Excavations and Trenching Accepted Alternative

The Commander, POJ, has determined that there are adequate provisions provided under the Government of Japan Ordinance to achieve the required protections for excavation and trenching systems including shoring, shielding, and engineered designed systems employed as part of contracted activities provided for USACE within the locality of Japan. Reference POJ-SO decision memorandum, subject, Alternate Requirement Analysis - Excavation and Trenching Review, dated 7 November 2019.

This determination only includes excavation and trenching systems including shoring, shielding, and engineered designed systems where the manufactures or a registered professional engineer's tabulated data has

been employed.

Provisions within JISHA ORDINANCE NO. 47, Articles 368-371 provide sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j). JISHA ORDINANCE NO. 47 instructions included in Articles 368-371 conform with the provisions for manufactured shoring/shielding and engineered designed support systems including material serviceability, structural design, and erection/installation. These provisions provided sufficient protection required to achieve the provisions of EM 385-1-1, Chapter 25-2, paragraph j (25-2.j).

3.8.2 Utility Locations

The Contractor shall positively identify underground utilities in the work area and coordinate with the installation utility department. The Contractor shall obtain digging permits prior to start of excavation by contacting the Contracting Officer at least fifteen (15) calendar days in advance.

3.8.3 Utility Location Verification

The Contractor shall scan the construction site with electromagnetic or sonic equipment, and mark the surface of the ground where existing underground utilities are discovered. Verify the elevations of existing piping, utilities, and any type of underground or encased obstruction not indicated to be specified or removed, but indicated or discovered during scanning in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made. Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter of the underground system.

3.8.4 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with installation utility departments in addition to location and depth verification by the Contractor. The Contractor shall locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.8.5 Munitions and Explosives of Concern (MEC)

Munitions and Explosives of Concern (MEC): Unexploded Ordinance (UXO), Material Presenting a Potential Explosive Hazard (MPPEH), Chemical Agents (CA), or Discarded Military Munitions (DMM) on jobsites shall be treated as extremely dangerous and must be reported immediately. Follow the 3Rs: RECOGNIZE, RETREAT, and REPORT.

3.8.5.1 Identification and Notification

If you encounter or suspect you have encountered MEC, the on-site supervisor shall immediately suspend all operations put at risk due to the suspected MEC identified. STOP WORK, and DO NOT TOUCH, mark the location, keep people out of the area, and report to the Government Designated Authority. Project personnel will withdraw along cleared path upwind 300 meters from the discovery or as determined by emergency responders.

The on-site supervisor shall contact the installation emergency services who will intern contact the appropriate Explosives Ordinance Disposal (EOD) unit. EOD will determine the threat and mitigate any immediate explosives hazard(s).

Where suspected MEC are encountered during construction operations and other activities, notify the Contracting Officer's Representative/Contracting Officer immediately. If no hazard is present or poses no danger, the Government will direct the Contractor to proceed without change after given the all clear from the emergency responders. If MEC are found the condition or probability of encountering MEC may also change. The responsible government authority will conduct a new probability assessment, this determination will be used to plan the level of support required and determine further requirements. Pursuant to FAR 52.243-4, "Changes" the Government may issue a modification.

3.9 EQUIPMENT

3.9.1 Use of Explosives

Explosives must not be used or brought to the project site without prior written approval from the Contracting Officer. Such approval does not relieve the Contractor of responsibility for injury to persons or for damage to property due to blasting operations.

Storage of explosives, when permitted on Government property, must be only where directed and in approved storage facilities. These facilities must be kept locked at all times except for inspection, delivery, and withdrawal of explosives.

3.10 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Chapters 11 and 12.

3.10.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear

of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety footwear, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910 and 29 CFR 1926.

3.10.1.1 Energized Electrical Work Permit (EERP)

All equipment and circuits must be placed into an Establishing an Electrically Safe Working Condition (ESWC) must be de-energized before work is started unless energized work can be justified. A QP must complete and verify the following to establish electrically safe working condition in accordance with EM 385-1-1, paragraph 11-8.d unless the work meets the EERP exemption located under EM 385-1-1, paragraph 11-8.e(5) or the host employer or system owner can show that there are additional hazards or an increased risk from de-energizing.

Hazardous Energy: Any energy including but not limited to mechanical (for example, power transmission apparatus, counterbalances, springs, pressure, gravity), pneumatic, hydraulic, electrical, stored electrical energy exists such as Uninterrupted Power Systems (UPS), chemical, nuclear, and thermal (for example, high or low temperature) energies, which could cause injury to employees.

Once it has been determined that equipment must be worked on in an energized condition, an EERP must be submitted to the USACE supervisor/KO or COR and AHJ for acceptance. Do not perform energized work without prior authorization. All permits must be prepared, signed, and authorized in advance of performing any electrical work. A non-mandatory ENG Form 6277 (Energized Electrical Work Permit) is provided in EM 385-1-1, Chapter 11-10.

3.10.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with Host Nation requirements applicable to where work is being performed.

3.10.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors

and circuit parts. Training must be administered by an electrically qualified source and documented.

3.10.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

3.10.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

ACOUSTICAL SOCIETY OF AMERICA (ASA)
1305 Walt Whitman Road, Suite 110
Melville, NY 11747-4300
Ph: 516-576-2360
Fax: 631-923-2875
E-mail: asa@acousticalsociety.org
Internet: <https://acousticalsociety.org/>

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)
30 West University Drive
Arlington Heights, IL 60004 U.S.A.
Ph: +1 847-394-0150
E-mail: communications@amca.org
Internet: <https://www.amca.org/>

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)
2111 Wilson Blvd, Suite 400
Arlington, VA 22201
Ph: 703-524-8800
Internet: <http://www.ahrinet.org>

ALUMINUM ASSOCIATION (AA)
1400 Crystal Drive
Suite 430
Arlington, VA 22202
Ph: 703-358-2960
E-Mail: info@aluminum.org
Internet: <https://www.aluminum.org/>

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)
1900 E Golf Rd, Suite 1250
Schaumburg, IL 60173
Ph: 847-303-5664

E-mail: customerservice@aamanet.org
Internet: <https://aamanet.org/>

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)
1001 N. Fairfax Street Suite 500
Alexandria, VA 22314
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Internet: <https://www.americanbearings.org/>

AMERICAN CONCRETE INSTITUTE (ACI)
ACI World Headquarters
38800 Country Club Drive
Farmington Hills, MI
48331-3439 USA
Ph: 248-848-3800
Internet: <https://www.concrete.org/>

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-5444
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org
Internet: <https://www.aisc.org/>

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)
7470 New Technology Way, Suite F
Frederick, MD 21703
Ph: 301-972-1700
Fax: 301-540-8004
E-mail: alsc@alsc.org
Internet: <http://www.alsc.org>

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
1899 L Street, NW, 11th Floor
Washington, DC 20036
Ph: 202-293-8020
Fax: 202-293-9287
E-mail: storemanager@ansi.org
Internet: <https://www.ansi.org/>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)
1801 Alexander Bell Drive
Reston, VA 20191
Ph: 800-548-2723; 703-295-6300
Internet: <https://www.asce.org/>

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: <https://www.ashrae.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
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Park Ridge, IL 60068
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E-mail: customerservice@assp.org
Internet: <https://www.assp.org/>

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)
18927 Hickory Creek Drive, Suite 220
Mokena, IL 60448
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Fax: 708-479-6139
Internet: <http://www.asse-plumbing.org>

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: <https://www.awwa.org/>

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Internet: <https://www.aws.org/>

AMERICAN WOOD COUNCIL (AWC)
222 Catoctin Circle SE, Suite 201
Leesburg, VA 20175
Ph: 800-890-7732
Fax: 412-741-0609
E-mail: publications@awc.org
Internet: <https://www.awc.org/>

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)
P.O. Box 361784
Birmingham, AL 35236-1784
Ph: 205-733-4077
Fax: 205-733-4075
Internet: <http://www.awpa.com>

APA - THE ENGINEERED WOOD ASSOCIATION (APA)
7011 South 19th St.
Tacoma, WA 98466-5333
Ph: 253-565-6600
Fax: 253-565-7265
Internet: <https://www.apawood.org/>

ARCHITECTURAL INSTITUTE OF JAPAN (AIJ)
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Internet: <https://www.aij.or.jp/aijhome.htm>

ASSOCIATED AIR BALANCE COUNCIL (AABC)
1220 19th St NW, Suite 410
Washington, DC 20036
Ph: 202-737-0202
Fax: 202-315-0285
E-mail: info@aabc.com
Internet: <https://www.aabc.com/>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
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E-mail: service@astm.org
Internet: <https://www.astm.org/BACNET> TESTING LABORATORIES (BTL)
BACnet Testing Laboratories
2900 Delk Road
Suite 700, PMB 321
Marietta, GA 30067
Ph: 770-971-6003
Fax: 678-229-2777

Internet: <https://www.btl.org/>

CARPET AND RUG INSTITUTE (CRI)
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Dalton, GA 30722-2048
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COMPRESSED GAS ASSOCIATION (CGA)
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CSA GROUP (CSA)
178 Rexdale Blvd.
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Internet: <https://www.csagroup.org/>

DOOR AND ACCESS SYSTEM MANUFACTURERS ASSOCIATION (DASMA)
1300 Sumner Avenue
Cleveland, OH 44115-2851
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Fax: 216-241-0105
Email dasma@dasma.com
Internet: <https://www.dasma.com/>

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)
CEN-CENELEC Management Centre
Rue de la Science 23
B - 1040 Brussels, Belgium
Ph: 32-2-550-08-11
Fax: 32-2-550-08-19
Internet: <https://www.cencenelec.eu/>

ETL TESTING LABORATORIES (ETL)
Intertek
Ph: 800-967-5352
Internet: <http://www.intertek.com/>

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)
615 Chestnut Street
One Independence Mall, Sixth Floor
Philadelphia, PA 19106-4404
Ph: 215-931-5597
E-mail: femar3newsdesk@fema.dhs.gov
Internet: <https://www.fema.gov/>

FLUID CONTROLS INSTITUTE (FCI)
1300 Sumner Avenue
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E-mail: fcf@fluidcontrolsinstitute.org
Internet: <https://fluidcontrolsinstitute.org/>

FM GLOBAL (FM)
270 Central Avenue
Johnston, RI 02919-4949
Ph: 401-275-3000
Fax: 401-275-3029
Internet: <https://www.fmglobal.com/>

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH
(FCCCHR)
USC Foundation Office
Research Annex 219
Los Angeles, CA 90089-7700
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Internet: <https://fccchr.usc.edu/>

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Internet: <https://www.green SEAL.org/>

GYPSUM BOARD ASSOCIATION OF JAPAN
5th floor, Yoshino Gypsum Toranomon Building
2-13-10 Nishi-Shinbashi
Minato-ku, Tokyo 105-0003, JAPAN
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HYDRAULIC INSTITUTE (HI)
300 Interpace Parkway
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ICC EVALUATION SERVICE, INC. (ICC-ES)
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Internet: <https://www.ieee.org/>

INTERNATIONAL CODE COUNCIL (ICC)
500 New Jersey Avenue, NW
6th Floor, Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348
E-mail: order@iccsafe.org
Internet: <https://www.iccsafe.org/>

INSTITUTE OF INSPECTION, CLEANING, AND RESTORATION CERTIFICATION
(IICRC)
IICRC Global Resource Center
4043 S. Eastern Ave.
Las Vegas, NV 89119
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E-mail: Marketing@iicrcnet.org
Internet: <https://www.iicrc.org/>

INTERNATIONAL CAST POLYMER ASSOCIATION (ICPA)
12195 Highway 92 Suite 114 #176
Woodstock, GA 30188
Ph: 470-219-8139
Internet: <https://theicpa.com/>

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)
3, rue de Varembe, 1st floor
P.O. Box 131
CH-1211 Geneva 20, Switzerland
Ph: 41-22-919-02-11
Fax: 41-22-919-03-00
E-mail: info@iec.ch
Internet: <https://www.iec.ch/>

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO Central Secretariat
BIBC II
Chemin de Blandonnet 8
CP 401 - 1214 Vernier, Geneva
Switzerland
Ph: 41-22-749-01-11
E-mail: central@iso.ch
Internet: <https://www.iso.org>

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)
1901 North Moore Street
Arlington, VA 22209-1762
Ph: 703-525-1695
Fax: 703-528-2148
Internet: <https://safetyequipment.org/>

THE JAPANESE ELECTRIC WIRE & CABLE MAKERS' ASSOCIATION (JCMA)
Konwa Building
2F, 1-12-22 Tsukiji
Chuo-ku, Tokyo 104-0045, JAPAN
Internet: <https://www.jcma2.jp/index.html>

JAPAN CAST IRON COVER & WASTE FITTING ASSOCIATION (JCW)
1-5-14 Kawaguchi
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JAPANESE ELECTROTECHNICAL COMMITTEE (JEC)
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THE JAPAN ELECTRICAL MANUFACTURERS' ASSOCIATION (JEMA)
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JAPAN PIPE FITTINGS ASSOCIATION (JPFA)
Saito Building 8F, 3-14-6 Kyobashi
Chuo-ku, Tokyo 104-0031
Ph: 03-3564-2035
Internet: <http://www.tsugite.jp/>

JAPAN VALVE MANUFACTURERS' ASSOCIATION (JVMA)
5th Floor, Kikai-Shinko Bldg.
3-5-8, Shiba-Koen, Minato-Ku
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Internet: <https://j-valve.or.jp/>

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Internet: <https://www.jraia.or.jp/english/>

JAPANESE STANDARDS ASSOCIATION (JSA)
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E-mail: po@jsa.or.jp
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MINISTRY OF AGRICULTURE, FORESTRY AND FISHERIES (MAFF)
1-2-1 Kasumigaseki
Chiyoda-ku, Tokyo 100-8950, JAPAN
Ph: 03-3502-8111
Internet: <https://www.maff.go.jp/>

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)
330 N. Wabash Avenue
Chicago, IL 60611
Ph: 312-644-6610
E-mail: MFMAstats@smithbucklin.com
Internet: <http://www.metalframingmfg.org/>

MINISTRY OF HEALTH, LABOUR AND WELFARE, GOVERNMENT OF JAPAN (MHLW)
1-2-2 Kasumigaseki
Chiyoda-ku Tokyo, 100-8916, JAPAN
Ph: 03-5253-1111
E-mail: www-admin@mhlw.go.jp
Internet: <https://www.mhlw.go.jp/index.html>

MINISTRY OF LAND, INFRASTRUCTURE, TRANSPORT AND TOURISM (MLIT)
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Internet: <https://www.mlit.go.jp/en/>

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Vienna, VA 22180-4602
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Internet: <http://msshq.org>

NATIONAL AIR DUCT CLEANERS ASSOCIATION (NADCA)
1120 Route 73, Suite 200
Mt. Laurel, NJ 08054
Toll Free: 855-GO-NADCA
Ph: (856) 380-6810
Fax: (856) 439-0525
Email: info@nadca.com
Internet: <https://nadca.com/>

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
1300 North 17th Street, Suite 900
Arlington, VA 22209
Ph: 703-841-3200
Internet: <https://www.nema.org>

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)
8575 Grovemont Circle
Gaithersburg, MD 20877
Ph: 301-977-3698
Fax: 301-977-9589
Internet: <http://www.nebb.org>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
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Fax: 800-593-6372
Internet: <https://www.nfpa.org>

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)
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Washington, DC 20201
Ph: 800-232-4636
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Email: sfitzgerald-reddd@naima.org
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Fax: 734-769-0109
E-mail: info@nsf.org
Internet: <http://www.nsf.org>

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Glen Ellyn, IL 60137
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Fax: 630-790-3095
Internet: <https://www.ppfahome.org/>

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800 Turnpike Street, Suite 300
North Andover, MA 01845
Ph: 978-557-0720 or 800-589-8956
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SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)
4201 Lafayette Center Drive
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Internet: <https://www.smacna.org/>

SOCIETY FOR PROTECTIVE COATINGS (SSPC)
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Fax: 412-444-3591
E-mail: customerservice@sspc.org
Internet: <http://www.sspc.org>

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)
21865 Copley Drive
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E-mail: webinquiry@aqmd.gov
Internet: <http://www.aqmd.gov>

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THE SOCIETY OF HEATING, AIR-CONDITIONING AND SANITARY ENGINEERS OF
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Shinjuku-ku, Tokyo 162-0825, JAPAN
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Building 714
Yokota Air Base
Fussa-shi, Tokyo 197-0001, JAPAN
Internet: <https://www.usfj.mil/>

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Washington, DC 20004
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Internet: <https://www.epa.gov>
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VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION
(VISCMA)
994 Old Eagle School Road
Suite 1019
Wayne, PA 19087-1866
Ph: 610-971-4850
E-mail: info@viscma.com
Internet: <http://www.viscma.com>

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

DRAFT

SECTION 01 42 15

METRIC MEASUREMENTS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM SI10 (2010) American National Standard for Use of the International System of Units (SI): The Modern Metric System

1.2 GENERAL

This project includes metric units of measurements. The metric units used are the International System of Units (SI) developed and maintained by the General Conference on Weights and Measures (CGPM); the name International System of Units and the international abbreviation SI were adopted by the 11th CGPM in 1960. The following criteria allow both metric and English inch-pound (I-P) measurements to be shown for a product:

- a. To substitute products that are manufactured to an industry recognized rounded metric (hard metric) dimension with I-P products.
- b. To indicate industry and/or Government standards, test values or other controlling factors (such as the code requirements where I-P values are needed for clarity).
- c. To trace back to the referenced standard(s), test value(s) or code(s).

1.3 USE OF MEASUREMENTS IN DRAWINGS AND SPECIFICATIONS

Measurements in drawings and specifications shall be in SI units, with I-P units in parenthesis in cases as indicated in Paragraph GENERAL and as otherwise authorized by the Contracting Officer. Procure the product in SI units unless otherwise authorized by the Contracting Officer. The Contractor is responsible for all associated labor and materials when authorized to substitute one system of units for another and for the final assembly and performance of the specified work and/or products.

1.3.1 Hard Metric

Hard metric measurements are often used for field data such as distance from one point to another or distance above the floor. Products are considered to be hard metric when they are manufactured to metric dimensions or have an industry recognized metric designation.

1.3.2 Soft Metric

- a. A soft metric measurement is a non-mathematical, industry related conversion. Soft metric measurements are used for measurements pertaining to products, test values, and other situations where the

I-P units are the standard for manufacture, verification, or other controlling factor.

- b. A soft metric measurement is also indicated for products that are manufactured in industry designated metric dimensions but are required by law to allow substitute I-P products.

1.3.3 Neutral

A neutral measurement is indicated by an identifier which has no expressed relation to either an SI or an I-P value (e.g., American Wire Gage (AWG) which indicates thickness but in itself is neither SI nor I-P).

1.4 COORDINATION

Bring discrepancies, such as mismatches or product unavailability, arising from use of both metric and non-metric measurements and discrepancies between the measurements in the specifications and the measurements in the drawings to the attention of the Contracting Officer for resolution.

1.5 RELATIONSHIP TO SUBMITTALS

Submittals for Government approval or for information only covers the SI products actually being furnished for the project. Submit the required drawings and calculations in the same units used in the Contract documents describing the product or requirement unless otherwise instructed or approved. Use ASTM SI10 as the basis for establishing metric measurements required to be used in submittals.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

-- End of Section --

SECTION 01 45 00.00 10

QUALITY CONTROL
11/16

PART 1 GENERAL

1.1 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program, all associated costs shall be included in the applicable CLIN Schedule unit or job prices.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

SD-06 Test Reports

Verification Statement

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC consist of plans, procedures, and organization necessary to produce an end product which complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and be keyed to the proposed construction sequence. The project superintendent will be held responsible for the quality of work and is subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. In this context the highest level manager responsible for the overall construction activities at the site, including quality and production is the project superintendent. The project superintendent maintains a physical presence at the site at all times and is responsible for all construction and related activities at the site, except as otherwise acceptable to the Contracting Officer.

3.2 CONTRACTOR QUALITY CONTROL (CQC) PLAN

Submit no later than 15 days after receipt of notice to proceed, the Contractor Quality Control (CQC) Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Include the submittal register with CQC Plan submissions. The Government will consider an interim plan for the first 14 days of operation. Construction will be permitted to begin only after acceptance of the CQC Plan or acceptance of

an interim plan applicable to the particular feature of work to be started. Work outside of the accepted interim plan will not be permitted to begin until acceptance of the CQC Plan.

3.2.1 Content of the CQC Plan

Include, as a minimum, the following to cover all construction-operations, both onsite and offsite, including work by subcontractors fabricators, suppliers and purchasing agents:

- a. A description of the quality control organization, including a chart showing lines of authority and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified. The CQC System Manager shall report to the assigned Project Manager.
- b. The name, qualifications (in resume format), duties, responsibilities, and authorities of each person assigned a CQC function.
- c. A copy of the letter to the CQC System Manager signed by an authorized official of the firm which describes the responsibilities and delegates sufficient authorities to adequately perform the functions of the CQC System Manager, including authority to stop work which is not in compliance with the Contract. Letters of direction to all other various quality control representatives outlining duties, authorities, and responsibilities will be issued by the CQC System Manager. Furnish copies of these letters to the Contracting Officer.
- d. Procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. These procedures must be in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.
- e. Control, verification, and acceptance testing procedures for each specific test to include the test name, specification paragraph requiring test, feature of work to be tested, test frequency, and person responsible for each test. (Laboratory facilities approved by the Contracting Officer are required to be used.)
- f. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.
- g. Procedures for tracking construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected.
- h. Reporting procedures, including proposed reporting formats.
- i. A list of the definable features of work. A definable feature of work is a task which is separate and distinct from other tasks, has separate control requirements, and is identified by different trades or disciplines, or it is work by the same trade in a different environment. Although each section of the specifications can generally be considered as a definable feature of work, there are frequently more than one definable features under a particular section. This list will be agreed upon during the coordination meeting.

3.2.2 Acceptance of Plan

Acceptance of the Contractor's plan is required prior to the start of construction. Acceptance is conditional and will be predicated on satisfactory performance during the construction. The Government reserves the right to require the Contractor to make changes in the Contractor Quality Control(CQC) Plan and operations including removal of personnel, as necessary, to obtain the quality specified.

3.2.3 Notification of Changes

After acceptance of the CQC Plan, notify the Contracting Officer in writing a minimum of seven (7) days prior of any proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

3.3 COORDINATION MEETING

After the Preconstruction Conference, before start of construction, and prior to acceptance by the Government of the CQC Plan, meet with the Contracting Officer and discuss the Contractor's quality control system. Submit the CQC Plan a minimum of 15 calendar days prior to the Coordination Meeting. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations,, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's Management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the Government, signed by both the Contractor and the Contracting Officer and will become a part of the contract file. There can be occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

3.4 QUALITY CONTROL ORGANIZATION

3.4.1 Personnel Requirements

The requirements for the CQC organization are a Site Safety and Health Officer (SSHO), CQC System Manager, and sufficient number of additional qualified personnel to ensure safety and Contract compliance. The SSHO reports directly to a senior project (or corporate) official independent from the CQC System Manager. The SSHO will also serve as a member of the CQC Staff. Personnel identified in the technical provisions as requiring specialized skills to assure the required work is being performed properly will also be included as part of the CQC organization. The Contractor's CQC staff maintains a presence at the site at all times during progress of the work and have complete authority and responsibility to take any action necessary to ensure Contract compliance. The CQC staff will be subject to acceptance by the Contracting Officer. Provide adequate office space, filing systems and other resources as necessary to maintain an effective and fully functional CQC organization. Promptly complete and furnish all letters, material submittals, shop drawing submittals, schedules and all other project documentation to the CQC organization. The CQC organization is responsible to maintain these documents and records at the site at all times, except as otherwise acceptable to the Contracting Officer.

3.4.2 CQC System Manager

Identify as CQC System Manager an individual within the onsite work

organization that is responsible for overall management of CQC and has the authority to act in all CQC matters for the Contractor. This CQC System Manager is on the site at all times during construction and is employed by the prime Contractor. Identify in the plan an alternate to serve in the event of the CQC System Manager's absence. The requirements for the alternate are the same as the CQC System Manager. The CQC System Manager shall report to the assigned Project Manager. Refer to Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control System Manager" for experience requirements.

3.4.3 CQC Personnel

In addition to CQC personnel specified elsewhere in the contract, provide as part of the CQC organization specialized personnel to assist the CQC System Manager for the required areas listed in Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control Personnel". All personnel listed in this table may not be required on this Contract. These individuals or specialized technical companies must be responsible to the CQC System Manager; must be physically present at the construction site during work on the specialized personnel's areas of responsibility; must have the necessary education or experience in accordance with the experience matrix listed herein. The table below defines only the qualification requirements. Refer to the Section 01 11 00.00 10 GENERAL CONTRACT REQUIREMENTS, Paragraph "Contractor Quality Control Personnel" for actual required personnel.

Experience Matrix	
Area	Qualifications
Civil	Graduate of an accredited university with an Civil Engineering or Construction Manager degree with 3 years experience in the type of work being performed on this project or technician with 5 yrs related experience. Or Non-US degree plus 1 Kyu Doboku Sekou Kanrigishi (1st Class Civil Engineering Works Management Engineer).
Mechanical	Graduate of an accredited university with an Mechanical Engineer degree with 3 years experience or person with 5 years of experience supervising mechanical features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer).

Experience Matrix	
Area	Qualifications
Electrical	Graduate of an accredited university with an Electrical Engineering degree with 3 years related experience or person 5 years of experience supervising electrical features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kankouji Sekou Kanrigishi (1st Class Building Mechanical and Electrical Engineer) or 1 Kyu Denikouji Sekou Kanrigishi (1st Class Electric Construction Management Engineer).
Structural	Graduate of an accredited university with an Civil Engineering degree (with Structural Track or Focus) or Construction Management degree with 3 years experience or technician 5 years of experience supervising structural features of work in the field with a construction company. Or Non-US degree plus 1 Kyu Kenchikushi (1st Class Qualified Architect) or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construction Management Engineer).
Architectural	Graduate of an accredited university with an Architecture degree with 3 years experience or person with 5 years related experience. Or Non-US degree plus 1 Kyu Kenchikushi (1st Class Qualified Architect) or 1 Kyu Kenchiku Sekou Kanrigishi (1st Class Building Construction Management Engineer).
Fire Protection	3 to 5 years of documented experience of similar projects. Registered U.S. Professional Engineer who has passed the fire protection engineering written examination administered by the
Communications	Graduate of an accredited university with an Electrical Engineering degree with 3 yrs related experience or technician with 5 years of experience supervising communications features of work in the field with a construction company. Non-US degree plus AI*DD Sougoushu(AI*DD Construction Engineer).
Environmental	Refer to Section [01 57 19][01 57 19.01] ENVIRONMENTAL MANAGER.
Submittals	Submittal Clerk with 1 year experience
Occupied Family Housing	Person, customer relations type, coordinator experience
Concrete, Pavements and Soils	Materials Technician with 2 years experience for the appropriate area

Experience Matrix	
Area	Qualifications
Testing, Adjusting and Balancing (TAB) Personnel	Specialist must be a member of AABC or an experienced technician of the firm certified by the NEBB

3.4.4 Site Safety and Health Officer (SSHO)

See Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS.

3.4.5 Additional Requirement

In addition to the above experience and education requirements, the Contractor Quality Control(CQC) System Manager and Alternate CQC System Manager are required to have completed the Construction Quality Management (CQM) for Contractors course. If the CQC System Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the CQC System Manager's certificate has expired, retake the course to remain current.

3.4.6 Organizational Changes

Maintain the CQC staff at full strength at all times. When it is necessary to make changes to the CQC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

3.5 SUBMITTALS AND DELIVERABLES

Submittals, if needed, have to comply with the requirements in Section 01 33 00 SUBMITTAL PROCEDURES. The CQC organization is responsible for certifying that all submittals and deliverables are in compliance with the contract requirements. When Section 01 91 00.15 10 TOTAL BUILDING COMMISSIONING is included the submittals required by those sections must be coordinated with Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

3.6 CONTROL

CQC is the means by which the Contractor ensures that the construction, to include that of subcontractors and suppliers, complies with the requirements of the contract. At least three phases of control are required to be conducted by the CQC System Manager for each definable feature of the construction work as follows:

3.6.1 Preparatory Phase

This phase is performed prior to beginning work on each definable feature of work, after all required plans/documents/materials are approved/accepted, and after copies are at the work site. This phase includes:

- a. A review of each paragraph of applicable specifications, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review of the Contract drawings.
- c. Check to assure that all materials and equipment have been tested, submitted, and approved.
- d. Review of provisions that have been made to provide required control inspection and testing.
- f. Examination of the work area to assure that all required preliminary work has been completed and is in compliance with the Contract.
- g. Examination of required materials, equipment, and sample work to assure that they are on hand, conform to approved shop drawings or submitted data, and are properly stored.
- h. Review of the appropriate activity hazard analysis to assure safety requirements are met.
- i. Discussion of procedures for controlling quality of the work including repetitive deficiencies. Document construction tolerances and workmanship standards for that feature of work.
- j. Check to ensure that the portion of the plan for the work to be performed has been accepted by the Contracting Officer.
- k. Discussion of the initial control phase.
- l. The Government needs to be notified at least 72 hours in advance of beginning the preparatory control phase. Include a meeting conducted by the CQC System Manager and attended by the superintendent, other CQC personnel (as applicable), and the foreman responsible for the definable feature. Document the results of the preparatory phase actions by separate minutes prepared by the CQC System Manager and attach to the daily CQC report. Instruct applicable workers as to the acceptable level of workmanship required in order to meet contract specifications.

3.6.2 Initial Phase

This phase is accomplished at the beginning of a definable feature of work. Accomplish the following:

- a. Check work to ensure that it is in full compliance with contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full contract compliance. Verify required control inspection and testing are in compliance with the contract.
- c. Establish level of workmanship and verify that it meets minimum acceptable workmanship standards. Compare with required sample panels as appropriate.

- d. Resolve all differences.
- e. Check safety to include compliance with and upgrading of the safety plan and activity hazard analysis. Review the activity analysis with each worker.
- f. The Government needs to be notified at least 24 hours in advance of beginning the initial phase for definable feature of work. Prepare separate minutes of this phase by the CQC System Manager and attach to the daily CQC report. Indicate the exact location of initial phase for definable feature of work for future reference and comparison with follow-up phases.
- g. The initial phase for each definable feature of work is repeated for each new crew to work onsite, or any time acceptable specified quality standards are not being met.

3.6.3 Follow-up Phase

Perform daily checks to assure control activities, including control testing, are providing continued compliance with contract requirements, until completion of the particular feature of work. Record the checks in the CQC documentation. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work. Do not build upon nor conceal non-conforming work.

3.6.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same definable features of work if: the quality of on-going work is unacceptable; if there are changes in the applicable CQC staff, onsite production supervision or work crew; if work on a definable feature is resumed after a substantial period of inactivity; or if other problems develop.

3.7 TESTS

3.7.1 Testing Procedure

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of a Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all of the test documentation requirements, have been prepared.

- e. Record results of all tests taken, both passing and failing on the CQC report for the date taken. Specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, actual test reports are submitted later with a reference to the test number and date taken. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

3.7.2 Onsite Laboratory

The Government reserves the right to utilize the Contractor's control testing laboratory and equipment to make assurance tests, and to check the Contractor's testing procedures, techniques, and test results at no additional cost to the Government.

3.8 COMPLETION INSPECTION

3.8.1 Punch-Out Inspection

Conduct an inspection of the work by the CQC System Manager near the end of the work, or any increment of the work established by a time stated in FAR 52.211-10 Commencement, Prosecution, and Completion of Work. Prepare and include in the CQC documentation a punch list of items which do not conform to the approved drawings and specifications, as required by paragraph DOCUMENTATION. Include within the list of deficiencies the estimated date by which the deficiencies will be corrected. Make a second inspection the CQC System Manager or staff to ascertain that all deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government Pre-Final inspection.

3.8.2 Pre-Final Inspection

The Government will perform the pre-final inspection to verify that the facility is complete and ready to be occupied. A Government Pre-Final Punch List may be developed as a result of this inspection. Ensure that all items on this list have been corrected before notifying the Government, so that a Final inspection with the customer can be scheduled. Correct any items noted on the Pre-Final inspection in a timely manner. These inspections and any deficiency corrections required by this paragraph need to be accomplished within the time slated for completion of the entire work or any particular increment of the work if the project is divided into increments by separate completion dates.

3.8.3 Final Acceptance Inspection

The Contractor's Quality Control Inspection personnel, plus the superintendent or other primary management person, and the Contracting Officer's Representative is required to be in attendance at the final acceptance inspection. Additional Government personnel including, but not limited to, those from Base/Post Civil Facility Engineer user groups, and major commands can also be in attendance. The final acceptance inspection will be formally scheduled by the Contracting Officer based upon results of the Pre-Final inspection. Notify the Contracting Officer at least 14 days prior to the final acceptance inspection and include the Contractor's

assurance that all specific items previously identified to the Contractor as being unacceptable, along with all remaining work performed under the Contract, will be complete and acceptable by the date scheduled for the final acceptance inspection. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance FAR 52.246-12 Inspection of Construction.

3.9 DOCUMENTATION

3.9.1 Quality Control Activities

Maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractor/Subcontractor.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day, giving location, description, and by whom. When Network Analysis (NAS) is used, identify each phase of work performed each day by NAS activity number.
- d. Test and control activities performed with results and references to specifications/drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action.
- e. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications/drawings requirements.
- f. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- g. Offsite surveillance activities, including actions taken.
- h. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- i. Instructions given/received and conflicts in plans and specifications.

3.9.2 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract. Furnish the original and one copy of these records in report form to the Government daily within 24 hours after the date covered by the report, except that reports need not be submitted for days on which no work is performed. As a minimum, prepare and submit one report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the contract. The first report following a day of no work will be for that day only. Reports need to be signed and dated by the Contractor

Quality Control(CQC) System Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the CQC System Manager Report.

3.10 SAMPLE FORMS

Sample forms can be obtained from the Resident Office or found through Contractor link of the RMS website <https://rms.usace.army.mil/>.

3.11 NOTIFICATION OF NONCOMPLIANCE

The Contracting Officer will notify the Contractor of any detected noncompliance with the foregoing requirements. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, will be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer can issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of claim for extension of time or for excess costs or damages by the Contractor.

-- End of Section --

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SECTION 01 45 00.15 10

RESIDENT MANAGEMENT SYSTEM CONTRACTOR MODE (RMS CM)
11/16

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this section to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

Safety and Health Requirements Manual

1.2 MEASUREMENT AND PAYMENT

The work of this section is not measured for payment. The Contractor is responsible for the work of this section, without any direct compensation other than the payment received for contract items.

1.3 CONTRACT ADMINISTRATION

The Government will use the Resident Management System (RMS) to assist in its monitoring and administration of this contract. The Government accesses the system using the Government Mode of RMS (RMS GM) and the Contractor accesses the system using the Contractor Mode (RMS CM). The term RMS will be used in the remainder of this section for both RMS GM and RMS CM. The joint Government-Contractor use of RMS facilitates electronic exchange of information and overall management of the contract. The Contractor accesses RMS to record, maintain, input, track, and electronically share information with the Government throughout the contract period in the following areas:

- Administration
- Finances
- Quality Control
- Submittal Monitoring
- Scheduling
- Closeout
- Import/Export of Data

1.3.1 Correspondence and Electronic Communications

For ease and speed of communications, exchange correspondence and other documents in electronic format to the maximum extent feasible. Some correspondence, including pay requests and payrolls, are also to be provided in paper format with original signatures. Paper documents will govern, in the event of discrepancy with the electronic version.

1.3.2 Other Factors

Other portions of this document have a direct relationship to the reporting accomplished through RMS. Particular attention is directed to FAR 52.236-15 Schedules for Construction Contracts; FAR 52.232-27 Prompt Payment for Construction Contracts; FAR 52.232-5 Payments Under

Fixed-Priced Construction Contracts; Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES; Section 01 33 00 SUBMITTAL PROCEDURES; Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS; and Section 01 45 00.00 10 QUALITY CONTROL.

1.4 RMS SOFTWARE

RMS is a Windows-based program that can be run on a Windows-based PC meeting the requirements as specified in paragraph SYSTEM REQUIREMENTS. Download, install and be able to utilize the latest version of the RMS software within 7 calendar days of receipt of the Notice to Proceed. RMS software, user manuals, access and installation instructions, program updates and training information are available from the RMS website (<https://rms.usace.army.mil>). The Government and the Contractor will have different access authorities to the same contract database through RMS. The common database will be updated automatically each time a user finalizes an entry or change.

1.5 SYSTEM REQUIREMENTS

The following is the recommended system configuration to run the Contractor Mode RMS for full utilization of all features for all types and sizes of contracts. Smaller, less complicated, projects may not require the configuration levels described below. Required configuration also noted below.

Recommended RMS System Requirements	
Hardware	
Windows-based PC	1.7 GHz i3; AMD A6 3650 GHz or higher processor (REQUIRED)
RAM	8 GB
Hard drive disk	100 GB space for sole use by RMS system
Monitor	Screen resolution 1366 x 768
Mouse or other pointing device	
Windows compatible printer	Laser printer must have 4 MB+ of RAM
Connection to the Internet	minimum 4 Mbs per user
Software	
MS Windows	Windows 10 x 64 bit (RMS requires 64 bit O/S) or newer (REQUIRED)
Word Processing software	Viewer for MS Word 2013, MS Excel 2013 or newer (REQUIRED)
E-mail	MAPI compatible (REQUIRED)

Recommended RMS System Requirements	
Virus protection software	Regularly upgraded with all issued Manufacturer's updates and is able to detect most zero day viruses (REQUIRED)

1.6 CONTRACT DATABASE - GOVERNMENT

The Government will enter the basic contract award data in RMS prior to granting the Contractor access. The Government entries into RMS will generally be related to submittal reviews, correspondence status, and Quality Assurance(QA)comments, as well as other miscellaneous administrative information.

1.7 CONTRACT DATABASE - CONTRACTOR

Contractor entries into RMS establish, maintain, and update data throughout the duration of the contract. Contractor entries generally include prime and subcontractor information, daily reports, submittals, RFI's, schedule updates and payment requests. RMS includes the ability to import attachments and export reports in many of the modules, including submittals. The Contractor responsibilities for entries in RMS typically include the following items:

1.7.1 Administration

1.7.1.1 Contractor Information

Enter all current Contractor administrative data and information into RMS within 7 calendar days of receiving access to the contract in RMS. This includes, but is not limited to, Contractor's name, address, telephone numbers, management staff, and other required items.

1.7.1.2 Subcontractor Information

Enter all missing subcontractor administrative data and information into RMS CM within 7 calendar days of receiving access to the contract in RMS or within 7 calendar days of the signing of the subcontractor agreement for agreements signed at a later date. This includes name, trade, address, phone numbers, and other required information for all subcontractors. A subcontractor is listed separately for each trade to be performed.

1.7.1.3 Correspondence

Identify all Contractor correspondence to the Government with a serial number. Prefix correspondence initiated by the Contractor's site office with "S". Prefix letters initiated by the Contractor's home (main) office with "H". Letters are numbered starting from 0001. (e.g., H-0001 or S-0001). The Government's letters to the Contractor will be prefixed with "C" or "RFP".

1.7.1.4 Equipment

Enter and maintain a current list of equipment planned for use or being used on the jobsite, including the most recent and planned equipment

inspection dates.

1.7.1.5 Reports

Track the status of the project utilizing the reports available in RMS. The value of these reports is reflective of the quality of the data input. These reports include the Progress Payment Request worksheet, Quality Control (QC) comments, Submittal Register Status, and Three-Phase Control worksheets.

1.7.1.6 Request For Information (RFI)

Create and track all Requests For Information (RFI) in the RMS Administration Module for Government review and response.

1.7.2 Finances

1.7.2.1 Pay Activity Data

Develop and enter a list of pay activities in conjunction with the project schedule. The sum of pay activities equals the total contract amount, including modifications. Each pay activity must be assigned to a Contract Line Item Number (CLIN). The sum of the activities assigned to a CLIN equals the amount of each CLIN.

1.7.2.2 Payment Requests

Prepare all progress payment requests using RMS. Update the work completed under the contract at least monthly, measured as percent or as specific quantities. After the update, generate a payment request and prompt payment certification using RMS. Submit the signed prompt payment certification and payment request as well as supporting data either electronically or by hard copy. Unless waived by the Contracting Officer, a signed paper copy of the approved payment certification and request is also required and will govern in the event of discrepancy with the electronic version.

1.7.3 Quality Control (QC)

Enter and track implementation of the 3-phase QC Control System, QC testing, transferred and installed property and warranties in RMS. Prepare daily reports, identify and track deficiencies, document progress of work, and support other Contractor QC requirements in RMS. Maintain all data on a daily basis. Insure that RMS reflects all quality control methods, tests and actions contained within the Contractor Quality Control (CQC) Plan and Government review comments of same within 7 calendar days of Government acceptance of the CQC Plan.

1.7.3.1 Quality Control (QC) Reports

The Contractor's Quality Control (QC) Daily Report in RMS is the official report. The Contractor can use other supplemental formats to record QC data, but information from any supplemental formats are to be consolidated and entered into the RMS QC Daily Report. Any supplemental information may be entered into RMS as an attachment to the report. QC Daily Reports must be finalized and signed in RMS within 24 hours after the date covered by the report. Provide the Government a printed signed copy of the QC Daily Report, unless waived by the Contracting Officer.

1.7.3.2 Deficiency Tracking.

Use the QC Daily Report Module to enter and track deficiencies. Deficiencies identified and entered into RMS by the Contractor or the Government will be sequentially numbered with a QC or QA prefix for tracking purposes. Enter each deficiency into RMS the same day that the deficiency is identified. Monitor, track and resolve all QC and QA entered deficiencies. A deficiency is not considered to be corrected until the Government indicates concurrence in RMS.

1.7.3.3 Three-Phase Control Meetings

Maintain scheduled and actual dates and times of preparatory and initial control meetings in RMS. Worksheets for the three-phase control meetings are generated within RMS.

1.7.3.4 Labor and Equipment Hours

Enter labor and equipment exposure hours on a daily basis. Roll up the labor and equipment exposure data into a monthly exposure report.

1.7.3.5 Accident/Safety Reporting

Both the Contractor and the Government enter safety related comments in RMS as a deficiency. The Contractor must monitor, track and show resolution for safety issues in the QC Daily Report area of the RMS QC Module. In addition, follow all reporting requirements for accidents and incidents as required in EM 385-1-1, Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS and as required by any other applicable Federal, State or local agencies.

1.7.3.6 Definable Features of Work

Enter each feature of work, as defined in the approved CQC Plan, into the RMS QC Module. A feature of work may be associated with a single or multiple pay activities, however a pay activity is only to be linked to a single feature of work.

1.7.3.7 Activity Hazard Analysis

Import activity hazard analysis electronic document files into the RMS QC Module utilizing the document package manager.

1.7.4 Submittal Management

Enter all current submittal register data and information into RMS within 7 calendar days of receiving access to the contract in RMS. The information shown on the submittal register following the specification Section 01 33 00 SUBMITTAL PROCEDURES will already be entered into the RMS database when access is granted. Group electronic submittal documents into transmittal packages to send to the Government, except very large electronic files, samples, mock ups, color boards, or where hard copies are specifically required. Track transmittals and update the submittal register in RMS on a daily basis throughout the duration of the contract. Submit hard copies of all submittals unless waived by the Contracting Officer.

1.7.5 Schedule

Enter and update the contract project schedule in RMS by either manually entering all schedule data or by importing the Standard Data Exchange Format (SDEF) file, based on the requirements in Section 01 32 16.00 20 SMALL PROJECT CONSTRUCTION PROGRESS SCHEDULES.

1.7.6 Closeout

Closeout documents, processes and forms are managed and tracked in RMS by both the Contractor and the Government. Ensure that all closeout documents are entered, completed and documented within RMS.

1.8 IMPLEMENTATION

Use of RMS as described in the preceding paragraphs is mandatory. Ensure that sufficient resources are available to maintain contract data within the RMS system. RMS is an integral part of the Contractor's required management of quality control.

1.9 MONTHLY COORDINATION MEETING

Update the RMS CM database each workday. At least monthly, generate and submit one schedule update. At least one week prior to submittal, meet with the Government representative to review the planned progress payment data submission for errors and omissions.

Make required corrections prior to Government acceptance of the export file and progress payment request. Payment requests accompanied by incomplete or incorrect data submittals will not be accepted. The Government will not process progress payments until all required corrections are processed.

1.10 NOTIFICATION OF NONCOMPLIANCE

Take corrective action within 7 calendar days after receipt of notice of RMS non-compliance by the Contracting Officer.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2007) Standard for Reduced-Pressure
Principle Backflow Prevention Assembly

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH
(FCCCHR)

FCCCHR List (continuously updated) List of Approved
Backflow Prevention Assemblies

FCCCHR Manual (10th Edition) Manual of Cross-Connection
Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National
Electrical Code

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 Safety and Health Requirements Manual

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction
Marking and Lighting

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Temporary Utility Connections Plan; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 BACKFLOW PREVENTERS CERTIFICATE

Certificate of Full Approval from FCCCHR List, attesting that the design, size and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval will not be acceptable.

1.4.1 Backflow Tester Certificate

Certification issued by the local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in any other phase of this Contract.

1.4.2 Backflow Prevention Training Certificate

Certification recognized by the local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current. Training and certification can be obtained from the following sources:

- a. Kyuusuikouji Gijutsu Koshin Zaidan; <https://www.kyuukou.or.jp/>
- b. Zenkoku Kensetsu Kenshu Center; <https://www.jctc.jp/>
- c. Tester Training Course, Foundation for Cross-Connection Control and Hydraulic Research, University of Southern California; <https://fccchr.usc.edu/tester.html>

1.5 TYPHOON AND STORM PREPARATIONS

In order to minimize damage to public properties and prevent personal injury, the following actions shall be taken upon declaration of the typhoon conditions described below. Ensure that the construction sites are well-prepared for protection from the damages of heavy rain and strong winds. Develop and establish the necessary procedures to be taken for storm preparation. In Part 1 of the Accident Prevention Plan, provide emergency phone numbers and addresses where at least three Contractor officials may be reached and notified in the event that an immediate typhoon alert is declared.

1.5.1 Tropical Cyclone Conditions of Readiness (TCCOR)

- a. TCCOR STORM WATCH: The winds are not forecast to exceed 50 knots (58 mph/26 m/s) sustained (the criteria for "destructive winds") but there still exists a probability of high winds due to the proximity of the

storm. High winds may include gusts exceeding 50 knots and/or sustained winds meeting TCCOR 1 Caution criteria. The storm is also close enough to the area that a heightened alert status is necessary in order to rapidly establish elevated TCCOR conditions should the storm deviate from the forecast track. Personnel should follow Standard Operating Procedures for TCCOR Storm Watch and stay alert for any changes to TCCOR status.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor shall be required to perform general area cleanup and monitor TCCOR levels.

When a local storm warning has been issued, the Contractor competent person and/or SSHO shall determine whether it is necessary to implement manufacturer recommendations for securing the equipment to include Weight Handling Equipment (WHE).

Contractor reviews project site storm preparation plans and discuss plan of action with Contracting Officer's Representative.

- b. TCCOR 4: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 72 hours.

When wind gusts of 30 knots (35 mph/16 m/s) or greater are forecast, the Contractor will be required to perform general area cleanup and monitor TCCOR levels.

Cranes shall not be operated when wind speeds at the site attain the maximum wind velocity based on the surface/load ratio recommendations of the manufacturer.

At winds greater than 20 mph (17 knots/9 m/s), the operator, rigger, lift supervisor, and SSHO shall cease all crane operations, evaluate conditions and determine if the lift can proceed with Contracting Officer's Representative acceptance.

All elevated work above 6 ft (1.8 m) shall be risk assessed by the Contractor's competent person and/or SSHO and discussed with the Contracting Officer's Representative for acceptance.

- c. TCCOR 3: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 48 hours.

- (1) Clean and remove all loose debris and waste including scrap wood and metal, empty barrels, and construction form materials no longer required on the job site, to a safe area for proper disposition.

- (2) Secure, tie down, and anchor construction field office and storage facilities, scaffolding, concrete forms and supports, doors, windows, opening covers, stored lumber and other materials, mechanized construction equipment, temporary power lines and supports, and other items that may be blown away or that might cause injury or damage.

- (3) Inspect all excavation and trenching work in process, and provide necessary temporary drainage and proper protection and shoring for excavation sides and openings, in order to prevent damage to public roads and facilities by slides or flooding. Accumulation

of water in the excavation of structure foundation work shall be controlled and minimized.

- (4) Inspect scaffolding or work platforms for loose materials, planking, etc. that could become airborne projectile hazards and secure scaffold netting, tarps, etc. from wind loads.

- d. TCCOR 2: Winds of 50 knots (58 mph/ 26 m/s) sustained or greater are possible within 24 hours.

Work required at remote areas such as off-shore facilities or high elevations shall cease immediately and the workers shall be evacuated to a safe area. During TCCOR 2, the Contractor shall continue the actions described in TCCOR 3 above and the construction site shall be inspected for storm preparation by the Contracting Officer's Representative. The Contractor shall request an inspection by calling the Contracting Officer's Representative at the appropriate Japan Engineer District field office.

- e. TCCOR 1: Winds of 50 knots (58 mph/26 m/s) sustained or greater are possible within 12 hours.

All work shall cease immediately and the Contractor's representative shall insure that all necessary storm preparations, including the items listed below, are completed.

- (1) All electrical circuits and equipment including temporary power lines are cut off and secured against unauthorized use.
- (2) Gas cylinders, hot work equipment, and flammable materials properly stored at a safe area.
- (3) No igniting source is present.
- (4) All workers have been evacuated from the construction site. When TCCOR 1 is declared without the normal progression through TCCOR 3 and/or 2, the Contractor shall take the actions listed in TCCOR 3 above, and also follow the procedures described herein.

- f. TCCOR 1 Caution: Winds of 35 to 49 knots (39/17 to 56/25 mph/ m/s) sustained are occurring.

Construction Sites Secured.

- g. TCCOR 1 Emergency: Winds of 50 knots (58 mph/26 m/s) sustained or greater are occurring.

Construction Sites Secured.

- h. TCCOR 1 Recovery: Winds of 50 knots (58 mph/26 m/s) sustained or greater are no longer forecast to occur. Strong winds may still exist.

Installation personnel begin storm damage assessments and clean up.

- i. TCCOR Storm Clear: The storm is over and not forecast to return. Used to inform personnel that the threat of the storm is over, but the storm damage could still present a danger.

Contractor may resume normal activities. The construction site shall

be investigated for all damage caused by the typhoon or high winds, and the result of the investigation shall be furnished in verbal or written form to the Contracting Officer's Representative as soon as practicable.

Complete and submit PODWP 134 Typhoon Damage Report for storm damages or negative report of damage to Emergency Management, Construction Division, and Safety Office.

- j. TCCOR All Clear: The storm is over and not forecast to return, and recovery efforts are complete.

Resume normal activities.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Immediately upon beginning of work, provide a weatherproof glass-covered bulletin board not less than 915 by 1220 mm in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, and other information approved by the Contracting Officer. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

2.1.2 Project and Safety Signs

Provide one (1) project sign and one (1) safety sign fabricated to size and design as shown in Attachment 01 50 00-A Signage Requirements. The signs shall be rigidly formed and erected at location designated by the Contracting Officer prior to commencement of work. A blue-line drawing of different letter sizes and style shall be made available by the Contracting Officer upon request. Prior to painting the sign, submit for approval a sketch, similar to diagram shown in Attachment 01 50 00-A, indicating the actual information for the project. The sketch shall indicate lettering dimensions and locations. The Corps of Engineers castle logo shall be painted red per Attachment 01 50 00-A. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. No separate payment shall be made for the sign, and all costs in connection therewith shall be included in the Contract price for the project. The sign shall be subject to the approval of the Contracting Officer. Erect signs within 15 days after receipt of the Notice to Proceed. Correct the data required by the safety sign daily, with light-colored metallic or non-metallic numerals. Upon completion of work under this Contract, the signs shall be removed from the job site and shall remain the property of the Contractor.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract. Construct with suitable grades and widths; sharp curves, blind corners, and dangerous cross traffic are to be avoided. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control

must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and hauling roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic barricades will be required. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.2.3 Temporary Wiring

Provide temporary wiring in accordance with NFPA 241 and NFPA 70. Include frequent inspection of all equipment and apparatus.

2.2.4 Backflow Preventers

Reduced pressure principle type conforming to the applicable requirements AWWA C511. Provide backflow preventers complete with 65 kg flanged, mounted gate valve and strainer, stainless steel or bronze, internal parts. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Contractor employees will park privately owned vehicles in an area designated by the Contracting Officer. This area shall be within reasonable walking distance of the construction site. Contractor employee parking must not interfere with existing and established parking requirements of the government installation.

3.2 TEMPORARY BULLETIN BOARD

Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, as approved by the Contracting Officer.

3.3 AVAILABILITY AND USE OF UTILITY SERVICES

3.3.1 Payment for Utility Services

The Government shall make all reasonably required utilities available to the Contractor, without charge, from existing outlets and supplies, as specified in the Contract. Carefully conserve any utilities.

3.3.2 Meters and Temporary Connections

At the Contractor's expense and in a manner satisfactory to the Contracting Officer, provide and maintain necessary temporary connections, distribution lines, and meter bases required to measure the amount of each utility used. Materials may be new or used, must be adequate for the

required usage, not create unsafe conditions, and not violate applicable codes and standards. Submit a Temporary Utility Connections Plan for approval to the Contracting Officer, fifteen (15) working days before making temporary utility connections.

The Contractor shall

- Furnish the water meter
- Furnish the electric meter
- Furnish the backflow preventer
- Install the water meter
- Install the electric meter
- Install the backflow preventer

3.3.3 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, five working days before termination is desired. A final reading shall be taken by the Government. Afterwards, remove all temporary distribution lines, meter bases, meters, and associated appurtenances.

3.3.4 Sanitation

Provide and properly maintain temporary toilet facility for use of Contractor personnel. Government toilet facilities are not available to Contractor personnel. Toilet facility shall be constructed as required in EM 385-1-1 and by local installation regulations, where directed. Include provisions for pest control and elimination of odors. Upon completion of the Contract, dispose facility outside the limits of Government-controlled land and at Contractor's expense.

3.3.5 Electricity

The Contractor shall be responsible for all temporary connections required, as well as the removal of these temporary connections upon completion of the project. The Contractor's plan for temporary connections shall be submitted to the Contracting Officer Representative for approval prior to making any temporary connections.

Electrical power system frequencies for different installations are as follows:

- a. 50 Hertz (Hz): Camp Zama, Sagami Depot, Sagamihara Housing Area, Yokota Air Base, Commander Fleet Activities Yokosuka, Naval Air Facility Atsugi, Misawa Air Base, Camp Fuji, and Naval Support Facility Kamiseya.
- b. 60 Hertz (Hz): Marine Corps Air Station Iwakuni, Commander Fleet Activities Sasebo, and all of Okinawa.

3.3.6 Telephone and Data

Make arrangements and pay all costs for telephone facilities and data connections.

3.3.7 Obstruction Lighting of Cranes

Provide a minimum of 2 aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 30 meter above

ground level. Light construction and installation shall comply with FAA AC 70/7460-1. Lights shall be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.3.8 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic except with written permission of the Contracting Officer at least 30 calendar days prior to the proposed modification date, and provide a Traffic Control Plan detailing the proposed controls to traffic movement for approval. The plan must be in accordance with installation regulations. Coordinate with installation officials. Contractor may move oversized and slow-moving vehicles to the worksite once the requirements of the highway authority have been met. Include a schedule of planned road closures in the Initial Project Schedule.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.
- c. At the Contractor's expense, provide, erect, and maintain: lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage overhead protection authority having jurisdiction.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment, the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of any damage to roads caused by construction operations.

3.4.3 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Treat dust abatement on access roads with applications of calcium chloride, water sprinklers, or similar methods or treatment.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

3.5.1 Safety

Protect the integrity of any installed safety systems or personnel safety devices. If entrance into systems serving safety devices is required, the Contractor must obtain prior approval from the Contracting Officer. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.2 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.5.3 Temporary Project Safety and Storage Area Fencing

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety and storage area fencing to control access by unauthorized people. Fencing shall be able to restrain a force of at least 114 kg. Safety and storage area fencing shall be 1.8 m high wire mesh fence with lockable gates and visibility screening (similar to tennis court screening) located around the perimeter of project work areas and the perimeter of Contractor lay-down areas. Use plastic panels or perforated steel sheets to improve visibility for vehicular traffic when erecting fencing within 3 meters of street corners. The intent is to block (screen) public view of the construction, and prevent unauthorized access to the site. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Maintain the safety and storage area fencing during the life of the Contract and, upon completion and acceptance of the work, shall become the property of the Contractor and be removed from the work site.

3.5.4 Storage Area

Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on any given day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

3.5.5 Supplemental Storage Area

Upon Contractor's request, the Contracting Officer will designate another or supplemental area for the Contractor's use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Fencing of materials or equipment will be required at this site. The Contractor is responsible for cleanliness and orderliness of the area used and for the security of any material or equipment stored in this area. Utilities will not be provided to this area by the Government.

3.5.6 Appearance of Trailers

Trailers utilized by the Contractor for administrative or material storage purposes must present a clean and neat exterior appearance and be in a state of good repair. Trailers which, in the opinion of the Contracting Officer, require exterior painting or maintenance will not be allowed on installation property.

3.5.7 Maintenance of Storage Area

Keep fencing in a state of good repair and proper alignment. Grassed or unpaved areas, which are not established roadways, will be covered with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways, should the Contractor elect to traverse them with construction equipment or other vehicles; gravel gradation will be at the Contractor's discretion. Mow and maintain grass to a maximum height of 100 mm located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, buildings, under trailers, and in areas not accessible to mowers will be edged or trimmed neatly.

3.5.8 Security Provisions

Provide adequate outside security lighting at the Contractor's temporary facilities. The Contractor will be responsible for the security of its own equipment.

3.5.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 PLANT COMMUNICATION

Whenever the Contractor has the individual elements of its plant so located that operation by normal voice between these elements is not satisfactory, the Contractor must install a satisfactory means of communication, such as telephone or other suitable devices and made available for use by Government personnel.

3.7 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto

paved or surfaced roadways must be cleaned away. Store any salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.8 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and any other temporary products from the site. All tools, equipment, and materials not the property of the Government shall be removed from the premises and properly disposed of off Government property. After removal of trailers, materials, and equipment from within the fenced area, remove the temporary project safety and storage area fencing. Restore areas used by the Contractor for the storage of equipment or material, or other use to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary. Upon completion of the work, notify the Contracting Officer for final inspection.

-- End of Section --

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Introduction: Construction Project Signs

EP 310-1-6a
01 Jun 06

The use of signs to identify Corps managed or supervised design, construction, and rehabilitation projects for military is an important part of efforts to keep the public informed of Corps work. For this purpose, a construction project sign package has been adopted. This package consists of two signs: one for project identification and the other to show on-the-job safety performance of the contractor.

These two signs are to be displayed side by side and mounted for reading by passing viewers. Exact placement location will be designated by the contracting officer's representative.

The panel sizes and graphic formats have been standardized for visual consistency throughout all Corps operations.

Panels are fabricated using HDO plywood or aluminum with dimensional lumber uprights and bracing. The sign faces are nonreflective vinyl.

All legends are to be die-cut or computer-cut in the sizes and typefaces specified and applied to the white panel background following the graphic formats shown on pages 16-2 and 16-3. The Communication Red panel on the left side of the construction project sign with Corps Signature (reverse version) is screen-printed onto the white background.

A display of these two signs is shown on the following two pages. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

EP 310-1-6a (Modified per ECB 2020-1)
01 Jun 06 (ECB modified 1/31/2020)

Construction Project Identification Sign

Below is one sample of the Construction Project Identification sign showing how this panel is to identify military projects. The graphic format for this 4'x 6' sign panel follows the legend guidelines and layout as specified below. The large 4'x 4' section of the panel on the right is to be white with black legend.

The 2'x 4' section of the sign on the left with the full Corps Signature (reverse version) is to be screen-printed Communication Red on the white background. The designation of a sponsor in the area indicated is optional with Military construction signs.

This sign is to be placed with the Safety Performance sign shown on the following page. Mounting and fabrication details are provided on page 16-4.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Post size is 4"x4".
Mounting height is 48".

Legend Group 1: One- to two-line description of Corps relationship to project.
Color: White
Typeface: 1.25" Helvetica Regular
Maximum line length: 19"

Legend Group 2: Division or District Name (optional). Placed below 10.5" reverse Signature (6" Castle).
Color: White
Typeface: 1.25" Helvetica Regular

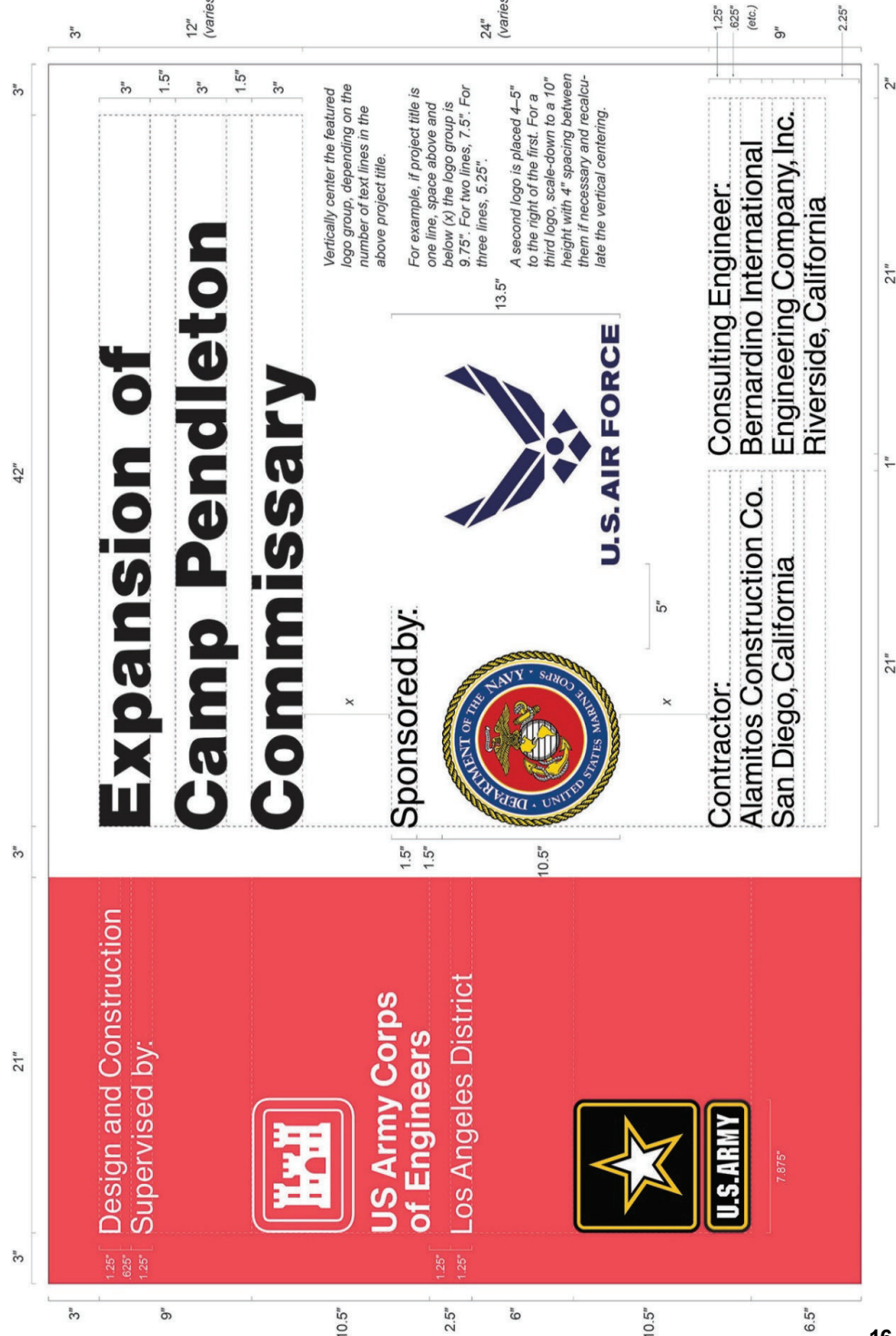
Legend Group 2a: Army Star emblem.
Color: Various

Legend Group 3: One- to three-line project title legend describes the work being done under this contract.
Color: Black
Typeface: 3" Helvetica Bold
Maximum line length: 42"

Legend Group 4: Logo of sponsoring department (military).
Color: Various

Legend Groups 5a-b: One- to five-line identification of prime contractors including: type (architect, general contractor, etc.), corporate or firm name, city, state. Use of Legend Group 5 is optional.
Color: Black
Typeface: 1.25" Helvetica Regular
Maximum line length: 21"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Safety Performance Sign

EP 310-1-6a
01 Jun 06

Each contractor's safety record is to be posted on Corps managed or supervised construction projects and mounted with the Construction Project Identification sign specified on page 16-2.

The graphic format, color, size and typefaces used on the sign are to be reproduced exactly as specified below. The

title with First Aid logo in the top section of the sign, and the performance record captions are standard for all signs of this type. Legend groups 2 and 3 below identify the project and the contractor and are to be placed on the sign as shown.

Safety record numbers are mounted on individual metal plates and are screw-

mounted to the background to allow for daily revisions to posted safety performance record.

Special applications or situations not covered in these guidelines should be referred to the district Sign Program Manager.

Legend Group 1: Standard two-line title "Safety is a Job Requirement" with 8" (outside diameter) Safety Green first aid logo.
Color: To match Pantone system 347
Typeface: 3" Helvetica Bold
Color: Black

Legend Group 2: One- to two-line project title legend describes the work being done under this contract and name of host project.
Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

Legend Group 3: One- to two-line identification: name of prime contractor and city, state address. Color: Black
Typeface: 1.5" Helvetica Regular
Maximum line length: 42"

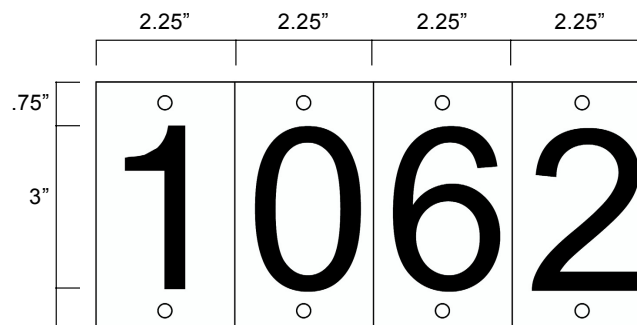
Legend Group 4: Standard safety record captions as shown.
Color: Black
Typeface: 1.25" Helvetica Regular

Replaceable numbers are to be mounted on white .060 aluminum plates and screw-mounted to background.
Color: Black
Typeface: 3" Helvetica Regular
Plate size: 2.5" x 4.5"

All typography is flush left and rag right, upper and lower case with initial capitals only as shown. Letter- and word-spacing to follow Corps standards as specified in Appendix D.



Sign Type	Legend Size (A)	Panel Size	Post Size	Specification Code	Mounting Height	Color Bkg/Lgd
CID-02	various	4'x4'	4"x4"	HDO-3	48"	WH/BK-SG



16-3

EP 310-1-6a
01 Jun 06

Fabrication and Mounting Guidelines

All Construction Project Identification signs and Safety Performance signs are to be fabricated and installed as described below. The signs are to be erected at a location designated by the contracting officer representative and shall conform to the size, format, and typographic standards shown on pages 16-2 and 16-3. Detailed specifications

for HDO plywood panel preparation are provided in Appendix B.

For additional information on the proper method to prepare sign panel graphics, contact the district Sign Program Manager.

Shown below the mounting diagram is a panel layout grid with spaces provided for project information. Photocopy this page and use as a worksheet when preparing sign legend orders.

The sign panels are to be fabricated from .75" High Density Overlay Plywood. Panel preparation to follow HDO specifications provided in Appendix B.

Sign graphics to be prepared on a white nonreflective vinyl film with positionable adhesive backing.

All graphics except for the Communication Red background with Corps Signature on the project sign are to be die-cut or computer-cut nonreflective vinyl, prespaced legends prepared in the sizes and typefaces specified and applied to the background panel following the graphic formats shown on pages 16-2 and 16-3.

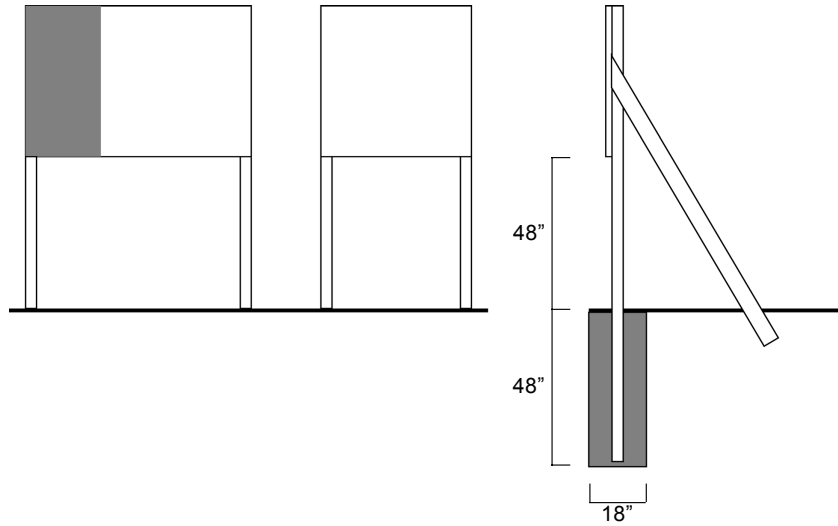
The 2'x 4' Communication Red panel (to match Pantone system 032) with full Corps Signature (reverse version) is to be screen-printed on the white background. Identification of the district or division may be applied under the signature with white cut vinyl letters prepared to Corps standards.

Drill and insert six (6) .375" T-nuts from the front face of the HDO sign panel. Position holes as shown. Flange of T-nut to be flush with sign face.

Apply graphic panel to prepared HDO plywood panel following manufacturers' instructions.

Sign uprights to be structural grade 4" x 4" treated Douglas Fir or Southern Yellow Pine, No.1 or better. Post to be 12' long. Drill six (6) .375" mounting holes in uprights to align with T-nuts in sign panel. Countersink (.5") back of hole to accept socket head cap screw (4" x .375").

Assemble sign panel and uprights. Imbed assembled sign panel and uprights in 4' hole. Local soil conditions and/or wind loading may require bolting additional 2" x 4" struts on inside face of uprights to reinforce installation as shown.



Construction Project Identification Sign
Legend Group 1: Corps Relationship

1. _____
2. _____

Legend Group 2: Division/District Name

1. _____
2. _____

Legend Group 2a: Military/Civil Works Sponsor

1. _____
2. _____

Legend Group 3: Project Title

1. _____
2. _____
3. _____

Legend Group 4: Facility Name

1. _____
2. _____

Legend Group 5: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Legend Group 5b: Contractor/A&E

1. _____
2. _____
3. _____
4. _____
5. _____

Safety Performance Sign

Legend Group 2: Project Title

1. _____
2. _____

Legend Group 3: Contractor/A&E

1. _____
2. _____

SECTION 01 57 19

TEMPORARY ENVIRONMENTAL CONTROLS
01/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

JEGS (Apr 2024) Japan Environmental Governing Standards

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 261 Identification and Listing of Hazardous Waste

1.2 DEFINITIONS

In some cases, definitions are given only for illustrative purposes. Prevailing JEGS definitions shall be used for environmental compliance requirements.

1.2.1 Aboveground Storage Tank

A portable or fixed POL aboveground storage container as defined in the JEGS with a capacity greater than 55 gallons.

1.2.2 Asbestos Containing Material (ACM)

Any material containing more than one-tenth of one percent (0.1%) asbestos by weight.

1.2.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.4 Chemical Wastes

Salts, acids, alkalis, herbicides, pesticides, organic chemicals, and spent products, which serve no purpose.

1.2.5 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS are listed in the JEGS.

1.2.6 Construction and Demolition Waste

The waste building materials, packaging, and rubble resulting from

construction, remodeling, repair and demolition operations on pavements, houses, commercial buildings, and other structures.

1.2.7 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e. methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

1.2.8 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.9 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.10 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.11 Food Waste

Organic residues generated by the handling, storage, sale, preparation, cooking, and serving of foods (commonly called garbage).

1.2.12 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with the JEGS. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with the JEGS.

1.2.13 Hazardous Materials

Hazardous material is any material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in the JEGS, "Typical Hazardous Materials Characteristics," or the material is listed in the "List of Hazardous Waste/Substances/Materials". Munitions are excluded

1.2.14 Hazardous Substances

Any substance having the potential to do serious harm to human health or the environment if spilled or released in reportable quantity. A list of these substances and the corresponding reportable quantities is contained in the JEGS, "Characteristics of Hazardous Waste and Lists of Hazardous Waste and Hazardous Material."

1.2.15 Hazardous Waste

Hazardous Waste is discarded material that may be solid, semi-solid, liquid, or contained gas, and either exhibits a characteristic of a hazardous waste as defined in the JEGS. Excluded from this definition are domestic sewage sludge, household wastes, and medical wastes.

1.2.16 Installation Pest Management Consultant

Installation Pest Management Consultant (IPMC) is the professional DoD pest management personnel located at component headquarters, field operating agencies, major commands, facilities engineering field divisions or activities, or area support activities who provide technical and management guidance for the conduct of installation pest management operations. Some pest management consultants may be designated by their component as certifying officials.

1.2.17 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, or discharge into defined drainage areas (includes drainage ditches, streams, rivers, ocean, etc.) must occur. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.18 Lead-based Paint (LBP)

Paint or other surface coatings that contain lead greater than or equal to 1.0 milligram per square centimeter(cm), or 0.5 percent by weight, or 5,000 ppm by weight.

1.2.19 Oily Wastes

Oily wastes are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from those POLs. Oily wastes also means materials, including wastewaters, centrifuge solids, filter residues or sludges, bottom sediments, tank bottoms, and sorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements..

This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. These materials may be land filled provided that: It is not prohibited in the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements; the amount generated is "de minimus" (a small amount); it is the result of minor leaks or spills resulting from normal process operations; and free-flowing oil has been removed to the practicable extent possible. Large quantities of this material, generated

as a result of a major spill or in lieu of proper maintenance of the processing equipment, are a solid waste. As a solid waste, perform a hazardous waste determination prior to disposal. As this can be an expensive process, it is recommended that this type of waste be minimized through good housekeeping practices and employee education.

1.2.20 Petroleum, Oil, and Lubricants

Refined petroleum, oils, and lubricants, including, but not limited to, petroleum, fuel, lubricant oils, synthetic oils, mineral oils, animal fats, vegetable oil, sludge, and POL mixed with wastes other than dredged spoil.

1.2.21 Polychlorinated Biphenyl (PCB)

Any PCB article, PCB article container, PCB container, or PCB equipment that deliberately or unintentionally contains or has as a part of it any detectable concentration of PCB.

1.2.22 Regulated Waste

Regulated waste are solid wastes that have specific additional JEGS, Federal, GOJ national, and prefectural controls for handling, storage, or disposal.

1.2.23 Rubbish

A general term for solid waste, excluding food wastes and ashes, taken from residences, commercial establishments, and institutions.

1.2.24 Sanitary Waste

a. Sewage: Wastes characterized as domestic sanitary sewage.

b. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

1.2.25 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

1.2.26 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or their significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants. Types of solid waste typically generated at construction sites may include:

1.2.26.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed

concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may not be reinforced with or contain ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.26.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.26.3 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.26.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, characteristic of a hazardous waste as defined in the JEGS or is listed as a hazardous waste in the JEGS. Excluded from this definition is medical wastes.

1.2.26.5 Recyclables

Recyclables are materials, equipment and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing and mirrors that are recovered and sold as recyclable, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable if sold to a scrap metal company.

1.2.26.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

1.2.26.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of

hazardous material or hazardous waste is not included.

1.2.26.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.2.27 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks, or waters of Japan. Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

1.2.28 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant.

1.2.28.1 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.29 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.3 PROHIBITED PRODUCTS

The following items are forbidden for use by the JEGS or other criteria. Details of each are included in the text of each chapter of the Environmental Protection Plan (EPP) described in this section.

- a. Asbestos Containing Materials (ACM).
- b. Lead-Containing Paint. Paint containing greater than 0.009 percent lead by weight.
- c. Polychlorinated Biphenyls (PCBs). Materials containing PCBs greater than 0.5mg/kg shall not be used.
- d. Class I Ozone Depleting Substances (ODS). Class 1 ODS listed in the JEGS shall not be used.
- e. Lead-containing Drinking-Water Pipes, Solders, Flux, and Fittings. Comply with JEGS requirements where applicable.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation;

submittals not having a "G" designation are for information only. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Environmental Protection Plan; G

Environmental Manager Qualifications; G

Written Assessment Of Friable Asbestos Disturbance

SD-06 Test Reports

Solid Waste Management Report; G

Nonhazardous Solid Waste Diversion Report

SD-07 Certificates

Asbestos Certification; G

Lead Certification; G

ECATTS Certificate Of Completion; G

PCB Certifications; G

SD-11 Closeout Submittals

Disposal Documentation for Hazardous and Regulated Waste; G

Assembled Employee Training Records; G

Solid Waste Management Report; G

Hazardous Waste/Debris Management; G

Environmental Records Binder

1.5 PAYMENT

No separate payment shall be made for work covered under this section. Payment of fees associated with environmental permits, application, and/or notices obtained by the Contractor, and payment of all fines/fees for violation or non-compliance with GOJ, Federal, and local laws and regulations are the Contractor's responsibility. All costs associated with this section shall be included in the Contract price.

1.6 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected

outside the limits of permanent work during the entire duration of this Contract. Comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with applicable Environmental Laws may be required. Laboratory analyses necessary to implement the JEGS shall be conducted in a laboratory certified by a U.S. or GOJ regulatory authority for the applicable test method, and where required by law, the laboratories shall be certified. In the absence of a certified laboratory, contact the Contracting Officer for further guidance.

Contractor shall be responsible to ensure that subcontractors comply with all environmental protection requirements of this section.

The Contractor shall record any problems in complying with laws, regulations, permit requirements, ordinances, and corrective actions taken. The Contractor shall immediately inform the Contracting Officer of any environmental problems.

1.6.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

1.6.1.1 Personnel Requirements

The Environmental Manager is responsible for environmental compliance on projects. The Environmental Manager and other staff, must complete applicable ECATTS training modules (installation specific or general) prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete applicable ECATTS training within 14 days of assignment to the project.

1.6.1.2 Certification

Provide a copy of the ECATTS certificate of completion for personnel who have completed the required ECATTS training in the Environmental Records Binder. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Training and Tracking System, by logging on to <https://environmentaltraining.ecatts.com/>. Obtain the password for registration from the Contracting Officer.

All Contractor personnel who perform work at CFAS must complete the EMS Awareness Training using the Environmental Compliance, Training and Tracking System (ECATTS). The Contracting Officer will verify completion of training through the ECATTS database.

Go to <https://environmentaltraining.ecatts.com/> home page and click "enter". Register as a new user by entering "navfac" (all lower case with no quotation marks) in the Registration Password dialog box. Click the "Create an Account" link to establish an account. Enter the employee's name, e-mail, login ID (pick any ID you choose to use), password (pick any password you choose to use), training type (i.e. Other Construction Contractor or Contract Employee Working on Installation) and work location (Japan, COMFLEACT Sasebo JA). Click the "register" button and the website

will generate a password to log into the training site.

Go to the home page and login with the user name and password generated by the website. Click "Japanese language" button, click "Go To Your Training", click "Go To My Training Modules". Open "Environmental Management System Awareness Training", complete the training, take the test and print a certificate for your records. The training takes approximately 20 minutes to complete.

1.6.1.3 Refresher Training

This training has been structured to allow Contractor personnel to receive credit under this Contract and to carry forward credit to future contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work.

ECATTS is available for use by all Contractor and subcontractor personnel associated with this project. These other personnel are encouraged (but not required) to take the training and may do so at their discretion.

1.6.2 Conformance with the Environmental Management System

Perform work under this Contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the Contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.7 SPECIAL ENVIRONMENTAL REQUIREMENTS

Comply with the special environmental requirements listed within this section.

1.7.1 Asbestos Prohibition and Certification

- a. Materials or products containing more than one-tenth of one percent (0.1 percent) by total weight, of the material or product, of asbestos shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products

that contain asbestos in excess of one-tenth of one percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if asbestos (exceeding 0.1 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform asbestos abatement and restoration work, as required, at the Contractor's sole cost. Asbestos abatement work (removal and disposal of asbestos-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable asbestos-free substitutes do not exist for a material or product, the Contractor may use a material or product containing asbestos in the excess of 0.1 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of asbestos present in the material or product. When available, laboratory analysis of the material or product for asbestos content shall be included with the submittal.

- b. The Government may conduct asbestos testing on suspected asbestos-containing materials and products excluding items permitted by the "exception", and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of asbestos exceeding 0.1 percent by weight. In the event test results indicate 0.1 percent or less asbestos content or complete absence of asbestos, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.
- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are asbestos-free and do not contain asbestos in excess of 0.1 percent by weight, as applicable. However, when presence of asbestos is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Asbestos certification shall be required for the items applicable to this project only.
 - (1) Vinyl sheet/vinyl tile flooring, including accessories and adhesives.
 - (2) Insulation materials, including facing.
 - (3) Gaskets for piping and duct work.
 - (4) Acoustical tiles.
 - (5) Firestopping materials.
 - (6) Fireproofing materials.

- (7) Special coating, including factory applied coatings, on sheet metal roofing and siding.
 - (8) Wallboard for all interior and exterior applications, including joint compounds.
 - (9) Adhesives (other than item 1) used in the project.
 - (10) Tape materials used in the project.
 - (11) Roofing and siding, nonmetallic.
 - (12) Felt materials and cushion materials.
 - (13) Pre-mixed mortars, grouts, leveling compounds, fillers, and other cementitious materials.
 - (14) Caulking and sealing materials.
- d. All submittals shall be accompanied by a certification from the manufacturer of the material or product that the material or product is asbestos-free; or shall set forth, in specific detail, the amount of asbestos present in the material or product. Documentary evidence of laboratory analysis of the material or product for asbestos content, conducted by a qualified independent testing laboratory, shall be included with the submittal.

1.7.2 Lead Prohibition and Certification

- a. Paint or product coating containing more than 0.009 percent) by total weight of lead shall not be used in this project. The Contracting Officer, at any time prior to acceptance of the work, or during the period designated for warranty of the work, if any, may reject materials and products that contain lead in excess of 0.009 percent by weight, and direct the removal of such materials and products from the job site, at the sole expense of the Contractor, and without additional time granted for performance of the work. After completion of this Contract, if lead (exceeding 0.009 percent by weight) is discovered in the products or materials (excluding items permitted by the exception) installed by the Contractor, the Government reserves the right to direct the Contractor to perform lead abatement and restoration work, as required, at the Contractor's sole cost. Lead abatement work (removal and disposal of lead-containing materials and products) shall be accomplished in accordance with currently applicable Government standards for such work.

Exception: Where suitable lead-free substitutes do not exist for a paint or product coating, the Contractor may use a material or product containing lead in the excess of 0.009 percent by weight, with prior written approval of the Contracting Officer. Submit a written request for substitution, accompanied by a certification from the manufacturer of the material or product that shall set forth, in specific detail, the amount of lead present in the material or product. When available, laboratory analysis of the material or product for lead content shall be included with the submittal.

- b. The Government may conduct lead testing on suspected lead-containing materials and products excluding items permitted by the "exception",

and such testing shall be conducted at the expense of the Government. However, wherever destructive testing is required, or a material or product must be utilized by the Government for testing, the Contractor shall, at its own expense, repair or replace the material or product, or the item of work that has been disturbed by testing, if the results confirm presence of lead exceeding 0.009 percent by weight. In the event test results indicate 0.009 percent or less lead content or complete absence of lead, the Contractor shall restore the test site to its original condition and the cost of restoration work, as approved by the Contracting Officer, shall be borne by the Government.

- c. As a minimum, furnish manufacturer's certification for the items listed below, excluding items permitted by the "exception", certifying that the items are lead-free and do not contain lead in excess of 0.009 percent by weight, as applicable. However, when presence of lead is suspected in other products and materials used in this project, the Contractor shall be required to provide such certification for those additional items when so directed by the Contracting Officer. Lead certification shall be required for the items applicable to this project only.

1. Paints and Coatings
2. Any product or material with a factory applied coating

1.7.3 Class I and Class II Ozone Depleting Chemicals (ODC) or Substances (ODS)

Class I and II Ozone Depleting Substances listed in the JEGS are prohibited from being used. Contractor must provide certifications that materials utilized do not contain Class I and Class II ODC/ODS.

1.7.4 Polychlorinated Biphenyls (PCB)

Materials (ballasts, capacitors, transformers, dielectric fluid, switches, etc.) that contain PCBs are prohibited. Contractor must provide PCB Certifications that materials utilized do not contain PCBs, in accordance with the JEGS.

1.7.5 Hazardous Material Survey

The Contractor shall review the Hazmat Materials Inspection Report - FY25 Repair HVAC Wideband Satcom OPS B100 by ESC dated January 2025, attached at the end of this section, to familiarize themselves with the materials that have been sampled and tested for this project. The Contractor shall utilize the information contained within the report to develop their work and compliance plans with regards to Hazardous Materials. During construction, should potentially hazardous material be discovered, which has not been previously tested, the Contractor shall take action to assure that the untested material is not disturbed and contact the Contracting Officer. The quantities provided in the report represents a rough order of magnitude estimate of materials and shall not be used for bidding purposes as conditions may have changed from the date of the survey and the present existence of the material(s).

1.8 QUALITY ASSURANCE

1.8.1 Regulatory Notifications

Provide regulatory notification requirements in accordance with GOJ national and prefectural laws and regulations and installation requirements. Include copies of regulatory notifications in the approved EPP prior to commencement of work activities translated to English.

1.8.2 Environmental Brief

Attend an environmental brief to be included in the preconstruction meeting. Provide the following information: types, quantities, and use of hazardous materials that will be brought onto the installation; and types and quantities of wastes/wastewater that may be generated during the Contract. Discuss the results of the Preconstruction Survey at this time.

Prior to initiating any work on site, meet with the Contracting Officer and installation Environmental Office to discuss the proposed Environmental Protection Plan (EPP). Develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural and cultural resources, required reports, required permits, permit requirements (such as mitigation measures), and other measures to be taken.

Permits, licenses, or other forms of official approvals are not required by DoD activities and installations. Permits, licenses, or other forms of official approvals may, however, be required under GOJ law for certain contracted activities. When required, all such permits, licenses and other forms of official approval shall be obtained by the Contractor from the appropriate GOJ authorities and included in the Environmental Protection Plan. DoD Components shall assist Contractors when they are applying for a required permit, license or other form of official approval by providing necessary information only.

1.8.3 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for Environmental Quality Control and coordinating Contractor compliance with the JEGS, Federal, GOJ national and prefectural laws and regulations, and installation requirements. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB transformers). This can be a collateral position; however, the person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in JEGS requirements in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance and close-out. Insert Environmental Manager Qualifications into the Environmental Records Binder to indicate the training and past experience which meets the requirements of this

position as described in this section.

1.8.4 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the Contract meeting applicable JEGS requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations and installation requirements. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet the JEGS and all applicable Federal, GOJ national or prefectural laws or regulations, and Installation requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Japan, and endangered species and their habitat that are known to be in the area.

1.8.5 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with the JEGS, applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements, and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the Contract, or in accordance with the Federal Acquisition Regulation or Federal Law.

1.9 ENVIRONMENTAL PROTECTION PLAN

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction. Incorporate construction related objectives and targets from the installation's EMS into the EPP. Include in the EPP measures for protecting natural and cultural resources, required reports, and other measures to be taken. Meet with the Contracting Officer or Contracting Officer Representative to discuss the EPP and develop a mutual understanding relative to the details for environmental protection including measures for protecting natural resources, required reports, and other measures to be taken. Submit the EPP within 15 days after receiving notice to proceed. Commencement of work will not begin until the Environmental Protection Plan has been approved.

Revise the EPP throughout the project to include any reporting

requirements, changes in site conditions, or Contract modifications that change the project scope of work in a way that could have an environmental impact. No requirement in this section will relieve the Contractor of the JEGS, and applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. During Construction, identify, implement, and submit for approval any additional requirements to be included in the EPP. Maintain the current version of the EPP onsite.

The EPP includes, but is not limited to, the following elements as they apply to this project:

1.9.1 General Overview and Purpose

1.9.1.1 Descriptions

A brief description of each specific plan required by environmental permit or elsewhere in this Contract such as stormwater pollution prevention plan, spill control plan, solid waste management plan, wastewater management plan, air pollution control plan, contaminant prevention plan, traffic control plan Hazardous, Toxic and Radioactive Waste (HTRW) Plan Non-Hazardous Solid Waste Disposal Plan borrowing material plan.

Include a list of applicable Federal, GOJ, JEGS, prefectural laws, regulations, and permits concerning environmental protection, pollution control, and abatement that are applicable to the Contractor's proposed operations and requirements imposed by those laws, regulations, and permits. Whenever there is conflict between Federal, GOJ, JEGS, or prefectural laws, regulations, and permit requirements, the strictest applicable rule applies.

1.9.1.2 Duties

The duties and level of authority assigned to the person(s) on the job site who oversee environmental compliance, such as who is responsible for adherence to the EPP, who is responsible for spill cleanup and training personnel on spill response procedures, who is responsible for manifesting hazardous waste to be removed from the site (if applicable), and who is responsible for training the Contractor's environmental protection personnel.

Provide the name, telephone number, and address of a Primary and Alternate Environmental Representative.

1.9.1.3 Procedures

A copy of any standard or project-specific operating procedures that will be used to effectively manage and protect the environment on the project site.

Provide schedule of digging or trenching actions to include date/time, location, purpose, method and depth.

1.9.1.4 Communications

Communication and training procedures that will be used to convey environmental management requirements to Contractor employees and subcontractors.

1.9.1.5 Contact Information

Emergency contact information contact information (office phone number, cell phone number, and e-mail address).

1.9.2 General Site Information

1.9.2.1 Drawings

Drawings showing locations of proposed temporary excavations or embankments for haul roads, stream crossings, jurisdictional wetlands, material storage areas, structures, sanitary facilities, storm drains and conveyances, and stockpiles of excess soil.

1.9.2.2 Work Area

Work area plan showing the proposed activity in each portion of the area and identify the areas of limited use or nonuse. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site, and a traffic control plan.

1.9.2.3 Documentation

A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this Contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

1.9.3 Management of Natural Resources

- a. Land resources
- b. Replacement of damaged landscape features
- c. Temporary construction
- d. Waters of Japan

1.9.4 Stormwater Management and Control

- a. Ground cover
- b. Erodible soils
- c. Temporary measures
 - (1) Structural Practices
 - (2) Temporary and permanent stabilization
- d. Effective selection, implementation and maintenance of Best Management Practices (BMPs)
- e. Erosion Control Plan. Prepared in accordance with the JEGS
- f. Storm Water Pollution Prevention Plan. Prepared in accordance with

paragraph Stormwater Pollution Prevention Plan and the JEGS (if required)

1.9.5 Hazardous Material List

Prepared in accordance with paragraph Hazardous Material and the JEGS (if required). Safety Data Sheets (SDS): SDS in English for all Hazardous materials to be used in accordance with the JEGS (if required).

List shall include all pesticides and herbicides that will be brought onto the Installation.

1.9.6 Protection of the Environment from Waste Derived from Contractor Operations

Control and disposal of solid and sanitary waste. Control and disposal of hazardous waste.

Include a Hazardous Waste Management and Disposal Plan prepared in accordance with paragraph Control and Management of Hazardous Waste and the JEGS (if required). This item consists of the management procedures for hazardous waste to be generated. The elements of those procedures will coincide with the Installation Hazardous Waste Management Plan. The Contracting Officer will provide a copy of the Installation Hazardous Waste Management Plan. As a minimum, include the following:

- a. List of the types of hazardous wastes expected to be generated
- b. Procedures to ensure a written waste determination is made for appropriate wastes that are to be generated
- c. Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications
- d. Methods and proposed locations for hazardous waste accumulation/storage (that is, in tanks or containers)
- e. Management procedures for storage, labeling, transportation, and disposal of waste (treatment of waste is not allowed unless specifically noted)
- f. Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal Restrictions, applicable Federal, GOJ and prefectural laws and regulations, JEGS, and Installation Hazardous Waste Management Plan
- g. Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar
- h. Used oil management procedures in accordance with the JEGS; Hazardous waste minimization procedures, and Installation Hazardous Waste Management Plan
- i. Plans for the disposal of hazardous waste by permitted facilities; and Procedures to be employed to ensure required employee training records are maintained
- j. Procedures to be employed to ensure required employee training records

are maintained

- k. Hazardous Waste Disposal local permits or licenses for hazardous waste disposal in accordance with the JEGS (if required)
- l. Hazardous Waste Disposal Statement of Agreement. From a treatment, storage, or disposal (TSD) facility that will accept the waste from the Contractor and also a statement from a certified hazardous waste transporter who will transport the waste to the TSD facility in accordance with the JEGS (if required)

1.9.7 Solid Waste Management Plan

Prepared in accordance with the Waste Management Plan requirements in paragraph Section 01 74 19 Solid Waste Management Plan and the JEGS.

1.9.8 Site Specific Spill Prevention Plan

Prepared in accordance with paragraph SPILLS and the JEGS (if required).

1.9.9 Prevention of Releases to the Environment

- a. Procedures to prevent releases to the environment
- b. Notifications in the event of a release to the environment

1.9.10 Regulatory Notification and Permits

List what notifications and permit applications must be made. Some permits require up to 180 days to obtain. Demonstrate that those permits have been obtained or applied for by including copies of applicable environmental permits. The EPP will not be approved until the permits have been obtained.

1.9.11 Air Quality Compliance

1.9.11.1 Haul Route

Include the truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways as a part of the EPP. As a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways.

1.9.11.2 Pollution Generating Equipment

Identify air pollution generating equipment or processes that may generate air emissions and require compliance with the JEGS or GOJ national or prefectural laws and regulations. Determine requirements based on any current JEGS and installation requirements and the impacts of the project. Provide a list of all fixed or mobile equipment, machinery or operations that could generate air emissions during the project to the Installation Environmental Office (Air Program Manager) in the EPP. Air emissions must comply with the JEGS.

1.9.11.3 Stationary Internal Combustion Engines

Identify portable and stationary internal combustion engines that will be supplied, used or serviced in the EPP. Comply with the JEGS and all

applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and EPA emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation.

1.9.11.4 Refrigerants

Identify management practices to ensure that heating, ventilation, and air conditioning (HVAC) work involving refrigerants and ozone depleting substances comply with the JEGS and installation requirements. Technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Japanese equivalent. Any refrigerant reclaimed is the property of the Government, coordinate with the Installation Environmental Office to determine the appropriate turn in location.

1.9.11.5 Air Pollution-engineering Processes

Identify planned air pollution-generating processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used.

1.9.11.6 Compliant Materials

Provide the Government a list of and SDSs for all hazardous materials proposed for use on site. Materials must be compliant with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for emissions including solvent and volatile organic compound contents, and applicable National Emission Standards for Hazardous Air Pollutants requirements. The Government may alter or limit use of specific materials as needed to meet installation permit requirements for emissions.

1.10 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7. A copy of all Licenses and Permits shall be included in the Environmental Protection Plan.

1.11 ENVIRONMENTAL RECORDS BINDER

Maintain on-site a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

1.12 SOLID WASTE MANAGEMENT PERMIT

Include as a part of the Environmental Protection Plan (EPP) written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or

recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

1.12.1 Solid Waste Management Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

PART 2 PRODUCTS

2.1 GREEN PROCUREMENT

The Contractor is required to buy items using recovered, or bio-based, materials listed at the following website:
<http://www.epa.gov/epp/pubs/products/index.htm> provided, however, the cost of buying and transporting these materials does not exceed the cost of buying the listed recovered, or bio-based, materials, and providing that these materials are competitively available and available within reasonable period. A reasonable period is the time required to buy, ship, receive, and use the materials that does not exceed the time allowed for completion of a delivery order. Recovered and bio-based materials include items listed by the EPA in the following categories:

- a. Bio-based.
- b. Construction Products.
- c. Non-Paper Office Products.
- d. Miscellaneous Products.
- e. Transportation Products.
- f. Vehicular Products.
- g. Landscape Products.
- h. Paper and Paper Products.

The Contractor shall provide a Recovered Materials Certification and estimate of Percentage of Recovered Material Content for EPA designated items.

2.1.1 Recycled Paper The Contractor shall use, as a minimum, paper that is Processed Chlorine Free (PCF) and contains at least 30 percent recycled content. All printed or copied products shall be double-sided and on recycled paper.

2.1.2 Energy Star and Energy Efficient Products

All applicable products delivered, used in the performance of the work, furnished for Government use, or specified in the design of a building or work shall be Energy Star and Energy Efficient Products.

2.1.3 Electronic Product Environmental Assessment Tool (EPEAT)

Purchased computers, notebook and monitors shall, as a minimum, meet the Electronic Product Environmental Assessment Tool (EPEAT) Bronze level of certification.

2.1.4 Leadership in Energy and Environmental Design (LEED)

All new construction Contracts or those performing major renovations shall incorporate the Leadership in Energy and Environmental Design (LEED) System guidelines with Silver being the goal, or shall use Comprehensive Assessment System for Building Environmental Efficiency (CASBEE).

PART 3 EXECUTION

3.1 STORMWATER

Obtain authorization in advance from the Installation Environmental Office for any release of contaminated water.

3.1.1 Erosion and Sediment Control Measures

Provide erosion and sediment control measures in accordance with the JEGS and applicable GOJ or local laws and regulations. Preserve vegetation to the maximum extent practicable.

Erosion control inspection reports may be compiled as part of a stormwater pollution prevention plan inspection reports.

3.1.1.1 Erosion Control

Prevent erosion by mulching, Compost Blankets, Geotextiles, temporary slope drains. Stabilize slopes by seeding. Use of hay bales is prohibited.

Any disturbed area with exposed soil that is not being worked on must be seeded or sodded no later than two weeks from the last disturbance regardless of the installation of any other erosion control measures in place. Remove any temporary measures after the area has been stabilized. No "red soil" shall be allowed to leave the site through runoff.

3.1.1.2 Sediment Control Practices

Implement sediment control practices to divert flows from exposed soils, temporarily store flows, or otherwise limit runoff and the discharge of pollutants from exposed areas of the site. Implement sediment control practices prior to soil disturbance and prior to creating areas with concentrated flow, during the construction process to minimize erosion and sediment laden runoff. Best Management Practices (BMPs) may include, but not limited to, vegetation cover, stream bank stabilization, slope stabilization, silt fences, construction of terraces, interceptor channels, sediment traps, inlet and outfall protection, diversion channels, and sedimentation basins. Include the following devices: storm drain inlet protection.

3.1.2 Work Area Limits

Mark the areas that need not be disturbed under this Contract prior to commencing construction activities. Mark or fence isolated areas within

the general work area that are not to be disturbed. Protect monuments and markers before construction operations commence. Where construction operations are to be conducted during darkness, any markers must be visible in the dark. Personnel must be knowledgeable of the purpose for marking and protecting particular objects.

3.1.3 Contractor Facilities and Work Areas

Place field offices, staging areas, stockpile storage, and temporary buildings in areas designated on the drawings or as directed by the Contracting Officer. Move or relocate the Contractor facilities only when approved by the Government. Provide erosion and sediment controls for onsite borrow and spoil areas to prevent sediment from entering nearby waters. Control temporary excavation and embankments for plant or work areas to protect adjacent areas.

3.2 AIR RESOURCES

Equipment operation, activities, or processes will be in accordance with the JEGS and all applicable Federal, GOJ national or prefectural air emission and performance laws and regulations.

3.2.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Office (Air Program Manager) through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input). Product data shall be provided prior to equipment installation.

3.2.2 Burning

Burning is prohibited on the Government premises.

3.2.3 Class I and II ODS Prohibition

Class I and II ODS are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the required recovery cylinders and the appropriate location for turn in of all reclaimed refrigerant. Class I and II ODS as defined and identified herein shall not be used in the performance of this Contract.

The Contractor shall coordinate with the HVAC shop on the quantity of ODS recovered from system for the purpose of recycle, reuse or dilution. In the event the ODS can be recycled, reused or diluted, the entire quantity will be turned over to the Refrigerant, AC and Heating shop.

In cases where the ODS is sufficiently contaminated and cannot be recycled, reused and/or diluted, the Contractor shall dispose of the ODS according to the JEGS and local regulations.

In the event of a release of ODS, base personnel and/or Contractor shall inform Fire and Emergency's Service Branch and Environmental Division.

3.2.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Comply with the JEGS.

3.2.5 Training/Certification Requirements

Heating and air conditioning technicians must be trained to meet requirements in the JEGS. Maintain copies of certifications at the employees' places of business; technicians must carry certification wallet cards, as provided by GOJ national or prefectural laws and regulations.

3.2.6 Dust Control

Keep dust down at all times, including during nonworking periods. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.2.6.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed the JEGS and applicable US Federal, GOJ national or prefectural air pollution standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs. Comply with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.2.6.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Air Program Manager. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with the JEGS, GOJ national or prefectural laws and regulations, and installation requirements.

3.2.7 Odors

Control odors from construction activities. The odors must be in compliance with applicable GOJ national or prefectural laws and regulations and may not constitute a health hazard.

3.3 WATER RESOURCES

Keep construction activities under surveillance, management, and control, and monitor all water areas affected by construction activities in order to prevent pollution of surface and ground waters. Toxic or hazardous chemicals shall not be applied to soil or vegetation when such application may cause contamination. Prevent oily wastes or other hazardous substances from entering the ground, drainage areas, or local bodies of water.

3.4 WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the EPP. Obtain a copy of the installation's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EPP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

3.4.1 Salvage, Reuse and Recycle

Identify anticipated materials and waste for salvage, reuse, and recycling. Describe actions to promote material reuse, resale or recycling. To the extent practicable, all scrap metal must be sent for reuse or recycling and will not be disposed of in a landfill.

Include the name, physical address, and telephone number of the hauler, if transported by a franchised solid waste hauler. Include the destination and, unless exempted, provide a copy of the GOJ, Prefectural, installation or local permit (cover) or license for recycling.

All solid wastes and materials that have been separated for the purpose of recycling shall be stored in such a manner that they do not constitute a fire, health or safety hazard, or provide food or harborage for vectors, and shall be contained or bundled so as to not result in spillage. Containers must be: leak proof, water proof, and vermin proof including sides, seams, tops and bottoms, durable enough to withstand anticipated usage, and stored on a firm, level, well-drained surface).

3.4.2 Nonhazardous Solid Waste Diversion Report

Maintain an inventory of nonhazardous solid waste diversion and disposal of construction and demolition debris. Submit a report to the Installation Environmental Office on the first working day after each fiscal year quarter, starting the first quarter that nonhazardous solid waste has been generated. Include the following in the report:

Construction and Demolition (C&D) Debris Disposed	(____) kilograms or metric tons as appropriate
C&D Debris Recycled	(____) kilograms or metric tons as appropriate
Total C&D Debris Generated	(____) kilograms or metric tons as appropriate
Waste Sent to Waste-To-Energy Incineration Plant (This amount should not be included in the recycled amount)	(____) kilograms or metric tons as appropriate

3.5 WASTE MANAGEMENT AND DISPOSAL

3.5.1 Waste Determination Documentation

Incorporate the Waste Determination Documentation for Contractor-derived wastes to be generated into the Environmental Records Binder. All potentially hazardous solid waste streams that are not subject to a specific exclusion or exemption from the hazardous waste regulations (e.g. scrap metal, domestic sewage) or subject to special rules, (lead-acid batteries and precious metals) must be characterized in accordance with the requirements of the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations. Base waste determination on user knowledge of the processes and materials used, and analytical data when necessary. Consult with the Installation environmental staff for guidance on specific requirements. Attach support documentation to the Waste Determination Documentation. As a minimum, provide Waste Determination Documentation for the following waste (this listing is not inclusive): oil- and latex -based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

3.5.1.1 Sampling and Analysis of Waste

3.5.1.1.1 Waste Sampling

Sample waste in accordance with the JEGS, Installation Hazardous Waste Management Plan, and appropriate Japanese or US EPA testing protocols that meet the purpose of the testing. Clearly mark each sampled drum or container with the Contractor's identification number, and cross reference to the chemical analysis performed.

Excess excavated soil shall become the property of the Contractor and shall be disposed of off base at a properly licensed facility in accordance with the latest version of the JEGS and applicable GOJ federal and local laws and regulations. Any excess soil shall be sampled and tested according to the applicable GOJ federal and local laws and regulations and disposal facility requirements prior to disposal.

3.5.1.1.2 Laboratory Analysis

Follow the analytical procedure and methods in accordance with the JEGS and all applicable GOJ national or prefectural laws and regulations and the Installation Hazardous Waste Management Plan. Provide analytical

results and reports performed in the Environmental Records Binder.

3.5.1.1.3 Analysis Type

Identify hazardous waste by analyzing for the following characteristics: ignitability, corrosivity, reactivity, toxicity based on TCLP results.

3.5.2 Solid Waste Management

3.5.2.1 Solid Waste Management Report

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, Contractor Certification and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, GOJ or prefectural registration number, along with the receiver's delivery and business addresses and telephone numbers.

3.5.2.2 Control and Management of Solid Wastes

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements for solid waste disposal. Solid waste disposal offsite must comply with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Include a copy of all licenses and permits secured in the Environmental Protection Plan

Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.3 Control and Management of Hazardous Waste

Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

3.5.3.1 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris. Provide a documented waste determination for resultant waste streams. Identify, label, handle, store, and dispose of hazardous waste or debris in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements

Manage hazardous waste in accordance with the approved Hazardous Waste

Management Section of the EPP. Store hazardous wastes in approved containers in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Do not bring hazardous waste onto Government property. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in the JEGS. For hazardous wastes spills, verbally notify the Contracting Officer immediately.

3.5.3.2 Hazardous Waste Records

The Contractor shall maintain sampling, analysis, and turn-in records for all hazardous waste generated during the project. These records shall include, but not be limited to: logs of sample locations or container identification data (including time and date of sample collection), analytical results, and quality control data provided by the analytical lab pertaining to the samples analyzed. Copies of this data shall be submitted to MCBB Environmental via the COR after the work is completed.

3.5.3.3 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such as computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements.

3.5.3.4 Disposal Documentation for Hazardous and Regulated Waste

Submit a copy of the applicable local permit(s), manifest(s), or license(s) for transportation, treatment, storage, and disposal of hazardous and regulated waste by permitted facilities in the Environmental Records Binder. Hazardous or toxic waste manifests must be reviewed, signed, and approved by the Contracting Officer before the Contractor may ship waste. To obtain specific disposal instructions, coordinate with the Installation Environmental Office.

3.5.4 Releases/Spills of Oil and Hazardous Substances

3.5.4.1 Response and Notifications

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with the JEGS and Installation plans. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the Fire Department, the Command Duty Officer, the MCBB Environmental Office at DSN 645-3425, and the Contracting Officer. The Contractor shall immediately report all POL or Hazardous Substances spills to the Contracting Officer. Submit verbal and written notifications as required by the JEGS and installation and service component instructions and plans, and local regulations. Provide copies of the written notification and documentation that a verbal notification was made within the timeframes required by the JEGS and installation instructions five working days. Spill response must be in

accordance with the JEGS and installation requirements. Contain and clean up these spills without cost to the Government.

3.5.4.2 Clean Up

Clean up hazardous and non-hazardous waste spills in accordance with the JEGS, local regulations and the installation's spill response procedures. Reimburse the Government for costs incurred including sample analysis materials, clothing, equipment, and labor if the Government will initiate its own spill cleanup procedures, for Contractor-responsible spills, when: Spill cleanup procedures have not begun within one hour of spill discovery/occurrence; or, in the Government's judgment, spill cleanup is inadequate and the spill remains a threat to human health or the environment.

3.5.5 Mercury Materials

Immediately report to the Environmental Office and the Contracting Officer instances of breakage or mercury spillage. Clean mercury spill area to the satisfaction of the Contracting Officer.

Do not recycle a mercury spill cleanup; manage it as a hazardous waste for disposal.

3.5.6 Wastewater

3.5.6.1 Disposal of wastewater must be as specified below.

3.5.6.1.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction-related waste water off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. or by collecting and placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water. The site for the retention pond must be coordinated and approved with the Contracting Officer. The residue left in the pond prior to completion of the project must be removed, tested, and disposed of off-Government property in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations, and installation requirements. Backfill the area to the original grade, top-soiled, and seeded or sodded.

3.5.6.1.2 Surface Discharge

For discharge of ground water. Surface discharge shall be done in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements.

3.5.6.1.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing hydrostatic testing must be discharged into the sanitary sewer with prior approval and notification to the owner of the sanitary sewer system.

3.6 HAZARDOUS MATERIAL MANAGEMENT

Include hazardous material control procedures in the Safety Plan, in accordance with Section 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. Do not bring hazardous material onto Government property that does not directly relate to requirements for the performance of this Contract. Submit an Safety Data Sheet in both English and Japanese and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the installation. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil and latex based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent.

Storage and handling of hazardous materials will adhere to the DoD Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction (DLAI) 4145.11, Army Technical Manual (TM) 38-410, Naval Supply Publication (NAVSUP PUB) 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order (MCO) 4450.12A, "Storage and Handling of Hazardous Materials," January 13, 1999 provide additional guidance on the storage and handling of hazardous materials. The International Maritime Dangerous Goods (IMDG) Code and appropriate DoD and Component instructions provide requirements for international maritime transport of hazardous materials originating from DoD installations. International air shipments of hazardous materials originating from DoD installations are subject to International Civil Aviation Organization Technical Instructions or DoD Component guidance, including Air Force Manual 24-204, (Interservice) TM 38-250, NAVSUP PUB 505, MCO P4030.19J, and DLAI 4145.3, DCMAD1, Ch3.4 (HM24), "Preparing Hazardous Materials for Military Air Shipments," 3 December 2012. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste, in accordance with the 40 CFR 261, JEGS, and installation requirements.

3.7 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

3.8 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos- containing waste in accordance with the JEGS and applicable Federal, GOJ national or prefectural laws and regulations. Refer to Section 02 82 00 ASBESTOS REMEDIATION. Manifest asbestos-containing waste and provide the manifest to the Contracting Officer. Notifications to the Contracting Officer and Installation Air Program Manager are required before starting any asbestos work.

Removal of mastics, necessary to achieve the Contract's objectives, shall be performed in a manner as if the material contains asbestos.

When applicable, a minimum of 14 calendar days prior to the demolition or renovation of a facility that involves removing or disturbing friable ACM,

the Contractor shall prepare a Written Assessment Of Friable Asbestos Disturbance and submit to the COR, who will, in turn submit to the Installation Commander in accordance with the JEGS.

For any previously untested material suspected to contain asbestos and located in areas impacted by the work, notify the Contracting Officer (CO) who will order up to 20 bulk samples to be obtained and analyzed at the Contractor's expense. Samples shall be collected in sufficient amounts such that a follow-on test utilizing TEM or 1000 point count analysis can be performed, should it be necessary. The Contractor shall deliver the sample(s) to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)" and analyzed by PLM. The laboratory shall have a working definition of "Trace" amounts of asbestos, and the laboratory shall report any detectable amount of asbestos in a bulk sample that is less than the PLM Limit of Quantification of 1% as a "Trace" concentration. If PLM does not detect the presence of asbestos (e.g. "non-detect"), the material shall be considered <0.1% asbestos. If PLM analysis detects asbestos in any discernible amount (to include "trace" or "less than 1%"), the material shall be considered >0.1% asbestos unless proven to be non-ACM by the use of quantification methods capable of achieving an analytical sensitivity of less than 0.1%, such as Transmission Electron Microscopy (TEM) or 1000 point counting.

The Contracting Officer (CO) will order, testing by TEM or 1000 point counting, for up to 20 samples, to be obtained at the Contractor's expense and delivered to a laboratory accredited under the National Institute of Standards and Technology (NIST) "National Voluntary Laboratory Accreditation Program (NVLAP)", for those samples with results of < 1% ACM as analyzed by PLM.

Any additional components identified as ACM that have been approved by the CO for removal shall be removed and will be paid for by an equitable adjustment to the Contract price under the CONTRACT CLAUSE titled "changes". Sampling shall be conducted by personnel who have successfully completed the EPA Model Accreditation Plan (MAP) "Building Inspector" training course and is AHERA certified as a "Building Inspector".

3.9 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with the JEGS and all applicable Federal, GOJ national or prefectural laws and regulations and installation requirements. Refer to Section 02 83 00 LEAD REMEDIATION. The Contractor shall assume all paints and coatings that will be disturbed during the course of meeting the contractual requirements, contain detectable levels of lead, requiring the Contractor to incorporate OSHA regulations for worker and occupant protection. The Contractor is to assume that all existing paint encountered in this Contract contain detectable levels of lead, and the OSHA regulations referenced in these specification applies. Any attached lead material survey information is provided for disposal purposes only. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

3.10 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with the JEGS and installation requirements. Contact JED Environmental for the most current version of Section 02 84 16 HANDLING OF LIGHTING BALLASTS AND LAMPS

CONTAINING PCBs AND MERCURY.

3.11 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

If petroleum-contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.12 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform with surrounding contours.

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FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100 Camp Buckner Communications Annex, Okinawa, Japan

ESC Project No. 9187

29 January 2025

HAZARDOUS MATERIALS INSPECTION REPORT



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- C ACM Sampling Chain of Custody & Laboratory Reports
- D LBP Sampling Chain of Custody & Laboratory Reports
- E Inspection Photos
- F Building Inspector and Laboratory Certifications
- G Past Report

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ACRONYMS & ABBREVIATIONS

ACM	Asbestos Containing Material
CFR	Code of Federal Regulations
COC	Chain of Custody
EPA	Environmental Protection Agency
ESC	Environmental Science Corporation
GOJ	Government of Japan
HAZMAT	Hazardous Materials
ICP	Inductively Coupled Plasma
JEGS	Japan Environmental Governing Standards
LBP	Lead Based Paint
LCP	Lead Containing Paint
LLR	Low-Level Radiation
MAP	Model Accreditation Program
NESHAP	National Emission Standard for Asbestos
ODS	Ozone Depleting Substances
PCB	Polychlorinated Biphenyl
PLM	Polarized Light Microscopy
PWL	Paint with Lead
TCLP	Toxicity Characteristic Leaching Procedure
US	United States of America
USFJ	United States Forces Japan



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EXECUTIVE SUMMARY

Environmental Science Corporation (ESC) conducted this Hazardous Materials (HAZMAT) inspection from 31 October to 1 November 2024. The materials surveyed include asbestos; lead-containing paint (LCP); and arsenic/cadmium gypsum board. During this inspection, a total of 82 bulk samples for asbestos, and 34 paint chip samples for lead, cadmium, and chromium analysis were collected. Additionally, this inspection included a review of historical sampling data provided by the Government.

Summary of Hazardous Materials Findings

Regulated Material	Material Description
ACM	Beige Mastic / White Paint on Vinyl Cove Base *
	Gray Plaster / Cementitious Material / White Paint on Interior Concrete Wall *
	White Joint Compound / Off-White Paint on Interior Concrete Wall *
	Gray Cementitious / Gray Paint on Floor Slab *
	Gray Cementitious / Gray Paint on Floor Mortar Repair Seal *
	Gray Cementitious on Concrete Wall *
	Gray Concrete on Ceiling Slab *
	Blue Joint Compound / White Joint Compound / White Paint on Wall Board *
	Gray Concrete / Yellow Adhesive *
LBP	Gray Paint on Metal Frame
	Off-White over Gray Paint on Metal Door Frame
	Off-White Paint on Metal Pipe

ACM: Asbestos Containing Material – LBP: Lead Based Paint

**Material identified in historical survey data. refer to appendix G.*

1. INTRODUCTION

Environmental Science Corporation (ESC) conducted a hazardous materials (HAZMAT) survey at building 100, Camp Buckner Communications Annex, Okinawa, Japan. The objective of the inspection was to verify the location of hazardous materials that are, or likely, to be disturbed during the total renovation on interior spaces at this facility. The following tasks were conducted as part of the HAZMAT Survey:

- Field-verify Asbestos Containing Materials (ACM) and Lead Base Paint (LBP) identified in existing Government data.
- Conduct a HAZMAT survey, bulk sampling, and analysis of suspect ACM and LBP. All samples collected were analyzed at a certified analytical laboratory in California, USA.
- Visual survey to identify the presence of gypsum board that may contain elevated levels of arsenic/cadmium.

2. SITE OVERVIEW

B100 is located on the west side of the camp. A site map illustrating the investigation boundary is provided as Figure 1 on the following page

FIGURE 1: Bldg. 100 Location



3. PAST RECORDS REVIEW

The survey team conducted a review of the architectural as built drawing dated 2014, and a previous HM inspection report dated 2022 (Appendix G), provided by the Government.

The as built drawings did not specifically identify any finishes as asbestos. The 2022 inspection report identified the presence of asbestos containing material (ACM) in concrete wall, floor slab, flooring and cove base mastic, as well as drywall joint compound. The previously identified ACM has been incorporated into this inspection report and will be carried into the design documents.



4. SURVEY & SAMPLING METHODS

4.1. ASBESTOS SURVEY

This asbestos survey was conducted in accordance with United States (U.S.) Environmental Protection Agency (EPA) National Emission Standard for Asbestos (NESHAP, 40 CFR 61, Subpart M.) and in accordance with the United States Forces Japan (USFJ) Japan Environmental Governing Standards (JEGS) (USFJ, 2024). Under the JEGS, ACM is defined as any material containing equal to or greater than 0.1 percent asbestos by weight (this is more stringent criteria than observed in the United States). The quantification of asbestos via Polarized Light Microscopy (PLM) was performed using EPA *Method for the Determination of Asbestos in Bulk Building Materials* (EPA, 1993).

4.1.1. Identification of Homogeneous Areas

Building materials uniform in color and texture found in the facility were grouped as homogeneous materials. Throughout the building, homogeneous materials should be considered to possess the same hazard characteristics, unless specifically noted otherwise in the report, or by supplemental sampling and analysis.

4.1.2. Suspect Asbestos-Containing Material

Suspect ACM is any material suspected as asbestos-containing based on appearance, usage, age of building, etc., but has not been proven conclusively as ACM. Homogeneous areas of suspect ACM were identified, and samples were collected from each homogeneous area, in accordance with regulatory guidance and the professional judgment of a qualified EPA Model Accreditation Program (MAP) building inspector.

4.1.3. Asbestos Sampling Methods/Analysis

During this survey a total of eighty-two (82) bulk samples were collected randomly from within the identified homogeneous areas and sent for analysis. Lab analysis was conducted via PLM following the EPA *Method for the Determination of Asbestos in Bulk Building Materials* (EPA, 1993). Appendix A Table 1 summarizes the homogeneous areas suspected to be asbestos containing materials, their locations, and the bulk samples collected from the homogeneous area. The sample locations are shown in the figures included in Appendix B. Sampling forms, chains-of-custody (COCs), and analytical reports for asbestos samples are included in Appendix C.

4.1.4. Asbestos Containing Building Materials

The JEGS define ACM as material that contains more than 0.1% asbestos (USFJ, 2024). Based on the analysis of samples using PLM, no ACM was identified in the samples collected during this survey. Appendix A Table 2 summarizes the locations, types, and quantities of ACM identified from past reports.

4.2. PAINT WITH LEAD, CADMIUM AND CHROMIUM

Paint with lead (PWL) can be categorized as lead based paint (LBP) or lead containing paint (LCP). LBP is defined as paint or other surface coatings that contain lead equal to or greater than one milligram per square centimeter, more than 0.5 percent lead by weight. Lead-containing paint (LCP) is paint containing detectable concentrations of lead, but not at sufficient



concentration to be classified as LBP. The JEGS define lead containing as paint or other similar surface coating materials containing lead or lead compounds in which the lead content (calculated as lead material) is more than 0.009 percent by weight of the total nonvolatile content of the paint or the weight of the dried paint film (USFJ, 2024). As with Lead, Cadmium and Chromium in paints and coatings are of concern due to the health risks posed to workers or occupants when such paints and coatings are disturbed, as well as their environmental impacts of improper disposal and handling.

4.2.1. Suspect Paints and Coatings

Painted surfaces were categorized into distinct areas of homogeneity, substrate material, building material, distinct paint type, and apparent time of paint application. Based on this grouping, full-depth samples of the paint or coating were collected. Appendix A Table 3 summarizes paints and coatings sampled during the field investigation. The sample locations are shown in Appendix B.

4.2.2. Paint with Lead Sampling Methods/Analysis

During this survey thirty-four (34) paint chip samples were collected in accordance with ASTM E1729-05, *Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination* (ASTM, 2005). Sample locations are shown in Appendix B. Upon receipt, the AIHA accredited laboratory analyzed each sample via EPA Method 6010B, Inductively Coupled Plasma-Atomic Emission Spectrometry. Laboratory reports are included in Appendix D.

4.2.3. Analytical Results

Lab analysis of the paint chip samples collected during this survey identified the presence of LBP in this facility. These materials have been highlighted in blue text. It has also been found that paints and coatings in this facility contain Cadmium and Chromium at detectable levels. Later in this report are recommendations for the safe handling and control of these paints and coatings during renovation of this facility.

4.2.4. LBP Paint as a GOJ Specially Controlled Industrial Waste

It should be noted that GOJ regulatory limit for lead as a specially controlled industrial waste is 0.3 milligrams per liter (mg/L) for waste sludge, acid, and alkali, as determined by toxicity characteristic leaching procedure (TCLP) laboratory analysis. Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on LCP-positive materials (particularly for those materials of high surface area) to determine whether the paint in question should be classified as a specially controlled industrial waste for disposal purposes.

4.3. OTHER POTENTIALLY HAZARDOUS BUILDING MATERIALS

To the extent possible, this inspection included survey for other sources of hazardous building materials and equipment were noted, including potential mercury-containing / potential low-level radiation (LLR) sources, mercury containing lighting, mold, ozone depleting substances (ODS), polychlorinated biphenyl (PCBs), and radon.



4.3.1. Arsenic Containing Gypsum Board

Some types of Japanese manufactured gypsum board have been documented as having elevated concentrations of leachable cadmium or arsenic. To the extent possible, the survey team inspected gypsum boards for the presence of stamps indicating arsenic gypsum board. Inspectors were limited to small portions above ceilings where access hatches were found.

ESC visually inspected the area above the ceilings for gypsum board markings. The team identified "Tiger Board," manufactured by the Onahama Yoshino Gypsum Co.; Ltd. The JIS Registration mark was identified as JIS A6901 – TC 0808003. This specific material has not been identified by the Government of Japan (GOJ) Ministry of Health and Welfare, Environmental Health Division, Notification No. 35, dated 13 June 1997, as gypsum board known to contain elevated levels of arsenic/cadmium. However, during construction phase, it is recommended that any board removed be inspected to confirm the lot number prior to disposal.

4.3.2. Low Level Radiation Sources

Smoke detectors were identified on the building. In total thirty-five (35) detectors were identified. During the visual inspection, there were no markings available to determine if these units contained a source of Low-Level Radiation (LLR).

4.3.3. Mercury

According to the Japan Lighting Manufacturers Association, straight tube fluorescent lamps manufactured in Japan and having a part number starting with "F," contain mercury. Of those lamps inspected during the initial site visit, each had a part number which began with "F." Mercury containing light bulbs were identified in fluorescent lighting in B100. It is estimated that 200 bulbs are likely found in this facility. Additionally, there is LED lighting throughout the facility. These bulbs are not suspected or known to contain mercury gas.

4.3.4. Mold

Visual confirmation of the existence of mold was identified in the B100. However, it was found in very limited areas. Specifically, the inspection team found mold on the ceiling air vent in the kitchen. Upon inspection it was identified that the mold is associated with condensation caused by temperature differences.

In Room 113 of the server area and the kitchen at B100, the team identified water staining on the ceilings. This was also observed to be a result of condensation on mechanical piping above the ceiling. There was no mold growth associated with this staining; however, it is likely that mold growth will occur should condensation continue to be an issue.

4.3.5. Ozone Depleting Substances

There were no Ozone Depleting Substances identified on site. The current chiller units at B100 contain R-410A, R-134A and R-32.

4.3.6. Polychlorinated Biphenyls

Polychlorinated Biphenyls (PCBs) are synthetic chemicals that were manufactured for use in various industrial and commercial applications – including oil in electrical and hydraulic equipment, and plasticizers in paints, plastics, and rubber products – because of their



nonflammability, chemical stability, high boiling point, and electrical insulation properties. Until the late 1970s, PCBs were commonly used as insulators in electrical equipment because they have high tolerance to heat, do not burn easily, and are non-explosive. In 1979, the EPA banned the manufacture and sales of PCBs.

PCBs are suspected in light ballasts that will be removed during the construction of B100. During the visual inspection, ESC documented approximately 250 lighting ballasts. The inspection team confirmed that the existing lighting fixtures are of Chinese manufacture.

5. GENERAL CONCLUSIONS & RECOMMENDATIONS

5.1. ASBESTOS CONTAINING MATERIALS

Materials confirmed and suspected as ACM have been identified in B100. Further action with respect to asbestos will be required when this material is to be disturbed or removed prior to renovation or demolition of the building. Those building materials identified as “assumed” in the tables at appendix A will need to be sampled and analyzed or assumed ACM and handled appropriately.

During construction, employees involved in renovation or demolition activities that disturb ACM must conduct work in accordance with the most current JEGS; 29 Code of Federal Regulations (CFR) 1926.1101 OSHA Asbestos Construction Standard; and applicable GOJ regulations. Work practices that would trigger these requirements include, but are not limited to, repair, maintenance, or renovation of structures containing asbestos, as well as removal or encapsulation of materials containing asbestos. For each project, the contractor must determine, in accordance with contract specifications, the appropriate safety measures based on the area to be disturbed, the type, volume, and condition of ACM.

5.2. LEAD CONTAINING PAINT

LBP and paints Containing Cadmium and Chromium are present in B100. The condition of the paint was generally found to be intact. Worker and environmental protection precautions, including but not limited to proper training, engineering controls, and monitoring, are required when this paint is disturbed. Should any lead-based paint be found to be peeling or in otherwise poor condition, it should be properly removed. Surrounding areas, including exposed soil, must be protected from any activities that disturb paint with lead.

Since much of the LBP was identified on doors, door frames, and handrails, the recommended means of removal is component replacement. This means removing the coated item intact without the need to abate/remove or otherwise disturb the coating. These steel items can then be disposed of as scrap metal for recycling via smelting.

Should removal or abatement become necessary, removed paints and coatings must be further tested in the context of local hazardous waste regulations to evaluate disposal options for the painted building material off-installation. Specially controlled waste criteria for lead concentration in demolition materials for disposal in Japan is 0.3 mg/L, tested by TCLP. If a material contains more than 0.3 mg/L of lead content, it must undergo stricter handling and disposal requirements in accordance with the Japan Waste Management and Public Cleansing



Law (GOJ, 1970, updated 2001). Although TCLP testing was not part of the scope of this survey, it is recommended that TCLP analysis be performed on lead-positive materials to determine whether the paint in question can be disqualified as a specially controlled industrial waste for disposal.

Employees involved in renovation or demolition activities that disturb lead paints must conduct work in accordance with JEGS, 29 CFR 1926.62 OSHA Lead Construction Standard, and applicable GOJ regulations. Work practices that would trigger these requirements include, but are not limited to, sanding, blasting, welding, cutting, or scraping. For each project, the contractor must determine the appropriate safety measures based on the area to be disturbed, the lead concentration, and the paint condition.

5.3. ARSENIC BOARD

During removal of gypsum board, the board must be checked for manufacturer markings typically found on the reverse side of the board. If boards are of lots known to contain arsenic or cadmium, they must be segregated and transported to an approved disposal facility in accordance with GOJ requirements.

5.4. LOW LEVEL RADIATION (LLR) SOURCES

During the construction phase, an additional survey of the smoke detectors found in the facility will be required to determine the presence of LLR sources. Generally, there is no cost impact associated with this action. Contractors will handle and turn in LLR equipment to the Government for disposal.

5.5. MOLD

Mold was noted on limited surfaces in the facility. During any post demolition survey, it is recommended that additional measures be taken to visually inspect areas behind walls.

5.6. PCBs

During removal of gypsum board, the board must be checked for manufacturer markings typically found on the reverse side of the board. If boards are of lots known to contain arsenic or cadmium, they must be segregated and transported to an approved disposal facility in accordance with GOJ requirements.



6. LIMITATIONS

This survey was conducted to identify potentially hazardous building materials, in accordance with the contractual scope of work. No attempt was made to identify whether materials in or around the building that were not part of, nor affixed to, the building structure included potentially hazardous materials. Any potentially hazardous substance present onsite that were not building materials were not evaluated and should be properly disposed of in accordance with applicable regulations.

Since the building is currently in use, destructive sampling that may compromise the integrity of structure was not conducted during this survey. This includes, but is not limited to, removing portions of walls to inspect the cavities, sampling inside of ducts, or other equipment, etc.

While ESC made every effort to locate and sample all suspect materials for lead and asbestos, there are limitations inherent to physical inspections, especially for materials located behind walls, under multiple layers of flooring, above ceilings, or plenum spaces, or inside of piping or equipment, etc. It is possible that materials may exist in the surveyed buildings that have not been identified or evaluated. Therefore, while ESC has performed this survey in accordance with the standard of care typical to the trade, ESC cannot confirm that no other potentially hazardous materials are present within the building or space surveyed. No other warranty expressed or implied is extended. This report should therefore not be construed as a final statement that all hazardous materials have been identified.

Should any potentially hazardous building materials be encountered which were not accessible or evaluated in this survey, they should be assumed hazardous and not be disturbed prior to evaluation by a licensed and competent inspector.

This survey was conducted to inspect the indicated building(s) or portions thereof, as indicated, for the presence of asbestos and lead in paint. Nothing in this report should be construed as recommending for or against any response or other activities or as endorsing any construction methods, techniques, procedures, or safety methods.



7. REFERENCES

40 CFR Part 745, Lead-Based Paint Poisoning Prevention in Certain Residential Structures.

ASTM, 2005. Standard Practice for Field Collection of Dried Paint Samples for Subsequent Lead Determination.

Envirotemp FR3 Fluid Testing Guide, 2008. Section R900-20-12 Reference Document, April 2008 (Supersedes July 2007).

Government of Japan, 2001. Japan Waste Management and Public Cleansing Law of 1970, updated 2001.

US EPA, Method 6010B, Inductively Coupled Plasma-Atomic Emission Spectrometry. December 1996.

US EPA, 1993. Method for the Determination of Asbestos in Bulk Building Materials

USFJ. Japan Environmental Governing Standards (JEGS). April 2024.



APPENDIX A: SUMMARY TABLES

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Appendix A, Table 1 Summary of Suspect Asbestos-Containing Materials

HA No	Description	HA Location	Friable	Type	Condition	Analysis Date	Sample ID	Result	Remarks
1	Gypsum Wall Board, Painted Off-White	Server Area (#106 to #113, Break Room, Kitchen), Admin Area (#101 to #110, #114 to #117)	No	Misc.	Good	11/20/2024	BUK-B100-A1	ND	
						11/20/2024	BUK-B100-A3	ND	
						11/20/2024	BUK-B100-A13	ND	
						11/20/2024	BUK-B100-A24	ND	
						11/20/2024	BUK-B100-A31	ND	
2	Yellow Insulation Dampeners on Duct	#118 (Mech Room)	Yes	TSI	Good	11/20/2024	BUK-B100-A49	ND	
						11/20/2024	BUK-B100-A50	ND	
3	Concrete Wall Coating	Server Area (#106 to #109, Break Room, Kitchen) Admin Area (#101, #102, #105 to #110)	No	Surface	Good	11/20/2024	BUK-B100-A4	ND	
						11/20/2024	BUK-B100-A41	ND	
4	Concrete Wall	Server Area (#106 to #109, Break Room, Kitchen) Admin Area (#101, #102, #105 to #110)	No	Misc.	Good	11/20/2024	BUK-B100-A5	ND	
						11/20/2024	BUK-B100-A42	ND	
5	Gypsum Ceiling Board, Painted Off-White	Server Area (#107)	No	Misc.	Good	11/20/2024	BUK-B100-A6	ND	
6	Yellow Coating on Floor Slab	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A7	ND	
						11/20/2024	BUK-B100-A18	ND	
7	Floor Slab Coating	Server Area (#108, #109)	No	Surface	Good	11/20/2024	BUK-B100-A8	ND	
						11/20/2024	BUK-B100-A16	ND	
						11/20/2024	BUK-B100-A19	ND	
8	Yellow Coating on Concrete Wall	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A9	ND	
						11/20/2024	BUK-B100-A20	ND	
9	Concrete Wall Coating	Server Area (#108, #109)	No	Surface	Good	11/20/2024	BUK-B100-A10	ND	
						11/20/2024	BUK-B100-A21	ND	
10	White Caulking around Pipe	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A11	ND	
						11/20/2024	BUK-B100-A12	ND	
11	Hard Wall Board	Server Area (#108)	No	Misc.	Good	11/20/2024	BUK-B100-A14	ND	
						11/20/2024	BUK-B100-A15	ND	
12	Floor Slab	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A17	ND	
						11/20/2024	BUK-B100-A23	ND	
13	Concrete Wall	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A22	ND	
						11/20/2024	BUK-B100-A27	ND	
14	Concrete Wall Coating	Mech Room	No	Surface	Good	11/20/2024	BUK-B100-A2	ND	
						11/20/2024	BUK-B100-A26	ND	
15	Concrete Wall	Mech Room	No	Misc.	Good	11/20/2024	BUK-B100-A25	ND	
						11/20/2024	BUK-B100-A28	ND	

HA No	Description	HA Location	Friable	Type	Condition	Analysis Date	Sample ID	Result	Remarks
16	Tan Mastic on Black Foam	Server Area (#108, #109)	No	TSI	Good	11/20/2024	BUK-B100-A29	ND	
						11/20/2024	BUK-B100-A30	ND	
17	Ceiling Slab Coating	Server Area (#108, #109)	No	Surface	Good	11/20/2024	BUK-B100-A32	ND	
						11/20/2024	BUK-B100-A34	ND	
18	Ceiling Slab	Server Area (#108, #109)	No	Misc.	Good	11/20/2024	BUK-B100-A33	ND	
						11/20/2024	BUK-B100-A35	ND	
19	Yellow Mastic on Black Vinyl Cove Base	Server Area (#108 to #113)	No	Misc.	Good	11/20/2024	BUK-B100-A36	ND	
						11/20/2024	BUK-B100-A37	ND	
20	Black Coating on Concrete Wall	Admin Area (#102, #106 to #109)	No	Misc.	Good	11/20/2024	BUK-B100-A38	ND	
						11/20/2024	BUK-B100-A39	ND	
						11/20/2024	BUK-B100-A40	ND	
21	Yellow Insulation under Silver Wrap on 2/3inch Plumbing Pipe, Elbow	#118 (Mech Room)	Yes	TSI	Good	11/20/2024	BUK-B100-A43	ND	
						11/20/2024	BUK-B100-A44	ND	
						11/20/2024	BUK-B100-A45	ND	
22	Yellow Insulation under Silver Wrap on 2/3inch Plumbing Pipe, Straight	#118 (Mech Room)	Yes	TSI	Good	11/20/2024	BUK-B100-A46	ND	
						11/20/2024	BUK-B100-A47	ND	
						11/20/2024	BUK-B100-A48	ND	
23	Ext. Concrete Wall Coating, Painted Off-White	Exterior	No	Surface	Good	11/20/2024	BUK-B100-A51	ND	
						11/20/2024	BUK-B100-A52	ND	
						11/20/2024	BUK-B100-A53	ND	
24	Ext. Concrete Wall	Exterior	No	Misc.	Good	11/20/2024	BUK-B100-A54	ND	
						11/20/2024	BUK-B100-A55	ND	
						11/20/2024	BUK-B100-A56	ND	
25	Ext. Siding Board on Concrete Wall, Painted Off-White	Exterior	No	Misc.	Good	11/20/2024	BUK-B100-A57	ND	
						11/20/2024	BUK-B100-A59	ND	
26	Ext. Concrete Wall	Exterior	No	Misc.	Good	11/20/2024	BUK-B100-A58	ND	
						11/20/2024	BUK-B100-A60	ND	
27	Ext. Siding Board on Concrete Eave, Painted Beige	Exterior	No	Misc.	Good	11/20/2024	BUK-B100-A61	ND	
						11/20/2024	BUK-B100-A63	ND	
28	Ext. Concrete Eave	Exterior	No	Misc.	Good	11/20/2024	BUK-B100-A62	ND	
						11/20/2024	BUK-B100-A64	ND	
29	Concrete Pad	Exterior Condenser	No	Misc.	Good	11/20/2024	BUK-B100-A65	ND	
						11/20/2024	BUK-B100-A66	ND	
30	Concrete Pad	Exterior Condenser	No	Misc.	Good	11/20/2024	BUK-B100-A67	ND	
						11/20/2024	BUK-B100-A68	ND	
31	Concrete Pad	Exterior Condenser	No	Misc.	Good	11/20/2024	BUK-B100-A69	ND	
						11/20/2024	BUK-B100-A70	ND	

HA No	Description	HA Location	Friable	Type	Condition	Analysis Date	Sample ID	Result	Remarks
32	Concrete Wall	Electrical Room	No	Misc.	Good	11/20/2024	BUK-B100-A71	ND	
						11/20/2024	BUK-B100-A73	ND	
33	Concrete Wall Coating, Painted White	Electrical Room	No	Surface	Good	11/20/2024	BUK-B100-A72	ND	
						11/20/2024	BUK-B100-A74	ND	
34	Concrete Base Coating	Electrical Room	No	Surface	Good	11/20/2024	BUK-B100-A75	ND	
						11/20/2024	BUK-B100-A77	ND	
35	Concrete Base	Electrical Room	No	Misc.	Good	11/20/2024	BUK-B100-A76	ND	
						11/20/2024	BUK-B100-A78	ND	
36	Floor Slab Coating	Mech Room	No	Surface	Good	11/20/2024	BUK-B100-A79	ND	
						11/20/2024	BUK-B100-A81	ND	
37	Floor Slab	Mech Room	No	Misc.	Good	11/20/2024	BUK-B100-A80	ND	
						11/20/2024	BUK-B100-A82	ND	

Appendix A, Table 2 Consolidated Summary of Asbestos-Containing Materials

HA No	Description	HA Location	Quantities	Type	Condition	Analysis Date	Lab ID	Sample ID	Result	Remarks
38	Vinyl Cove Base w/Mastic	Rm 101, Rm 105	55 LM	Misc	Good	5/24/2022	202236065	B100-A-50	Chrysotile (2%) in Beige Mastic / white paint	Assumed ACM; material identified on as past reports, see Appendix G
						5/24/2022	202236066	B100-A-51	Chrysotile (2%) in Beige Mastic (limited) / white paint	
						5/24/2022	202236087	B100-A-80	Chrysotile (2%) in Beige Mastic	
						5/24/2022	202236088	B100-A-81	Chrysotile (2%) in Beige Mastic	
						5/24/2022	202236089	B100-A-82	Chrysotile (2%) in Beige Mastic	
39	Interior Concrete Wall	Elec Room	172 SM	Misc	Good	5/24/2022	202236124	B100-A-120	Chrysotile (3%) in Gray plaster / white paint	Assumed ACM; material identified on as past reports, see Appendix G
						5/24/2022	202236126	B100-A-122	Chrysotile (1%) in Gray plaster/cementitious material / white paint	
40	Interior Concrete Wall	Mech Room	148 SM	Misc	Good	5/24/2022	202236141	B100-A-142	Chrysotile (1%) in White joint compound (limited) / off-white paint	Assumed ACM; material identified on as past reports, see Appendix G
41	Floor Slab	Mech Room	83 SM	Misc	Good	5/24/2022	202236148	B100-A-152	Chrysotile (1%) in Gray cementitious material	Assumed ACM; material identified on as past reports, see Appendix G
						5/24/2022	202236149	B100-A-153	Chrysotile (1%) in Gray cementitious material (1) / gray paint	
42	Floor Mortar Repair Seal	Mech Room	1 SM	Misc	Good	5/24/2022	202236152	B100-A-156	Chrysotile (2%) in Gray cementitious material (1) / gray paint	Assumed ACM; material identified on as past reports, see Appendix G
43	Concrete Wall	Exterior	770 SM	Misc	Good	5/24/2022	202236216	B100-A-227	Chrysotile (5%) in Gray cementitious material	Assumed ACM; material identified on as past reports, see Appendix G
44	Ceiling Slab	Server Rm	820 SM	Misc	Good	5/24/2022	202236284	B100-A-508	Chrysotile (2%) in Gray concrete	Assumed ACM; material identified on as past reports, see Appendix G
45	Wallboard	Server Rm	710 SM	Misc	Good	5/24/2022	202236286	B100-A-511	Chrysotile (1%) in Blue joint compound	Assumed ACM; material identified on as past reports, see Appendix G
						5/24/2022	202236289	B100-A-514	Chrysotile (1%) in White joint compound / white paint	
46	Floor Slab	Server Rm	820 SM	Misc	Good	5/24/2022	202236298	B100-A-523	Chrysotile (2%) in Gray concrete / yellow adhesive	Assumed ACM; material identified on as past reports, see Appendix G



Appendix A, Table 3 Summary of Metals Paint Chip Samples

Sample Description	Condition	Sample ID	Analysis Date	Analyte	Results (mg/kg)
Gray Paint on Metal Panel	Intact	BUK-B100-L1	11/22/2024	Cadmium	13.6
				Chromium	3480
				Pb (Lead)	2730
Gray Paint on Metal Frame	Intact	BUK-B100-L2	11/22/2024	Cadmium	5.5
				Chromium	42.3
				Pb (Lead)	7380
Off-White Paint on Board wall	Intact	BUK-B100-L3	11/22/2024	Cadmium	0.09
				Chromium	3.7
				Pb (Lead)	27.8
Off-white Paint on Metal Pipe	Intact	BUK-B100-L4	11/22/2024	Cadmium	0.8
				Chromium	12.5
				Pb (Lead)	5.1
Off-White Paint on Hard Board Wall	Intact	BUK-B100-L5	11/22/2024	Cadmium	ND
				Chromium	5.6
				Pb (Lead)	0.5
Off-white Paint on Metal Pane	Intact	BUK-B100-L6	11/22/2024	Cadmium	0.3
				Chromium	3.1
				Pb (Lead)	1.8
Blue Paint on Metal Cover	Intact	BUK-B100-L7	11/22/2024	Cadmium	ND
				Chromium	3.2
				Pb (Lead)	2.6
Dark Gray Paint on Metal Panel	Intact	BUK-B100-L8	11/22/2024	Cadmium	1.1
				Chromium	26.7
				Pb (Lead)	32.4
Off-White over Gray Paint on Metal Door Frame	Intact	BUK-B100-L9	11/22/2024	Cadmium	0.6
				Chromium	221
				Pb (Lead)	726
Blue over White over Orange Paint on Metal Door Frame	Intact	BUK-B100-L10	11/22/2024	Cadmium	0.9
				Chromium	90.3
				Pb (Lead)	552
Off-White over Gray Paint on Metal Door Frame	Intact	BUK-B100-L11	11/22/2024	Cadmium	0.7
				Chromium	4.4
				Pb (Lead)	30100



Sample Description	Condition	Sample ID	Analysis Date	Analyte	Results (mg/kg)
Red Paint on Metal Pipe	Intact	BUK-B100-L12	11/22/2024	Cadmium	1.0
				Chromium	26.5
				Pb (Lead)	652
Off-White Paint on Metal Pipe	Intact	BUK-B100-L13	11/22/2024	Cadmium	3.6
				Chromium	1.8
				Pb (Lead)	13900
Off-white Paint over on Concrete Wall	Intact	BUK-B100-L14	11/22/2024	Cadmium	0.4
				Chromium	2.2
				Pb (Lead)	11.9
Off-White Paint on Plastic Pipe	Intact	BUK-B100-L15	11/22/2024	Cadmium	ND
				Chromium	3.4
				Pb (Lead)	56.1
Beige over White Paint on Concrete Eave	Intact	BUK-B100-L16	11/22/2024	Cadmium	0.5
				Chromium	0.9
				Pb (Lead)	5.2
Off-White Paint on Siding Wall	Intact	BUK-B100-L17	11/22/2024	Cadmium	0.6
				Chromium	11.2
				Pb (Lead)	8.4
Beige over Blue Paint on Siding Wall	Intact	BUK-B100-L18	11/22/2024	Cadmium	0.1
				Chromium	3.3
				Pb (Lead)	2.7
Silver Paint on Metal Pipe Support	Intact	BUK-B100-L19	11/22/2024	Cadmium	0.1
				Chromium	4110
				Pb (Lead)	2.8
Red Paint on Metal Pipe	Intact	BUK-B100-L20	11/22/2024	Cadmium	ND
				Chromium	6.5
				Pb (Lead)	0.3
Light Gray over Gray Paint on Metal Frame	Damaged	BUK-B100-L21	11/22/2024	Cadmium	13.6
				Chromium	1240
				Pb (Lead)	904
Yellow over Brown Paint on Metal Pole	Damaged	BUK-B100-L22	11/22/2024	Cadmium	0.9
				Chromium	22.9
				Pb (Lead)	193

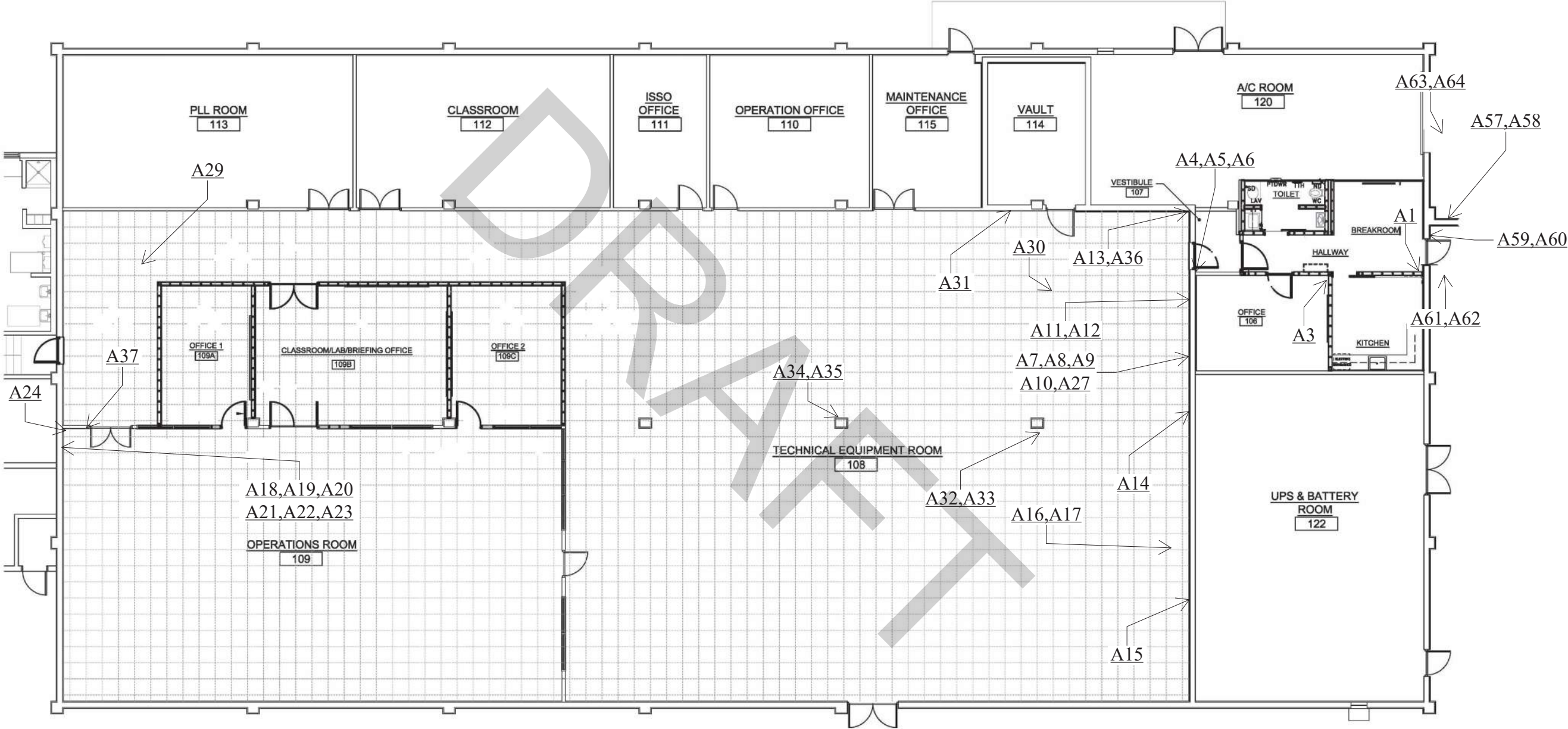


Sample Description	Condition	Sample ID	Analysis Date	Analyte	Results (mg/kg)
Silver and Gray Paint on Metal Pipe	Damaged	BUK-B100-L23	11/22/2024	Cadmium	0.2
				Chromium	9.1
				Pb (Lead)	16.5
Red Paint on Metal Pipe	Damaged	BUK-B100-L24	11/22/2024	Cadmium	1.2
				Chromium	843
				Pb (Lead)	4940
Off-White over Beige Paint on Concrete Wall	Intact	BUK-B100-L25	11/22/2024	Cadmium	0.1
				Chromium	13.2
				Pb (Lead)	30.5
Gray over Beige Paint on Concrete Wall	Intact	BUK-B100-L26	11/22/2024	Cadmium	0.08
				Chromium	5.8
				Pb (Lead)	36.1
White Paint on Concrete Wall	Intact	BUK-B100-L27	11/22/2024	Cadmium	0.7
				Chromium	49.3
				Pb (Lead)	19.0
Gray over Green Paint on Metal Frame	Intact	BUK-B100-L28	11/22/2024	Cadmium	106
				Chromium	2430
				Pb (Lead)	1790
Off-White Paint on Metal Duct	Intact	BUK-B100-L29	11/22/2024	Cadmium	0.5
				Chromium	20.7
				Pb (Lead)	33.6
Gray Paint on Metal Pipe	Intact	BUK-B100-L30	11/22/2024	Cadmium	0.6
				Chromium	3920
				Pb (Lead)	2060
Gray over Black Paint on Metal Support	Intact	BUK-B100-L31	11/22/2024	Cadmium	13.9
				Chromium	45.1
				Pb (Lead)	2120
Gray Paint on Metal Pipe	Intact	BUK-B100-L32	11/22/2024	Cadmium	46.1
				Chromium	128
				Pb (Lead)	2740
Gray over Dark Gray Paint on Metal Pipe	Damaged	BUK-B100-L33	11/22/2024	Cadmium	4.8
				Chromium	8.3
				Pb (Lead)	312
Off-white Paint on Metal Panel	Intact	BUK-B100-L34	11/22/2024	Cadmium	2.7
				Chromium	505
				Pb (Lead)	23.3



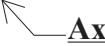
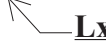
APPENDIX B: ASBESTOS / LEAD PAINT SAMPLING LOCATIONS

DRAFT



SOLICITATION NO.: TBD

LEGEND & NOTES

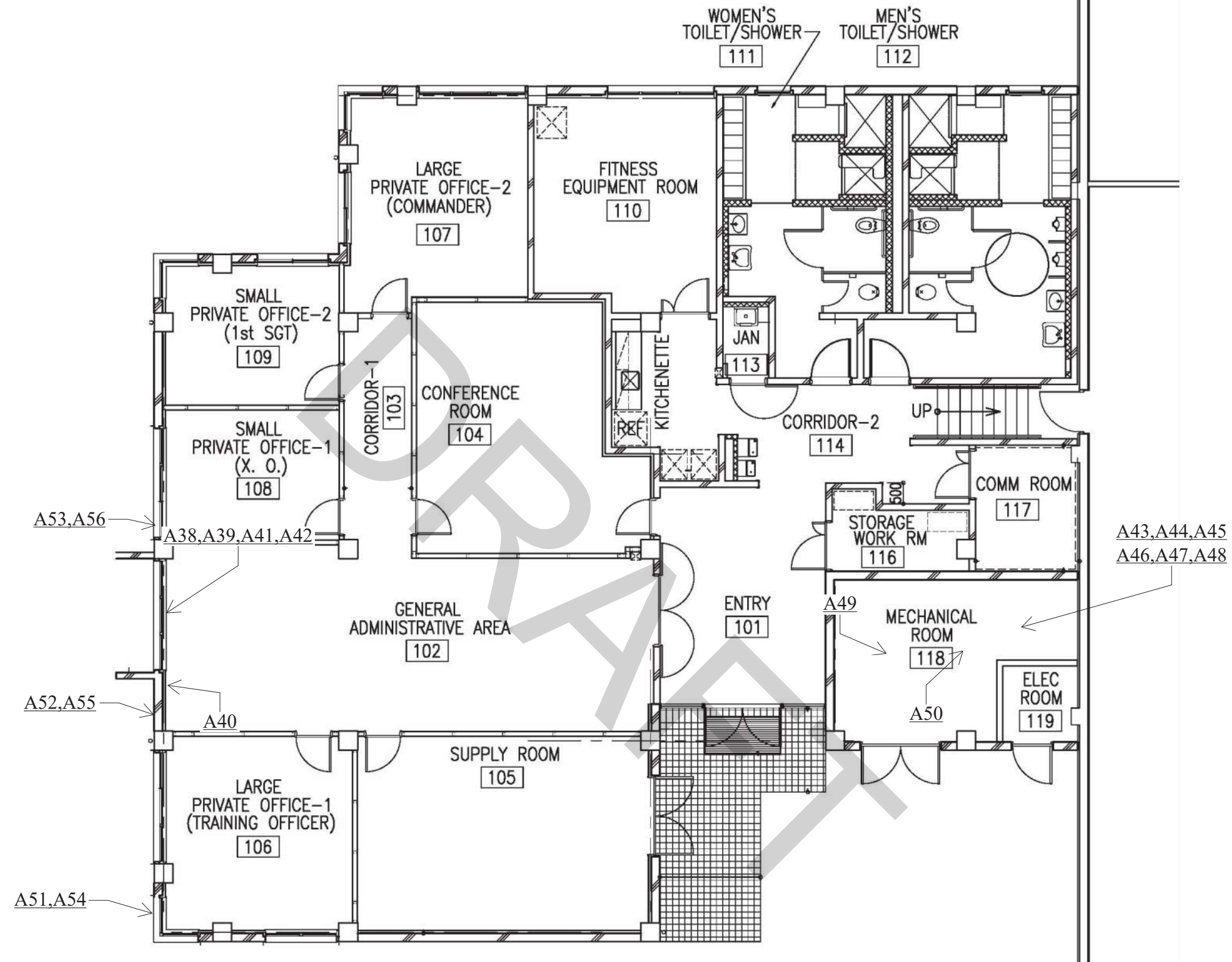
BULK ASBESTOS SAMPLE		LEAD SAMPLE	
NONE DETECT (ND)	ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%	LEAD BASED PAINT (LBP)	PAINT WITH LEAD (PWL) LEAD CONTENT < 0.5%
			
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.			

DWG TITLE: SAMPLE LOCATION MAP SERVER AREA

PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN

ESC PROJECT NO.: 9187



SOLICITATION NO.: TBD

LEGEND & NOTES

BULK ASBESTOS SAMPLE
NONE DETECT (ND)

ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%

← Ax

← Ax

LEAD SAMPLE

LEAD BASED PAINT (LBP)

PAINT WITH LEAD (PWL)
LEAD CONTENT < 0.5%

← Lx

← Lx

— ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.

DWG TITLE: SAMPLE LOCATION MAP ADMIN AREA

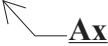
PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN

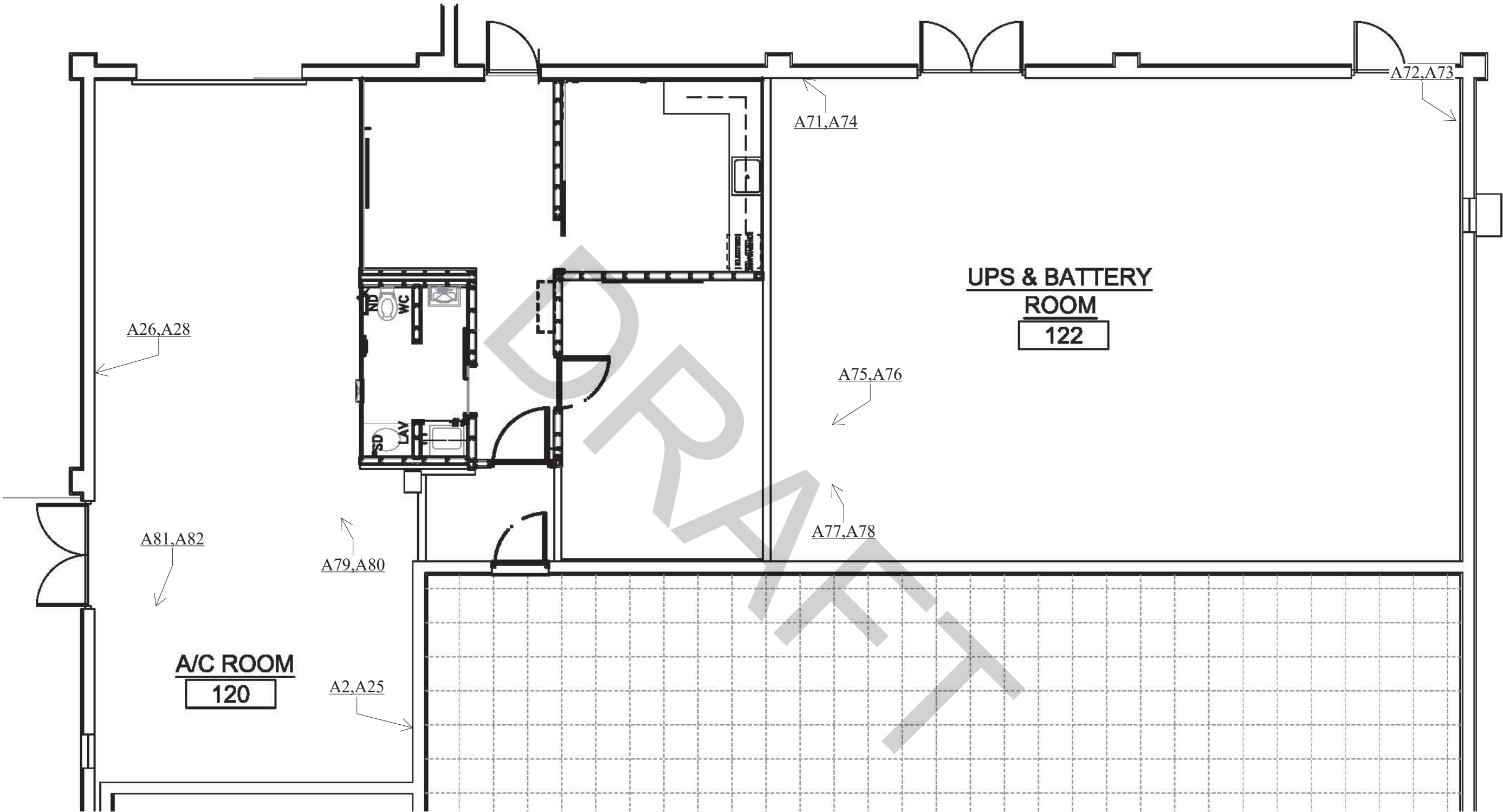
ESC PROJECT NO.: 9187



SOLICITATION NO.: TBD

LEGEND & NOTES			
BULK ASBESTOS SAMPLE	ASBESTOS CONTAINING MATERIAL (ACM)	LEAD SAMPLE	PAINT WITH LEAD (PWL)
NONE DETECT (ND)	TRACE OR >0.1%	LEAD BASED PAINT (LBP)	LEAD CONTENT < 0.5%
 Ax	 Ax	 Lx	 Lx
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.			

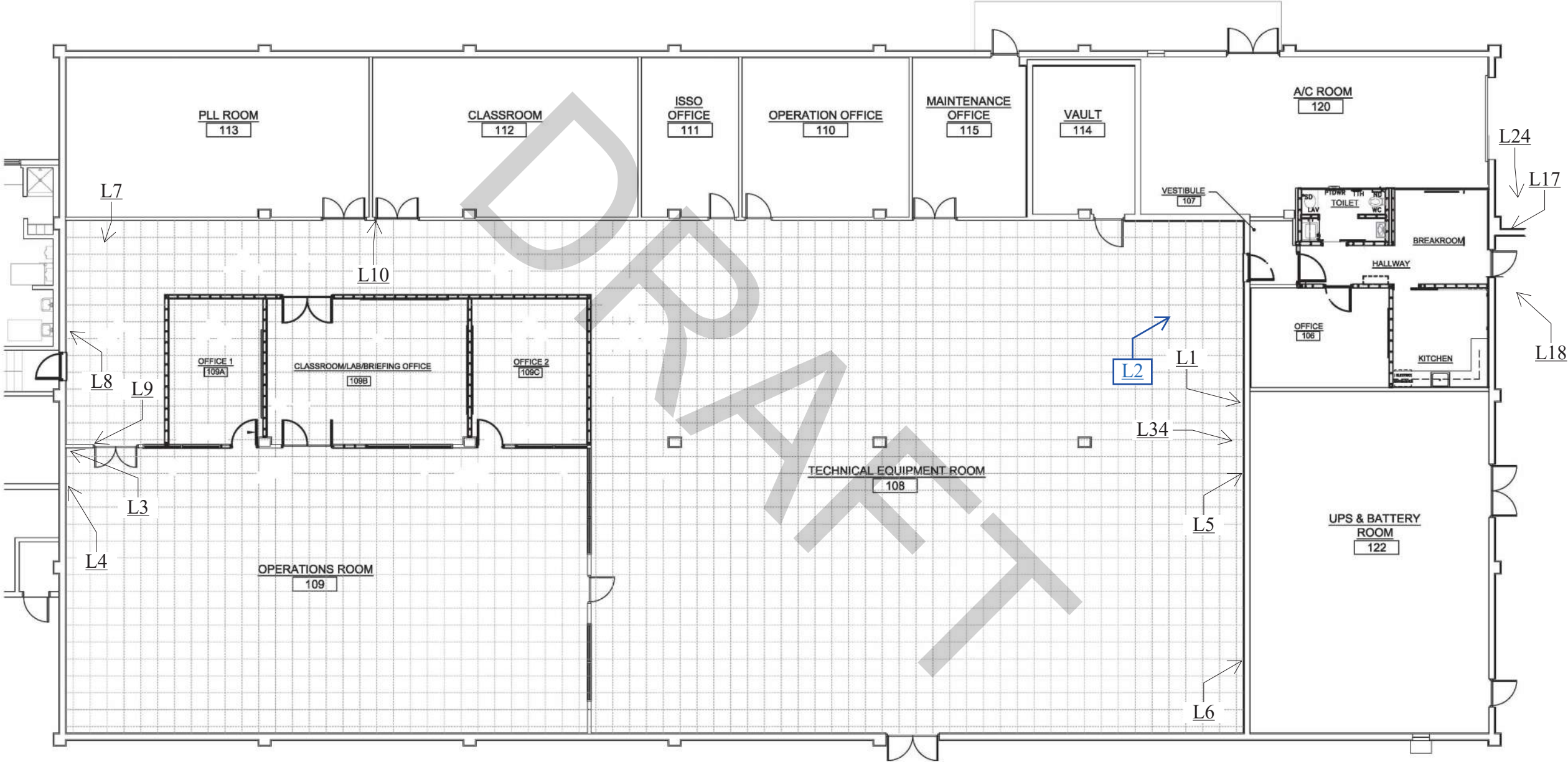
DWG TITLE: SAMPLE LOCATION MAP AHU AREA
PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ESC PROJECT NO.: 9187
BACKCHECK SUBMITTAL



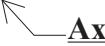
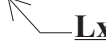
SOLICITATION NO.: TBD

LEGEND & NOTES			
BULK ASBESTOS SAMPLE		LEAD SAMPLE	
NONE DETECT (ND)	ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%	LEAD BASED PAINT (LBP)	PAINT WITH LEAD (PWL) LEAD CONTENT < 0.5%
Ax	Ax	Lx	Lx
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.			

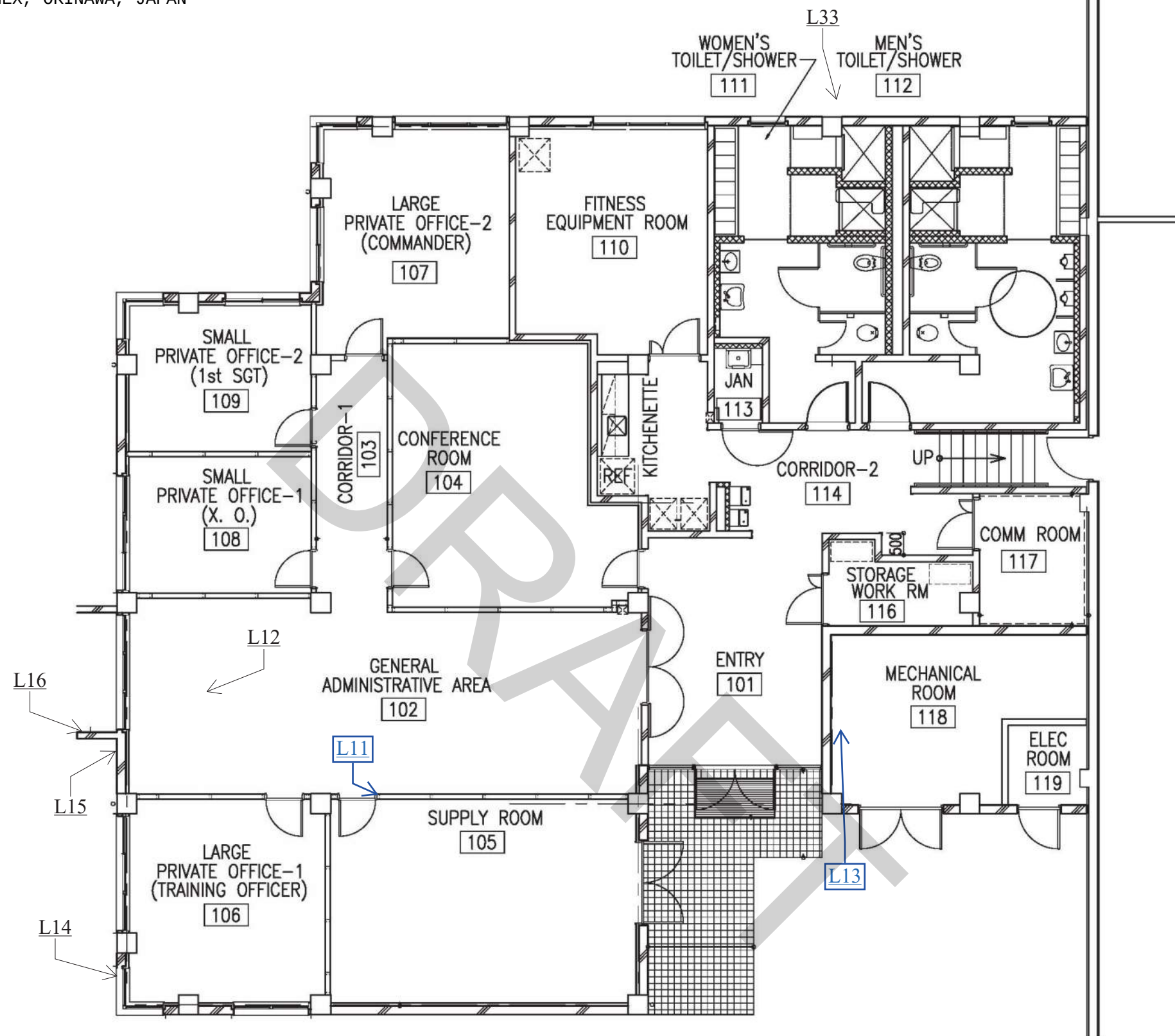
DWG TITLE: SAMPLE LOCATION MAP MECH RM
PROJECT TITLE: FY 2024 PROJECT RM 24-2702 REPAIR UH BLDG. 1721
LOCATION: YOKOSUKA, JAPAN
ESC PROJECT NO.: 9183
BACKCHECK SUBMITTAL



SOLICITATION NO.: TBD

LEGEND & NOTES			
BULK ASBESTOS SAMPLE		LEAD SAMPLE	
NONE DETECT (ND)	ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%	LEAD BASED PAINT (LBP)	PAINT WITH LEAD (PWL) LEAD CONTENT < 0.5%
 Ax	 Ax	 Lx	 Lx
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.			

DWG TITLE: SAMPLE LOCATION MAP SERVER AREA
PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ESC PROJECT NO.: 9187
BACKCHECK SUBMITTAL



SOLICITATION NO.: TBD

LEGEND & NOTES

BULK ASBESTOS SAMPLE
NONE DETECT (ND)

ASBESTOS CONTAINING MATERIAL
(ACM) TRACE OR >0.1%

— Ax

— Ax

LEAD SAMPLE

LEAD BASED PAINT (LBP)

PAINT WITH LEAD (PWL)
LEAD CONTENT < 0.5%

— Lx

— Lx

— ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.

DWG TITLE: SAMPLE LOCATION MAP ADMIN AREA

PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN

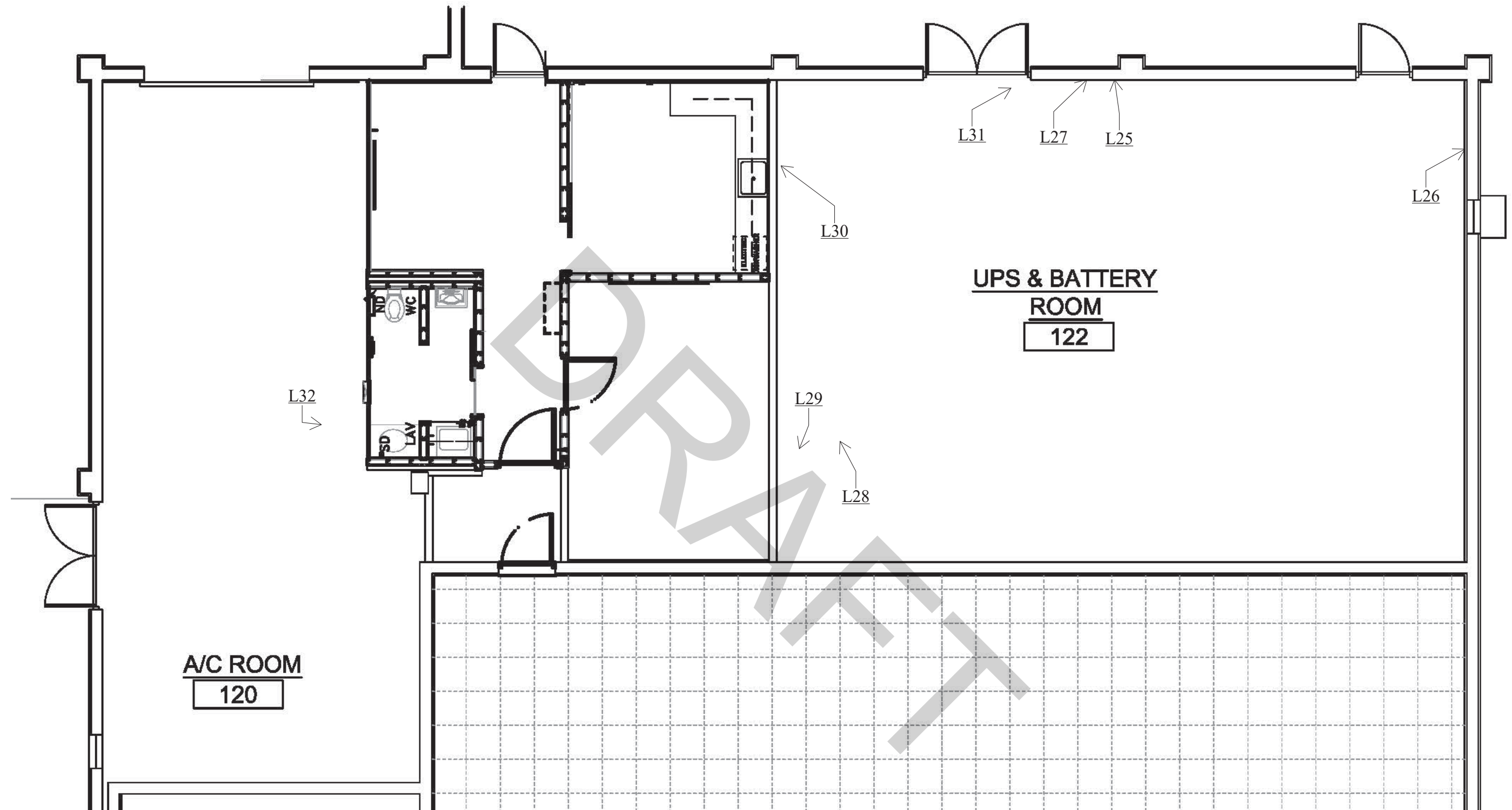
ESC PROJECT NO.: 9187



SOLICITATION NO.: TBD

LEGEND & NOTES			
BULK ASBESTOS SAMPLE NONE DETECT (ND)	ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%	LEAD SAMPLE LEAD BASED PAINT (LBP)	PAINT WITH LEAD (PWL) LEAD CONTENT < 0.5%
 Ax	 Ax	 Lx	 Lx
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.			

DWG TITLE: SAMPLE LOCATION MAP AHU AREA
PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ESC PROJECT NO.: 9187
BACKCHECK SUBMITTAL



SOLICITATION NO.: TBD

LEGEND & NOTES

BULK ASBESTOS SAMPLE		LEAD SAMPLE	
NONE DETECT (ND)	ASBESTOS CONTAINING MATERIAL (ACM) TRACE OR >0.1%	LEAD BASED PAINT (LBP)	PAINT WITH LEAD (PWL) LEAD CONTENT < 0.5%
 Ax	 Ax	 Lx	 Lx

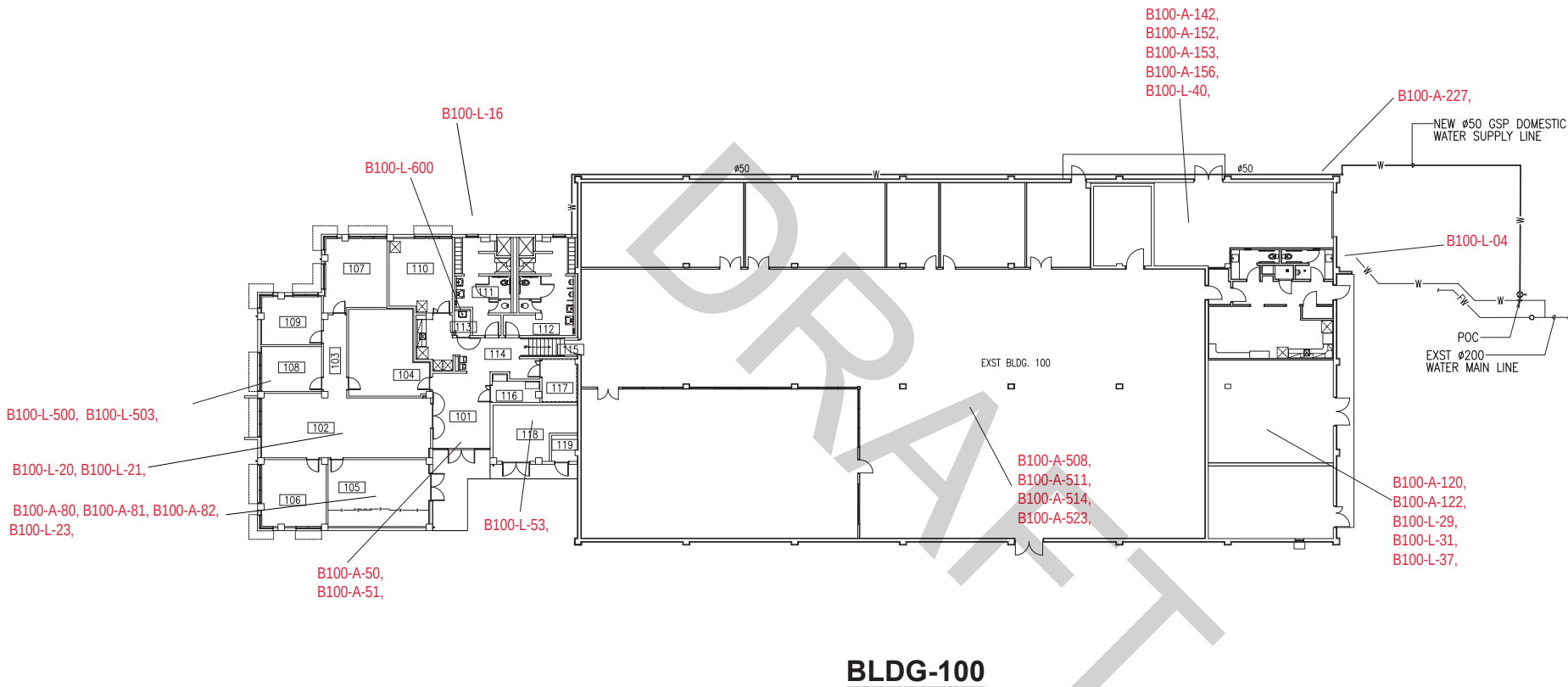
- ALL PAINTS AND COATINGS SHALL BE ASSUMED TO CONTAIN DETECTABLE LEVELS OF LEAD.

DWG TITLE: SAMPLE LOCATION MAP MECH RM

PROJECT TITLE: FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

LOCATION: CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN

ESC PROJECT NO.: 9187



BLDG-100

Bulk samples identified as ACM

B100-A-50	Vinyl Cove Base w/Mastic (Rm 101)	Chrysotile (2%)
B100-A-51	Vinyl Cove Base w/Mastic (Rm 101)	Chrysotile (2%)
B100-A-80	Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-81	Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-82	Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-120	Interior Concrete Wall (Elec Rm)	Chrysotile (3%)
B100-A-122	Interior Concrete Wall (Elec Rm)	Chrysotile (1%)
B100-A-142	Interior Concrete Wall (Mech)	Chrysotile (1%)
B100-A-152	Floor Slab (Mech)	Chrysotile (1%)
B100-A-153	Floor Slab (Mech)	Chrysotile (1%)
B100-A-156	Floor Mortar Repair Seal (Mech)	Chrysotile (2%)
B100-A-227	Concrete Wall (Exterior)	Chrysotile (5%)
B100-A-508	Ceiling Slab (Server Rm)	Chrysotile (2%)
B100-A-511	Wall Board (Server Rm)	Chrysotile (1%)
B100-A-514	Wall Board (Server Rm)	Chrysotile (1%)
B100-A-523	Floor Slab (Server Rm)	Chrysotile (2%)

Paint samples identified as LBP

B100-L-04	Pipe Paint (Red) (Exterior)	13000 mg/kg*
B100-L-16	Conduit Paint (Gray) (Exterior)	5100 mg/kg*
B100-L-20	Steel Door/Frame Paint (Tan) (Rm 102)	15000 mg/kg*
B100-L-21	Wood Door/Frame Paint (Ivory) (Rm 102)	12000 mg/kg*
B100-L-23	Wood Door/Frame Paint (Ivory) (Rm 105)	22000 mg/kg*
B100-L-29	Conduit Paint (Gray) (Elect Rm)	7800 mg/kg*
B100-L-29	Conduit Paint (Gray) (Elect Rm)	7800 mg/kg*
B100-L-31	Pipe Paint (Red) (Elect Rm)	25000 mg/kg*
B100-L-37	Duct Paint (Gray) (Elect Rm)	11000 mg/kg*
B100-L-40	Conduit Paint (Gray) (Mech)	5700 mg/kg*
B100-L-53	Pipe Paint (White) (Rm 118)	33000 mg/kg*
B100-L-500	Elect Box Paint (Gray) (Rm108)	10000 mg/kg*
B100-L-503	Frame Paint (Gray) (Rm108)	7900 mg/kg*
B100-L-600	Steel Door/ Frame Paint (Ivory) (Janitor)	8000 mg/kg*

* The definition of LBP with EPA is that it is greater than or equal to 0.5% (5,000 mg/kg) by weight



APPENDIX C: ACM SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS

DRAFT



ESC Project No: 9187

Date: November 4, 2024

Project Title: FY25 Repair HVAC Wideband SATCOM
Ops B100, Camp Buckner Communications
Annex, Okinawa, Japan

Location: B100, Camp Buckner, Okinawa

Material Description: 1		Gypsum Board Wall, Painted Off-White		Friable Non-friable	
Sample No.	Location				
BUK-B100-A1	Break Room				
BUK-B100-A3	106				
BUK-B100-A13	108				
BUK-B100-A24	109				
BUK-B100-A31	108				
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING ☐ Significant Damage ☐ Damage ☐ Minimal Damage					

Material Description: 2		Yellow Fibrous Dampeners on Duct		Friable Non-friable	
Sample No.	Location				
BUK-B100-A49	118				
BUK-B100-A50	118				
CONDITION: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING ☐ Significant Damage ☐ Damage ☐ Minimal Damage					

Sampled By: Shoma Yasutomi	Relinquished By/Date/Time: Shoma Yasutomi / November 4, 2024 / 16:00	Relinquished By/Date/Time:
Delivered to Lab By: JAPAN POST	Received By/Date/Time: Valmire Contreras 11-11-2024 3:29pm	Received By/Date/Time:

TURNAROUND TIME: ☐ < 12 Hours ☐ 24 Hours ☐ 3 Days ☒ 5 Days ☐

Surfacing	<1,000 ft ² = 3 Samples	1,000 – 5,000 ft ² = 5 Samples	>5,000 ft ² = 7 Samples
TSI	Minimum of 3 Samples UNLESS....	<6 in. or ft ² = 1 Sample	Minimum of 2 Samples (Cement/plaster valves, elbows & 'T')
Misc. Non-Friable	Minimum of 2 Samples (AHERA)		
Misc. Friable	Minimum of 2 Samples		

Surfacing	Sig. Damage = > 10% Dist. or 25% Local	Damaged = < 10% Dist. or 25% Local	Good = Very Limited Damage
TSI	Sig. Damage = >10% Missing Jacket OR >10% Dist. or 25% Local	Damaged = < 10% Missing Jacket OR < 10% Dist. or 25% Local	Good = Very Limited Damage
Misc.	Sig. Damage = > 10% Dist. or 25% Local	Damaged = < 10% Dist. or 25% Local	Good = Very Limited Damage



PLM Chain of Custody

ESC Project No: 9187

Page 2 of 12

Material Description: 3		Concrete Wall Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A4		107			
BUK-B100-A41		102			
CONDITION:		% Localized:	% Distributed:	Total Material Quantity:	
Surfacing Material				Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential		☐ High	☐ Moderate	☐ Low	
Vibration Potential		☐ High	☐ Moderate	☐ Low	
Air Erosion		☐ High	☐ Moderate	☐ Low	
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 4		Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A5		107			
BUK-B100-A42		102			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential		☐ High	☐ Moderate	☒ Low	
Vibration Potential		☐ High	☐ Moderate	☒ Low	
Air Erosion		☐ High	☐ Moderate	☒ Low	
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☒ Minimal Damage	

Material Description: 5		Gypsum ceiling board, Painted Off-White		Friable Non-friable	
Sample No.		Location			
BUK-B100-A6		107			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential		☐ High	☐ Moderate	☐ Low	
Vibration Potential		☐ High	☐ Moderate	☐ Low	
Air Erosion		☐ High	☐ Moderate	☐ Low	
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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BY: Sammie Contreras



ESC Project No: 9187

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Material Description: 6		Clear and Yellow Coating on Floor Slab		Friable Non-friable	
Sample No.		Location			
BUK-B100-A7		108			
BUK-B100-A18		109			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☒ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☒ Low		
Vibration Potential	☐ High	☐ Moderate	☒ Low		
Air Erosion	☐ High	☐ Moderate	☒ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☒ Minimal Damage	

Material Description: 7		Floor Slab Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A8		108			
BUK-B100-A19		109			
BUK-B100-A16		108			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 8		Clear and Yellow Coating on Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A9		108			
BUK-B100-A20		109			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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ESC Project No: 9187

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Material Description: 9		Concrete Wall Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A10		108			
BUK-B100-A21		109			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING ☐ Significant Damage ☐ Damage ☐ Minimal Damage					

Material Description: 10		White Caiking around Pipe		Friable Non-friable	
Sample No.		Location			
BUK-B100-A11		108			
BUK-B100-A12		108			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING ☐ Significant Damage ☐ Damage ☐ Minimal Damage					

Material Description: 11		Gypsum Hard Board Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A14		108			
BUK-B100-A15		108			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☒ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING ☐ Significant Damage ☐ Damage ☐ Minimal Damage					

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BY: [Signature]



ESC Project No: 9187

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Material Description: 12		Floor Slab		Friable Non-friable	
Sample No.		Location			
BUK-B100-A17		108			
BUK-B100-A23		109			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Damage ☐ Minimal Damage	

Material Description: 13		Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A22		109			
BUK-B100-A27		108			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Damage ☐ Minimal Damage	

Material Description: 14		Concrete Wall Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A2		Mech Room			
BUK-B100-A26		Mech Room			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Damage ☐ Minimal Damage	

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Material Description: 15		Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A25		Mech Room			
BUK-B100-A28		Mech Room			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Minimal Damage	

Material Description: 16		Tan Mastic on Black Form		Friable Non-friable	
Sample No.		Location			
BUK-B100-A29		108			
BUK-B100-A30		108			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Minimal Damage	

Material Description: 17		Ceiling Slab Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A32		108			
BUK-B100-A34		108			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage		☐ Minimal Damage	

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Material Description: 18		Ceiling Slab		Friable Non-friable	
Sample No.		Location			
BUK-B100-A33		108			
BUK-B100-A35		108			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 19		Beige Mastic behind Vinyl Black Cove Base		Friable Non-friable	
Sample No.		Location			
BUK-B100-A36		108			
BUK-B100-A37		108			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 20		Black Coating on Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A38		102			
BUK-B100-A39		102			
BUK-B100-A40					
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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Vermine Control



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Material Description: 21		Yellow Insulation under Silver Wrap on 2/3inch Plumbing duct, Elbow		Friable Non-friable	
Sample No.		Location			
BUK-B100-A43		118			
BUK-B100-A44		118			
BUK-B100-A45		118			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 22		Yellow Insulation under Silver Wrap on 2/3inch Plumbing duct, straight		Friable Non-friable	
Sample No.		Location			
BUK-B100-A46		118			
BUK-B100-A47		118			
BUK-B100-A48		118			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 23		Ext. Concrete Wall Coating, Painted Off-White		Friable Non-friable	
Sample No.		Location			
BUK-B100-A51		Exterior			
BUK-B100-A52		Exterior			
BUK-B100-A53		Exterior			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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Material Description: 24		Ext. Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A54		Exterior			
BUK-B100-A55		Exterior			
BUK-B100-A56		Exterior			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 25		Ext. Siding Board on Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A57		Exterior			
BUK-B100-A59		Exterior			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 26		Ext. Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A58		Exterior			
BUK-B100-A60		Exterior			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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BY: Garminne Contreras



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Material Description: 27		Ext. Siding Board on Concrete Eave		Friable Non-friable	
Sample No.		Location			
BUK-B100-A61		Exterior			
BUK-B100-A63		Exterior			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 28		Ext. Concrete Eave		Friable Non-friable	
Sample No.		Location			
BUK-B100-A62		Exterior			
BUK-B100-A64		Exterior			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 29		Concrete Base		Friable Non-friable	
Sample No.		Location			
BUK-B100-A65		Exterior Condenser			
BUK-B100-A66		Exterior Condenser			
CONDITION: % Damaged: % Localized: % Distributed: Total Material Quantity:					
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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Material Description: 30		Concrete Base		Friable Non-friable	
Sample No.		Location			
BUK-B100-A67		Exterior Condenser			
BUK-B100-A68		Exterior Condenser			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 31		Concrete Base		Friable Non-friable	
Sample No.		Location			
BUK-B100-A69		Exterior Condenser			
BUK-B100-A70		Exterior Condenser			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 32		Concrete Wall		Friable Non-friable	
Sample No.		Location			
BUK-B100-A71		Electrical Room			
BUK-B100-A73		Electrical Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 33		Concrete Wall Coating Painted Off-White		Friable Non-friable	
Sample No.		Location			
BUK-B100-A72		Electrical Room			
BUK-B100-A74		Electrical Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

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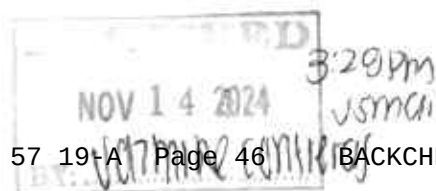
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Material Description: 34		Concrete Base Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A75		Electrical Room			
BUK-B100-A77		Electrical Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 35		Concrete Base		Friable Non-friable	
Sample No.		Location			
BUK-B100-A76		Electrical Room			
BUK-B100-A78		Electrical Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 36		Floor Slab Coating		Friable Non-friable	
Sample No.		Location			
BUK-B100-A79		Mech Room			
BUK-B100-A81		Mech Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	

Material Description: 37		Floor Slab		Friable Non-friable	
Sample No.		Location			
BUK-B100-A80		Mech Room			
BUK-B100-A82		Mech Room			
CONDITION:		% Damaged:	% Localized:	% Distributed:	Total Material Quantity:
Surfacing Material		TSI		Misc.	
☐ Sig. Damage	% Crumbling -	☐ Sig. Damage	% Gouge/Punct -	☐ Sig. Damage	% Crumbling -
☐ Damaged	% Delaminating -	☐ Damaged	% Crushed -	☐ Damaged	% Delaminating -
☐ Good Cond.	% H ₂ O/Gouges -	☐ Good Cond.	% H ₂ O Stains -	☐ Good Cond.	% H ₂ O/Gouges -
Contact Potential	☐ High	☐ Moderate	☐ Low		
Vibration Potential	☐ High	☐ Moderate	☐ Low		
Air Erosion	☐ High	☐ Moderate	☐ Low		
OVERALL POTENTIAL RATING		☐ Significant Damage	☐ Damage	☐ Minimal Damage	



Bulk Asbestos Analysis

(EPA Method 40CFR, Part 763, Appendix E to Subpart E and EPA 600/R-93-116, Visual Area Estimation)

Environmental Science Corporation
Shoma Yasutomi
ESC Building
1553-231 Ishikawa Iha
Uruma City, Okinawa, JP 904-1115

Client ID: L1232
Report Number: B365808
Date Received: 11/14/24
Date Analyzed: 11/20/24
Date Printed: 11/20/24
First Reported: 11/20/24

Job ID/Site: 9187; FY25 Repair HVAC Wideband SATCOM Ops B100, Camp Buckner
Communications Annex, Okinawa, Japan
Date(s) Collected: 11/04/2024

SGSFL Job ID: L1232
Total Samples Submitted: 82
Total Samples Analyzed: 82

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A1	51799413						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (10 %)	Fibrous Glass (Trace)						
BUK-B100-A3	51799414						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (5 %)	Fibrous Glass (1 %)						
BUK-B100-A13	51799415						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (5 %)	Fibrous Glass (1 %)						
BUK-B100-A24	51799416						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (3 %)	Fibrous Glass (Trace)						
BUK-B100-A31	51799417						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (20 %)							
BUK-B100-A49	51799418						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (95 %)							

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A50	51799419						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (95 %)							
BUK-B100-A4	51799420						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A41	51799421						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (1 %)							
BUK-B100-A5	51799422						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A42	51799423						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A6	51799424						
Layer: White Paint			ND				
Layer: Brown Fibrous Backing			ND				
Layer: White Drywall			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (5 %) Fibrous Glass (Trace)							
BUK-B100-A7	51799425						
Layer: Yellow Coating			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A18	51799426						
Layer: Yellow Coating			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A8	51799427						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A19	51799428						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A16	51799429						
Layer: Grey Cementitious Material			ND				
Layer: Yellow Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A9	51799430						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A20	51799431						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A10	51799432						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BULK-B100-A-21	51799433						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A11	51799434						
Layer: White Caulk			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A12	51799435						
Layer: White Caulk			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A14	51799436						
Layer: Grey Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (30 %) Fibrous Glass (2 %)							
BUK-B100-A15	51799437						
Layer: Grey Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (25 %) Fibrous Glass (2 %)							

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A17	51799438						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A23	51799439						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A22	51799440						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A27	51799441						
Layer: Grey Cementitious Material			ND				
Layer: Beige Paint			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A2	51799442						
Layer: Grey Cementitious Material			ND				
Layer: Beige Paint			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A26	51799443						
Layer: Grey Cementitious Material			ND				
Layer: Beige Paint			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A25	51799444						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A28	51799445						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A29	51799446						
Layer: Black Foam			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Comment: Hint of yellow discoloration on edges.							

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A30	51799447						
Layer: Black Foam			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
Comment: Hint of yellow discoloration on edges.							
BUK-B100-A32	51799448						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A34	51799449						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A33	51799450						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A35	51799451						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A36	51799452						
Layer: Black Baseboard			ND				
Layer: Yellow Mastic			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A37	51799453						
Layer: Black Baseboard			ND				
Layer: Yellow Mastic			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A38	51799454						
Layer: Black Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A39	51799455						
Layer: Black Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A40	51799456						
Layer: Black Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A43	51799457						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (95 %)							
BUK-B100-A44	51799458						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (90 %)							
BUK-B100-A45	51799459						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (90 %)							
BUK-B100-A46	51799460						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (95 %)							
BUK-B100-A47	51799461						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (90 %)							
BUK-B100-A48	51799462						
Layer: Silver Foil			ND				
Layer: Yellow Fibrous Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Fibrous Glass (90 %)							
BUK-B100-A51	51799463						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A52	51799464						
Layer: Beige Paint			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A53	51799465						
Layer: Beige Paint			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A54	51799466						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A55	51799467						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A56	51799468						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A57	51799469						
Layer: Beige Paint			ND				
Layer: Tan Non-Fibrous Material			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A59	51799470						
Layer: Beige Paint			ND				
Layer: Tan Non-Fibrous Material			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A58	51799471						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A60	51799472						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A61	51799473						
Layer: Beige Paint			ND				
Layer: Tan Non-Fibrous Material			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A63	51799474						
Layer: Beige Paint			ND				
Layer: Tan Non-Fibrous Material			ND				
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A62	51799475						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A64	51799476						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A65	51799477						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A66	51799478						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A67	51799479						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A68	51799480						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A69	51799481						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A70	51799482						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A71	51799483						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A73	51799484						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
BUK-B100-A72	51799485						
Layer: Beige Paint			ND				
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A74	51799486						
Layer: Beige Paint			ND				
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A75	51799487						
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A77	51799488						
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A76	51799489						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A78	51799490						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A79	51799491						
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A81	51799492						
Layer: Grey Cementitious Coating			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
BUK-B100-A80	51799493						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					
Cellulose (Trace)							
BUK-B100-A82	51799539						
Layer: Grey Cementitious Material			ND				
Total Composite Values of Fibrous Components:		Asbestos (ND)					

Client Name: Environmental Science Corporation

Report Number: B365808

Date Printed: 11/20/24

Sample ID	Lab Number	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer	Asbestos Type	Percent in Layer
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DRAFT



Karen Buehler, Laboratory Supervisor, Chicago Laboratory

Note: Limit of Quantification ('LOQ') = 1%. 'Trace' denotes the presence of asbestos below the LOQ. 'ND' = 'None Detected'.

Analytical results and reports are generated by SGS Forensic Laboratories (SGSFL) at the request of and for the exclusive use of the person or entity (client) named on such report. Results, reports or copies of same will not be released by SGSFL to any third party without prior written request from client. This report applies only to the sample(s) tested. Supporting laboratory documentation is available upon request. This report must not be reproduced except in full, unless approved by SGSFL. The client is solely responsible for the use and interpretation of test results and reports requested from SGSFL. This report must not be used by the client to claim product endorsement by NVLAP or any other agency of the U.S. Government. SGSFL is not able to assess the degree of hazard resulting from materials analyzed. SGS Forensic Laboratories reserves the right to dispose of all samples after a period of thirty (30) days, according to all state and federal guidelines, unless otherwise specified. All samples were received in acceptable condition unless otherwise noted.



APPENDIX D: LBP SAMPLING CHAIN OF CUSTODY & LABORATORY REPORTS

DRAFT



Subcontract Sample Release Form & Receipt

Sending To: SGS Excelchem
Attn: Becky Miller
1135 W. Sunset Blvd.
Suite A
Rocklin, CA 95765

Phone: 916-543-4445
Fax: 916-543-4449
Email: becky.miller@sgs.com

Date:

From:
Parent RN:

SGS Forensic Laboratories on this day releases custody and control of the following sample(s) presently in our custody:

FALI Job	P.O.	Analysis Requested	Due Date
L1232	U003448	Lead, Cadmium, Chromium by ICP	11/22/2024

PLEASE USE Report U003448 as the Purchase Order (PO).

Special Handling: Send results and invoice only to ENV.Hayward.SUBCONTRACT@sgs.com

SUBMISSION LOCATION (Please circle)

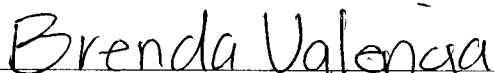
SGS
3777 Depot Road, Suite 409
Hayward, CA 94545

SGS
20535 S. Belshaw Avenue
Carson, CA 90746

SGS
3626 Sunset Rd., Suite 100
Las Vegas, NV 89120

Released By:


(SGS)


(Print Name)


(Date / Time)

Received By:

(Client / Vendor Representative)

(Print Name)

(Date / Time)



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232	PO / Job#: ESC.8249.06	Date: Nov, 2024			
Contact: Shoma Yasutomi		Phone: (098) 989-0426	Turn Around Time: ☐ Some Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day				
E-mail: shoma.yasutomi@esc-pacific.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Ratemeter ☐ PLM: ☐ Stondord / ☐ Paint Count 400 - 1000 / ☐ CARB 435					
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS		☐ TEM Air: ☐ AHERA / ☐ Yomate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Micravoc: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Site Location: Camp Buckner, Okinawa, Japan		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soat ☐ Porticle Identification (TEM LAB) ☐ Special Project					
Comments:		☒ Metals Analysis Matrix: Paint Chip Sam Method: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium ☐ Silico in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Tatol Time	
1. BUK-B100-L1	1 Nov 2024 +	Gray Paint on Metal Panel	A P C				
2. BUK-B100-L2	1 Nov 2024 +	Gray Paint on Metal Frame	A P C				
3. BUK-B100-L3	1 Nov 2024 +	Off-White Paint on Board wall	A P C				
4. BUK-B100-L4	1 Nov 2024 +	Off-white Paint on Metal Pipe	A P C				
5. BUK-B100-L5	1 Nov 2024 +	Off-White Paint on Hard Board Wall	A P C				
6. BUK-B100-L6	1 Nov 2024 +	Off-white Paint on Metal Panel	A P C				
7. BUK-B100-L7	1 Nov 2024 +	Blue Paint on Metal Cover	A P C				
8. BUK-B100-L8	1 Nov 2024 +	Dark Gray Paint on Metal Panel	A P C				
9. BUK-B100-L9	1 Nov 2024 +	Off-White over Gray Paint on Metal Door Frame	A P C				
10. BUK-B100-L10	1 Nov 2024 +	Blue over White over Orange Paint on Metal Door Frame	A P C				
Sampled By: Shoma Yasutomi		Date/Time: 1 Nov 24	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP				
Relinquished By: Shoma Yasutomi		Relinquished By: 4	Relinquished By: Bulat		Date / Time: 11-14-24 14:41		
Date / Time: Nov 5 2024		Date / Time:	Received By: Jermine Chirias		Date / Time: 11-14-2024 14:41pm		
Received By: Japan post		Received By:	Date / Time: 11-14-2024 14:41pm		Date / Time:		
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.

Son Francisco Office: 3777 Depot Road, Suite 409, Hoyward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Cammerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232		PO / Job#: ESC.8249.06		Date: Nov, 2024	
Contact: Shoma Yasutomi		Phone: (098) 989-0426		Turn Around Time: ☐ Some Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day			
E-mail: shoma.yasutomi@esc-pacific.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Ratameter ☐ PLM: ☐ Standard / ☐ Paint Caunt 400 - 1000 / ☐ CARB 435					
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chofield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Micravoc: ☐ Qual / ☐ D5755(str/oreo) / ☐ D5756(str/mass)					
Site Location: Camp Buckner, Okinawa, Japan		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opoques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project ☒ Metals Analysis Matrix: Paint Chip Sam Methad: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium					
Comments:						☐ Silica in Air ☐ w/Grovimetry ☐ Quartz Only	

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L11	1 Nov 2024 +	Off-White over Gray Paint on Metal Door Frame	A P C				
2. BUK-B100-L12	1 Nov 2024 +	Red Paint on Metal Pipe	A P C				
3. BUK-B100-L13	1 Nov 2024 +	Off-White Paint on Metal Pipe	A P C				
4. BUK-B100-L14	1 Nov 2024 +	Off-white Paint over on Concrete Wall	A P C				
5. BUK-B100-L15	1 Nov 2024 +	Off-White Paint on Plastic Pipe	A P C				
6. BUK-B100-L16	1 Nov 2024 +	Beige over White Paint on Concrete Eave	A P C				
7. BUK-B100-L17	1 Nov 2024 +	Off-White Paint on Siding Wall	A P C				
8. BUK-B100-L18	1 Nov 2024 +	Beige over Blue Paint on Sideing Wall	A P C				
9. BUK-B100-L19	1 Nov 2024 +	Silver Paint on Metal Pipe Support	A P C				
10. BUK-B100-L20	1 Nov 2024 +	Red Paint on Metal Pipe	A P C				

Sampled By: Shoma Yasutomi		Date/Time: 1 Nov 24		Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP	
Relinquished By: Shoma Yasutomi		Relinquished By:		Relinquished By: Bu Val	
Date / Time: Nov 5 2024		Date / Time:		Date / Time: 11-14-24 14:41	
Received By: Japan post		Received By: Jamine Contrera		Received By:	
Date / Time: Nov 5 2024		Date / Time: 11-14-24 1:44 pm		Date / Time:	
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No	

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 Las Angeles Office: 2959 Pacific Cammerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Los Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232	PO / Job#: ESC.8249.06	Date: Nov, 2024
Contact: Shoma Yasutomi		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒	
E-mail: shoma.yasutomi@esc-pacific.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotameter ☐ PLM: ☐ Standard / ☐ Paint Count 400 - 1000 / ☐ CARB 435		
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Micravac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)		
Site Location: Camp Buckner, Okinawa, Japan		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Saat ☐ Particle Identification (TEM LAB) ☐ Special Project		
Comments:		☒ Metals Analysis Matrix: Paint Chip Sample Methad: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium ☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only		

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L21	1 Nov 2024 ☒	Light Gray over Gray Paint on Metal Frame	A P C				
2. BUK-B100-L22	1 Nov 2024 ☒	Yellow over Brown Paint on Metal Pole	A P C				
3. BUK-B100-L23	1 Nov 2024 ☒	Silver and Gray Paint on Metal Pipe	A P C				
4. BUK-B100-L24	1 Nov 2024 ☒	Red Paint on Metal Pipe	A P C				
5. BUK-B100-L25	1 Nov 2024 ☒	Off-White over Beige Paint on Concrete Wall	A P C				
6. BUK-B100-L26	1 Nov 2024 ☒	Gray over Beige Paint on Concrete Wall	A P C				
7. BUK-B100-L27	1 Nov 2024 ☒	White Paint on Concrete Wall	A P C				
8. BUK-B100-L28	1 Nov 2024 ☒	Gray over Green Paint on Metal Frame	A P C				
9. BUK-B100-L29	1 Nov 2024 ☒	Off-White Paint on Metal Duct	A P C				
10. BUK-B100-L30	1 Nov 2024 ☒	Gray Paint on Metal Pipe	A P C				

Sampled By: Shoma Yasutomi		Date/Time: 1 Nov 24	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP	
Relinquished By: Shoma Yasutomi		Relinquished By:		Relinquished By: <i>Bren Walker</i>
Date / Time: Nov 5 2024		Date / Time:		Date / Time: 11-14-24 14:41
Received By: Japan post		Received By: <i>Uchiyama Contral</i>		Received By:
Date / Time: Nov 5 2024		Date / Time: 11-14-24 14:41		Date / Time:
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No

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Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040



FORENSIC
LABORATORIES

Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232	PO / Job#: ESC.8249.06	Date: Nov, 2024			
Contact: Shoma Yasutomi		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: shoma.yasutomi@esc-pacific.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotameter ☐ PLM: ☐ Standard / ☐ Paint Count 400 - 1000 / ☐ CARB 435					
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Micravac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Site Location: Camp Buckner, Okinawa, Japan		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Sat ☐ Particle Identification (TEM LAB) ☐ Special Project					
Comments:		☒ Metals Analysis Matrix: Paint Chip Sam Methad: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium ☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L31	1 Nov 2024 +	Gray over Black Paint on Metal Support	A P C				
2. BUK-B100-L32	1 Nov 2024 +	Gray Paint on Metal Pipe	A P C				
3. BUK-B100-L33	1 Nov 2024 +	Gray over Dark Gray Paint on Metal Pipe	A P C				
4. BUK-B100-L34	1 Nov 2024 +	Off-white Paint on Metal Panel	A P C				
5.	1 Nov 2024 +		A P C				
6.	1 Nov 2024 +		A P C				
7.	1 Nov 2024 +		A P C				
8.	1 Nov 2024 +		A P C				
9.	1 Nov 2024 +		A P C				
10.	1 Nov 2024 +		A P C				
Sampled By: Shoma Yasutomi		Date/Time: 1 Nov 24	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP				
Relinquished By: Shoma Yasutomi		Relinquished By:		Relinquished By: <i>Buenk Vabur</i>			
Date / Time: Nov 5 2024		Date / Time:		Date / Time: 11-14-24 1441			
Received By: Japan post		Received By: <i>Yasutomi</i>		Received By:			
Date / Time: Nov 5 2024		Date / Time: 11-14-2024 1441		Date / Time:			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No			

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 Las Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
 Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SGS EXCELCHEM
Laboratories, Inc.

1135 W Sunset Boulevard
Suite A
Rocklin, CA 95765
Phone# 916-543-4445



ELAP Certificate No. : 2119

22 November 2024

Brenda V

SGS Forensic Laboratories

20535 S Belshaw Ave

Carson, CA 90746

RE: L1232

Work order number:2411077

Enclosed are the results of analyses for samples received by the laboratory on 11/15/24 10:00. All Quality Control results are within acceptable limits except where noted as a case narrative. If you have any questions concerning this report, please feel free to contact the laboratory.

Sincerely,

A handwritten signature in grey ink that reads 'Doug Selby'. The signature is written in a cursive, flowing style.

Doug Selby, Technical Director

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

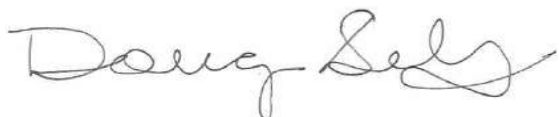
Date Reported:
11/22/24 15:42

ANALYTICAL REPORT FOR SAMPLES

Sample ID	Laboratory ID	Matrix	Date Sampled	Date Received
BUK-B100-L1	2411077-01	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L2	2411077-02	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L3	2411077-03	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L4	2411077-04	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L5	2411077-05	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L6	2411077-06	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L7	2411077-07	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L8	2411077-08	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L9	2411077-09	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L10	2411077-10	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L11	2411077-11	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L12	2411077-12	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L13	2411077-13	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L14	2411077-14	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L15	2411077-15	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L16	2411077-16	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L17	2411077-17	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L18	2411077-18	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L19	2411077-19	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L20	2411077-20	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L21	2411077-21	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L22	2411077-22	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L23	2411077-23	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L24	2411077-24	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L25	2411077-25	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L26	2411077-26	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L27	2411077-27	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L28	2411077-28	PAINT	11/01/24 12:00	11/15/24 10:00

SGS Excelchem Laboratories, Inc.

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Doug Sells, Technical Director

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Page 1 of 39

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

ANALYTICAL REPORT FOR SAMPLES

Sample ID	Laboratory ID	Matrix	Date Sampled	Date Received
BUK-B100-L29	2411077-29	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L30	2411077-30	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L31	2411077-31	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L32	2411077-32	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L33	2411077-33	PAINT	11/01/24 12:00	11/15/24 10:00
BUK-B100-L34	2411077-34	PAINT	11/01/24 12:00	11/15/24 10:00

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L1
2411077-01 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	13.6	9.2	0.7	1	mg/kg	AHK0138	11/20/24	11/20/24	EPA 6010B	
Chromium	3480	9.2	1.0	1	"	"	"	"	"	B
Lead	2730	9.2	1.1	1	"	"	"	"	"	

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L2
2411077-02 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	5.5	1.0	0.08	1	mg/kg	AHK0138	11/20/24	11/20/24	EPA 6010B	
Chromium	42.3	1.0	0.1	1	"	"	"	"	"	B
Lead	7380	1.0	0.1	1	"	"	"	"	"	

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories	Project:	L1232	
20535 S Belshaw Ave	Project Number:	U003448	Date Reported:
Carson, CA 90746	Project Manager:	Brenda V	11/22/24 15:42

BUK-B100-L3
2411077-03 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.09	1.0	0.08	1	mg/kg	AHK0138	11/20/24	11/20/24	EPA 6010B	J
Chromium	3.7	1.0	0.1	1	"	"	"	"	"	B
Lead	27.8	1.0	0.1	1	"	"	"	"	"	

SGS Excelchem Laboratories, Inc.



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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L4
2411077-04 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.8	2.6	0.2	1	mg/kg	AHK0138	11/20/24	11/21/24	EPA 6010B	J
Chromium	12.5	2.6	0.3	1	"	"	"	"	"	B
Lead	5.1	2.6	0.3	1	"	"	"	"	"	

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SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L5
2411077-05 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	ND	2.1	0.2	1	mg/kg	AHK0138	11/20/24	11/21/24	EPA 6010B	U
Chromium	5.6	2.1	0.2	1	"	"	"	"	"	B
Lead	0.5	2.1	0.2	1	"	"	"	"	"	J

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SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L6
2411077-06 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.3	1.4	0.1	1	mg/kg	AHK0138	11/20/24	11/21/24	EPA 6010B	J
Chromium	3.1	1.4	0.2	1	"	"	"	"	"	B
Lead	1.8	1.4	0.2	1	"	"	"	"	"	

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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L7
2411077-07 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	ND	4.4	0.3	1	mg/kg	AHK0138	11/20/24	11/21/24	EPA 6010B	U
Chromium	3.2	4.4	0.5	1	"	"	"	"	"	B, J
Lead	2.6	4.4	0.5	1	"	"	"	"	"	J

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L8
2411077-08 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	1.1	5.5	0.4	1	mg/kg	AHK0138	11/20/24	11/21/24	EPA 6010B	J
Chromium	26.7	5.5	0.6	1	"	"	"	"	"	B
Lead	32.4	5.5	0.6	1	"	"	11/19/24	"	"	

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SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L9
2411077-09 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.6	2.6	0.2	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	221	2.6	0.3	1	"	"	"	"	"	B
Lead	726	2.6	0.3	1	"	"	"	"	"	

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L10
2411077-10 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.9	2.6	0.2	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	90.3	2.6	0.3	1	"	"	"	"	"	B
Lead	552	2.6	0.3	1	"	"	"	"	"	

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SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

**BUK-B100-L11
2411077-11 (PAINT)**

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.7	0.9	0.07	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	4.4	0.9	0.1	1	"	"	"	"	"	B
Lead	30100	9.5	1.1	10	"	"	"	11/22/24	"	R-06

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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L12
2411077-12 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	1.0	11.4	0.9	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	26.5	11.4	1.3	1	"	"	"	"	"	B
Lead	652	11.4	1.3	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L13
2411077-13 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	3.6	2.2	0.2	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	
Chromium	1.8	2.2	0.2	1	"	"	"	"	"	B, J
Lead	13900	2.2	0.3	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L14
2411077-14 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.4	1.0	0.08	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	2.2	1.0	0.1	1	"	"	"	"	"	B
Lead	11.9	1.0	0.1	1	"	"	"	"	"	

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SGS Forensic Laboratories
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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L15
2411077-15 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	ND	3.7	0.3	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	U
Chromium	3.4	3.7	0.4	1	"	"	"	"	"	B, J
Lead	56.1	3.7	0.4	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L16
2411077-16 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.5	1.0	0.08	1	mg/kg	AHK0138	11/19/24	11/20/24	EPA 6010B	J
Chromium	0.9	1.0	0.1	1	"	"	"	"	"	B, J
Lead	5.2	1.0	0.1	1	"	"	"	"	"	

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SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L17
2411077-17 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.6	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	11.2	1.0	0.1	1	"	"	"	"	"	
Lead	8.4	1.0	0.1	1	"	"	"	"	"	

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SGS Forensic Laboratories	Project:	L1232	
20535 S Belshaw Ave	Project Number:	U003448	Date Reported:
Carson, CA 90746	Project Manager:	Brenda V	11/22/24 15:42

BUK-B100-L18
2411077-18 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.1	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	3.3	1.0	0.1	1	"	"	"	"	"	
Lead	2.7	1.0	0.1	1	"	"	"	"	"	

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SGS Forensic Laboratories
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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L19
2411077-19 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.1	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	4110	1.0	0.1	1	"	"	"	"	"	
Lead	2.8	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L20
2411077-20 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	ND	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	U
Chromium	6.5	1.0	0.1	1	"	"	"	"	"	
Lead	0.3	1.0	0.1	1	"	"	"	"	"	J

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L21
2411077-21 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	13.6	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	
Chromium	1240	1.0	0.1	1	"	"	"	"	"	
Lead	904	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L22
2411077-22 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.9	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	22.9	1.0	0.1	1	"	"	"	"	"	
Lead	193	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L23
2411077-23 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.2	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	9.1	1.0	0.1	1	"	"	"	"	"	
Lead	16.5	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L24
2411077-24 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	1.2	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B
Chromium	843	1.0	0.1	1	"	"	"	"	"
Lead	4940	1.0	0.1	1	"	"	"	"	"

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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L25
2411077-25 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.1	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	13.2	1.0	0.1	1	"	"	"	"	"	
Lead	30.5	1.0	0.1	1	"	"	"	"	"	

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SGS Forensic Laboratories
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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L26
2411077-26 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.08	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	5.8	1.0	0.1	1	"	"	"	"	"	
Lead	36.1	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L27
2411077-27 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	0.7	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	49.3	1.0	0.1	1	"	"	"	"	"	
Lead	19.0	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L28
2411077-28 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	106	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B
Chromium	2430	1.0	0.1	1	"	"	"	"	"
Lead	1790	1.0	0.1	1	"	"	"	"	"

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L29
2411077-29 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.5	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	20.7	1.0	0.1	1	"	"	"	"	"	
Lead	33.6	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L30
2411077-30 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
---------	--------	-----------------	-----	----	-------	-------	---------------	---------------	--------	-------

Total Recoverable Metals

Cadmium	0.6	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	J
Chromium	3920	1.0	0.1	1	"	"	"	"	"	
Lead	2060	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L31
2411077-31 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	13.9	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B
Chromium	45.1	1.0	0.1	1	"	"	"	"	"
Lead	2120	1.0	0.1	1	"	"	"	"	"

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L32
2411077-32 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	46.1	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B
Chromium	128	1.0	0.1	1	"	"	"	"	"
Lead	2740	1.0	0.1	1	"	"	"	"	"

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L33
2411077-33 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	4.8	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	
Chromium	8.3	1.0	0.1	1	"	"	"	"	"	
Lead	312	1.0	0.1	1	"	"	"	"	"	

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Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

BUK-B100-L34
2411077-34 (PAINT)

Analyte	Result	Reporting Limit	MDL	DF	Units	Batch	Date Prepared	Date Analyzed	Method	Notes
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Total Recoverable Metals

Cadmium	2.7	1.0	0.08	1	mg/kg	AHK0148	11/20/24	11/21/24	EPA 6010B	
Chromium	505	1.0	0.1	1	"	"	"	"	"	
Lead	23.3	1.0	0.1	1	"	"	"	"	"	

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Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

Total Recoverable Metals - Quality Control

Analyte	Result	Reporting Limit	MDL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes
---------	--------	-----------------	-----	-------	-------------	---------------	------	-------------	-----	-----------	-------

Batch AHK0138 - EPA 6010B

Blank (AHK0138-BLK1)

Prepared & Analyzed: 11/20/24

Cadmium	ND	1.0	0.08	mg/kg							U
Chromium	0.358	1.0	0.1	"							J
Lead	ND	1.0	0.1	"							U

LCS (AHK0138-BS1)

Prepared & Analyzed: 11/20/24

Cadmium	99.2	1.0	0.08	mg/kg	104		95.3	80-120			
Chromium	101	1.0	0.1	"	104		96.5	80-120			
Lead	101	1.0	0.1	"	104		97.1	80-120			

LCS Dup (AHK0138-BS1)

Prepared & Analyzed: 11/20/24

Cadmium	94.7	1.0	0.08	mg/kg	98.0		96.6	80-120	4.62	25	
Chromium	94.9	1.0	0.1	"	98.0		96.8	80-120	5.75	25	
Lead	95.4	1.0	0.1	"	98.0		97.3	80-120	5.77	25	

Batch AHK0148 - EPA 6010B

Blank (AHK0148-BLK1)

Prepared: 11/20/24 Analyzed: 11/21/24

Cadmium	ND	1.0	0.08	mg/kg							U
Chromium	ND	1.0	0.1	"							U
Lead	ND	1.0	0.1	"							U

LCS (AHK0148-BS1)

Prepared: 11/20/24 Analyzed: 11/21/24

Cadmium	94.2	1.0	0.08	mg/kg	100		94.2	80-120			
Chromium	95.1	1.0	0.1	"	100		95.1	80-120			
Lead	92.2	1.0	0.1	"	100		92.2	80-120			

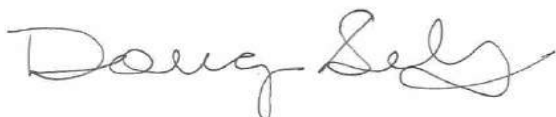
LCS Dup (AHK0148-BS1)

Prepared: 11/20/24 Analyzed: 11/21/24

Cadmium	98.1	1.0	0.08	mg/kg	100		98.1	80-120	4.04	25	
Chromium	98.0	1.0	0.1	"	100		98.0	80-120	3.03	25	
Lead	95.3	1.0	0.1	"	100		95.3	80-120	3.23	25	

SGS Excelchem Laboratories, Inc.

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.



SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

Total Recoverable Metals - Quality Control

Analyte	Result	Reporting Limit	MDL	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit	Notes
---------	--------	-----------------	-----	-------	-------------	---------------	------	-------------	-----	-----------	-------

Batch AHK0148 - EPA 6010B

Matrix Spike (AHK0148-MS1)

Source: 2411088-01

Prepared: 11/20/24 Analyzed: 11/21/24

Cadmium	61.3	1.0	0.08	mg/kg	100	0.920	60.4	75-125			QM-05
Chromium	100	1.0	0.1	"	100	46.2	53.8	75-125			QM-05
Lead	64.0	1.0	0.1	"	100	7.43	56.6	75-125			QM-05

Matrix Spike Dup (AHK0148-MSD1)

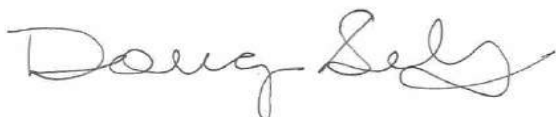
Source: 2411088-01

Prepared: 11/20/24 Analyzed: 11/21/24

Cadmium	61.2	1.0	0.08	mg/kg	100	0.920	60.3	75-125	0.0903	25	QM-05
Chromium	105	1.0	0.1	"	100	46.2	58.6	75-125	4.67	25	QM-05
Lead	65.3	1.0	0.1	"	100	7.43	57.8	75-125	1.96	25	QM-05

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

Notes and Definitions

U Undetected

R-06 This sample was diluted due to high analyte concentration, resulting in an elevated reporting limit.

QM-05 The spike recovery was outside acceptance limits for the MS and/or MSD due to matrix interference. The LCS and/or LCSD were within acceptance limits showing that the laboratory is in control and the data is acceptable.

J Detected but below the Reporting Limit; therefore, result is an estimated concentration (CLP J-Flag).

B Analyte is found in the associated blank as well as in the sample (CLP B-flag).

ND Analyte not detected at the reporting limit.

NR Not reported

SGS Excelchem Laboratories, Inc.

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Doug Sells, Technical Director

ATTACHMENT 01 57 19-A Page 102

BACKCHECK SUBMITTAL

Page 39 of 39

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

SGS

Subcontract Sample Release Form & Receipt

Sending To: SGS Excelchem
Attn: Becky Miller
1135 W. Sunset Blvd.
Suite A
Rocklin, CA 95765

Phone: 916-543-4445
Fax: 916-543-4449
Email: becky.miller@sgs.com

Date:

From:
Parent RN:

SGS Forensic Laboratories on this day releases custody and control of the following sample(s) presently in our custody:

FALI Job	P.O.	Analysis Requested	Due Date
L1232	U003448	Lead, Cadmium, Chromium by ICP	11/22/2024

PLEASE USE Report U003448 as the Purchase Order (PO).

Special Handling: Send results and invoice only to ENV.Hayward.SUBCONTRACT@sgs.com

SUBMISSION LOCATION (Please circle)

SGS
3777 Depot Road, Suite 409
Hayward, CA 94545

SGS
20535 S. Belshaw Avenue
Carson, CA 90746

SGS
3626 Sunset Rd., Suite 100
Las Vegas, NV 89120

Released By:

(SGS)

Brenda Valencia
(Print Name)

11-14-24 14:33
(Date / Time)

2411077

C-24

Received By:

(Client / Vendor Representative)

Becky Miller
(Print Name)

11/15/24 10:00
(Date / Time)

Printed: 11/14/24 14:32

Page 1 of 1

SGS Excelchem Laboratories, Inc.

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Doug Sells

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42



Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232		PO / Job#: ESC.8249.06		Date: Nov , 2024	
Contact: Shoma Yasutomi		Phone: (098) 989-0426		Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer	
E-mail: shoma.yasutomi@esc-pacific.com				☐ PLM: ☐ Standard / ☐ Point Count 400- / 1000 / ☐ CARB 435			
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS				☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)			
Site Location: Camp Buckner, Okinawa, Japan				☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project			
Comments:				☒ Metals Analysis Matrix: Paint Chip Sample Method: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only	
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L1	1 Nov 2024	Gray Paint on Metal Panel	A P C				
2. BUK-B100-L2	1 Nov 2024	Gray Paint on Metal Frame	A P C				
3. BUK-B100-L3	1 Nov 2024	Off-White Paint on Board wall	A P C				
4. BUK-B100-L4	1 Nov 2024	Off-white Paint on Metal Pipe	A P C				
5. BUK-B100-L5	1 Nov 2024	Off-White Paint on Hard Board Wall	A P C				
6. BUK-B100-L6	1 Nov 2024	Off-white Paint on Metal Panel	A P C				
7. BUK-B100-L7	1 Nov 2024	Blue Paint on Metal Cover	A P C				
8. BUK-B100-L8	1 Nov 2024	Dark Gray Paint on Metal Panel	A P C				
9. BUK-B100-L9	1 Nov 2024	Off-White over Gray Paint on Metal Door Frame	A P C				
10. BUK-B100-L10	1 Nov 2024	Blue over White over Orange Paint on Metal Door Frame	A P C				
Sampled By: Shoma Yasutomi Date/Time: 1 Nov 24 Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: jp							
Relinquished By: Shoma Yasutomi		Relinquished By: J		Relinquished By: Brenda			
Date / Time: Nov 5 2024		Date / Time:		Date / Time: 11-14-24 14:41			
Received By: Japan post		Received By: Carmine Contreras		Received By: Beckmiller			
Date / Time: Nov 5 2024		Date / Time: 11-14-2024 144pm		Date / Time: 11-15-24 1000			
Condition Acceptable? ☒ Yes ☐ No		Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No			

SGS Forensic Laboratories may subcontract client samples to other SGSFL locations to meet client requests.
San Francisco Office: 3777 Depot Road, Suite 409, Hayward, CA 94545-2761 • Phone: 510/887-8828 • 800/827-3274
Los Angeles Office: 2959 Pacific Commerce Drive, Rancho Dominguez, CA 90221 • Phone: 310/763-2374 • 888/813-9417
Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SGS Excelchem Laboratories, Inc.

Doug Sells

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42



Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232	PO / Job#: ESC.8249.06	Date: Nov , 2024			
Contact: Shoma Yasutomi		Phone: (098) 989-0426	Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒				
E-mail: shoma.yasutomi@esc-pacific.com		☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer					
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS		☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Location: Camp Buckner, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments:		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project ☒ Metals Analysis Matrix: Paint Chip Sample Method: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L11	1 Nov 2024	Off-White over Gray Paint on Metal Door Frame	A P C				
2. BUK-B100-L12	1 Nov 2024	Red Paint on Metal Pipe	A P C				
3. BUK-B100-L13	1 Nov 2024	Off-White Paint on Metal Pipe	A P C				
4. BUK-B100-L14	1 Nov 2024	Off-white Paint over on Concrete Wall	A P C				
5. BUK-B100-L15	1 Nov 2024	Off-White Paint on Plastic Pipe	A P C				
6. BUK-B100-L16	1 Nov 2024	Beige over White Paint on Concrete Eave	A P C				
7. BUK-B100-L17	1 Nov 2024	Off-White Paint on Siding Wall	A P C				
8. BUK-B100-L18	1 Nov 2024	Beige over Blue Paint on Siding Wall	A P C				
9. BUK-B100-L19	1 Nov 2024	Silver Paint on Metal Pipe Support	A P C				
10. BUK-B100-L20	1 Nov 2024	Red Paint on Metal Pipe	A P C				
Sampled By: Shoma Yasutomi		Date/Time: 1 Nov 24	Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP				
Relinquished By: Shoma Yasutomi		Date / Time: Nov 5 2024	Relinquished By:		Relinquished By: Bu Val		
Received By: Japan post		Date / Time: Nov 5 2024	Received By: Yasmine Contreras		Received By: Bu Val		
Condition Acceptable? ☒ Yes ☐ No			Date / Time: 11-14-24 1:44 pm		Date / Time: 11-15-24 1000		
			Condition Acceptable? ☐ Yes ☐ No		Condition Acceptable? ☒ Yes ☐ No		

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Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

SGS Excelchem Laboratories, Inc.

Doug Sells

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SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

SGS FORENSIC LABORATORIES Analysis Request Form (COC)

Client Name & Address: Environmental Science Corporation
ESC Building 1553-231, Ishikawa Iha, Uruma City,
Okinawa, Japan. Postal Code 904-1115 Client No.: L1232

PO / Job#: ESC.8249.06 Date: Nov, 2024

Turn Around Time: Same Day / 1Day / 2Day / 3Day / 4Day / 5Day ☒ 5Day

☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer

☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435

Contact: Shoma Yasutomi Phone: (098) 989-0426

E-mail: shoma.yasutomi@esc-pacific.com

Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS

Site Location: Camp Buckner, Okinawa, Japan

Comments:

☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402
☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield
☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight %
☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)
☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot
☐ Particle Identification (TEM LAB) ☐ Special Project
☒ Metals Analysis Matrix: Paint Chip Sample Method: EPA3050B/6010B
Analytes: Lead, Cadmium, Chromium ☐ Silica in Air ☐ w/Gravimetry
☐ Quartz Only

Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L21	1 Nov 2024	Light Gray over Gray Paint on Metal Frame	A P C				
2. BUK-B100-L22	1 Nov 2024	Yellow over Brown Paint on Metal Pole	A P C				
3. BUK-B100-L23	1 Nov 2024	Silver and Gray Paint on Metal Pipe	A P C				
4. BUK-B100-L24	1 Nov 2024	Red Paint on Metal Pipe	A P C				
5. BUK-B100-L25	1 Nov 2024	Off-White over Beige Paint on Concrete Wall	A P C				
6. BUK-B100-L26	1 Nov 2024	Gray over Beige Paint on Concrete Wall	A P C				
7. BUK-B100-L27	1 Nov 2024	White Paint on Concrete Wall	A P C				
8. BUK-B100-L28	1 Nov 2024	Gray over Green Paint on Metal Frame	A P C				
9. BUK-B100-L29	1 Nov 2024	Off-White Paint on Metal Duct	A P C				
10. BUK-B100-L30	1 Nov 2024	Gray Paint on Metal Pipe	A P C				

Sampled By: Shoma Yasutomi Date/Time: 1 Nov 24 Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP

Relinquished By: Shoma Yasutomi Date / Time: Nov 5 2024

Relinquished By: *Brenda V* Date / Time: 11-14-24 14:41

Received By: Japan post Date / Time: Nov 5 2024 Condition Acceptable? ☒ Yes ☐ No

Received By: *Charmaine Contreras* Date / Time: 11-14-24 14:41 Condition Acceptable? ☒ Yes ☐ No

Received By: *Brenda Miller* Date / Time: 11-15-24 10:00 Condition Acceptable? ☒ Yes ☐ No

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Las Vegas Office: 6765 S. Eastern Avenue, Suite 3, Las Vegas, NV 89119 • Phone: 702/784-0040

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Doug Sells

Doug Sells, Technical Director

ATTACHMENT 01 57 19-A Page 106

BACKCHECK SUBMITTAL

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories 20535 S Belshaw Ave Carson, CA 90746	Project: L1232 Project Number: U003448 Project Manager: Brenda V	Date Reported: 11/22/24 15:42
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SGS FORENSIC LABORATORIES		Analysis Request Form (COC)					
Client Name & Address: Environmental Science Corporation ESC Building 1553-231, Ishikawa Iha, Uruma City, Okinawa, Japan. Postal Code 904-1115		Client No.: L1232 PO / Job#: ESC.8249.06 Date: Nov , 2024					
Contact: Shoma Yasutomi Phone: (098) 989-0426 E-mail: shoma.yasutomi@esc-pacific.com		Turn Around Time: ☒ Same Day / ☐ 1Day / ☐ 2Day / ☐ 3Day / ☐ 4Day / ☒ 5Day ☐ PCM: ☐ NIOSH 7400A / ☐ NIOSH 7400B ☐ Rotometer ☐ PLM: ☐ Standard / ☐ Point Count 400 - 1000 / ☐ CARB 435					
Site Name: FY25 Repair HVAC WIDEBAND SATCOM OPS Site Location: Camp Buckner, Okinawa, Japan		☐ TEM Air: ☐ AHERA / ☐ Yamate2 / ☐ NIOSH 7402 ☐ TEM Bulk: ☐ Quantitative / ☐ Qualitative / ☐ Chatfield ☐ TEM Water: ☐ Potable / ☐ Non-Potable / ☐ Weight % ☐ TEM Microvac: ☐ Qual / ☐ D5755(str/area) / ☐ D5756(str/mass)					
Comments:		☐ IAQ Particle Identification (PLM LAB) ☐ PLM Opaques/Soot ☐ Particle Identification (TEM LAB) ☐ Special Project ☒ Metals Analysis Matrix: Paint Chip Sample Method: EPA3050B/6010B Analytes: Lead, Cadmium, Chromium					
		☐ Silica in Air ☐ w/Gravimetry ☐ Quartz Only					
Sample ID	Date / Time	Sample Location / Description	FOR AIR SAMPLES ONLY				Sample Area / Air Volume
			Type	Time On/Off	Avg LPM	Total Time	
1. BUK-B100-L31	1 Nov 2024	Gray over Black Paint on Metal Support	A P C				
2. BUK-B100-L32	1 Nov 2024	Gray Paint on Metal Pipe	A P C				
3. BUK-B100-L33	1 Nov 2024	Gray over Dark Gray Paint on Metal Pipe	A P C				
4. BUK-B100-L34	1 Nov 2024	Off-white Paint on Metal Panel	A P C				
5.	1 Nov 2024		A P C				
6.	1 Nov 2024		A P C				
7.	1 Nov 2024		A P C				
8.	1 Nov 2024		A P C				
9.	1 Nov 2024		A P C				
10.	1 Nov 2024		A P C				
Sampled By: Shoma Yasutomi Date/Time: 1 Nov 24		Shipped Via: ☐ Fed Ex ☐ UPS ☐ US Mail ☐ Courier ☐ Drop Off ☒ Other: JP					
Relinquished By: Shoma Yasutomi Date / Time: Nov 5 2024		Relinquished By: <i>Buenk Valen</i> Date / Time: 11-14-24 1441					
Received By: Japan post Date / Time: Nov 5 2024 Condition Acceptable? ☒ Yes ☐ No		Received By: <i>Yamamoto</i> Date / Time: 11-14-2024 144pm Condition Acceptable? ☐ Yes ☐ No					
Received By: <i>Bucky Miller</i> Date / Time: 11-15-24 1000 Condition Acceptable? ☒ Yes ☐ No							

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SGS Excelchem Laboratories, Inc.

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Doug Sells

SGS Excelchem Laboratories, Inc.

SGS Forensic Laboratories
20535 S Belshaw Ave
Carson, CA 90746

Project: L1232
Project Number: U003448
Project Manager: Brenda V

Date Reported:
11/22/24 15:42

SGS

SGS - Rocklin
Sample Integrity
FN: F-ROC-SC-001-00
Rev. Date: 11/01/2023
Page 1 of 1

WORK ORDER: 2411077

Sample Integrity

Date Received: 11/15/24

Company Name: SGS Carson
New Client: Y (N)

Section 1 - Sample Arrival Information

Sample Transport: ONTRAC UPS USPS Walk-In EXCELCHEM Courier Fed-Ex Other: _____

Transported In: Ice Chest Box Hand

Packing materials: Bubble Wrap Foam Packing Peanuts Paper Other: _____

Has chilling process begun? Y (N) Samples Received: Chilled to Touch / Ambient / On Ice

Temperature of Samples (°C): 21.1 Ice Chest Temperature(s) (°C): _____ (Correct Temp: *)

Thermometer ID: DT-19 CF (°C): -0.3

Section 2 - Bottle/Analysis Info.

	Yes	No	N/A	Comments
Did all bottles arrive unbroken and intact?				
Did all bottle labels agree with COC?				
Were correct containers used for the tests requested?				
Were correct preservations used for the tests requested?				
Was a sufficient amount of sample sent for tests indicated?				
Were bubbles present in VOA Vials?: (Volatile Methods Only)				
Is there head space in the VOA vials? (Volatile Methods Only)				

Section 3 - COC Information

	Yes	No	Comments	Yes	No
COC Received				Analysis Requested	
Date Sampled				Samples arrived within holding time	
Time Sampled				Hold times less than 72 hours	
Sample ID				Client Name	
Rush Turn Around Time				Client Contact Information	

SHORT HOLD LIST (<72 hours)

pH	Chlorine	Corrosivity	Coliform	Dissolved Oxygen	Odor	Nitrate	Nitrite	Ortho-phosphate
MB	Asbestos	Settable Solids	Turbidity	Biochemical Oxygen Demand	HPC	Color	Tedlars	Ammonia/TKN (unpreserved)
As								

Section 4 - Comments / Discrepancies

Client notified of discrepancies: Yes / No Notified by: _____

Comments: Client notified about our server/delay

Bin Number/ Location: C-26

COC Scanned/Attached by: AS

Samples labeled by: AS

Sample labels reviewed by: AS

Filled out by: [Signature]

Date: 11-15-24

Time: 1000

SGS Excelchem Laboratories, Inc.

The results in this report apply to the samples analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety.

[Signature]



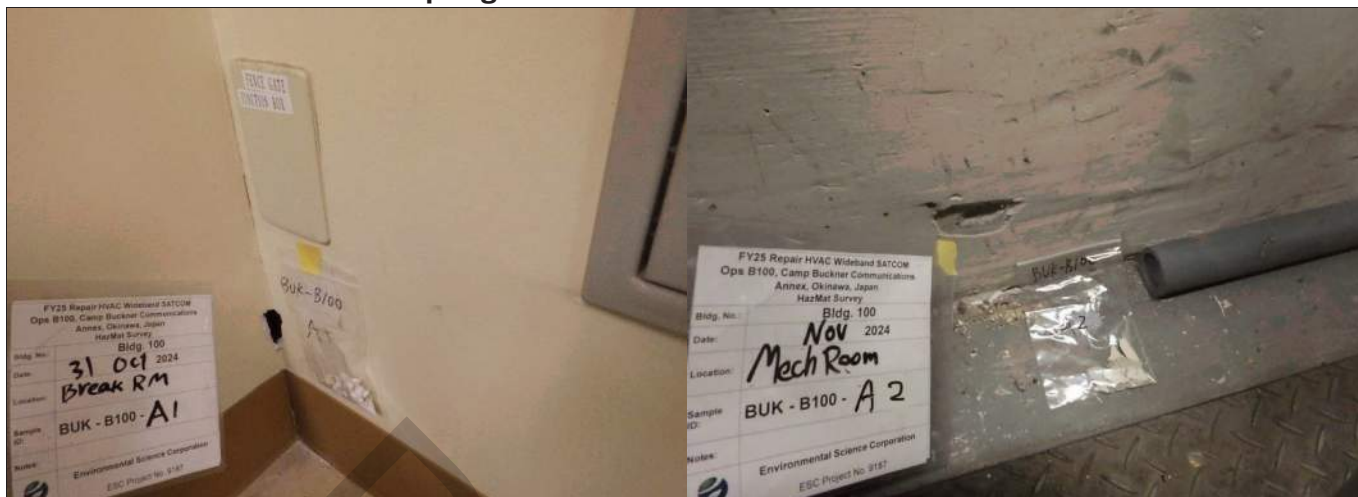
APPENDIX E: INSPECTION PHOTOS

DRAFT



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



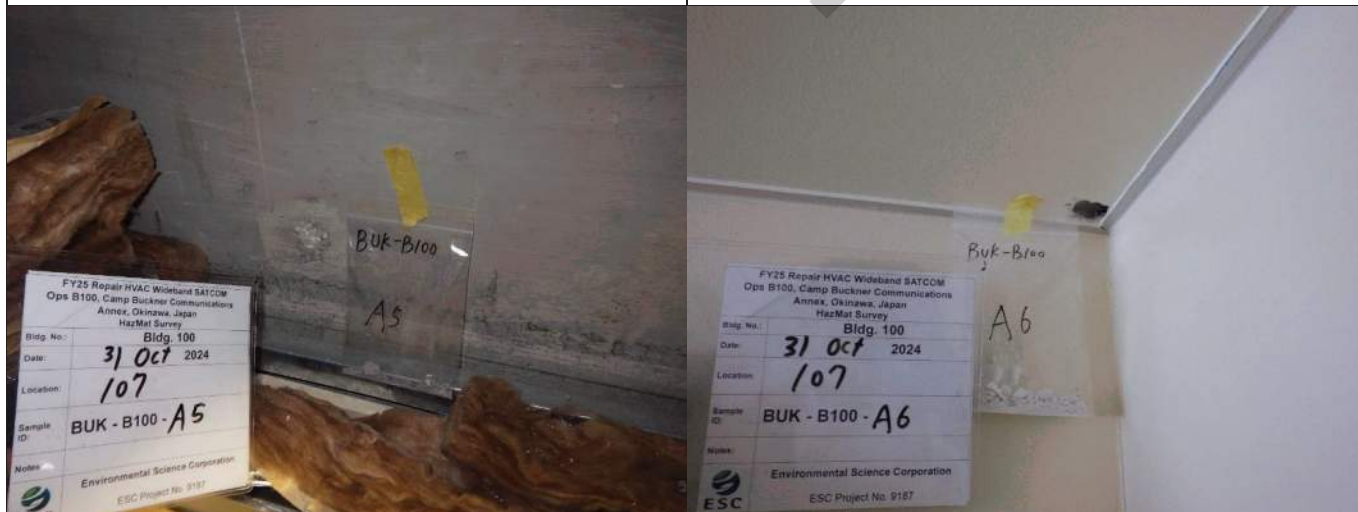
No. 1: BUK-B100-A1

No. 2: BUK-B100-A2



No. 3: BUK-B100-A3

No. 4: BUK-B100-A4



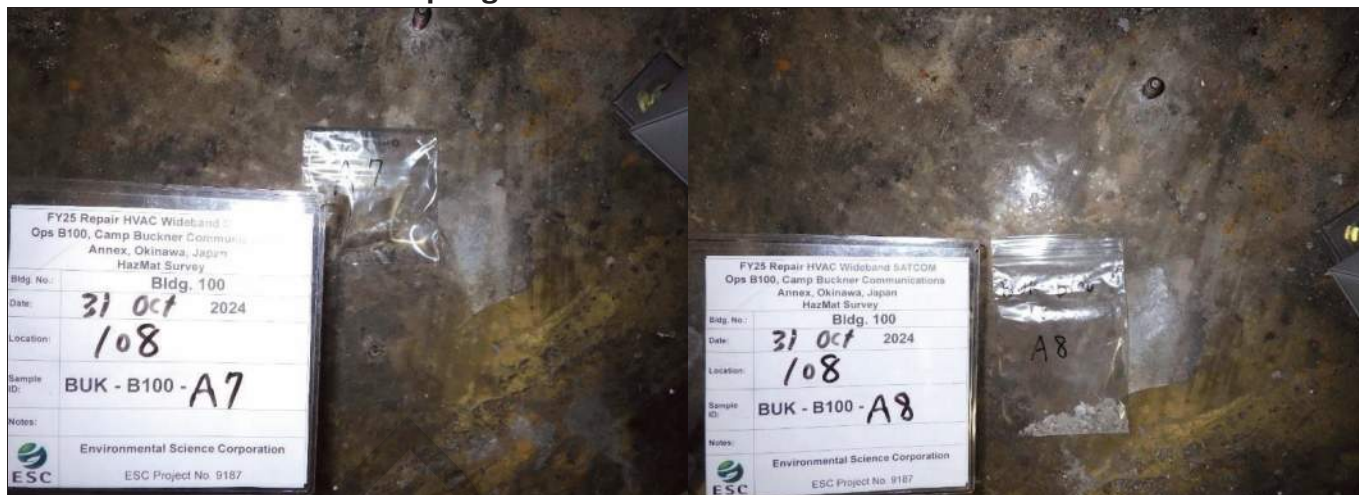
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No. 6: BUK-B100-A6



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 7: BUK-B100-A7

No. 8: BUK-B100-A8



No. 9: BUK-B100-A9

No. 10: BUK-B100-A10



No. 11: BUK-B100-A11

No. 12: BUK-B100-A12



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



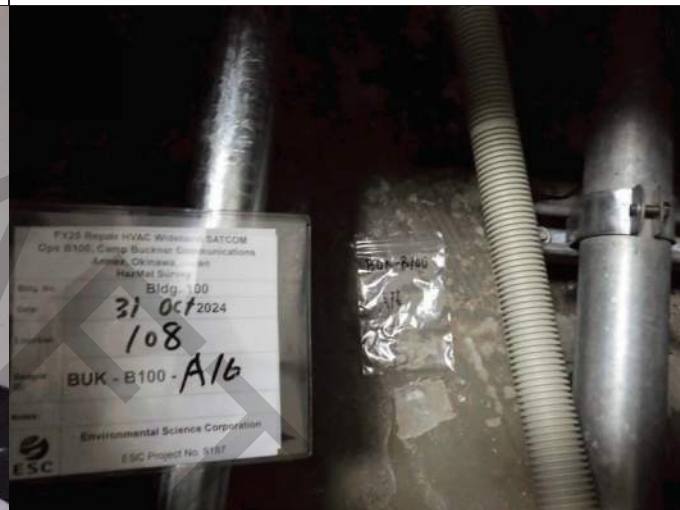
No. 13: BUK-B100-A13



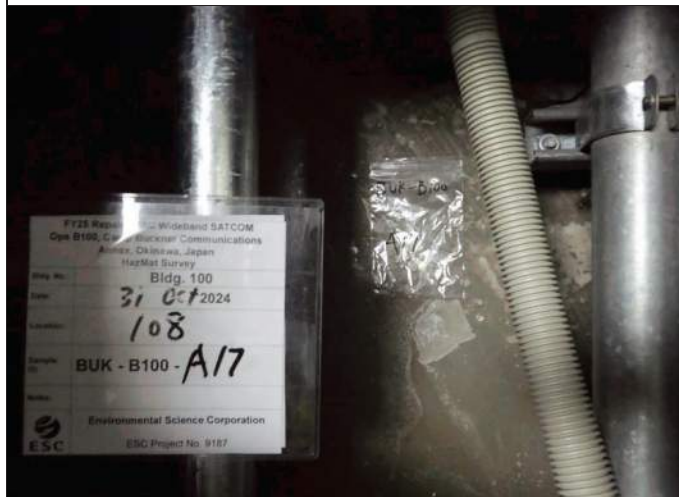
No. 14: BUK-B100-A14



No. 15: BUK-B100-A15



No. 16: BUK-B100-A16



No. 17: BUK-B100-A17



No. 18: BUK-B100-A18



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 19: BUK-B100-A19



No. 20: BUK-B100-A20



No. 21: BUK-B100-A21



No. 22: BUK-B100-A22



No. 23: BUK-B100-A23



No. 24: BUK-B100-A24

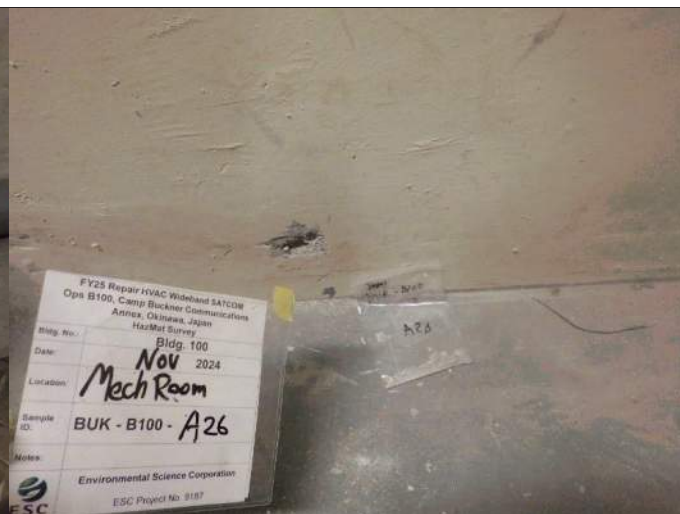


FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

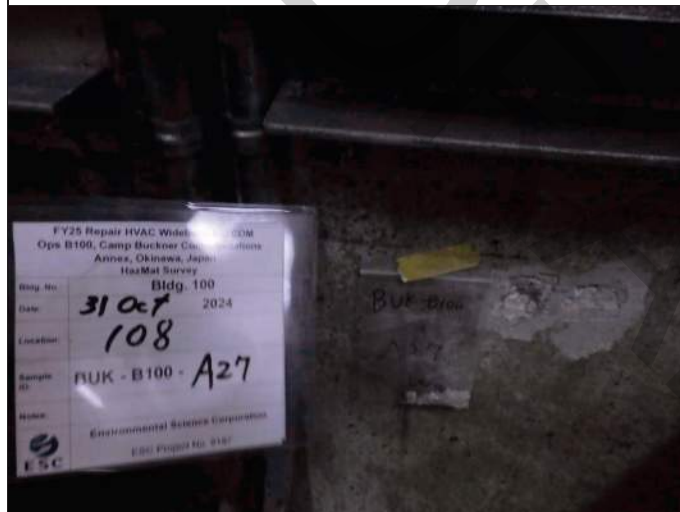
Photo Record: Asbestos Sampling



No. 25: BUK-B100-A25



No. 26: BUK-B100-A26



No. 27: BUK-B100-A27



No. 28: BUK-B100-A28



No. 29: BUK-B100-A29

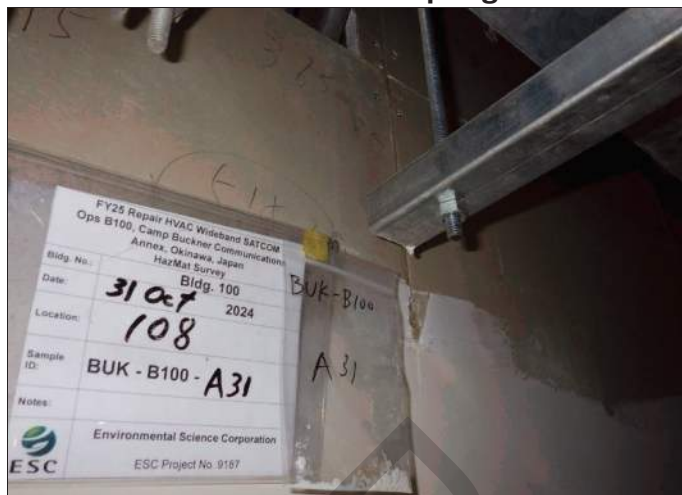


No. 30: BUK-B100-A30

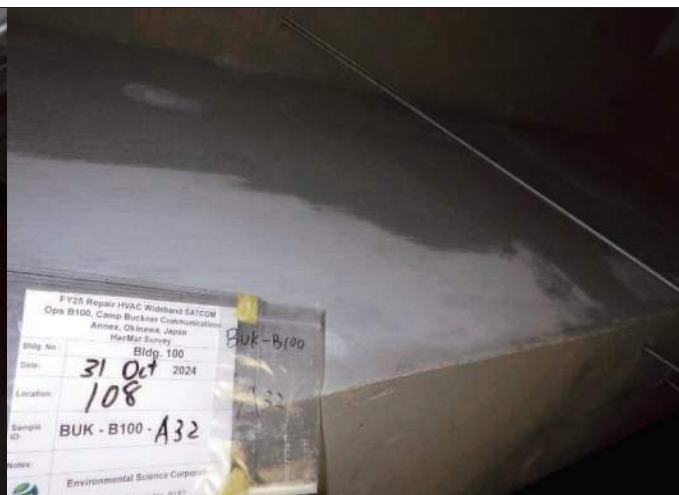


FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 31: BUK-B100-A31



No. 32: BUK-B100-A32



No. 33: BUK-B100-A33



No. 34: BUK-B100-A34



No. 35: BUK-B100-A35



No. 36: BUK-B100-A36



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 37: BUK-B100-A37



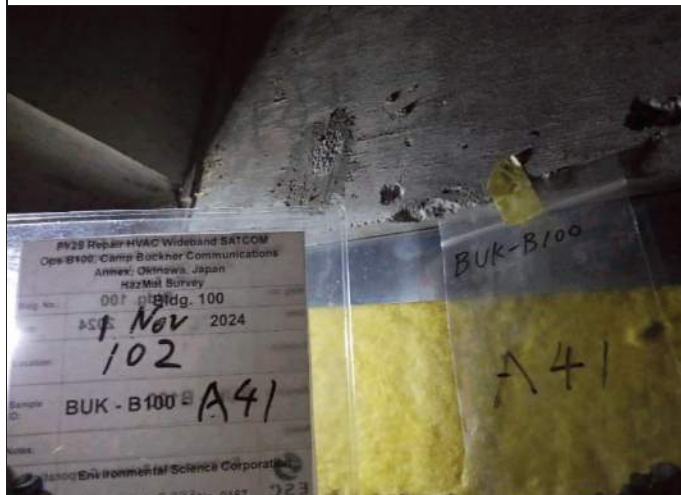
No. 38: BUK-B100-A38



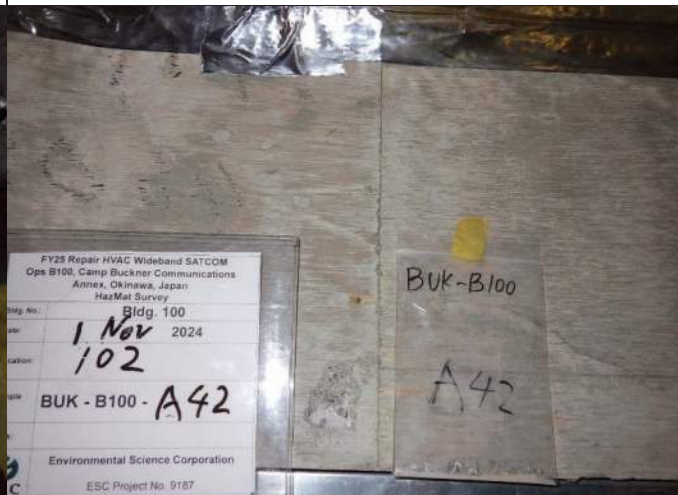
No. 39: BUK-B100-A39



No. 40: BUK-B100-A40



No. 41: BUK-B100-A41

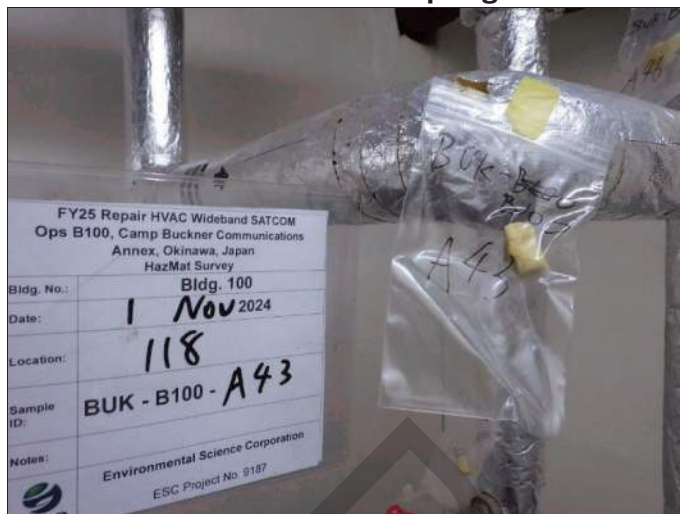


No. 42: BUK-B100-A42

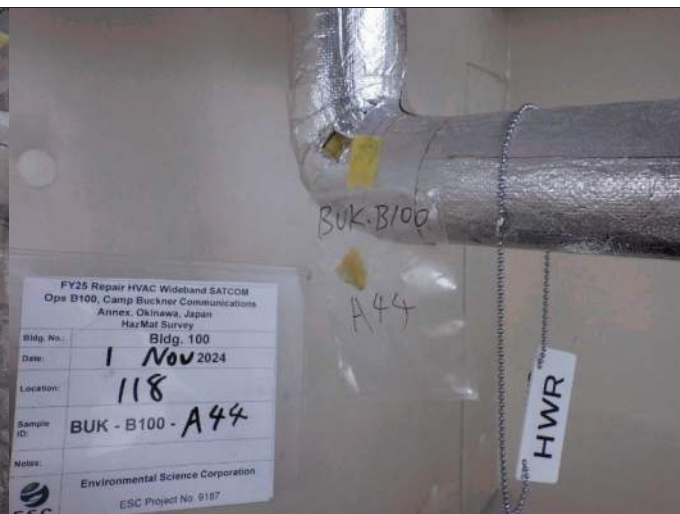


FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 43: BUK-B100-A43



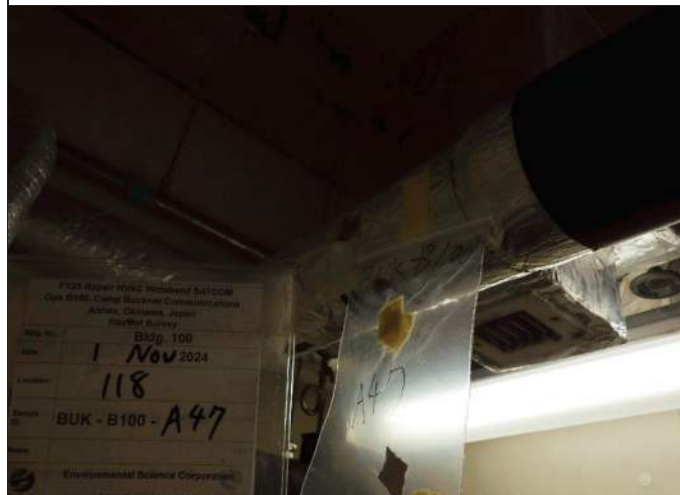
No. 44: BUK-B100-A44



No. 45: BUK-B100-A45



No. 46: BUK-B100-A46



No. 47: BUK-B100-A47

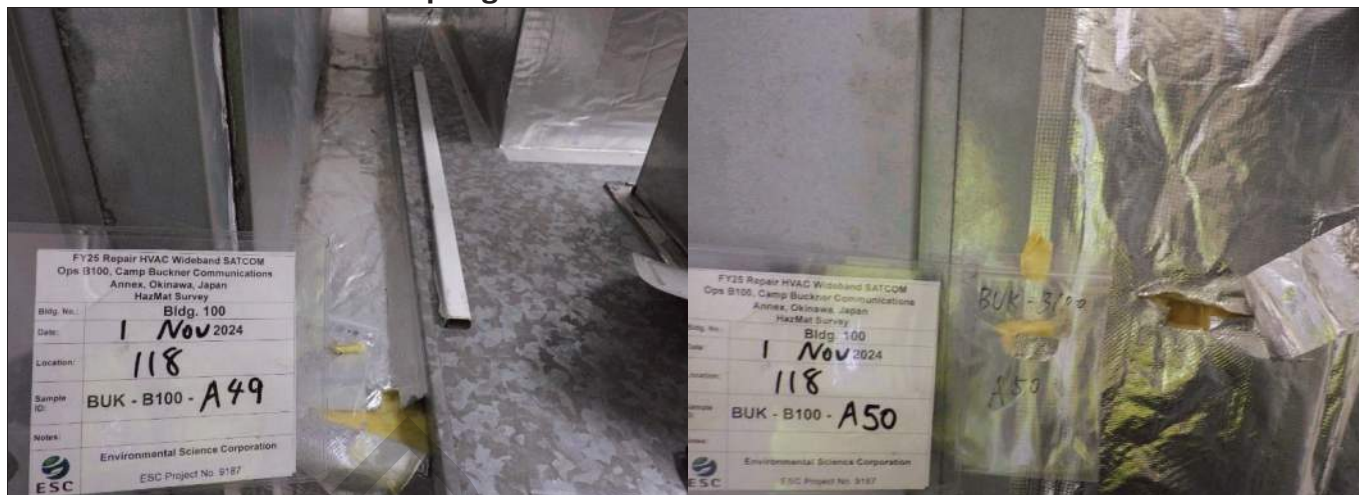


No. 48: BUK-B100-A48



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 49: BUK-B100-A49

No. 50: BUK-B100-A50



No. 51: BUK-B100-A51

No. 52: BUK-B100-A52



No. 53: BUK-B100-A53

No. 54: BUK-B100-A54



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



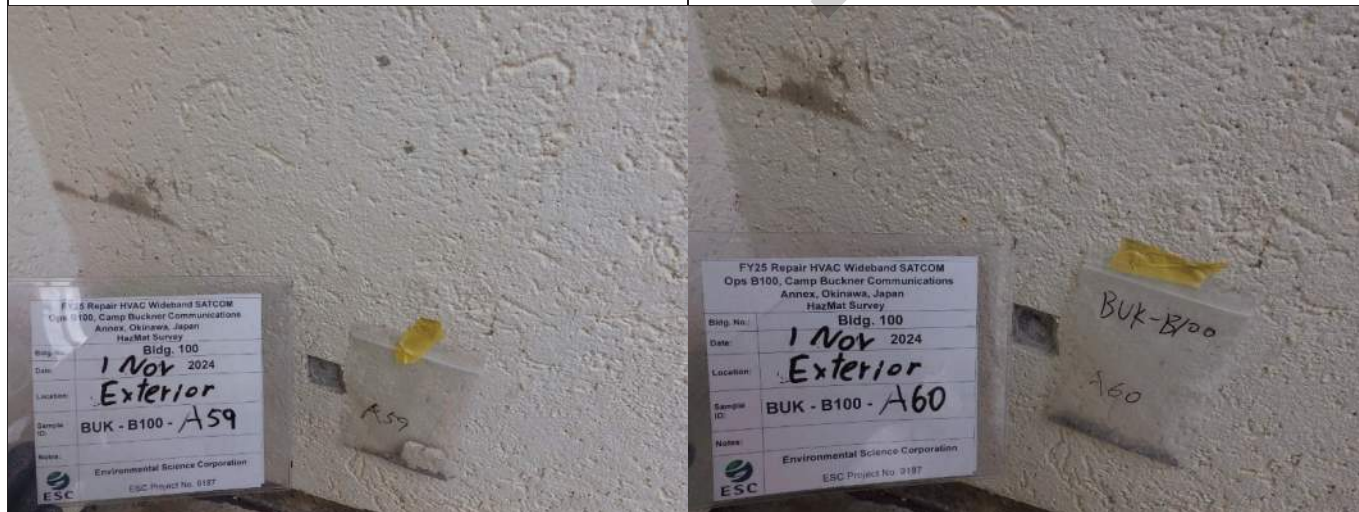
No. 55: BUK-B100-A55

No. 56: BUK-B100-A56



No. 57: BUK-B100-A57

No. 58: BUK-B100-A58



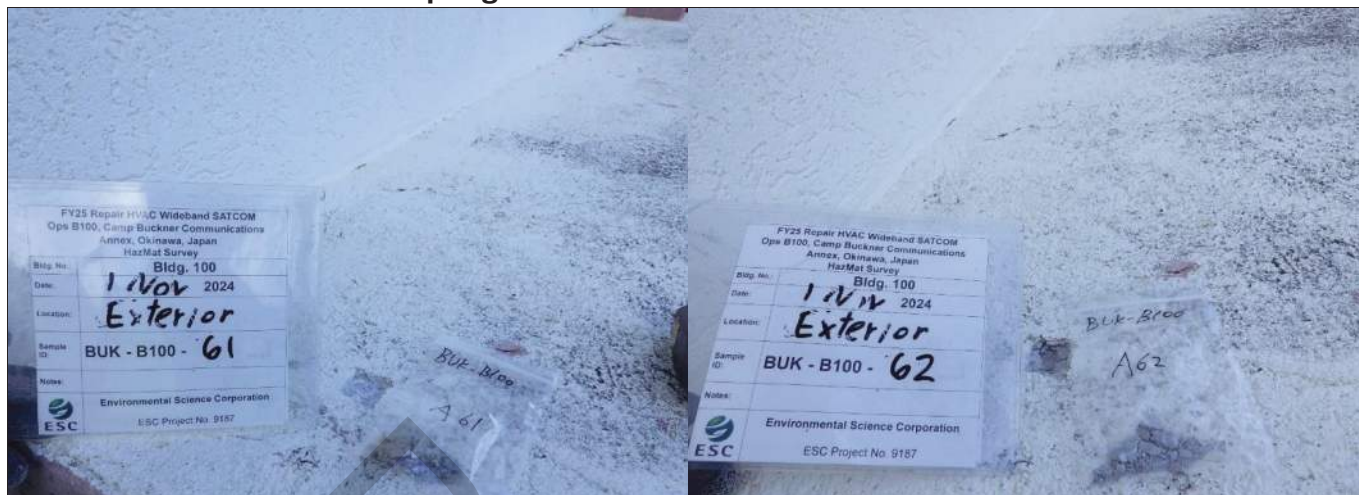
No. 59: BUK-B100-A59

No. 60: BUK-B100-A60



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 61: BUK-B100-A61

No. 62: BUK-B100-A62



No. 63: BUK-B100-A63

No. 64: BUK-B100-A64



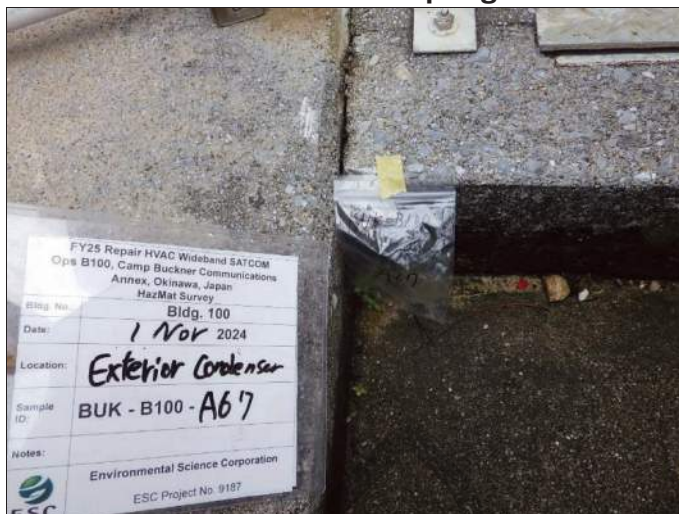
No. 65: BUK-B100-A65

No. 66: BUK-B100-A66



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 67: BUK-B100-A67



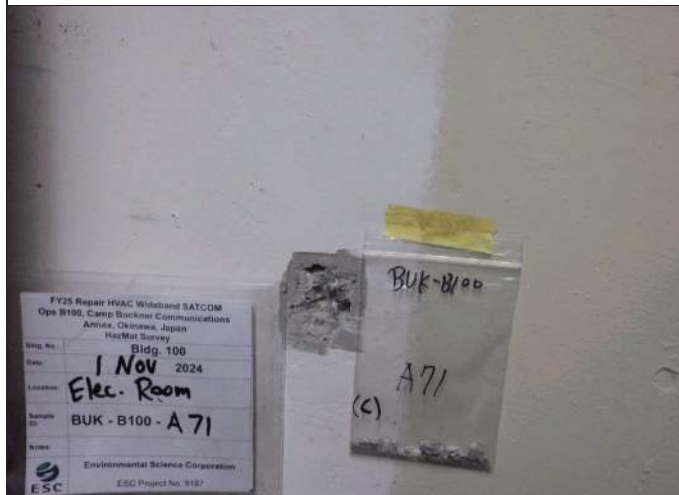
No. 68: BUK-B100-A68



No. 69: BUK-B100-A69



No. 70: BUK-B100-A70



No. 71: BUK-B100-A71



No. 72: BUK-B100-A72



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 73: BUK-B100-A73



No. 74: BUK-B100-A74



No. 75: BUK-B100-A75



No. 76: BUK-B100-A76



No. 77: BUK-B100-A77

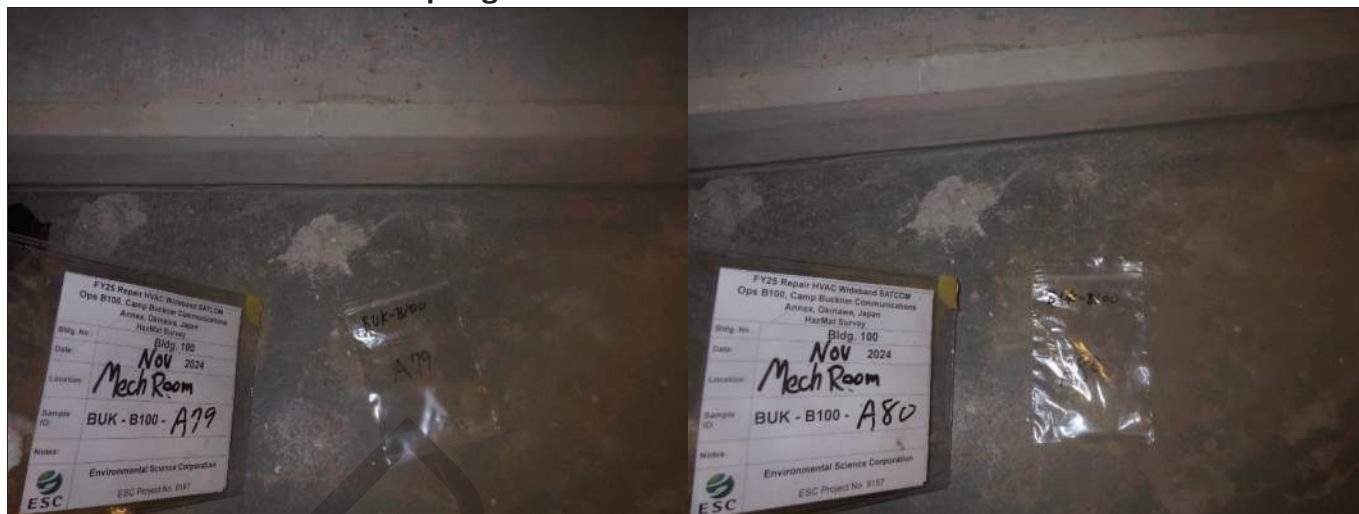


No. 78: BUK-B100-A78



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Asbestos Sampling



No. 79: BUK-B100-A79

No. 80: BUK-B100-A80



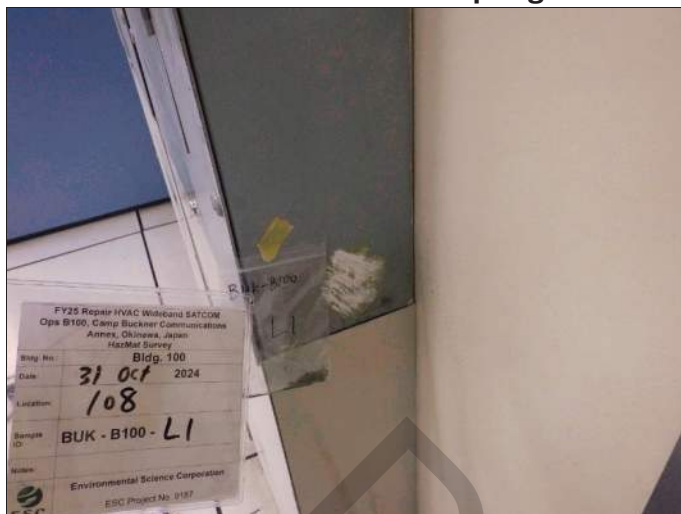
No. 81: BUK-B100-A81

No. 82: BUK-B100-A82



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Lead/Metals Sampling



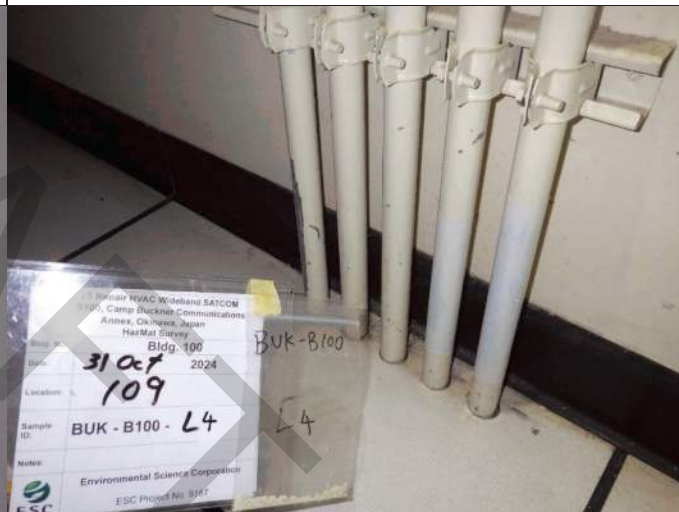
No. 83: BUK-B100-L1



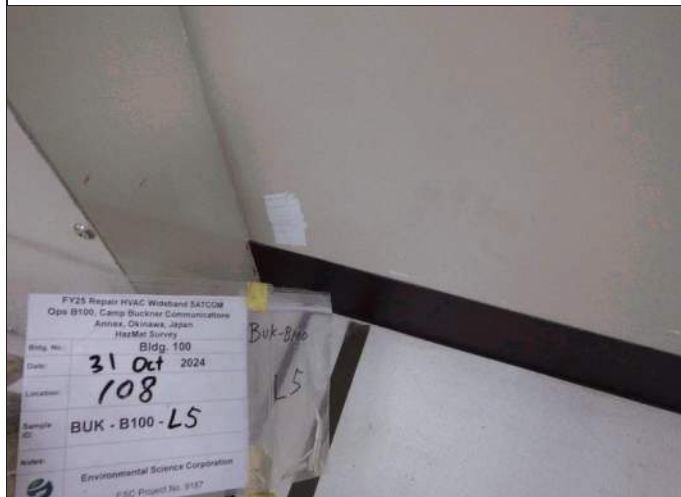
No. 84: BUK-B100-L2



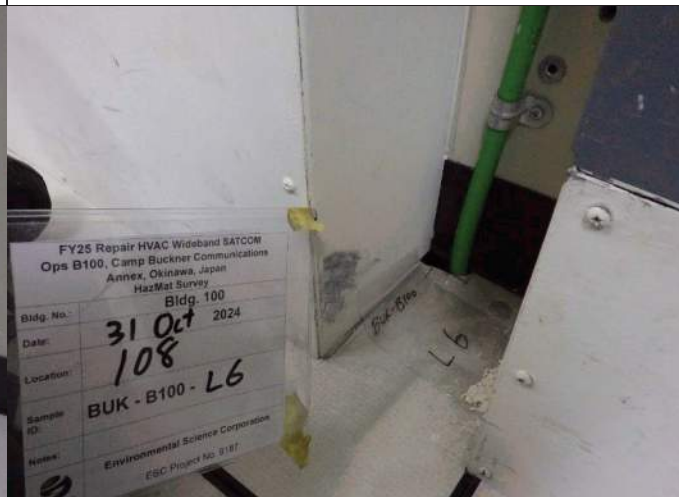
No. 85: BUK-B100-L3



No. 86: BUK-B100-L4



No. 87: BUK-B100-L5



No. 88: BUK-B100-L6



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Lead/Metals Sampling



No. 89: BUK-B100-L7



No. 90: BUK-B100-L8



No. 91: BUK-B100-L9



No. 92: BUK-B100-L10



No. 93: BUK-B100-L11

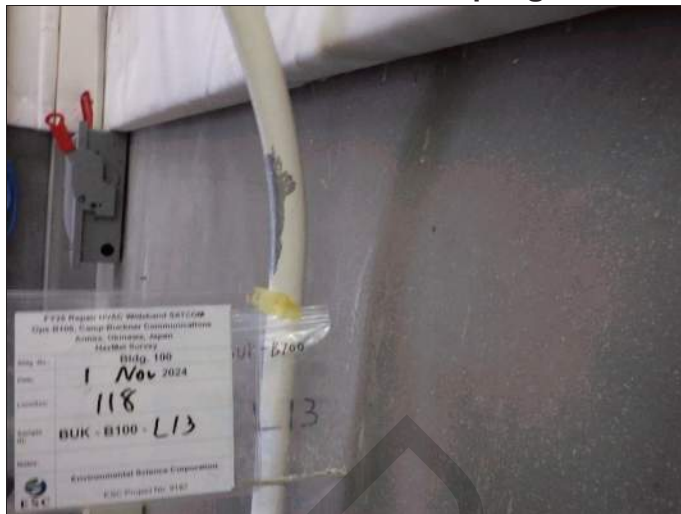


No. 94: BUK-B100-L12



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Lead/Metals Sampling



No. 95: BUK-B100-L13



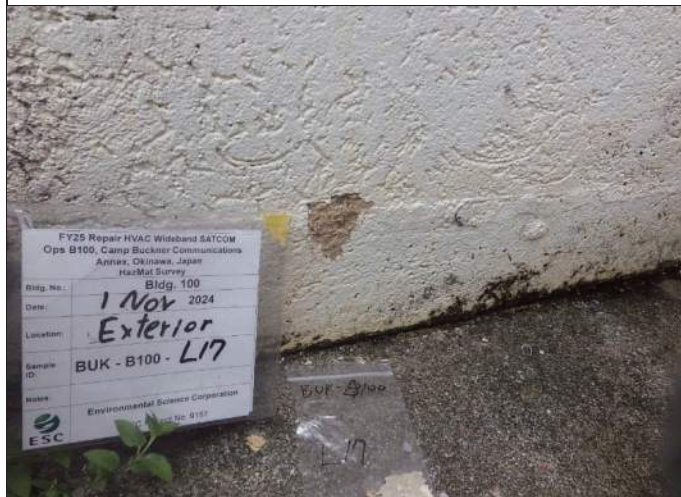
No. 96: BUK-B100-L14



No. 97: BUK-B100-L15



No. 98: BUK-B100-L16



No. 99: BUK-B100-L17



No. 100: BUK-B100-L18



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Lead/Metals Sampling



No. 101: BUK-B100-L19



No. 102: BUK-B100-L20



No. 103: BUK-B100-L21



No. 104: BUK-B100-L22



No. 105: BUK-B100-L23

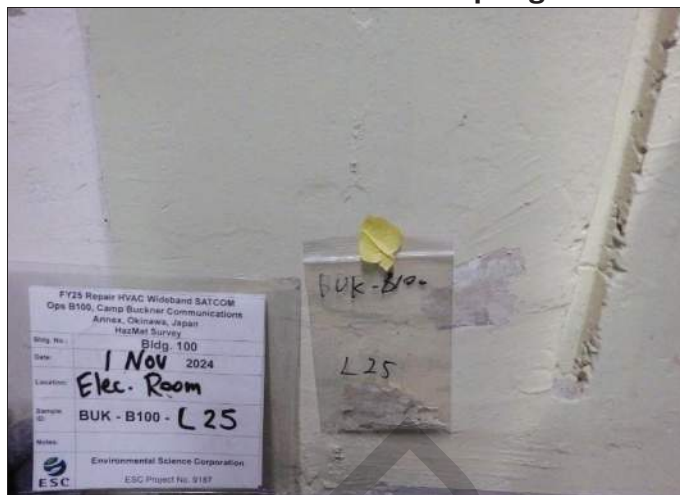


No. 106: BUK-B100-L24



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

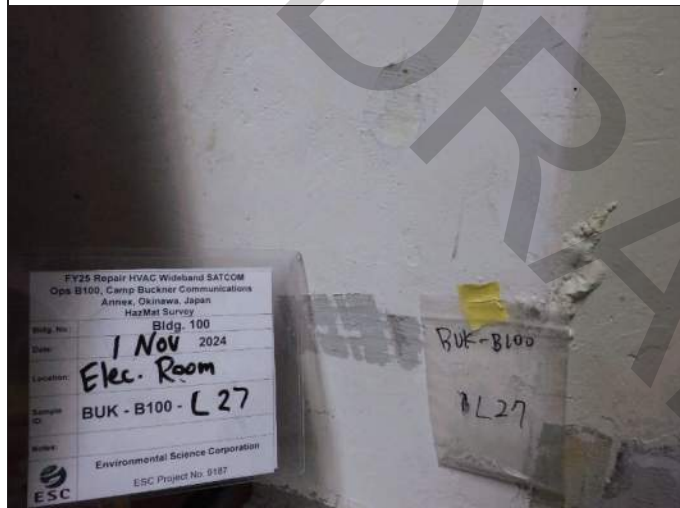
Photo Record: Lead/Metals Sampling



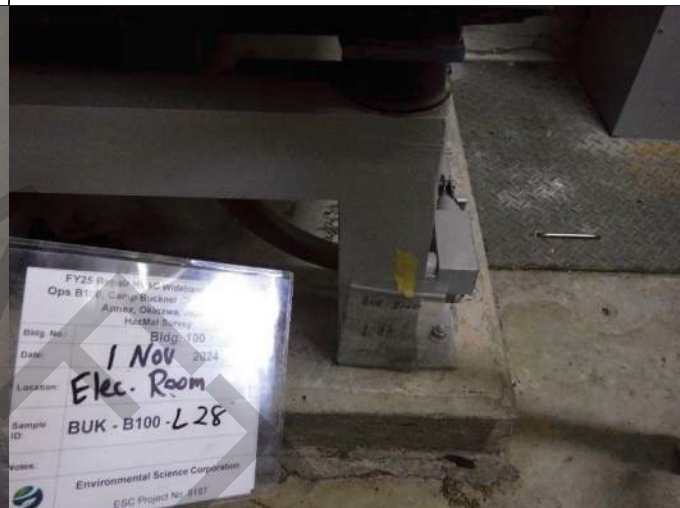
No. 107: BUK-B100-L25



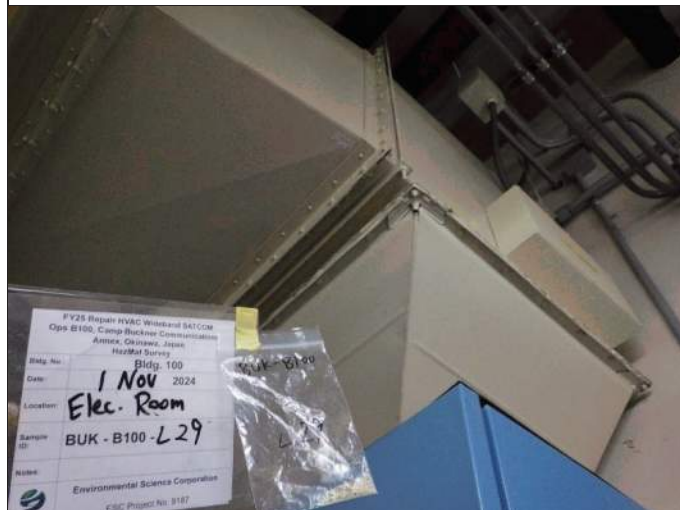
No. 108: BUK-B100-L26



No. 109: BUK-B100-L27



No. 110: BUK-B100-L28



No. 111: BUK-B100-L29



No. 112: BUK-B100-L30



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: Lead/Metals Sampling



No. 113: BUK-B100-L31



No. 114: BUK-B100-L32



No. 115: BUK-B100-L33



No. 116: BUK-B100-L34



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: HAZMAT Visual Survey – Arsenic Board, LLR, Mold, ODS. PCB, Mold



No. 117: Gypsum Board 0808003 Non-Arsenic



No. 118: Ballast (Kitchen)



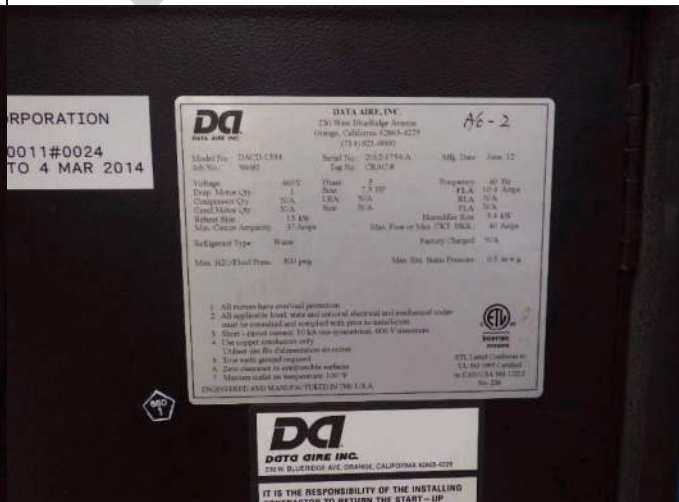
No. 119: Ballast (Server Room)



No. 120: Ballast



No. 121: Chiller No. 1



No. 122: Chiller No. 2



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: HAZMAT Visual Survey – Arsenic Board, LLR, Mold, ODS. PCB, Mold



No. 123: Chiller No. 3



No. 124: Chiller No. 4



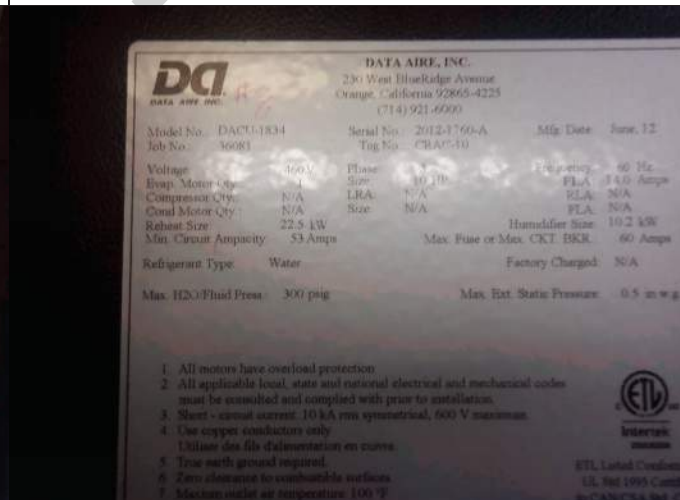
No. 125: Chiller No. 5



No. 126: Chiller No. 6



No. 127: Chiller No. 7



No. 128: Chiller No. 8



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: HAZMAT Visual Survey – Arsenic Board, LLR, Mold, ODS. PCB, Mold



No. 129: Chiller R32



No. 130: Chiller R134A No. 1



No. 131: Chiller R134A No. 2



No. 132: Chiller R410 Carrier



No. 133: Chiller R410 Daikin



No. 134: Ext. Light Fixture Potential Mercury



FY25 REPAIR HVAC WIDE BAND SATCOM OPS B100
Camp Buckner Communications Annex, Okinawa, Japan

Photo Record: HAZMAT Visual Survey – Arsenic Board, LLR, Mold, ODS. PCB, Mold



No. 135: Light Fixture Potential Mercury 2



No. 136: Light Fixture Potential Mercury 3



No. 137: Light Fixture Potential Mercury



No. 138: Mold (Kitchen)



No. 139: Smoke Detector Potential LLR



No. 140: Water Stain



APPENDIX F: BUILDING INSPECTOR & LABORATORY CERTIFICATIONS

DRAFT



**United States Department of Commerce
National Institute of Standards and Technology**

Certificate of Accreditation to ISO/IEC 17025:2017

NVLAP LAB CODE: 101459-1

SGS Forensic Laboratories
Carson, CA

*is accredited by the National Voluntary Laboratory Accreditation Program for specific services,
listed on the Scope of Accreditation, for:*

Asbestos Fiber Analysis

*This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2017.
This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality
management system (refer to joint ISO-ILAC-IAF Communiqué on ISO/IEC 17025).*

2024-07-01 through 2025-06-30
Effective Dates

 
For the National Voluntary Laboratory Accreditation Program

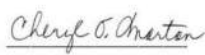
AIHA Laboratory Accreditation Programs, LLC
acknowledges that
SGS Forensic Laboratories
20535 S. Belshaw Ave Carson, CA 90746
Laboratory ID: LAP-101629

along with all premises from which key activities are performed, as listed above, has fulfilled the requirements of the AIHA Laboratory Accreditation
Programs (AIHA LAP), LLC accreditation to the ISO/IEC 17025:2017 international standards, General Requirements for the Competence of Testing
and Calibration Laboratories in the following:

LABORATORY ACCREDITATION PROGRAMS

☒	INDUSTRIAL HYGIENE	Accreditation Expires: February 01, 2025
☒	ENVIRONMENTAL LEAD	Accreditation Expires: February 01, 2025
☒	ENVIRONMENTAL MICROBIOLOGY	Accreditation Expires: February 01, 2025
☐	FOOD	Accreditation Expires:
☐	UNIQUE SCOPES	Accreditation Expires:

Specific Field(s) of Testing (FoT)/Method(s) within each Accreditation Program for which the above named laboratory maintains accreditation
is outlined on the attached Scope of Accreditation. Continued accreditation is contingent upon successful on-going compliance with ISO/IEC
17025:2017 and AIHA LAP, LLC requirements. This certificate is not valid without the attached Scope of Accreditation. Please review the AIHA LAP,
LLC website (www.aihaaccreditedlabs.org) for the most current Scope.


Cheryl O. Morton
Managing Director, AIHA Laboratory Accreditation Programs, LLC

Revision 20: 06/07/2022
Date Issued: 01/31/2023



APPENDIX G: PAST REPORTS

DRAFT

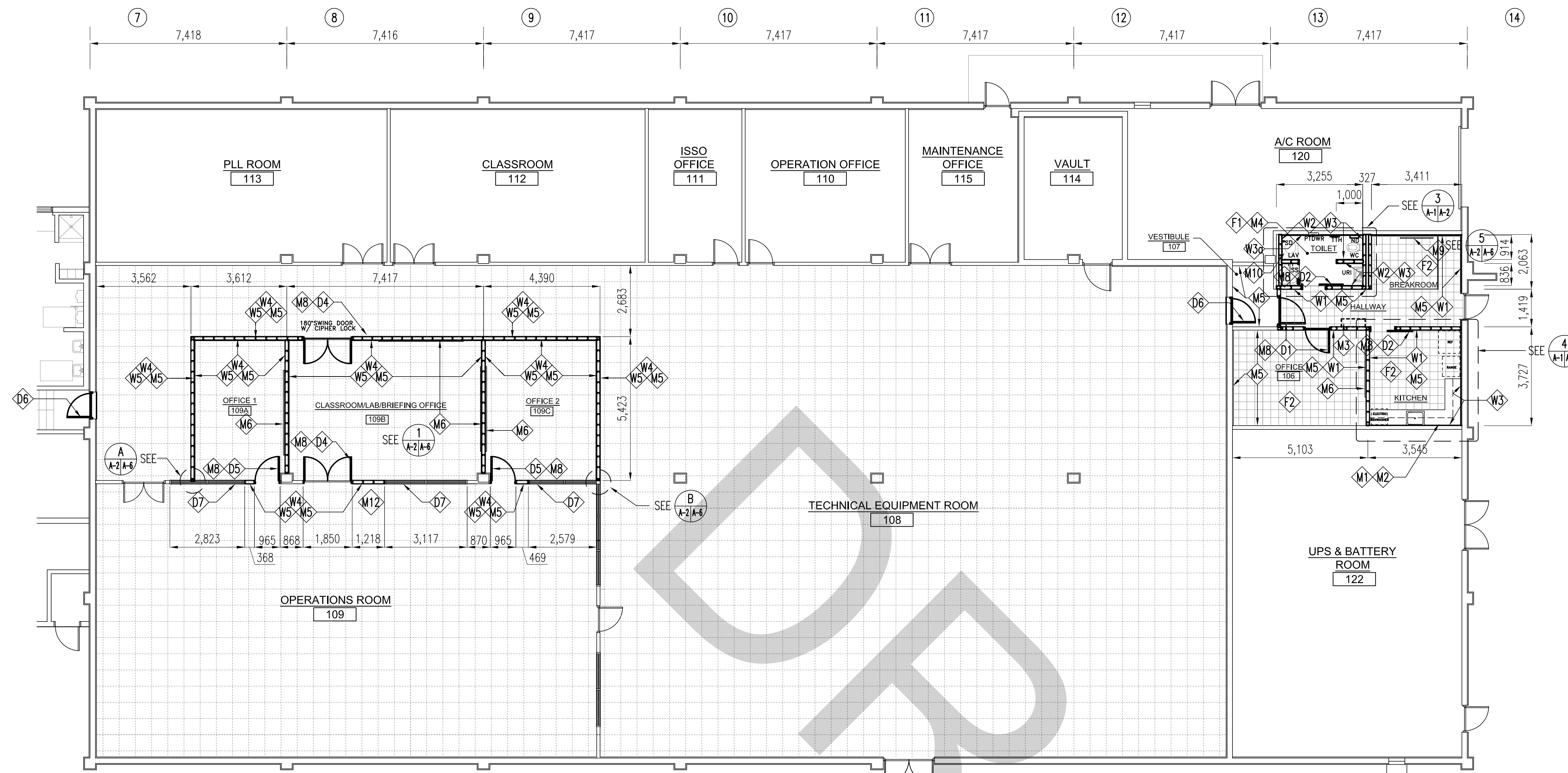
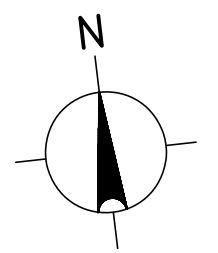
AS-BUILT DRAWINGS
JOK34982

RENOVATE BLDG 100,
EAST ENTRANCE & OFFICE AREA

FORT BUCKNER

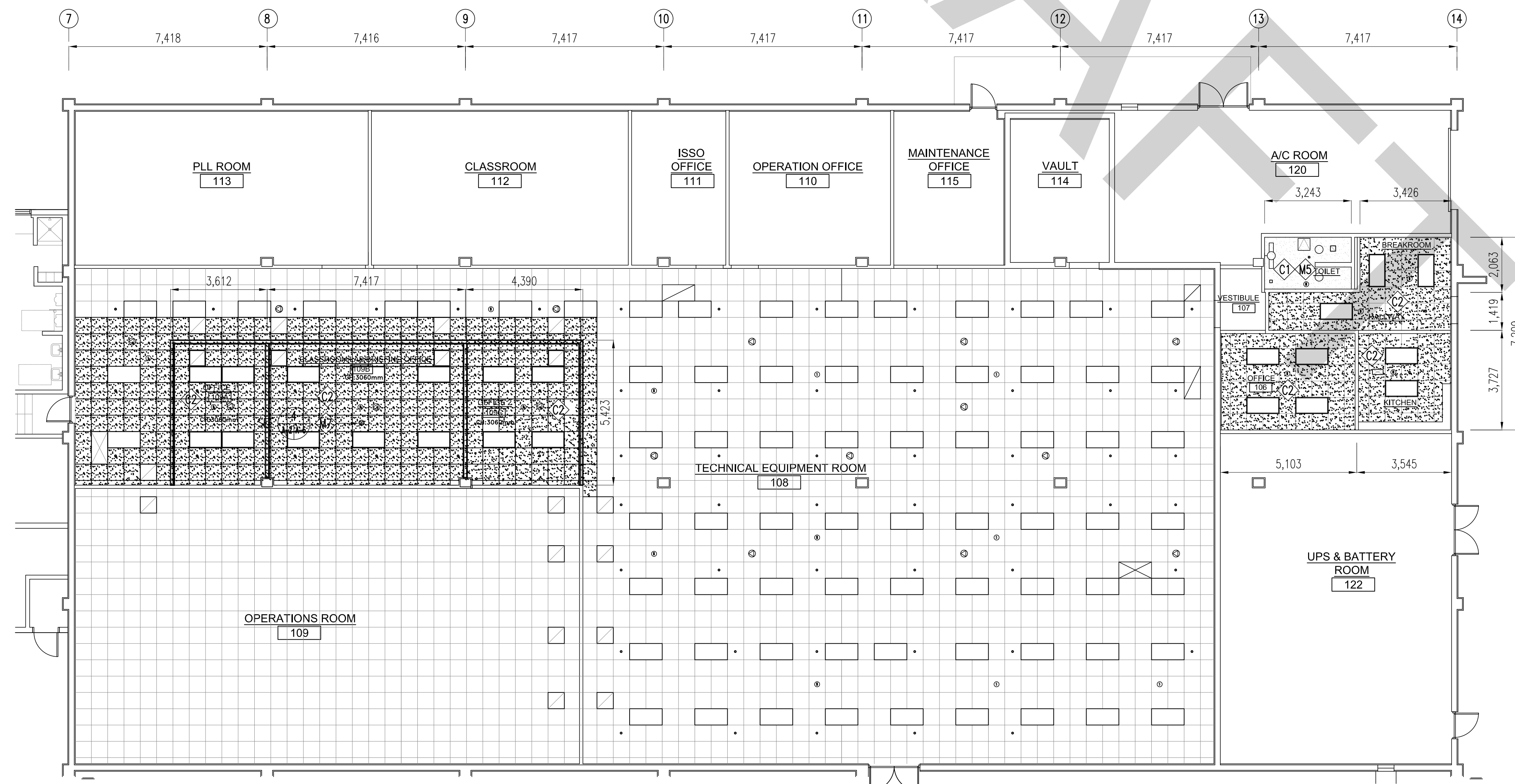
OKINAWA, JAPAN

SCHEDULE OF DRAWINGS			VICINITY MAP	LOCATION MAP	SYMBOLS
SEQ. NO.	SHT. NO.	DRAWING TITLE			
TS-1	1 OF 18	TITLE, SCHEDULE OF DRAWINGS, ABBREVIATIONS, VICINITY MAP, LOCATION MAP AND SYMBOLS			
A-1	2 OF 18	FLOOR & CEILING PLAN AND LEGEND			
A-2	3 OF 18	INTERIOR ELEVATION, KITCHEN PLAN, ELEVATION, SECTIONS AND KEY PLAN			
A-3	4 OF 18	WALL SECTIONS & DETAILS AND KEY PLAN			
A-4	5 OF 18	DOOR SCHEDULE, HARDWARE, DOOR TYPES, DOOR DETAILS AND KEY PLAN			
A-5	6 OF 18	INTERIOR FINISH SCHEDULE AND MISCELLANEOUS DETAILS			
M-1	7 OF 18	PARTIAL HVAC PLAN, HVAC EQUIPMENT SCHEDULE, LEGEND AND KEY PLAN			
M-2	8 OF 18	ENLARGED HVAC PLAN (EAST ENTRANCE) AND MISCELLANEOUS DETAILS			
P-1	9 OF 18	PARTIAL PLUMBING PLAN, PARTIAL PLUMBING CW & HW PLAN, DIAGRAMS, PLUMBING FIXTURE SCHEDULE AND KEY PLAN			
P-2	10 OF 18	ENLARGED DWV PLAN (EAST ENTRANCE), ENLARGED CW & HW PLAN (EAST ENTRANCE) AND MISCELLANEOUS DETAILS			
E-1	11 OF 18	POWER PLAN, LEGEND AND NOTES			
E-2	12 OF 18	LIGHTING PLAN AND TYP OCCUPANCY SENSOR CONTROL DIAGRAM (FOR TOILET)			
E-3	13 OF 18	RECEPTACLE PLAN			
E-4	14 OF 18	ACCESS CONTROL, INTERCOM, TV & TEL PLAN, WIRING SCHEDULE AND SECTION			
E-5	15 OF 18	PA SYSTEM PLAN AND WIRING SCHEDULE			
E-6	16 OF 18	PANELBOARD LOAD SCHEDULES, LIGHTING FIXTURE DETAILS AND SINGLE LINE DIAGRAM			
FP-1	17 OF 18	PARTIAL FIRE PROTECTION PLAN, SPRINKLER ISOMETRIC PIPING DIAGRAM, LEGEND, NOTES AND KEY PLAN			
FP-2	18 OF 18	FIRE ALARM SYSTEM PLAN AND WIRING SCHEDULE			
			ABBREVIATIONS		
			REVIEWED FOR CONSTRUCTIBILITY :		
			U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
			RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
			TITLE SHEET		
			TITLE, SCHEDULE OF DRAWINGS, ABBREVIATIONS, VICINITY MAP, LOCATION MAP AND SYMBOLS		
			JOK34982 TO#7206(RFP#0027) TS-1		
			DATE: 8 JUL '13 SCALE: NONE SHEET: 1 OF 18		
			BLDG. NO. DRAWING NO. FILE SEQ. NO.		



NOTES : RESTORE ANY DAMAGES CAUSED BY CONSTRUCTION IF ANY.

1 FLOOR PLAN
SCALE 1/100



2 CEILING PLAN
SCALE 1/100

LEGEND :

- NEW 610x610mm ACST CEILING TILE
- NEW 12mm WATER RESISTANT GYP BD, PAINT FIN
- ACCESS PANEL, 450x450mm
- LIGHTING FIXTURE, SEE ELEC DWGS
- TRANSFER DUCT GRILLES, SEE MECH DWGS
- SUPPLY AIR GRILLE, SEE MECH DWGS
- RETURN AIR GRILLE, SEE MECH DWGS
- SPEAKER, SEE ELEC DWGS
- SMOKE DETECTOR, SEE ELEC DWGS
- HEAT DETECTOR, SEE ELEC DWGS
- SPRINKLER HEAD, SEE MECH DWGS
- PROJECTOR

STATEMENT OF WORK :

NEW WORKS

EAST ENTRANCE :

FLOOR : F

- F1 PROVIDED QUARRY FLOOR TILES WITH MORTAR BED
- F2 PROVIDED VINYL FLOOR TILES AND COVE BASE

WALL : W

- W1 PROVIDED GYPSUM BOARD WALL WITH METAL STUD/FURRING AND INSULATION, PAINT FINISH
- W2 PROVIDED WATER RESISTANT GYPSUM BOARD WALL WITH METAL STUD AND INSULATION, PAINT FINISH
- W3 PROVIDED CERAMIC TILES WITH ADHESIVE ON WR GYP BOARD
- W30 PROVIDED CERAMIC TILES WITH BOND COAT AND MORTAR BED ON METAL LATH

CEILING : C

- C1 PROVIDED WATER RESISTANT GYPSUM BOARD CEILING INCLUDING METAL SUSPENSION SYSTEM AND ACCESS PANEL, PAINT FIN.
- C2 PROVIDED ACOUSTIC CEILING TILE INCLUDING METAL SUSPENSION SYSTEM

DOORS AND WINDOWS : D

- D1 PROVIDED DOORS WITH FRAME AND HARDWARE
- D2 PROVIDED SLIDING WOOD DOOR, STEEL CHANNEL GUIDE TRACK AND STEEL FRAME.

MISCELLANEOUS : M

- M1 PROVIDED WOOD BASE CABINET WITH SINK
- M2 PROVIDED WOOD WALL CABINET
- M3 PROVIDED CELL PHONE LOCKERS (LOCKERS FOR 50 CELLPHONES AND 8-10 LAPTOPS)
- M4 PROVIDED TOILET ACCESSORIES
- M5 PAINTED SURFACE
- M6 PROVIDED SLIDING WHITE BOARD
- M7 INSTALLED PROJECTOR MOUNT. (EXIST MOUNT AVAILABLE ON SITE, SEE ELEC DRWG)
- M8 PROVIDED ROOM NUMBER/SIGNAGE
- M9 PROVIDED TV WALL MOUNT
- M10 PROVIDED NON-POROUS SYNTHETIC COUNTERTOP WITH BASE CABINET
- M11 PROVIDED ENCLOSURE FOR JANITOR SINK. PROVIDE SHELF ABOVE AND MOP SINK NEAR JANITOR SINK.

OFFICE :

- C2 PROVIDED ACOUSTIC CEILING TYPE WITH SUSPENSION CEILING
- W4 PROVIDED GYPSUM WALL BOARD WITH METAL STUD/FURRING AND INSULATION, PAINT FINISH
- W5 PROVIDED VINYL COVE BASE
- D3 PROVIDE SLIDING DOOR WITH CIPHER LOCK (NOT USED)
- D4 PROVIDED DOUBLE WOOD DOOR WITH FRAME AND HARDWARE
- D5 PROVIDED WOOD DOOR WITH FRAME AND HARDWARE
- D6 REPLACED TWO (2) EA STAINLESS STEEL DOOR & FRAME AT BOTH ENDS OF ROOM 108 AND PROVIDE X09 HIGH SECURITY ELECTRONIC LOCK.
- D7 REPAIRED WINDOW GLASS AND FRAME AFFECTED BY NEW DOOR
- M12 RESTORED ANY DAMAGES ON EXISTING CONDITION.

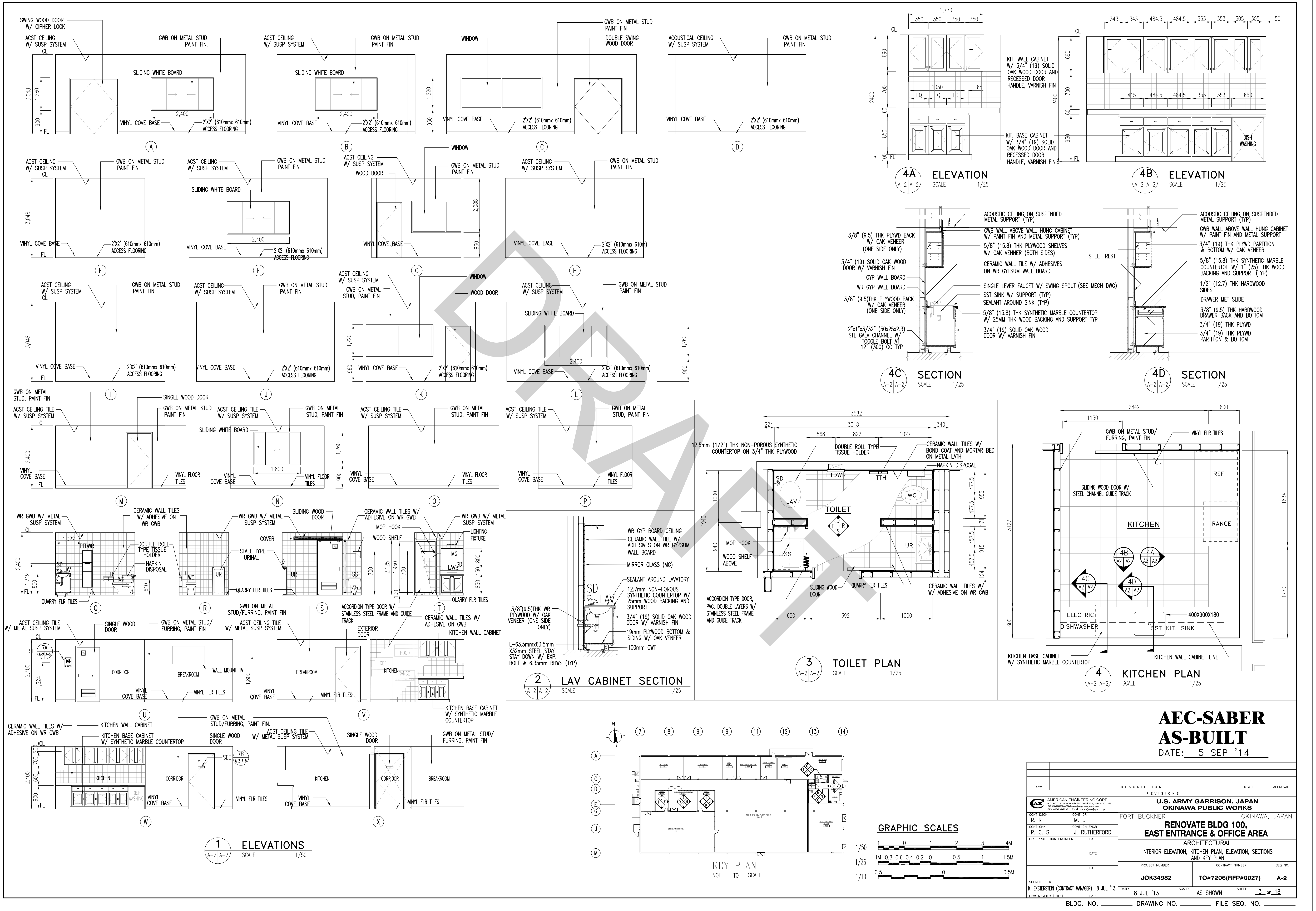
**AEC-SABER
AS-BUILT**

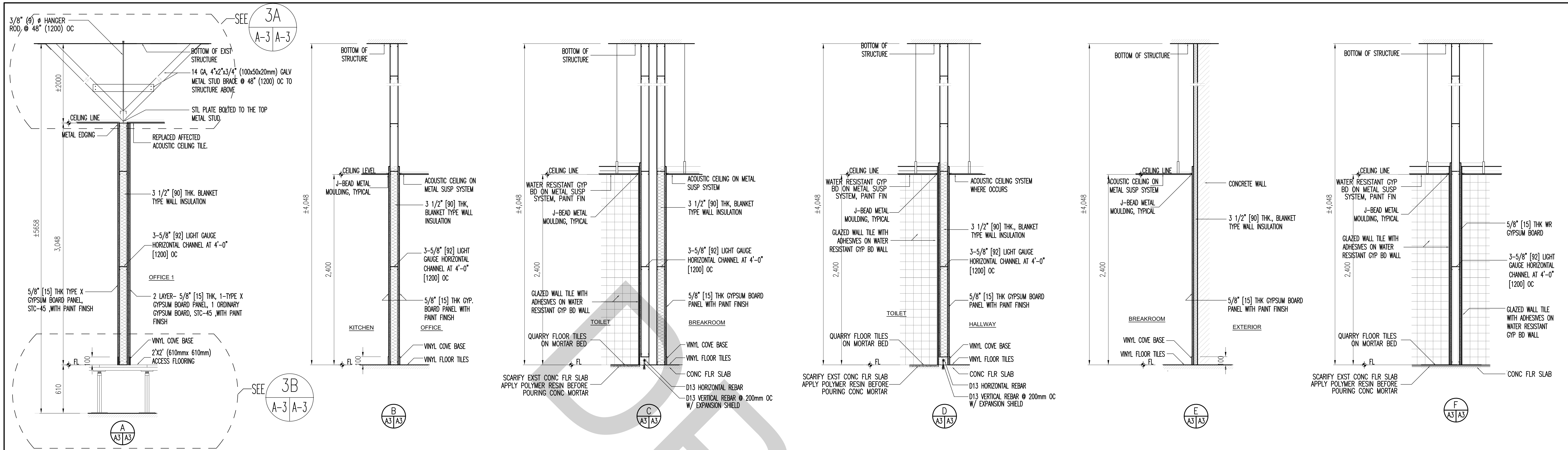
DATE: 5 SEP '14

GRAPHIC SCALES

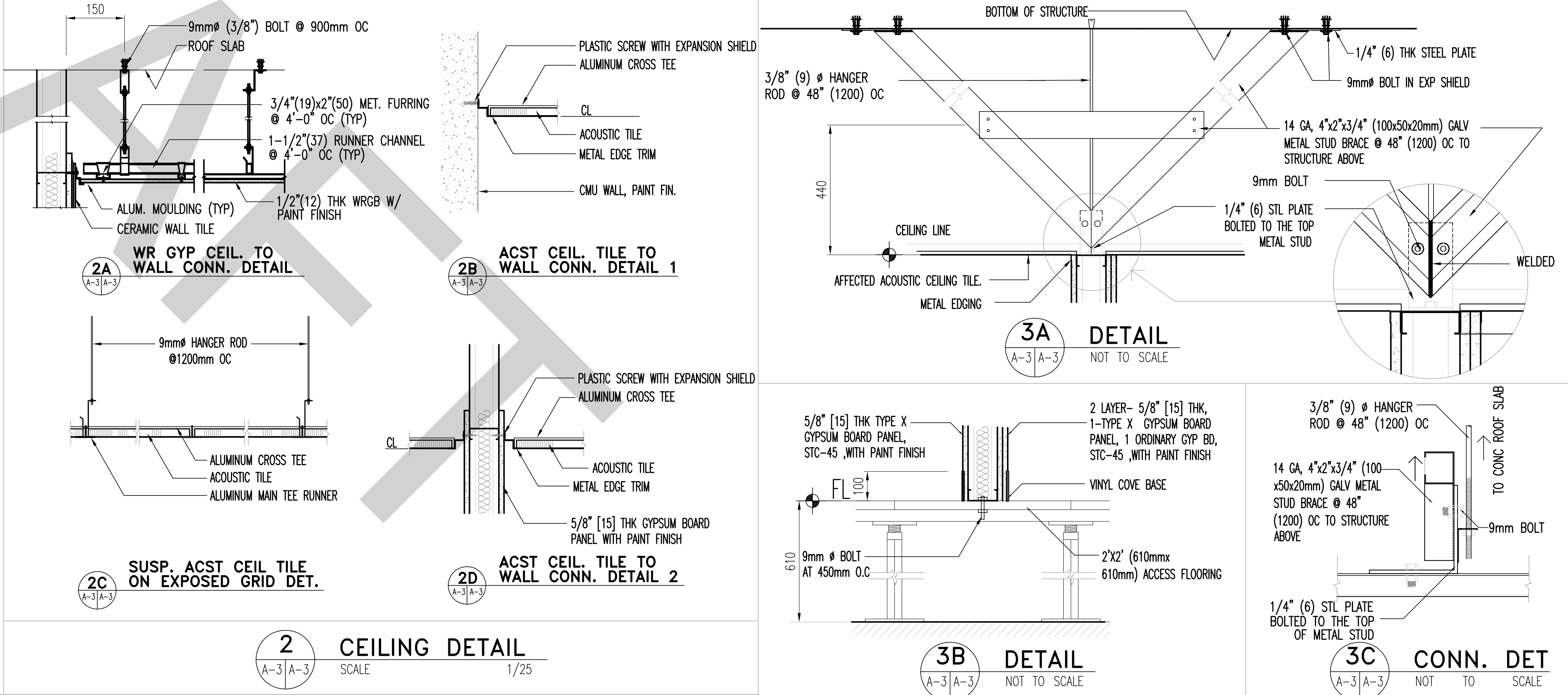


SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. W. 11th Ave., Suite 100, Fort Lauderdale, FL 33309 TEL: (954) 584-1234 FAX: (954) 584-1235		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS FORT BUCKNER, OKINAWA, JAPAN		
CONT. DESIG. R. R.		CONT. OR. ENGR. M. U.		
CONT. CIVIL P. C. S.		CONT. OR. ENGR. J. RUTHERFORD		
FIRE PROTECTION ENGINEER		DATE		
DATE		DATE		
DATE		DATE		
SUBMITTED BY K. EXETERSTEIN (CONTRACT MANAGER)		8 JUL '13		
FIRM NUMBER (TITLE)		DATE		
8 JUL '13		SCALE: AS SHOWN		
DATE		SHEET: 2 of 18		
BLDG. NO.		DRAWING NO.		
		FILE SEQ. NO.		



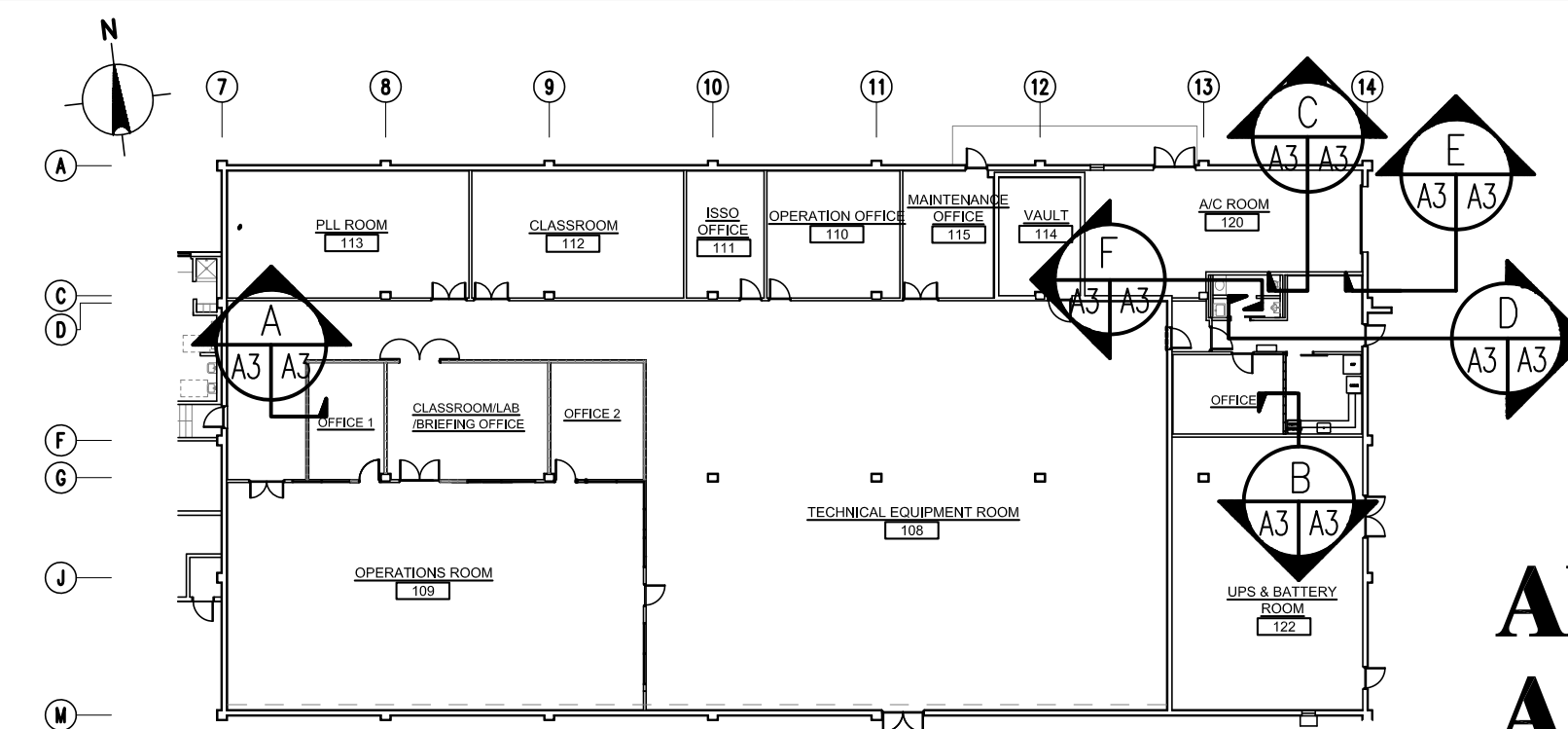


1 WALL SECTIONS
SCALE 1/20



2 CEILING DETAIL
SCALE 1/25

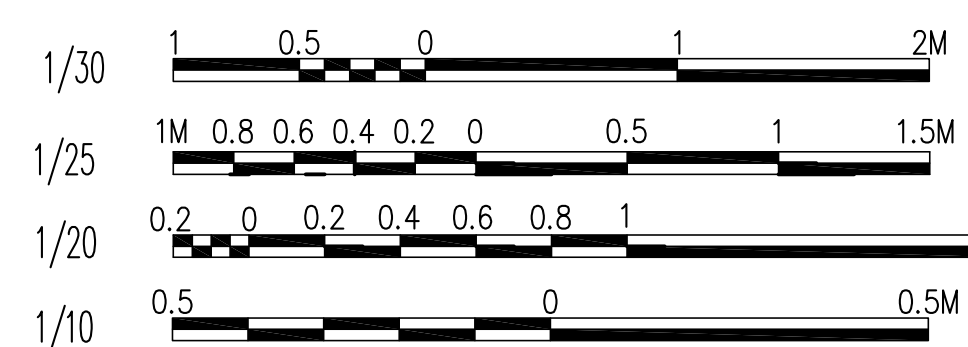
3 WALL ON RAISED FLOOR DET.
SCALE



KEY PLAN
NOT TO SCALE

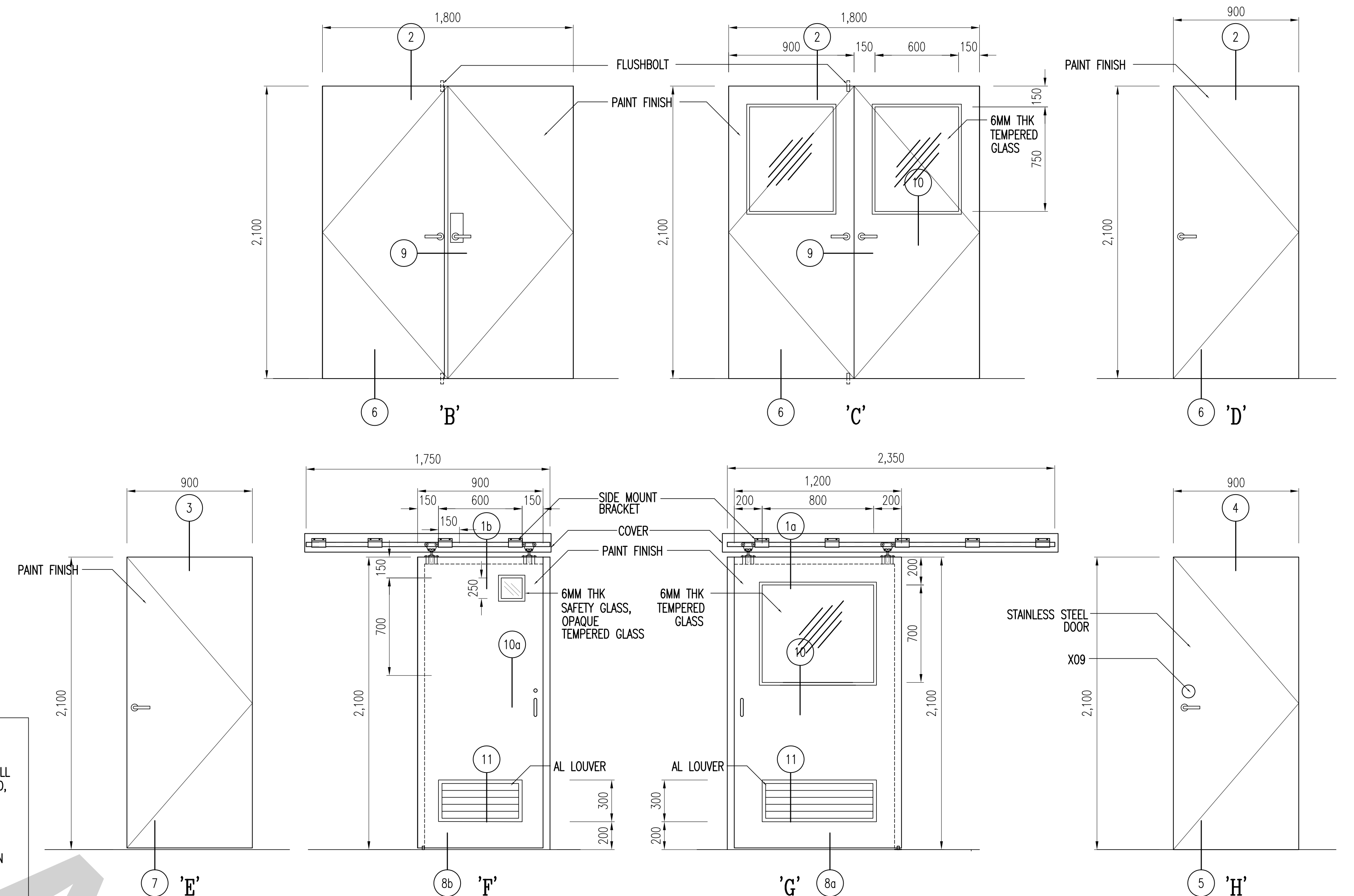
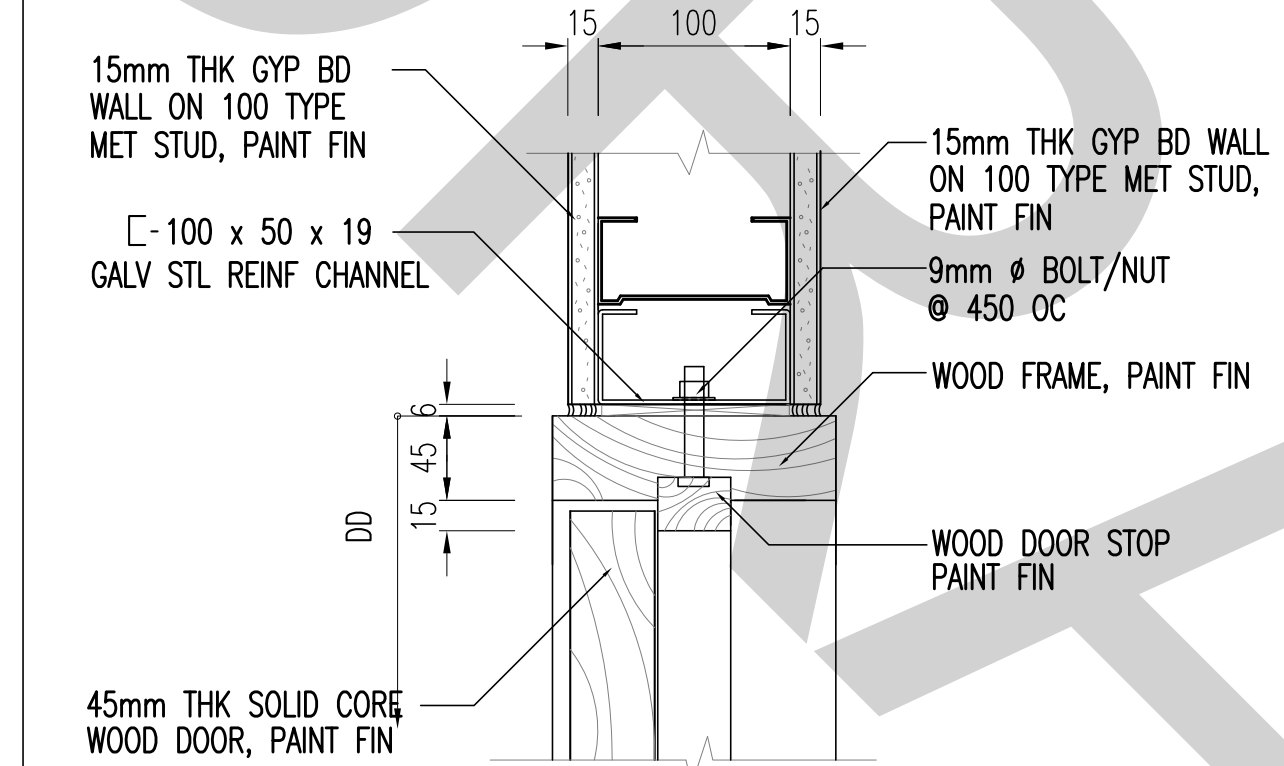
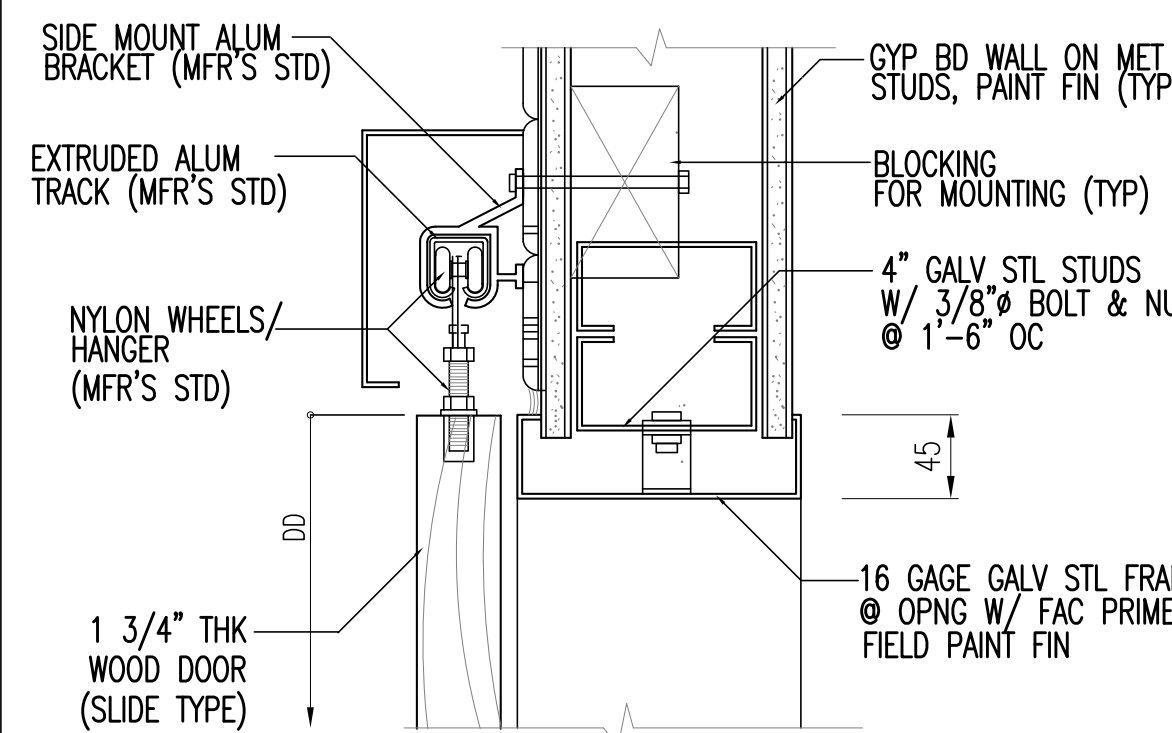
AEC-SABER
AS-BUILT
DATE: 5 SEP '14

GRAPHIC SCALES

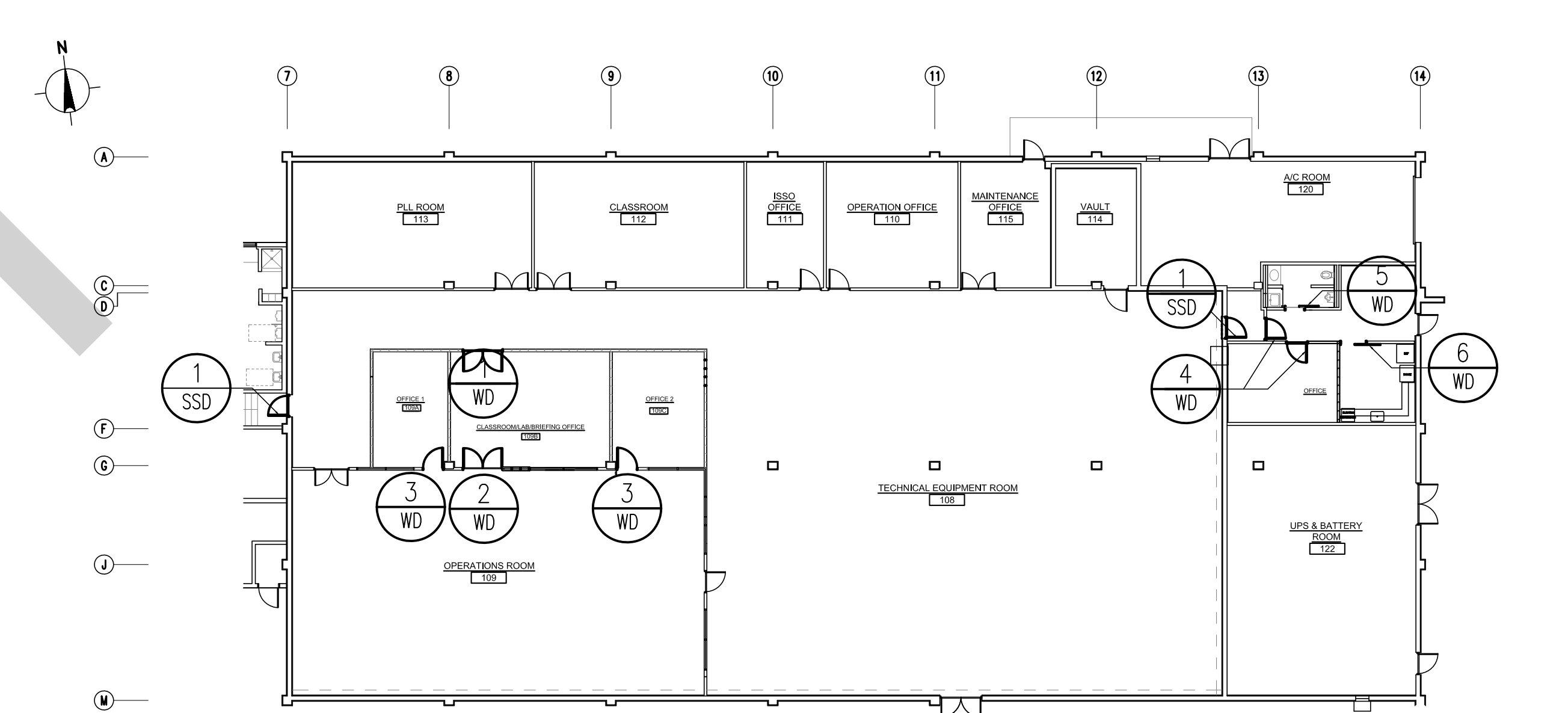
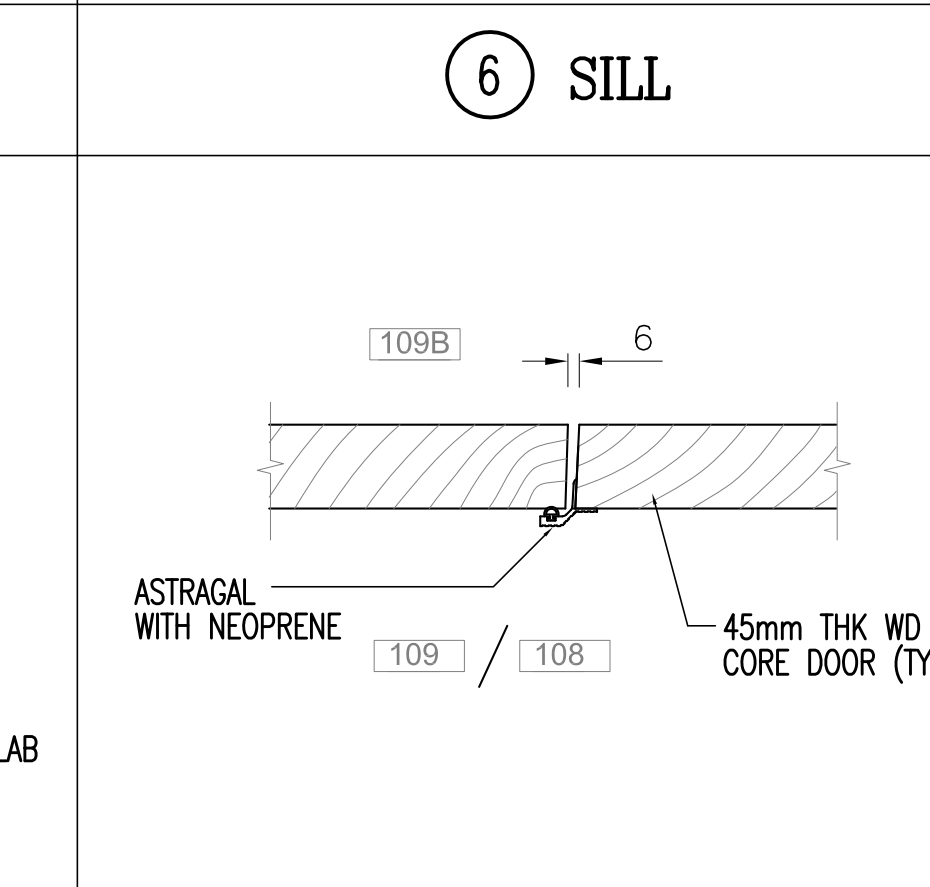
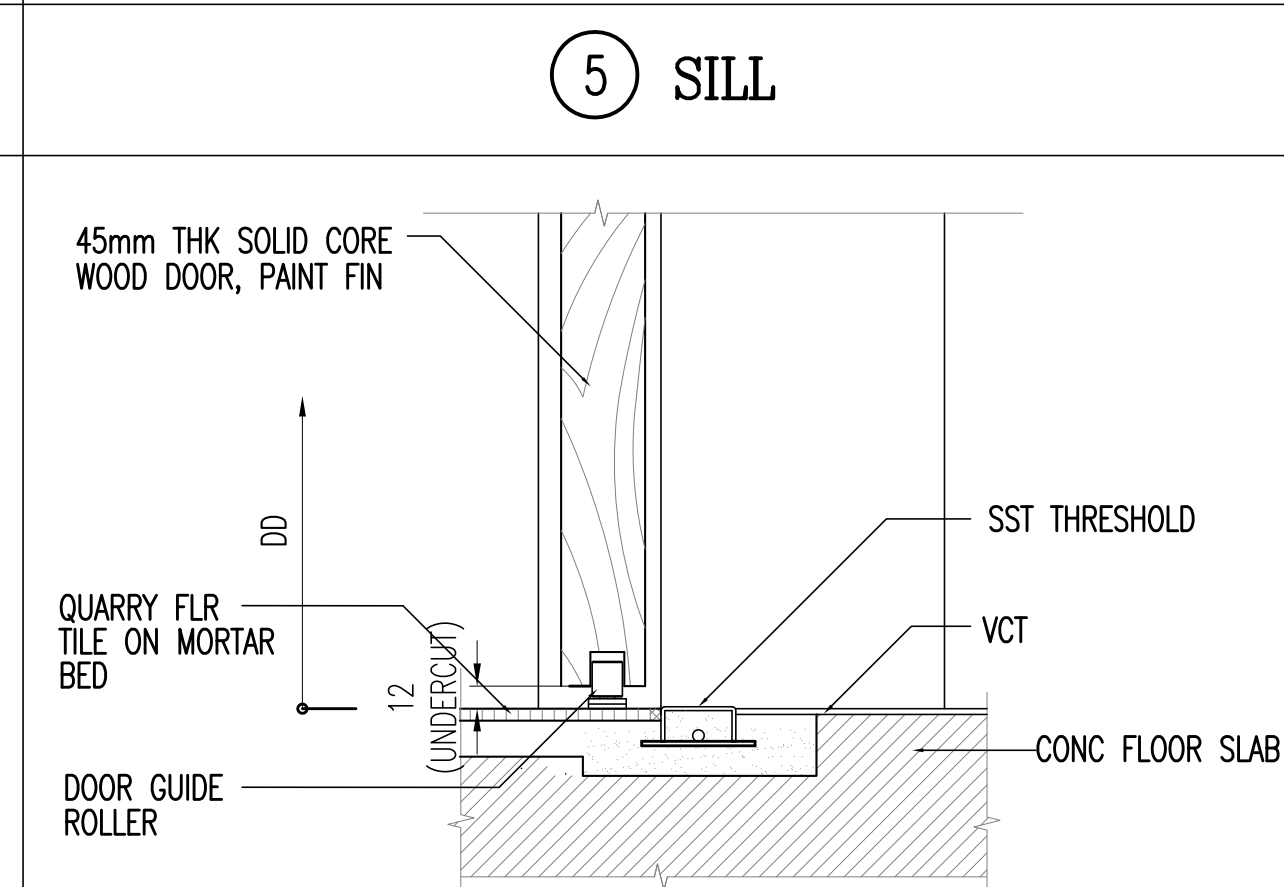
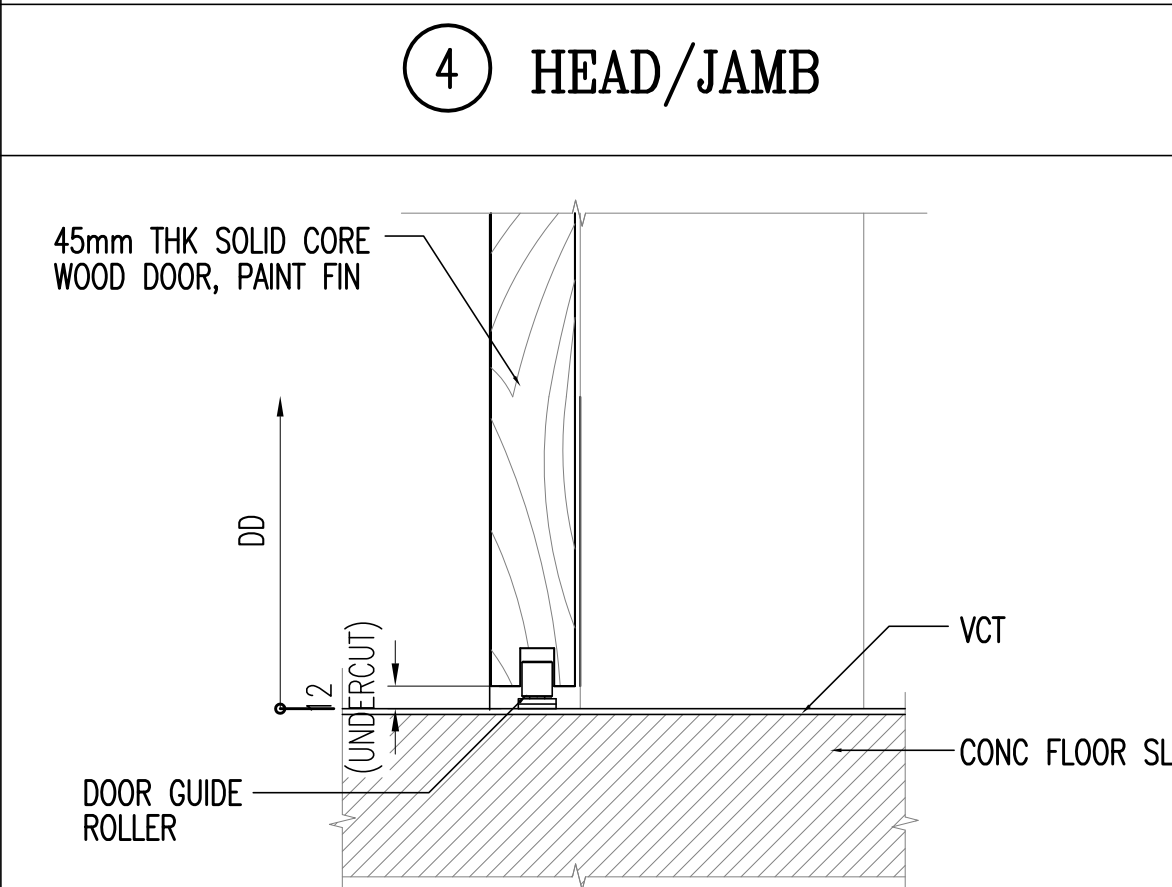
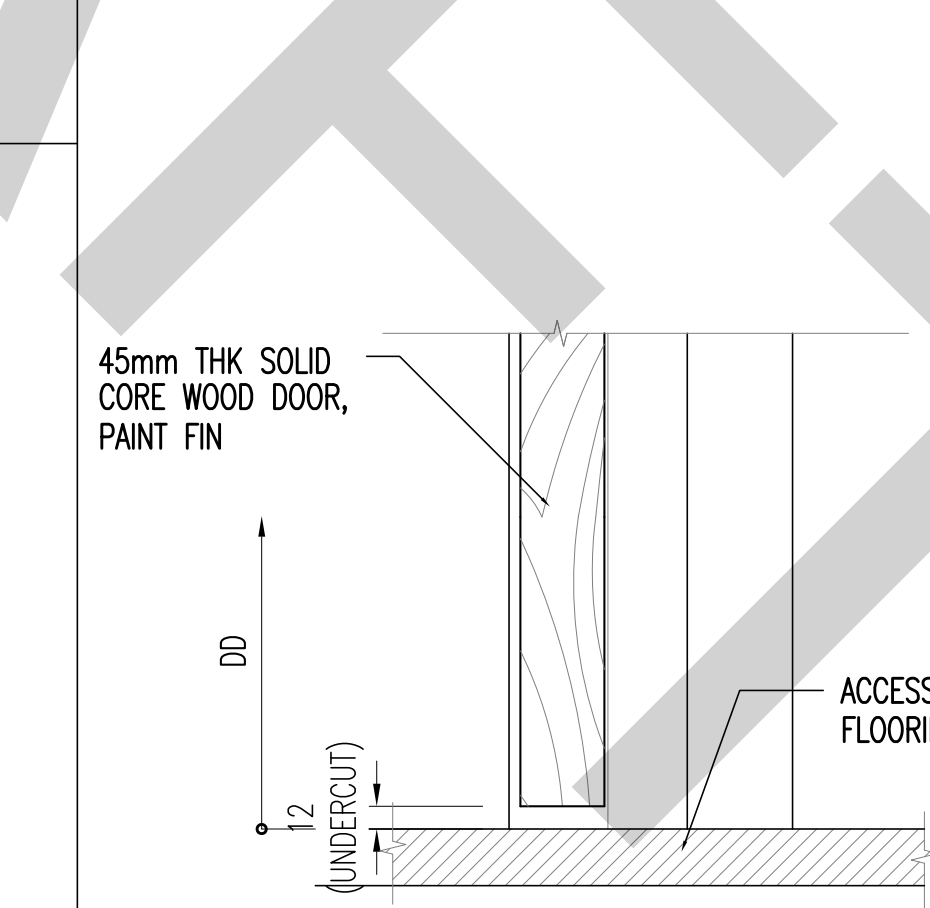
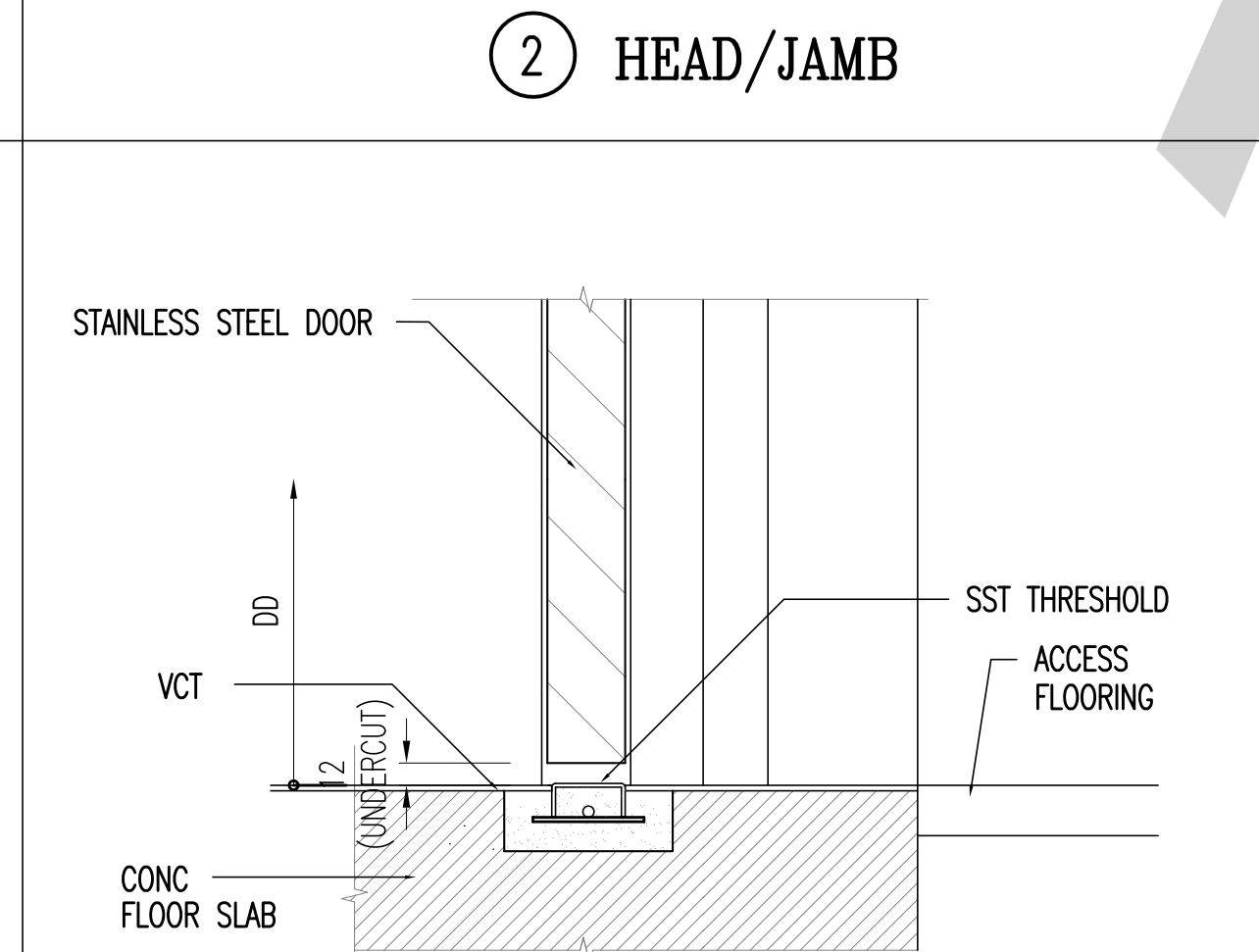
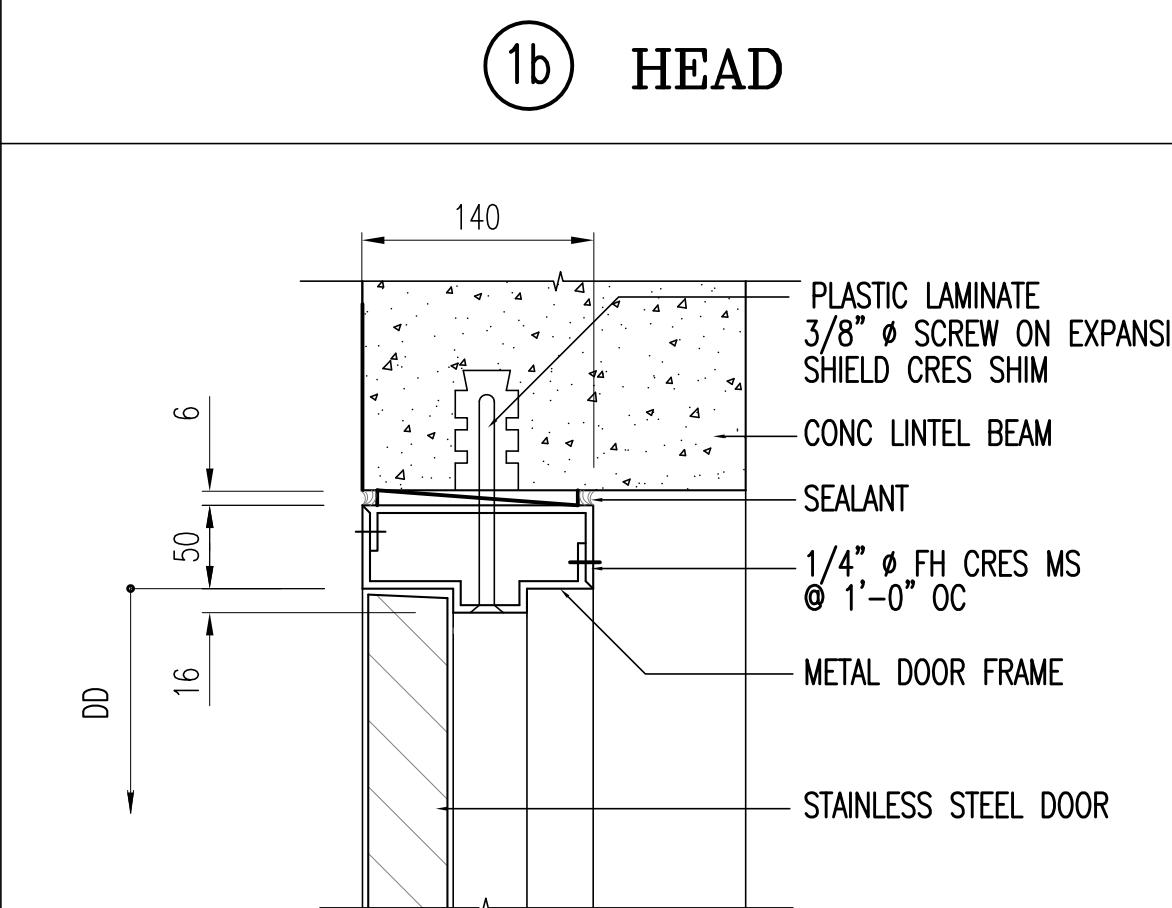


SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP.				
U.S. ARMY GARRISON, JAPAN				
OKINAWA PUBLIC WORKS				
FORT BUCKNER, OKINAWA, JAPAN				
RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA				
ARCHITECTURAL				
WALL SECTIONS & DETAILS AND KEY PLAN				
PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.
JOK34982		TO#7206(RFP#0027)		A-3
SUBMITTED BY		DATE		SCALE
K. EXETERSTEN (CONTRACT MANAGER)		8 JUL '13		AS SHOWN
BLDG. NO.		DRAWING NO.		FILE SEQ. NO.

DOOR HARDWARE				
HARDWARE	LOCKSET	CLOSER	HINGES	STOPPER
HW-1	10 PIN MECH. CIPHER LOCK, OTHER DOOR LEAF NORMALLY FIX WITH EXTENSION FLUSHBOLT	C02061, 180° HELD OPEN	3 PAIRS-4"x3" 2 BALL BEARING HINGES SATIN SST FINISH	ALUM TUBE, WALL BUMPER/HOOK & EYEBOLT TYPE HOLD OPEN
HW-2	F82, OFFICE DOOR LOCK FUNCTION, LEVER HANDLE, GRADE 2 OTHER DOOR LEAF NORMALLY FIX WITH EXTENSION FLUSHBOLT	C02021, PARALLEL ARM TYPE	3 PAIRS-4"x3" 2 BALL BEARING HINGES SATIN SST FINISH	ALUM TUBE, WALL BUMPER/HOOK & EYEBOLT TYPE HOLD OPEN
HW-3	F82, OFFICE DOOR LOCK FUNCTION, LEVER HANDLE, GRADE 2	C02021, PARALLEL ARM TYPE	1-1/2 PAIRS-4"x3" 2 BALL BEARING HINGES SATIN SST FINISH	ALUM TUBE, WALL BUMPER/HOOK & EYEBOLT TYPE HOLD OPEN
HW-4	PIVOTED HOOK TYPE DEAD BOLT BY KEY OUTSIDE THUMBTURN INSIDE W/ LEVER HANDLE INSIDE AND OUTSIDE	_____	_____	END GUIDE/STOP, BUMPER SHOE
HW-5	HOOK TYPE LATCH BOLT BY LEVER INSIDE AND OUTSIDE	_____	_____	END GUIDE/STOP, BUMPER SHOE
HW-6	X09, HIGH SECURITY ELECTRONIC LOCK	C02021, PARALLEL ARM TYPE	1-1/2 PAIRS-4"x3" 2 BALL BEARING HINGES SATIN SST FINISH	_____



1 DOOR TYPES
SCALE 1/20



KEY PLAN

NOT TO SCALE

[illegible]

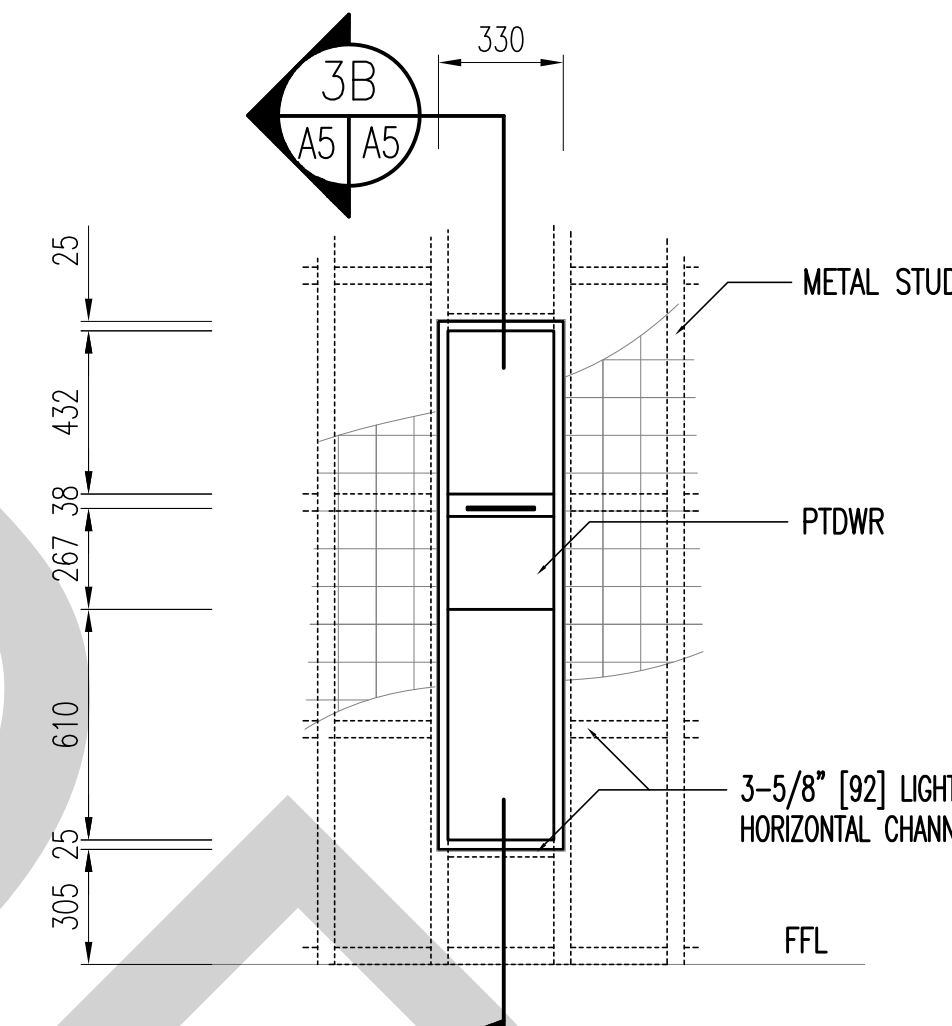
50 6mm THK TEMPERED GLASS NEOPRENE BED AL. MOULDING W/ 4.8mm Ø FHMS @ 200mm O.C. 45mm THK WOOD DOOR 15 12 10 GLASS	50 6mm THK SAFETY GLASS, LAMINATED NEOPRENE BED AL. MOULDING W/ 4.8mm Ø FHMS @ 200mm O.C. 45mm THK WOOD DOOR 15 12 10a GLASS	29 45mm THK WOOD DOOR ANODIZED BRONZE EXTRUDED ALUMINUM LOUVER ALUM INSECT SCREEN ANODIZED BRONZE EXTRUDED ALUMINUM LOUVER FRAME W/ 6mm Ø FHMS AT 250 OC 29 11 LOUVER
---	---	---

Technical drawing of a TV mounting rack showing dimensions and components:

- Overall width: 762
- Overall height: 1800
- TV mounting rack (MFR STANDARD) dimensions: 100-750mm (width) and 240 (height).
- TV mounting rack (MFR STANDARD) dimensions: 100-480mm (height).
- TV mounting rack (MFR STANDARD) dimensions: 3-5/8" [92] LIGHT GAUGE HORIZONTAL CHANNEL (ADDITIONAL SUPPORT).
- TV mounting rack (MFR STANDARD) dimensions: 75 (height).
- TV mounting rack (MFR STANDARD) dimensions: 5° (angle).
- TV mounting rack (MFR STANDARD) dimensions: METAL STUD.
- TV mounting rack (MFR STANDARD) dimensions: GWB.
- TV mounting rack (MFR STANDARD) dimensions: FL.

5A ELEVATION **5B** SECTION

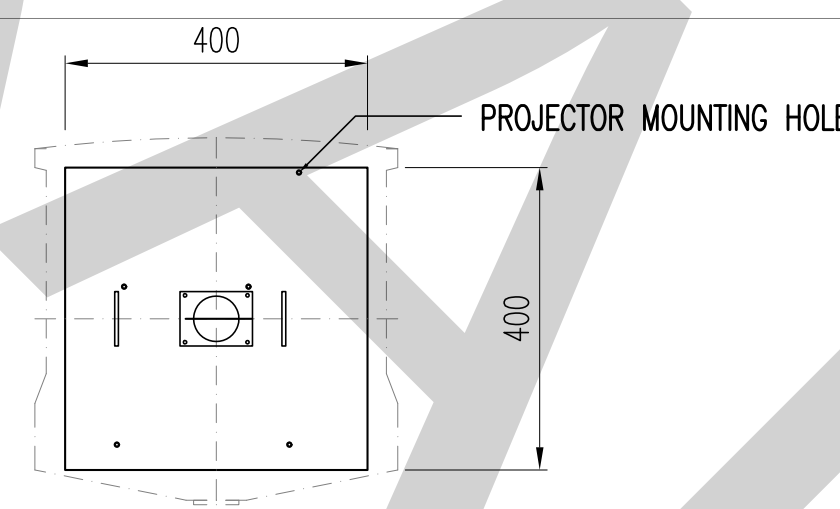
5 CEILING MOUNTED TV
A-2 | A-5 SCALE 1/10



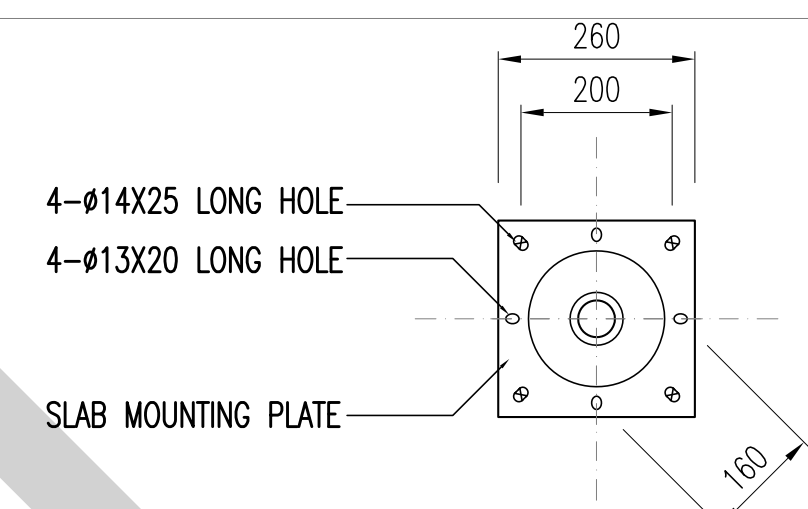
3A PLAN

3B SECTION
A-5 A-5

PTDWR INSTALLATION DETAIL
 SCALE 1/20

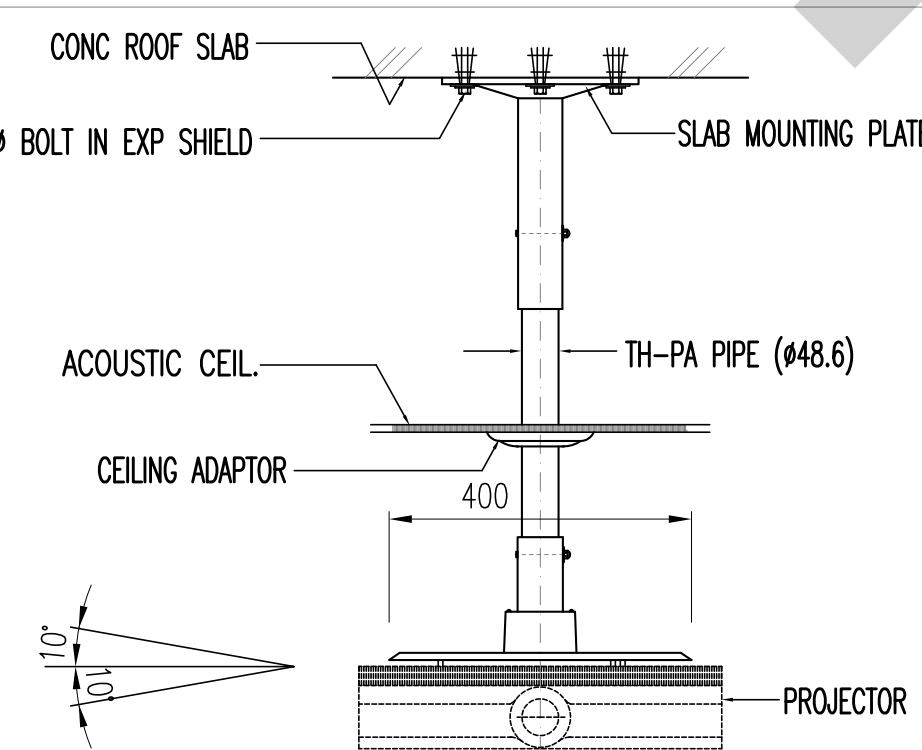


4A SECTION

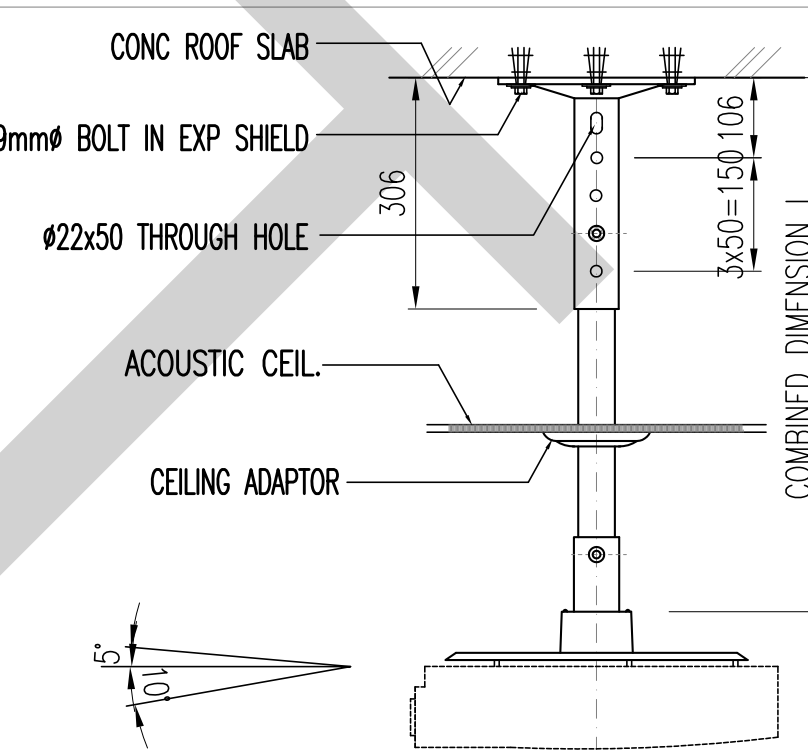


4B SECTION

PROJECTOR MOUNTED INSTALLATION

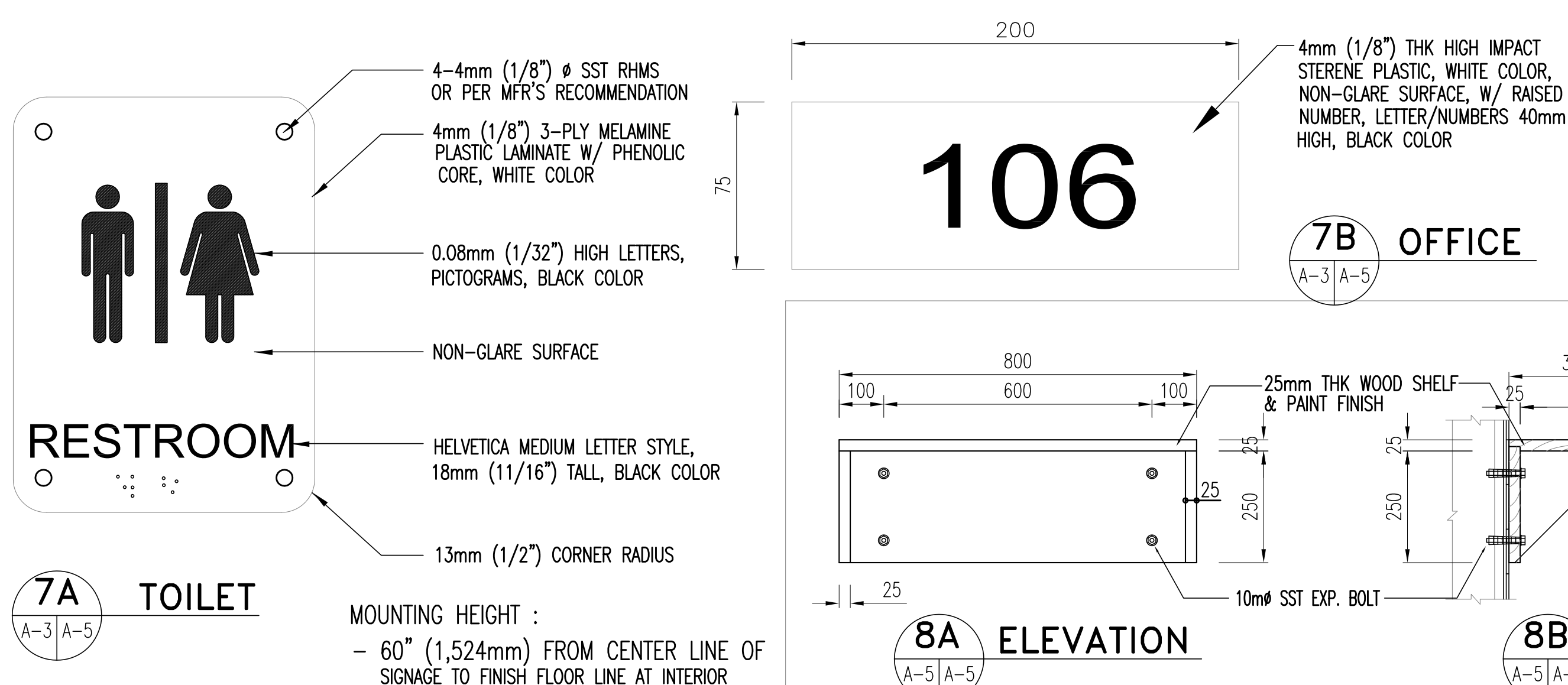
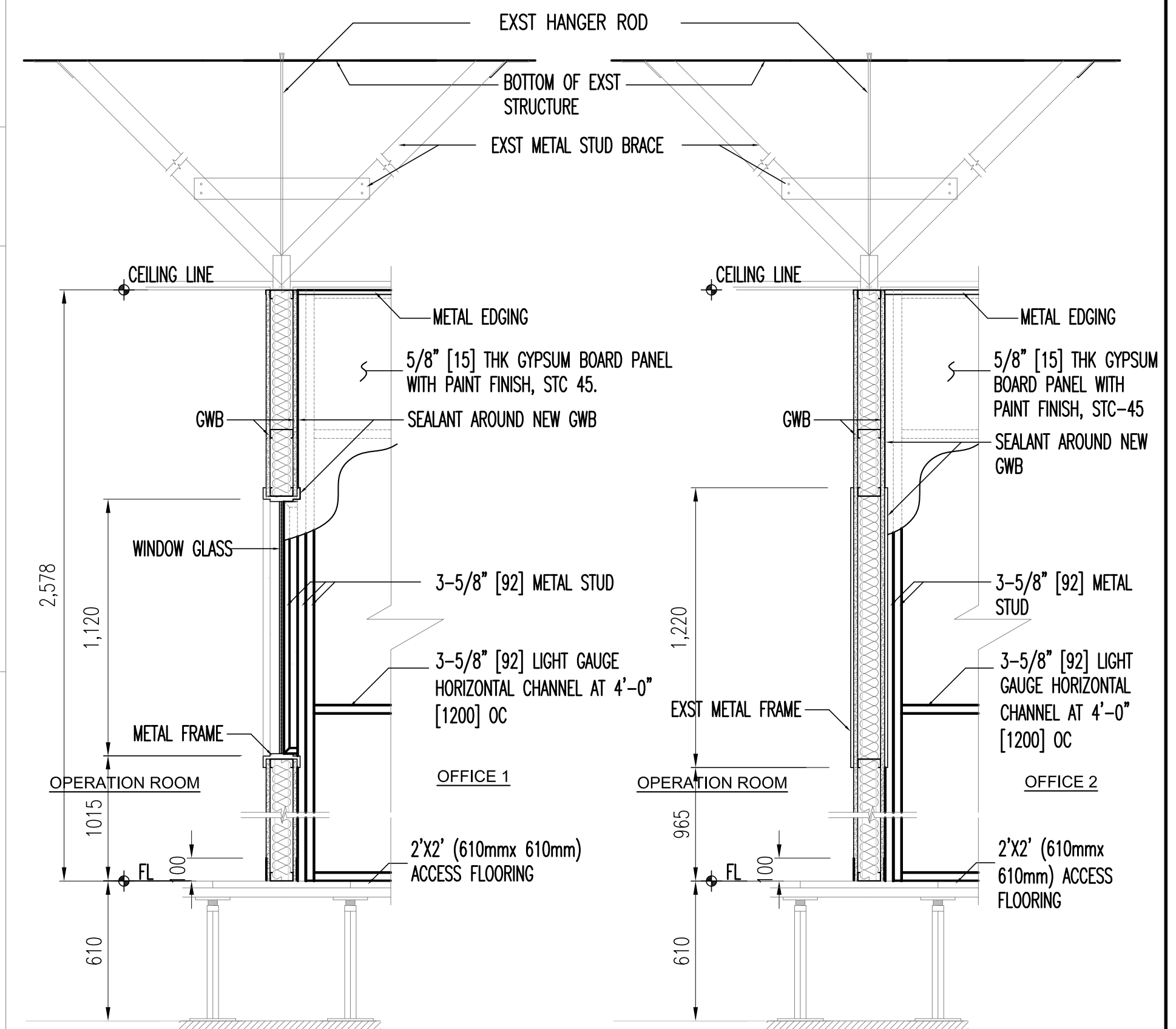


4C ELEVATION



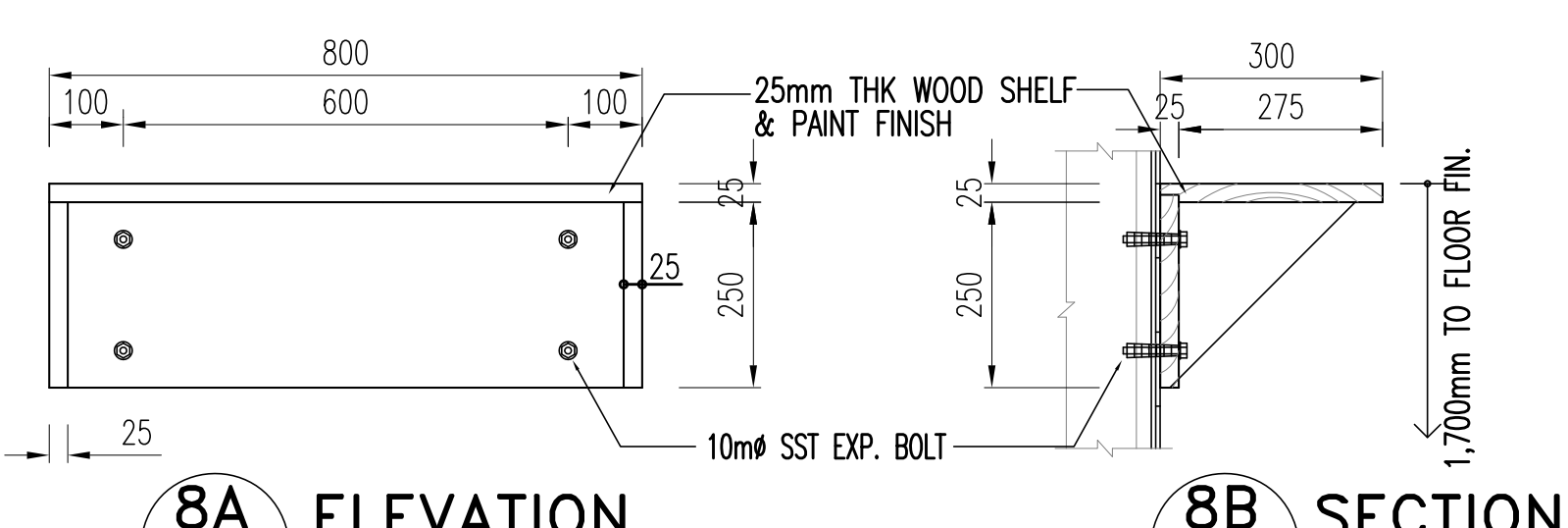
4D ELEVATION

6 EXST TO NEW WALL CONNECTION
A-5 | A-5 SCALE 1/20



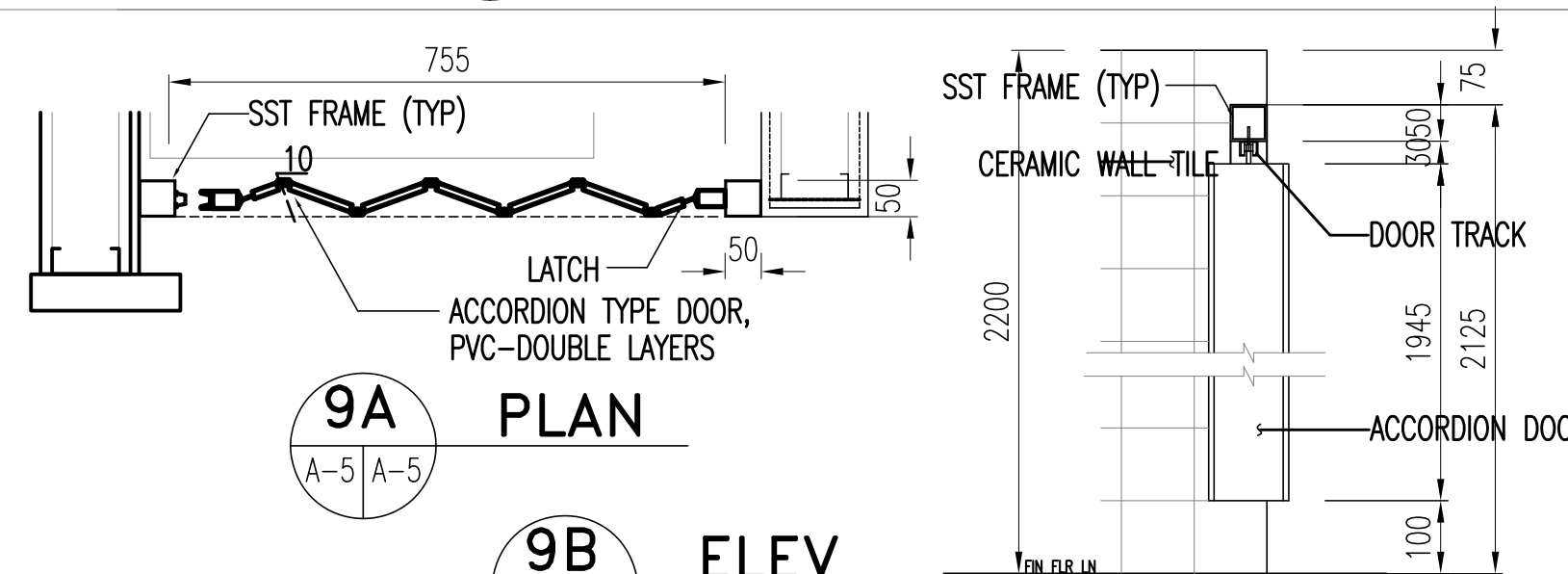
7A TOILET

7 SIGNAGE DETAIL
A-5 | A-5 SCALE 1/10



8A ELEVATION

8 WOOD SHEF DET.
A-5 A-5 NOT TO SCALE



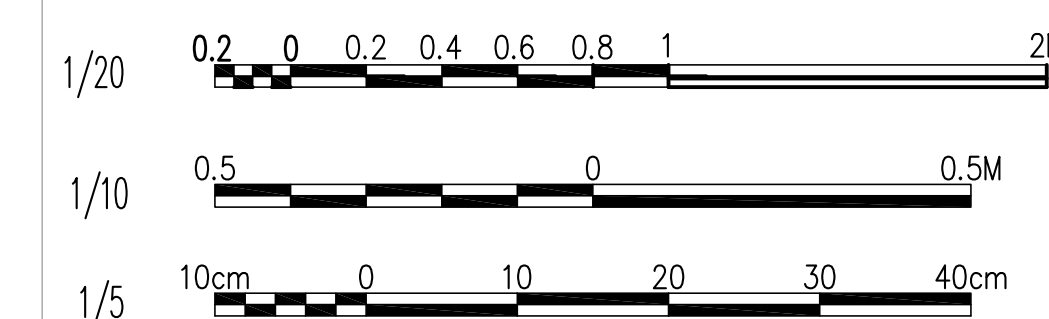
9A PLAN

9B ELEV

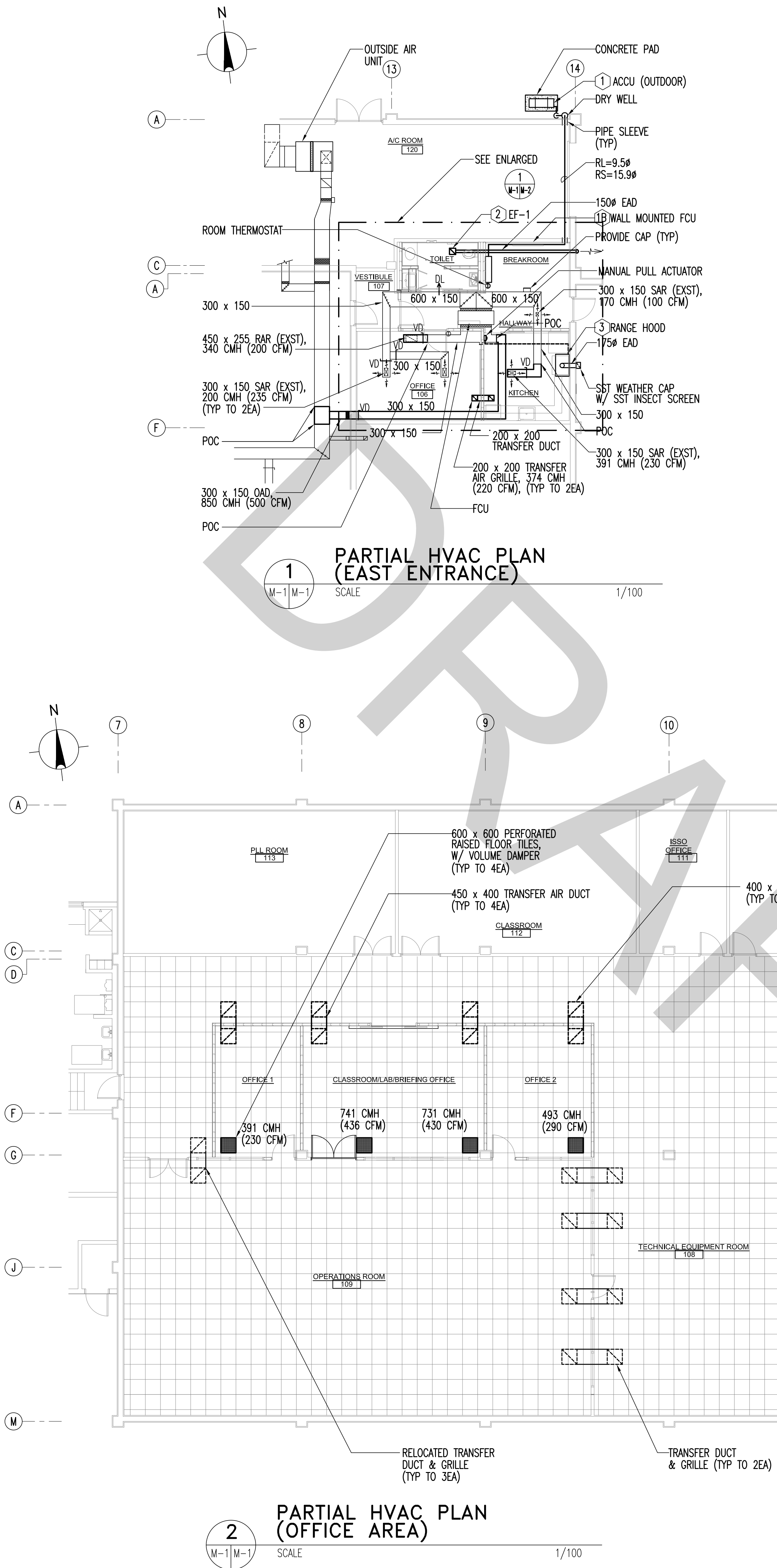
9 ACCORDION DOOR DET.
A-5 | A-5 NOT TO SCALE

AEC-SABER
AS-BUILT
DATE: 5 SEP '14

GRAPHIC SCALES

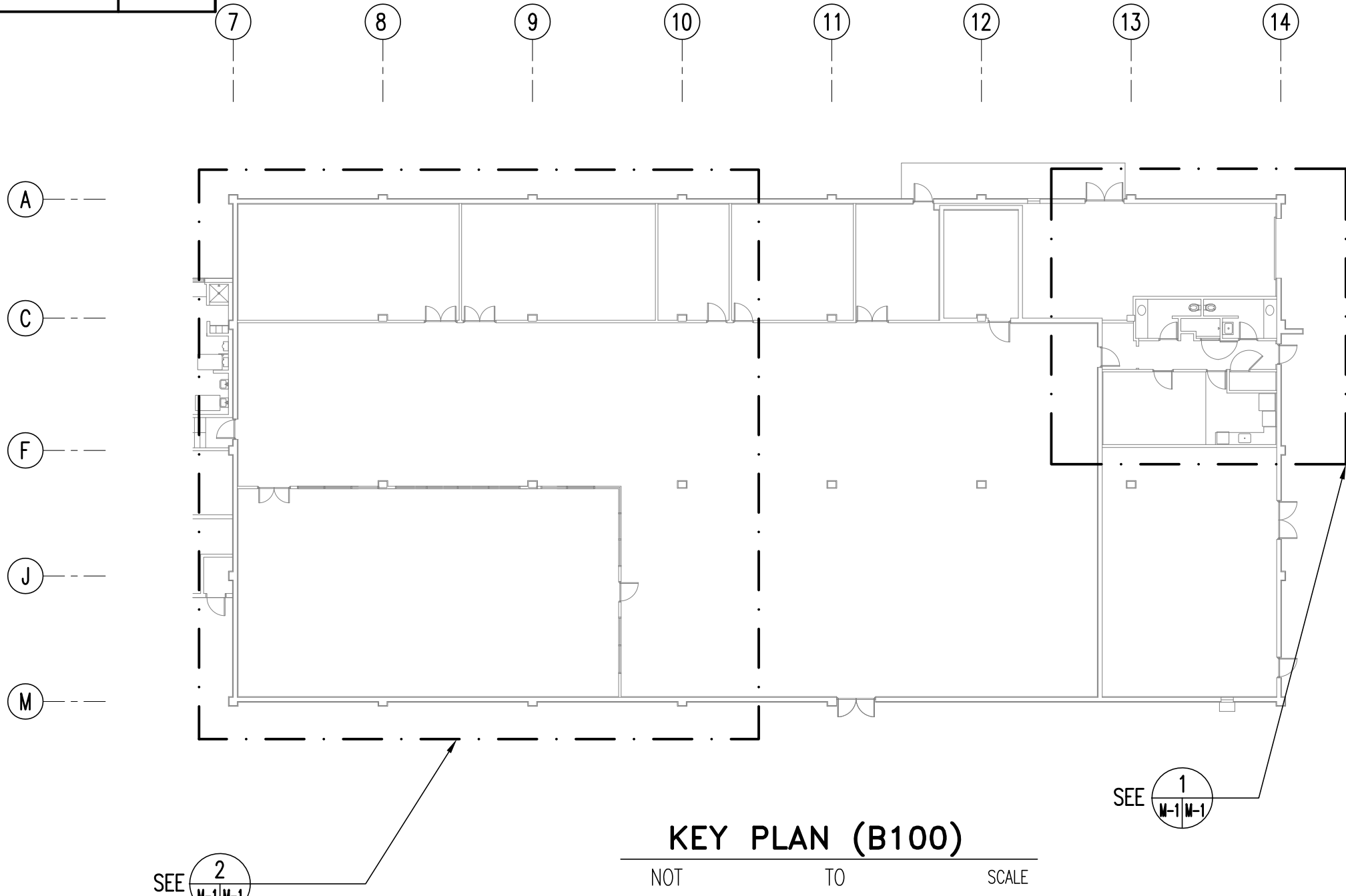


SYM		DESCRIPTION				DATE	
REVISIONS							
 AMERICAN MILITARY ENGINEERING CORP. 1415 S. 17TH ST. (2ND FL.) FORT BELLE, MISSISSIPPI 39244-0001 FAX: (601) 684-0027 (601) 684-0010		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS					
		FORT BUCKNER OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA					
CONT' DESGN R. R.		CONT' DR M. U.					
CONT' CHK P. C. S.		CONT' CH ENGR J. RUTHERFORD					
FIRE PROTECTION ENGINEER		DATE:					
		DATE:					
		DATE:					
		PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.	
		JOK34982		TO#7206(RFP#0027)		A-5	
SUBMITTED BY K. EXTERSTEN (CONTRACT MANAGER)		DATE: 8 JUL '13		SCALE:		SHEET: 6 OF 18	
(FIRM MEMBER TITLE)		DATE					
BLDG. NO.		DRAWING NO.		FILE NO.		NO.	



HVAC EQUIPMENT SCHEDULE		
SYMBOL	DESCRIPTION	Q'TY
①	SPLIT TYPE AIR CONDITIONING UNIT - (AC-1) SPLIT TYPE, PACKAGED AIR CONDITIONING UNIT COMPLETE WITH AIR COOLED CONDENSING UNIT (ACU) & WALL MOUNTED TYPE DUCT-FREE FAN COIL UNIT (FCU) W/ WALL MOUNTED PROGRAMMABLE THERMOSTAT/CONTROLLER. UNIT SHALL HAVE AUTO RECOVERY SYSTEM. CONDENSER COIL SHALL HAVE COPPER TUBES WITH ALUMINUM FINS W/ ANTI-CORROSIVE COATING. REFRIGERANT : R-410A MFR : DAIKIN MODEL : RF80CAT SERIAL : C001572B107 AIR COOLED CONDENSING UNIT : 1) COOLING CAPACITY ; 24,230 BTU/HR (7.1 KW) AT 75 °F (24.0 °C) DB ENTERING & 95 °F(35 °C) AMBIENT TEMPERATURE. 2) ELECTRICAL RATING; a) COMPRESSOR, MOTOR: 1 EA, 200 V, 3 PH, 60 HZ, 1.6 KW b) CONDENSER FAN MOTOR: 1 EA, 200 V, 3 PH, 60 HZ, 60 W MFR : DAIKIN MODEL : FAP80BB SERIAL : E01360U13.04 FAN COIL UNIT (FCU) : WALL MOUNTED TYPE 1) TOTAL COOLING CAPACITY; 24,230 BTU/HR (7.1 KW) 2) ELECTRICAL RATING; a) FAN MOTOR; 1 EA, 200 V, 1 PH, 60 HZ, 48 W	1
②	CEILING EXHAUST FAN (EF-1) : CEILING MOUNTED EXHAUST FAN, COMPLETE WITH EXHAUST GRILLE, DUCT CONNECTOR, PLASTIC IMPELLER AND MOTOR WITH DISCONNECT SWITCH. 1) CAPACITY ; 170 CMH AT 75 Pa STATIC PRESSURE 2) ELECTRICAL RATING ; 115V, 1PH, 60HZ, 80W MFR : COOK MODEL : GC144	1
③	RANGE HOOD : MANUFACTURER'S STANDARD, WALL MOUNTED SST RANGE HOOD COMPLETE WITH INLINE DUCT CENTRIFUGAL FAN, BUILT-IN WET CHEMICAL FIRE SUPPRESSION SYSTEM W/ REMOTE MANUAL PULL STATION, SPRAY NOZZLES, FUSIBLE LINKS, PRESSURIZED WET CHEMICAL AGENT TANK. RANGE HOOD SHALL BE UL LISTED. 1) ELECTRICAL RATING ; 115V, 1PH, 60HZ, 5Amps MFR : DENLAR MODEL : D1036-FE SERIAL : 1000SERIES	1

LEGEND	
SYMBOL	DESCRIPTION
①	ROOM THERMOSTAT
⊠	EXST SUPPLY AIR REGISTER
⊞	EXST RETURN AIR REGISTER
▨	TRANSFER AIR DUCT & GRILLE
VD	MANUAL VOLUME DAMPER

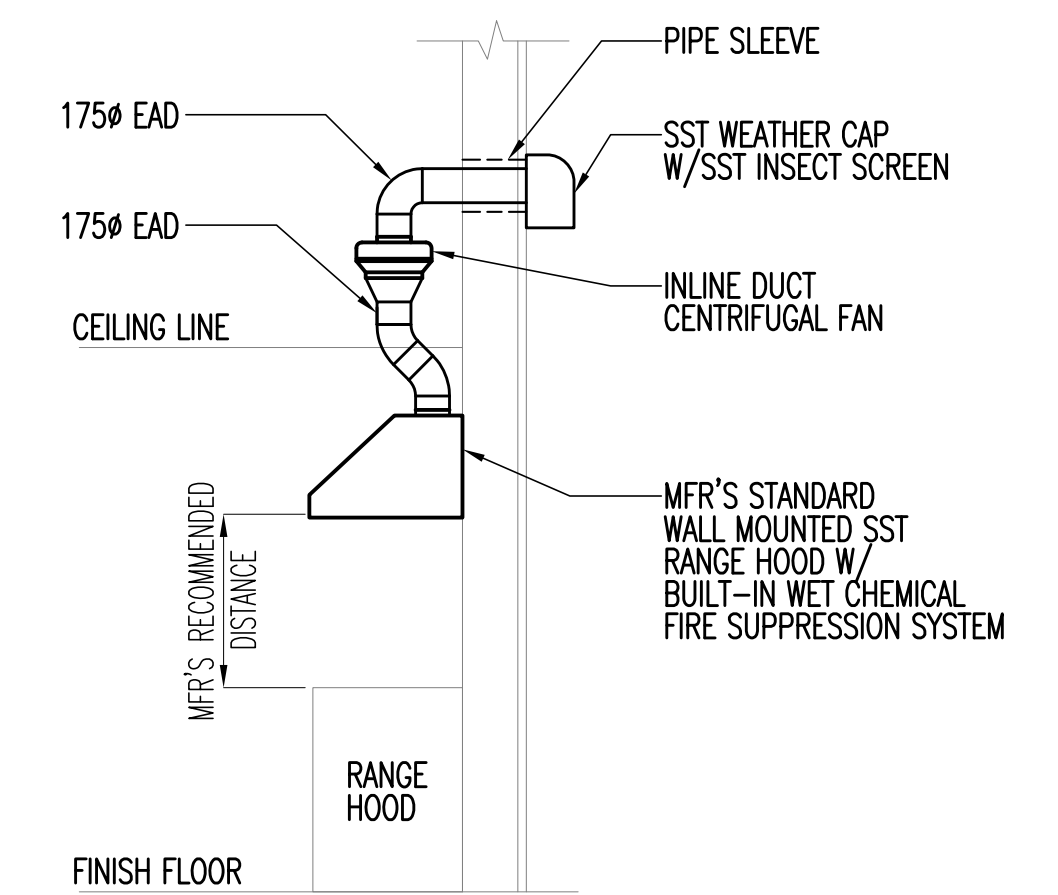
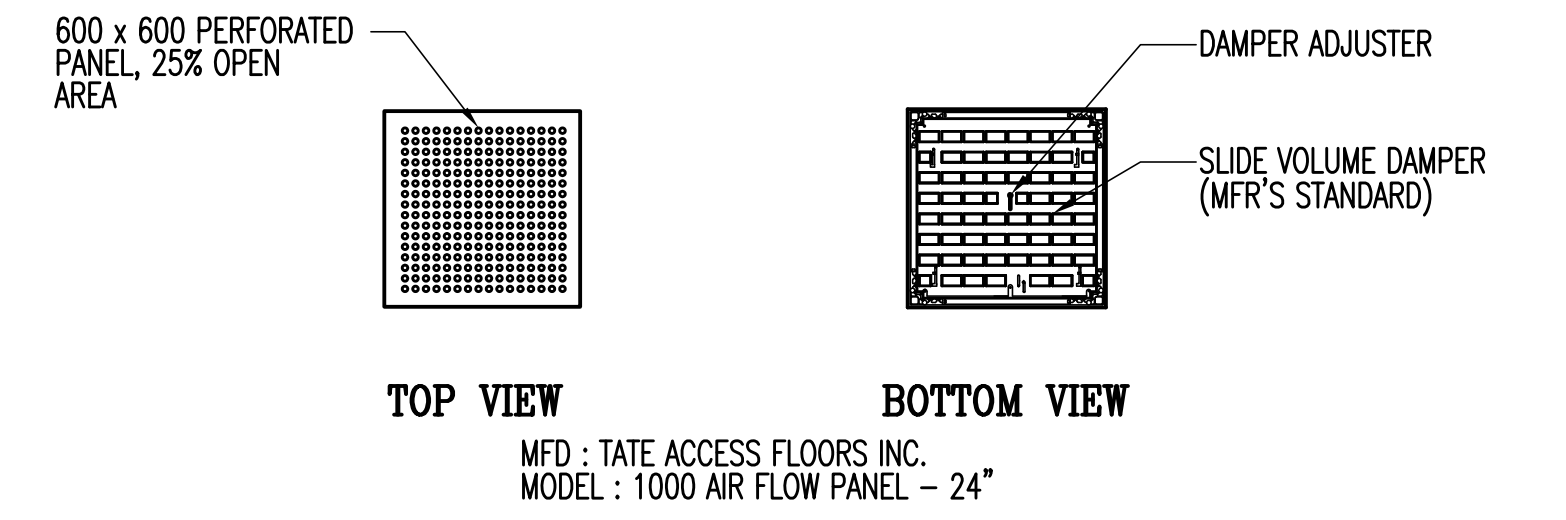
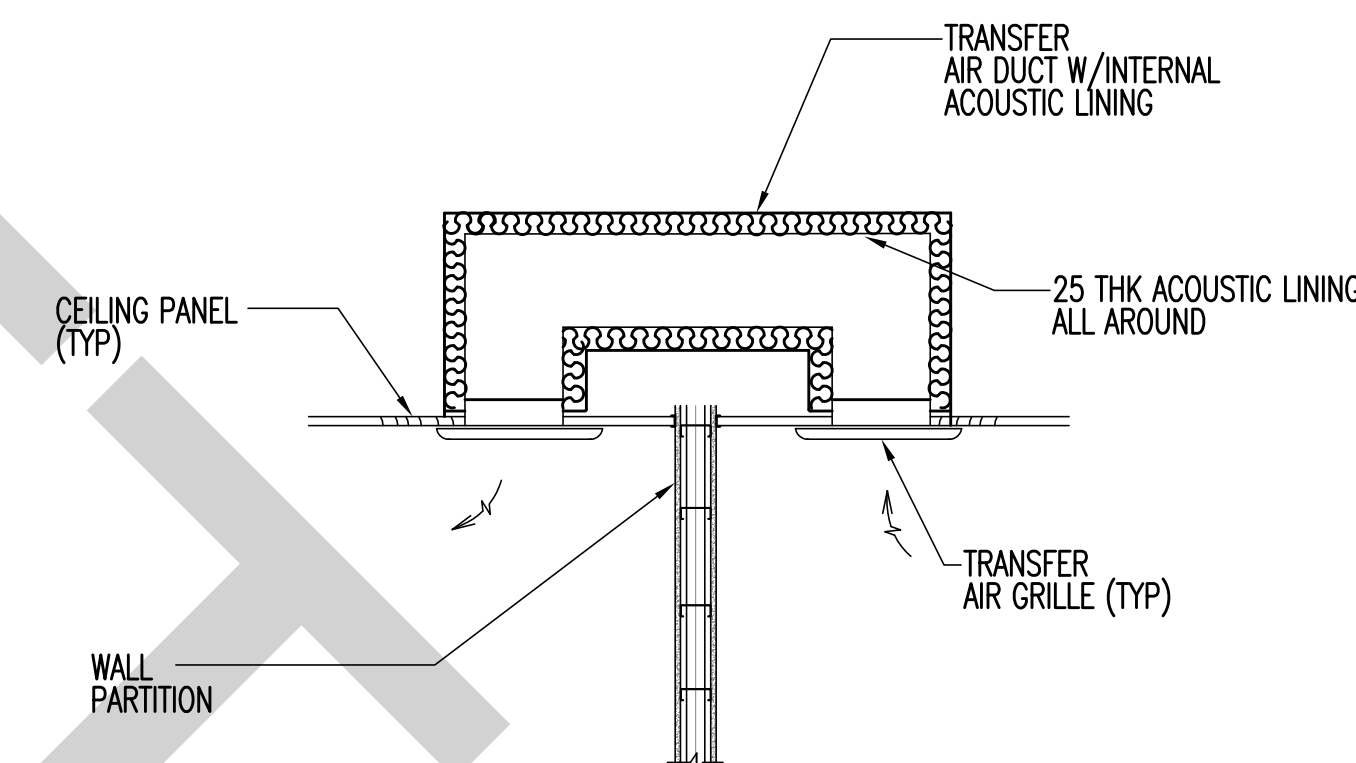
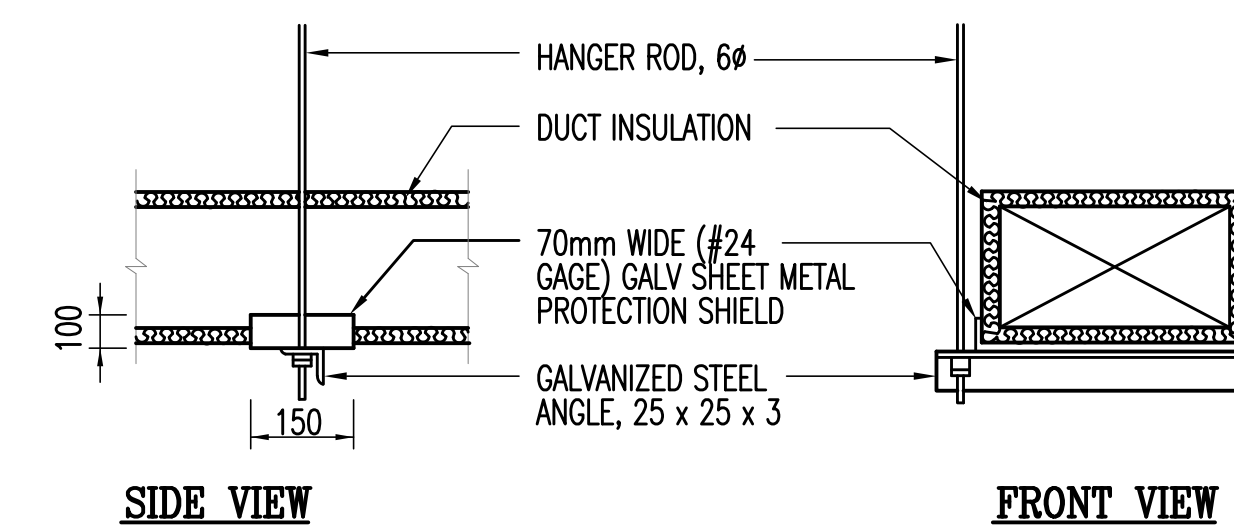
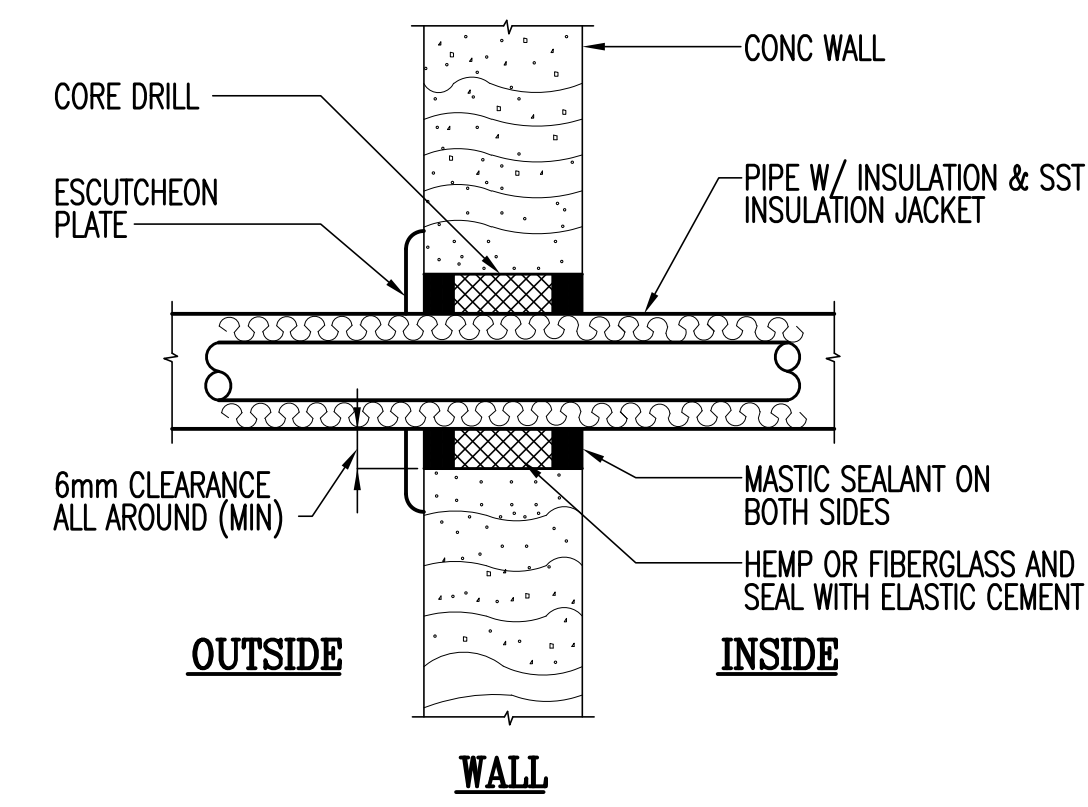
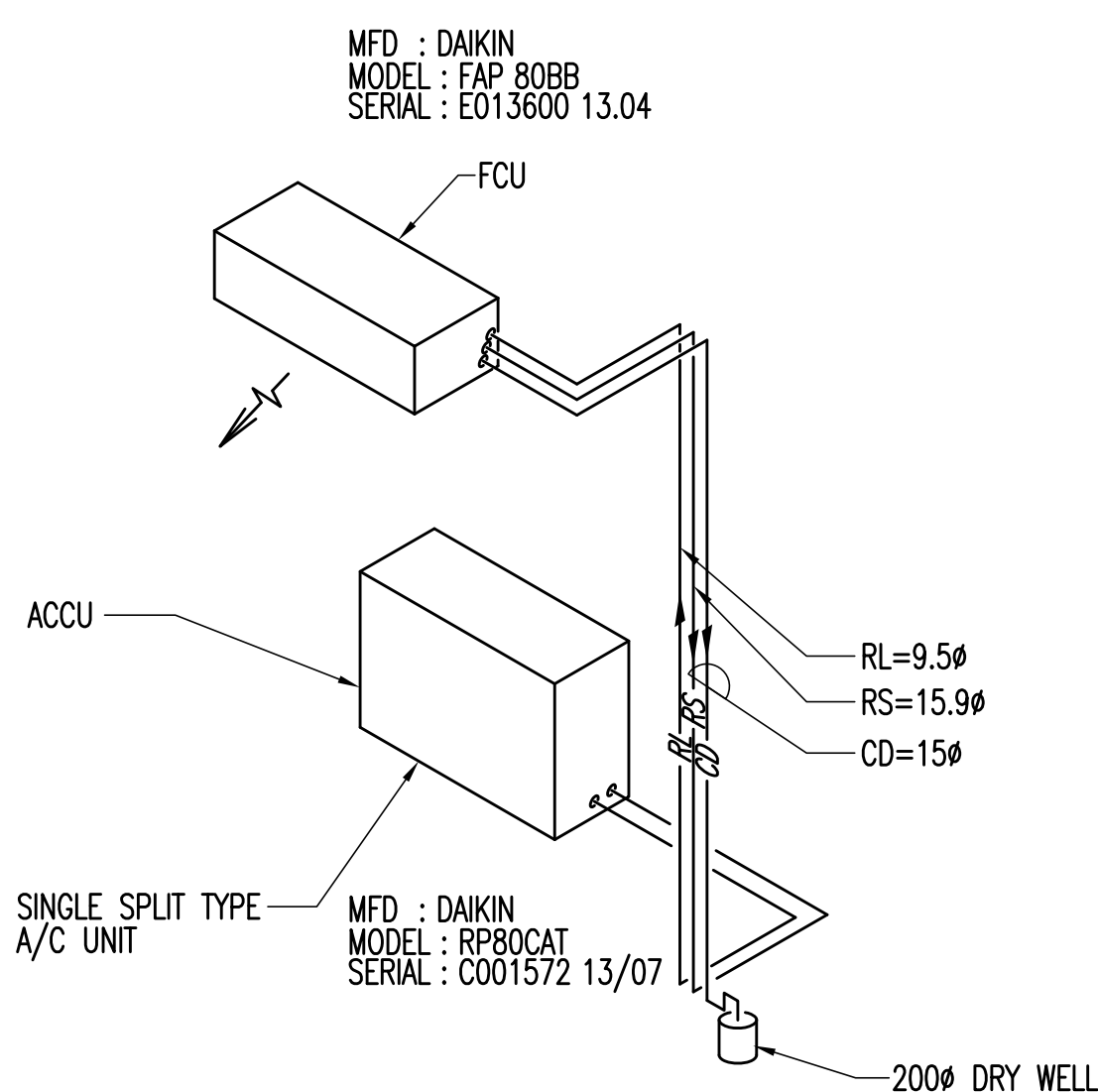
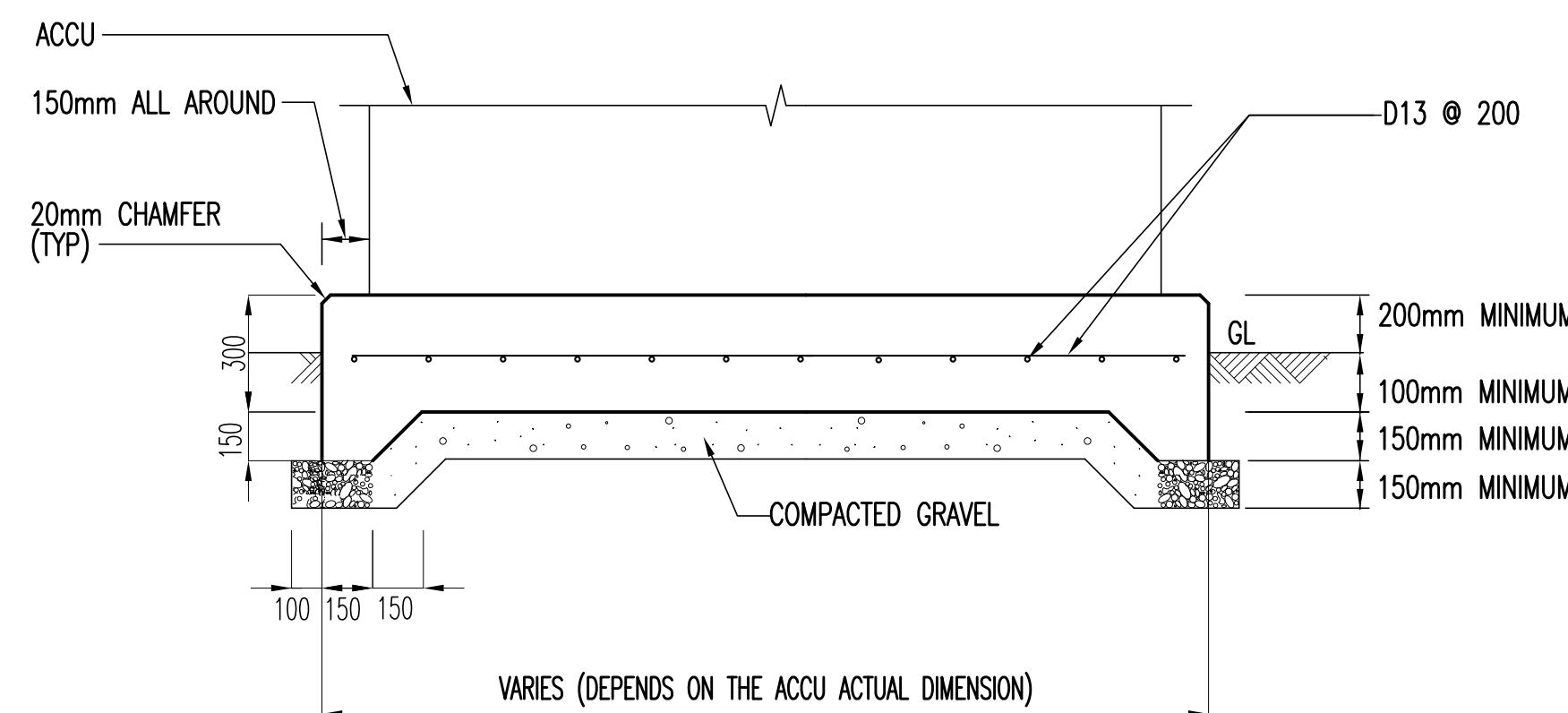
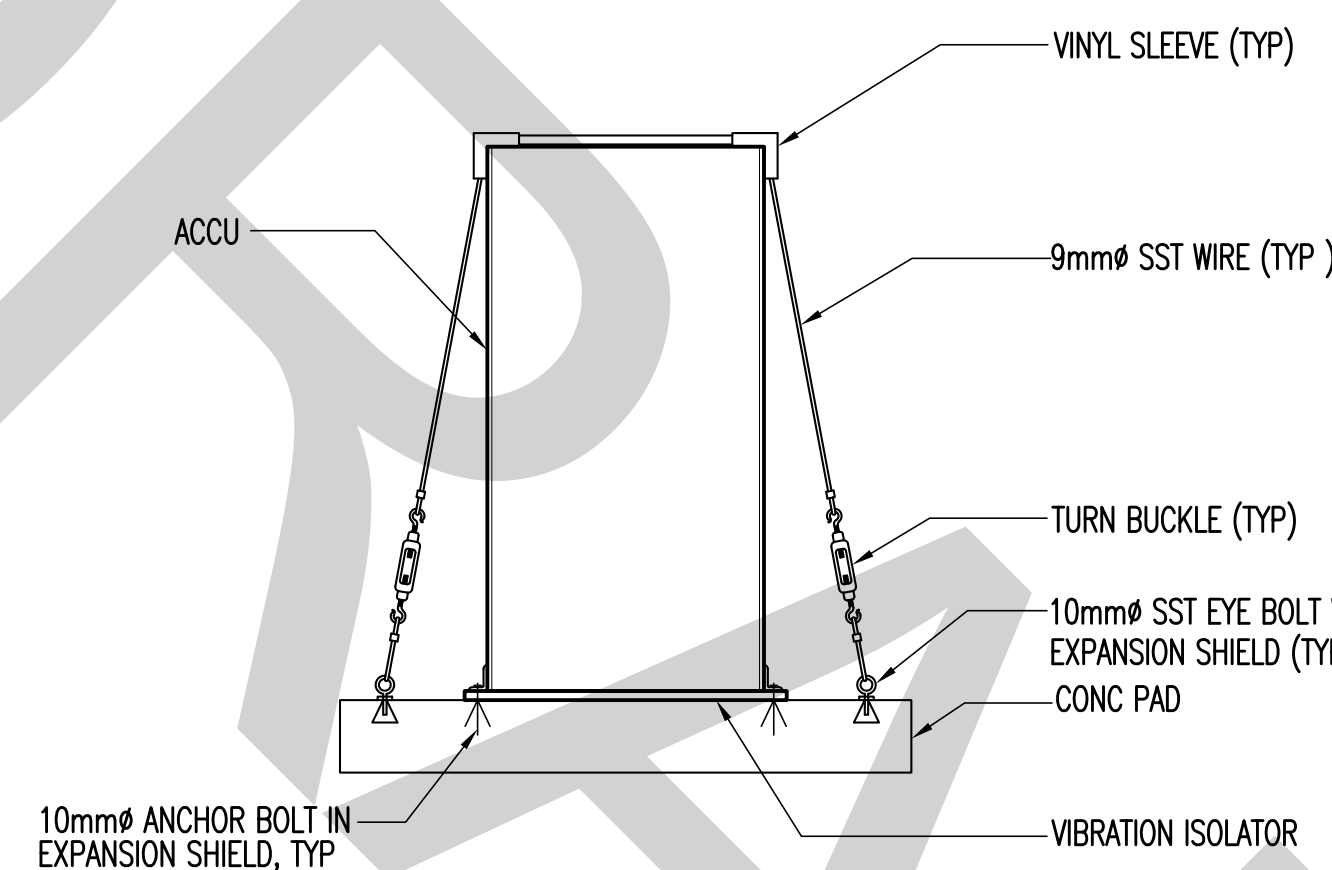
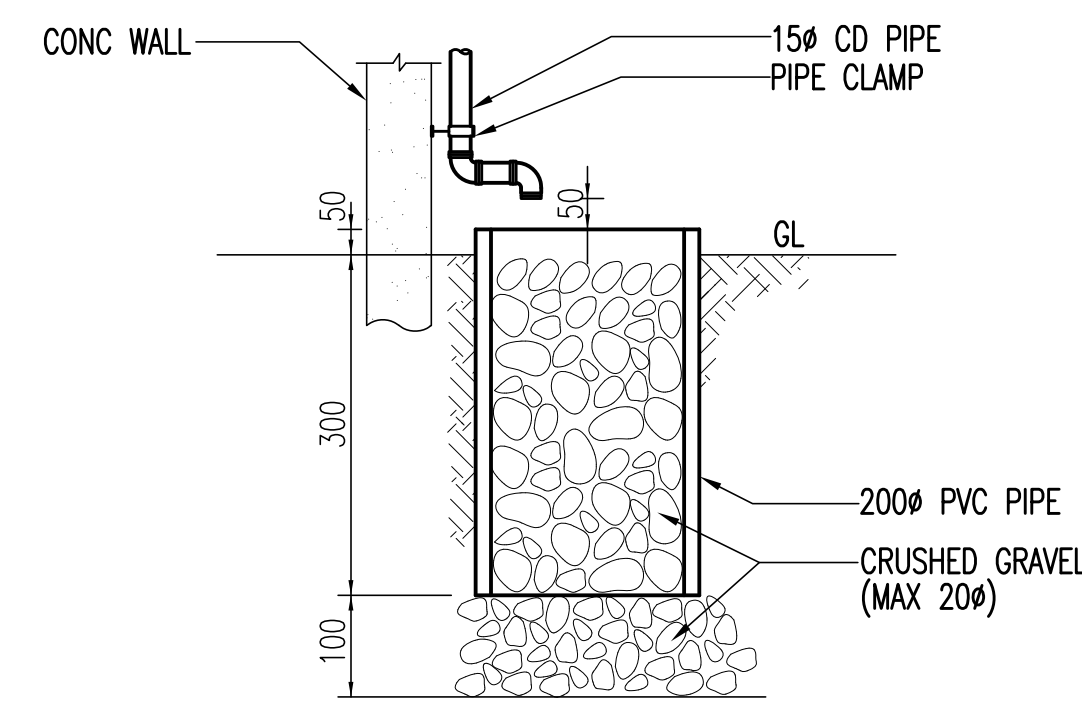
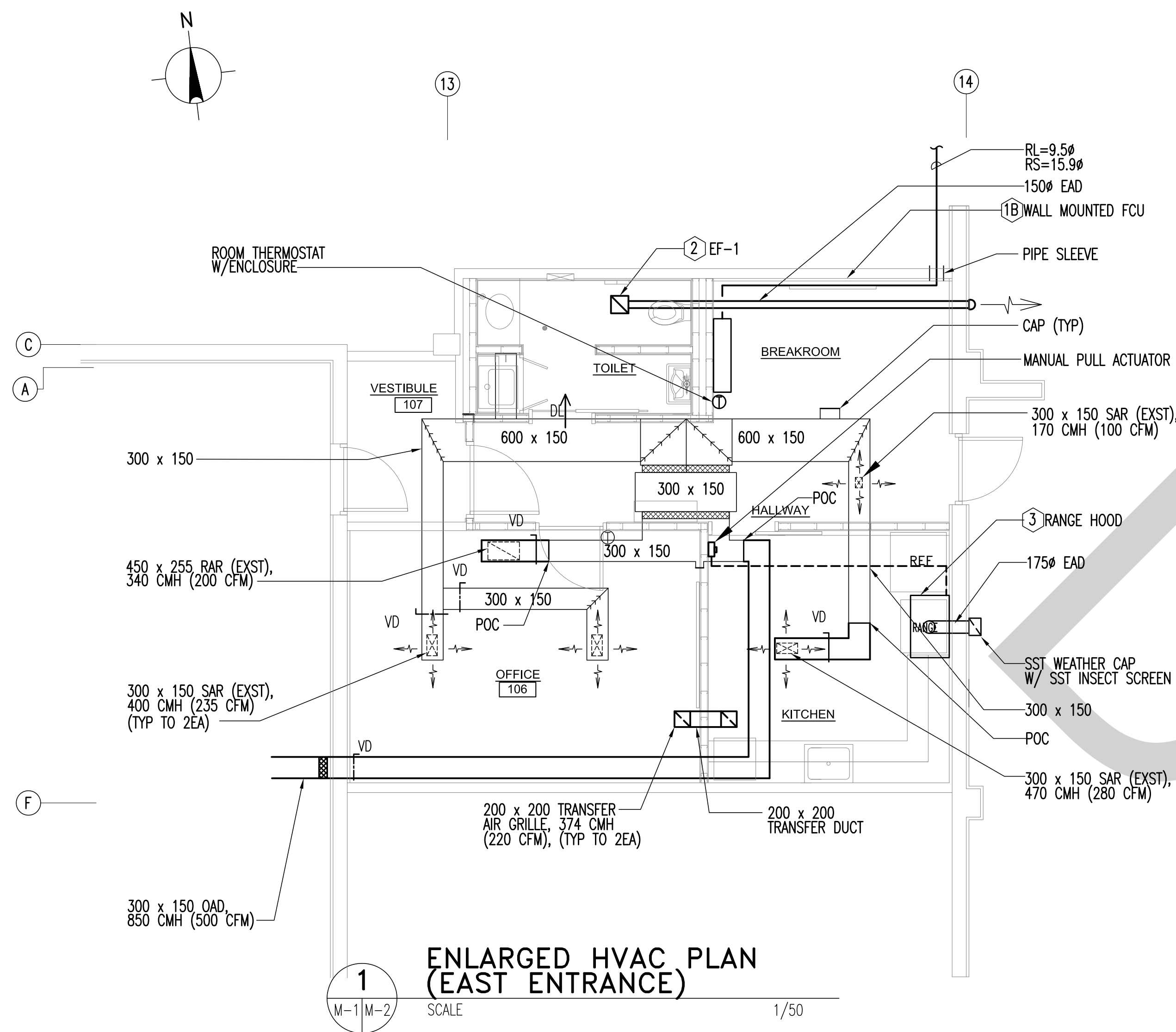


AEC-SABER
AS-BUILT
DATE: 5 SEP '14

GRAPHIC SCALES



SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 ROCK HILL DRIVE, SUITE 100, ROCK HILL, SC 29730 TEL: 803-963-1234 FAX: 803-963-1235 WWW.AEC-SABER.COM				
CONT. DESIG. J. M. S.	CONT. DR. H. O.	U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
CONT. CIV. A. S. C.	CONT. CIV. ENGR. J. RUTHERFORD	FORT BUCKNER OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
FIRE PROTECTION ENGINEER		MECHANICAL		
DATE		PARTIAL HVAC PLAN, HVAC EQUIPMENT SCHEDULE, LEGEND AND KEY PLAN		
DATE		PROJECT NUMBER JOK34982	CONTRACT NUMBER TO#7206(RFP#0027)	SEQ. NO. M-1
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER) 8 JUL '13		DATE 8 JUL '13	SCALE AS SHOWN	SHEET 7 of 18
FIRM MEMBER (TITLE)		DATE	FILE	SEQ. NO.
BLDG. NO.		DRAWING NO.		FILE SEQ. NO.

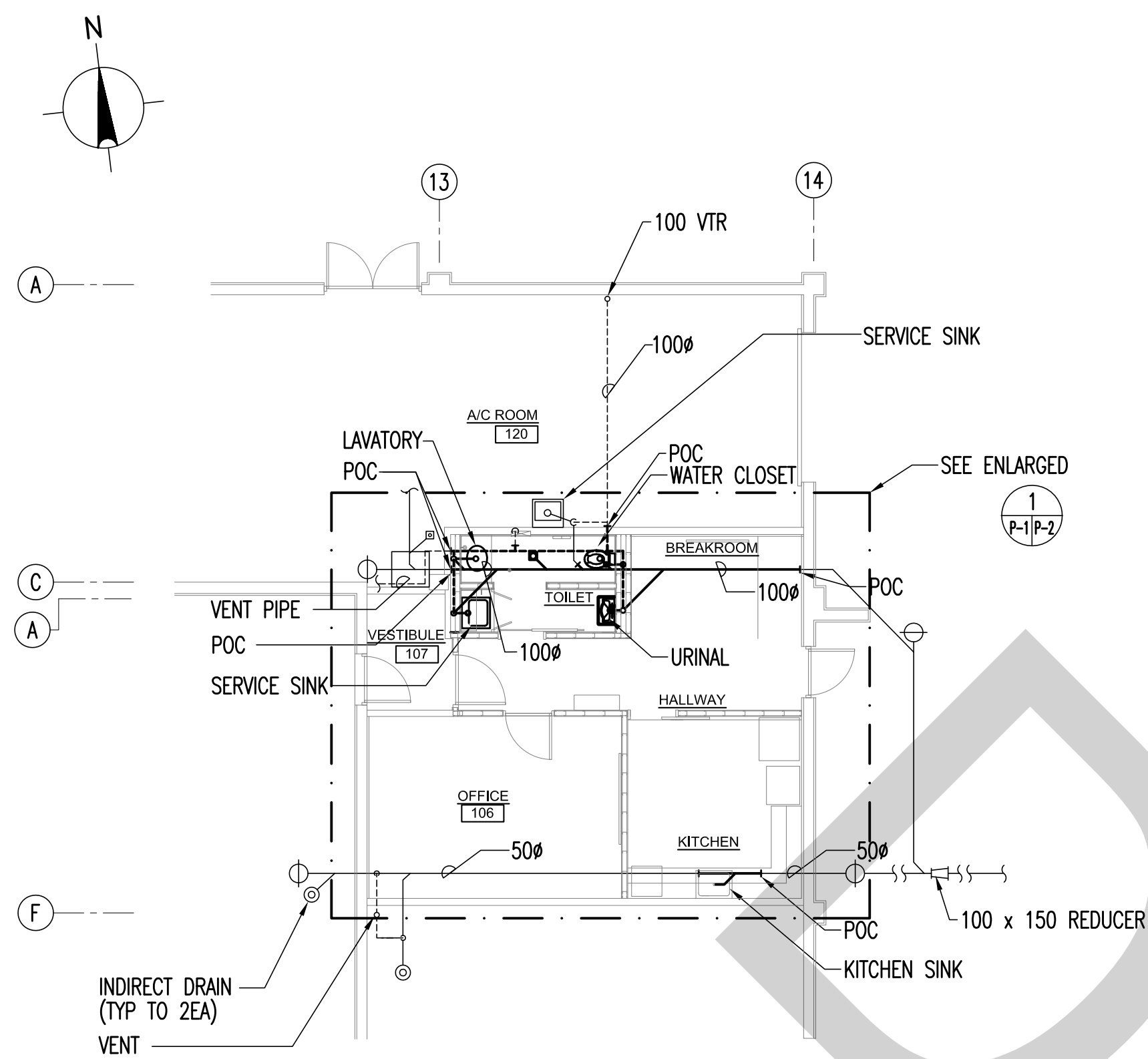


**AEC-SABER
AS-BUILT**
DATE: 5 SEP '14

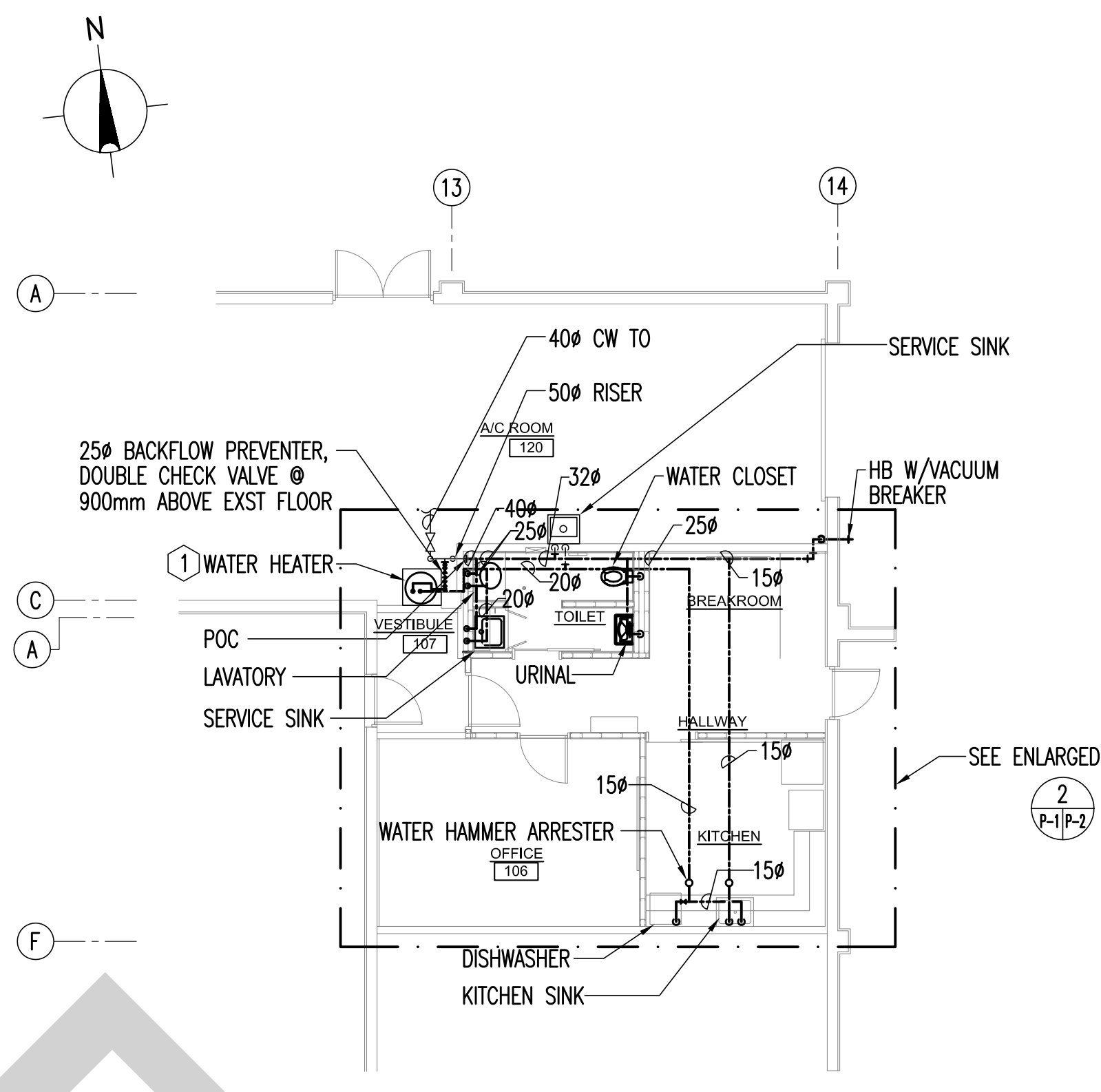
GRAPHIC SCALES



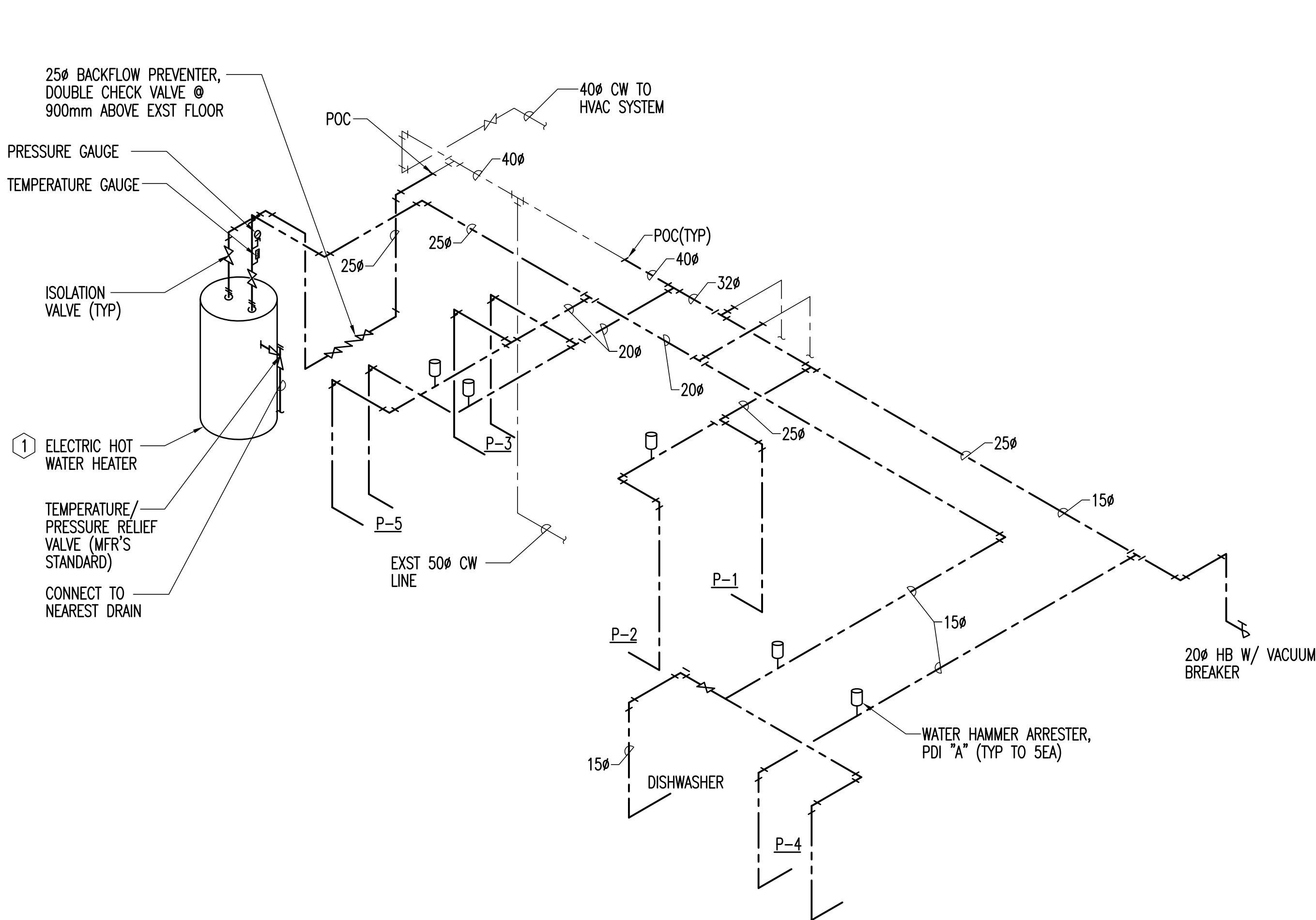
SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 1100 N. 17TH AVE. SUITE 100 FORT WORTH, TEXAS 76102 TEL: (817) 335-1000 FAX: (817) 335-1001 WWW.AEC-SABER.COM		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS FORT BUCKNER OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA MECHANICAL ENLARGED HVAC PLAN (EAST ENTRANCE) AND MISCELLANEOUS DETAILS		
CONT. DESIG. J. M. S.	CONT. DRG. H. O.	PROJECT NUMBER JOK34982	CONTRACT NUMBER TO#7206(RFP#0027)	SEQ. NO. M-2
CONT. CHK. A. S. C.	CONT. OF. ENGR. J. RUTHERFORD	SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER)	DATE 8 JUL '13	SCALE AS SHOWN
FIRE PROTECTION ENGINEER		DATE	DATE	SHEET 8 OF 18
BLDG. NO.		DRAWING NO.		FILE SEQ. NO.



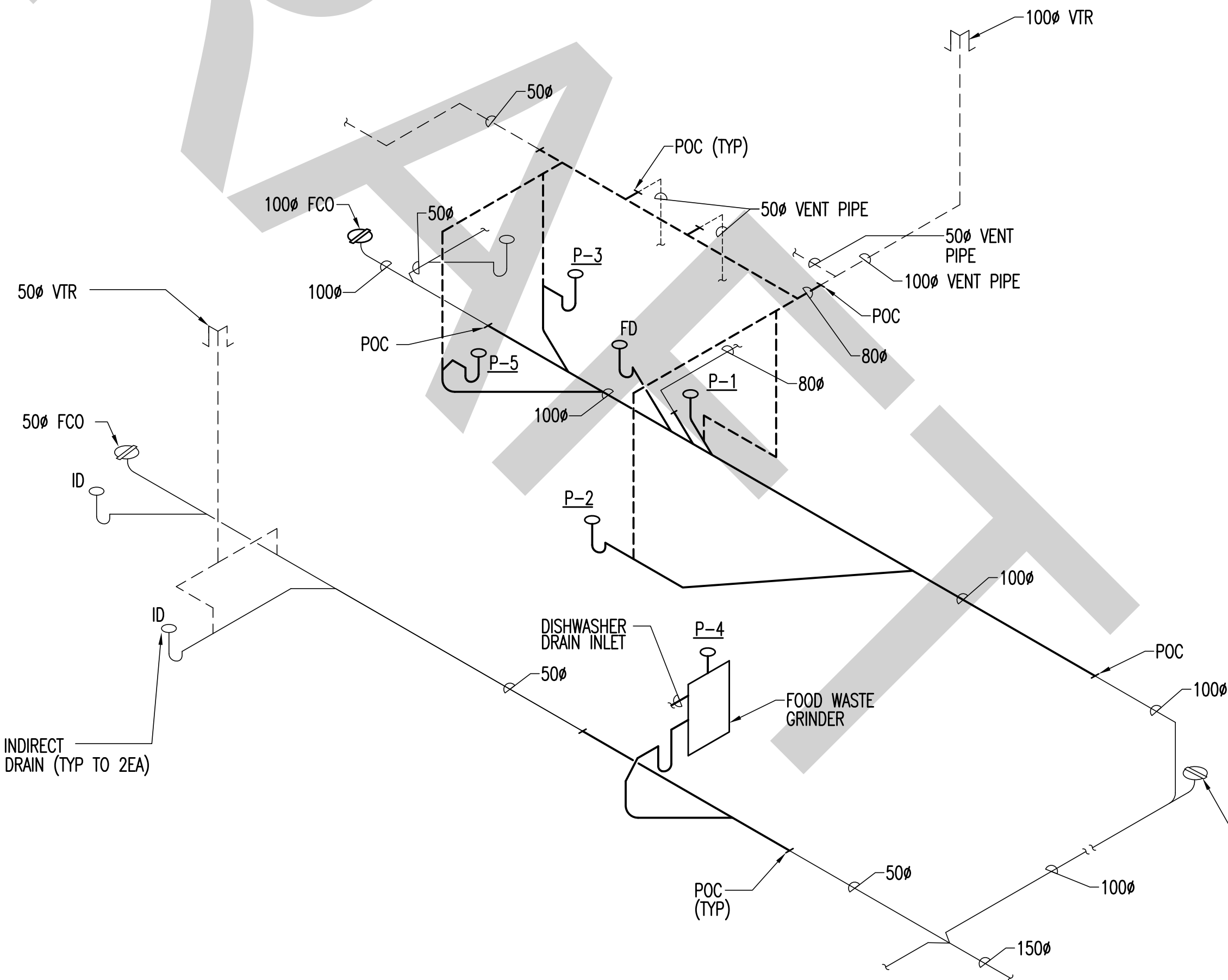
1 PARTIAL PLUMBING DWV
PLAN (EAST ENTRANCE)
SCALE 1/100



2 PARTIAL PLUMBING CW & HW
PLAN (EAST ENTRANCE)
SCALE 1/100



3 CW & HW PIPING DIAGRAM
(EAST ENTRANCE)
SCALE

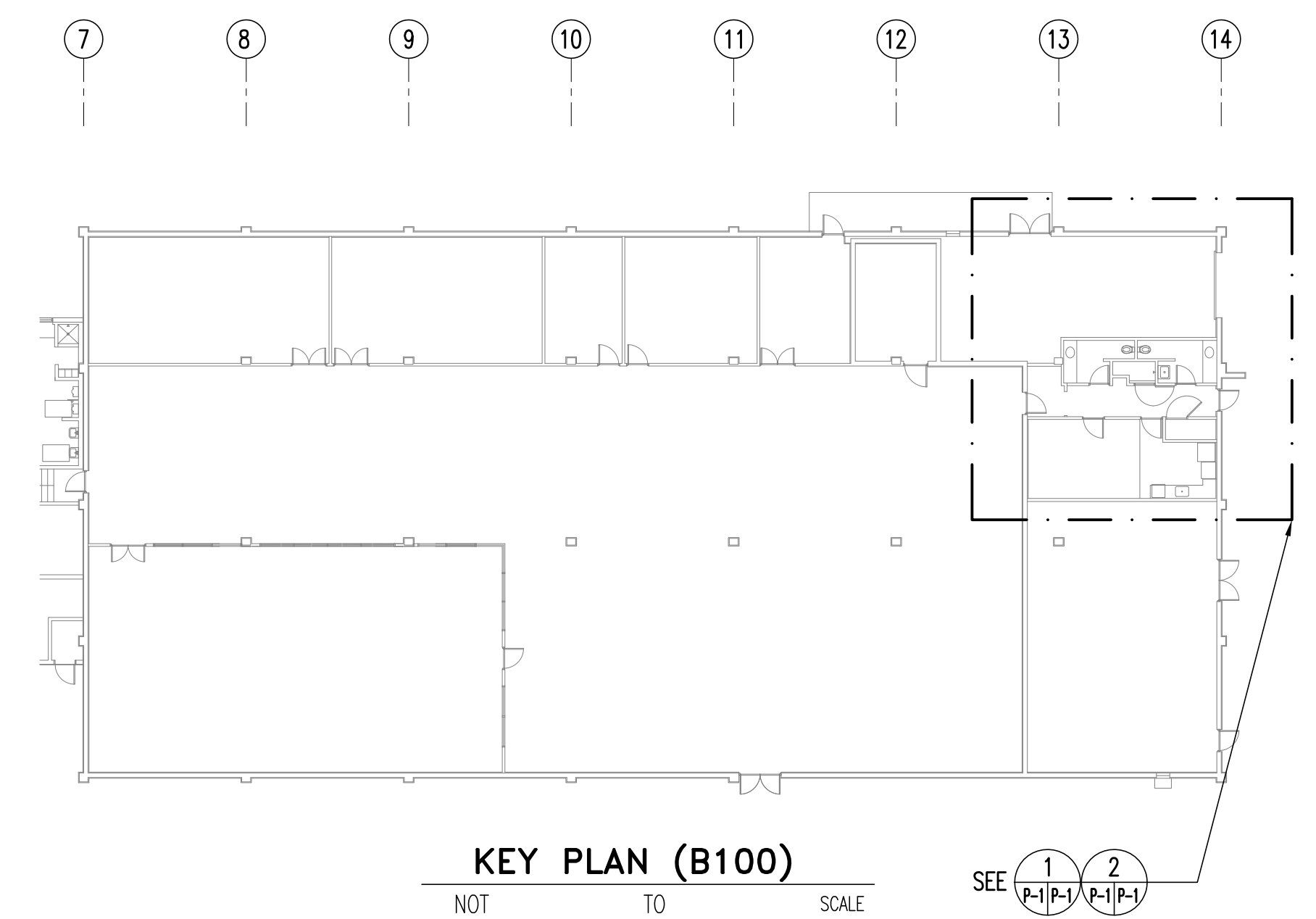


4 DWV PIPING DIAGRAM
(EAST ENTRANCE)
SCALE

PLUMBING FIXTURE SCHEDULE							
SYMBOL	FIXTURES	DESCRIPTION	PIPE CONNECTION SIZES (mm)				
			CW	HW	WASTE	VENT	QTY
P - 1	WATER CLOSET	WHITE VITREOUS CHINA, FLOOR MOUNTED, FLOOR OUTLET, SIPHON JET, ELONGATED BOWL WITH SEAT FLUSH VALVE TYPE, LOW FLOW TYPE, 4.8 LPF	25ø	-	100ø	50	1
P - 2	URINAL	WHITE VITREOUS CHINA, STALL TYPE, SIPHON JET WITH INTEGRAL TRAP, FLUSH VALVE, LOW FLOW TYPE.	20ø	-	50ø	40	1
P - 3	LAVATORY	SAME MATERIAL AS THE COUNTERTOP AND SUPPLIED BY THE COUNTER TOP MANUFACTURER COMPLETE WITH SINGLE LEVER CENTER SET FAUCET, LOW FLOW TYPE, 1.9 LPM	15ø	15ø	40ø	40	1
P - 4	KITCHEN SINK	SINGLE BOWL STAINLESS STEEL WITH SINGLE LEVER CENTERSET FAUCET, FOOD WASTE GRINDER AND DRAIN ASSEMBLY. FOOD WASTE GRINDER : CONTINUOUS FEED, GALVANIZED STEEL GRINDING ELEMENTS WITH TWO STAINLESS STEEL 360 DEGREES SWIVEL LUGS, AND SELF-SERVICE WRENCH. ELEC RATING : 120V, 1PH, 60HZ, 250W.	15ø	15ø	40ø	40	1
P - 5	SERVICE SINK	WHITE ENAMELED CAST-IRON, WALL-MOUNTED, WITH CAST-IRON P-TRAP, CW AND HW FAUCETS WITH VACUUM BREAKER, SPLASH BACK AND RIM GUARD.	15ø	15ø	80ø	40	1
①	ELECTRIC WATER HEATER : ELECTRIC RESISTANT TYPE, GLASS-LINED, INSULATED JACKET, COMPLETE WITH CASING, ALUMINUM ANODE, ADJUSTABLE WATER TEMPERATURE CONTROLLER AND PRESSURE/TEMPERATURE RELIEF VALVE. 1) CAPACITY : 50 GALLONS 2) ELECTRICAL ELEMENT : 240V, 1PH, 60HZ, 6KW.						1

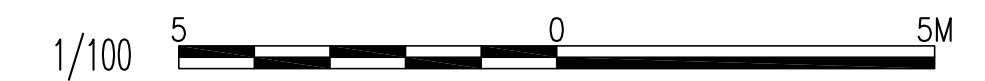
LEGEND :

DESCRIPTION	
	GATE VALVE
	COLD WATER
	DOMESTIC HOT WATER SUPPLY
	FLOOR DRAIN
	FLOOR CLEANOUT/GROUND CLEANOUT
	INDIRECT DRAIN
	WASTE LINE
	VENT LINE
	VENT THRU ROOF
	WATER HAMMER ARRESTER



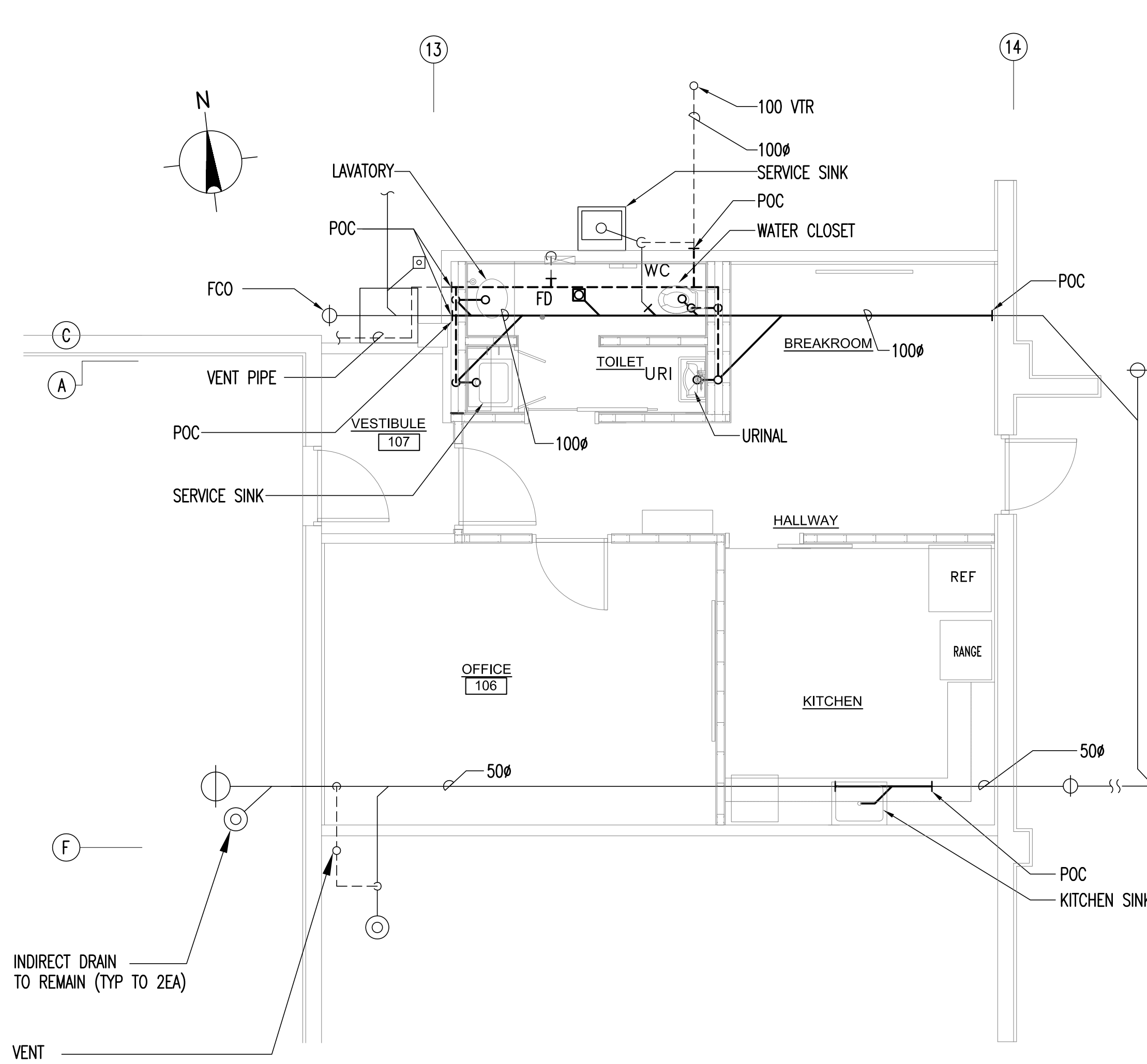
KEY PLAN (B100)
SCALE

GRAPHIC SCALES

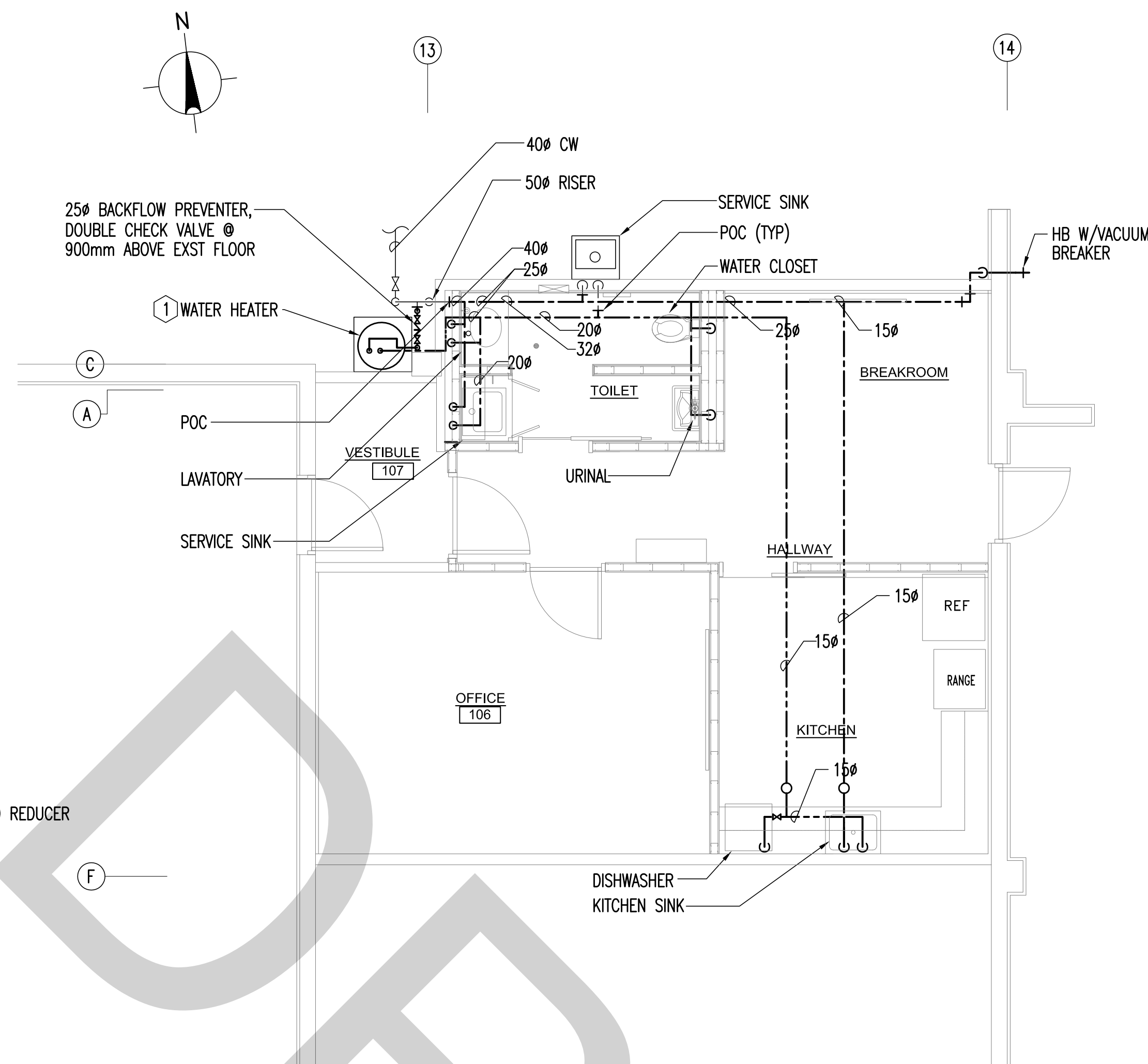


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DATE: 5 SEP '14

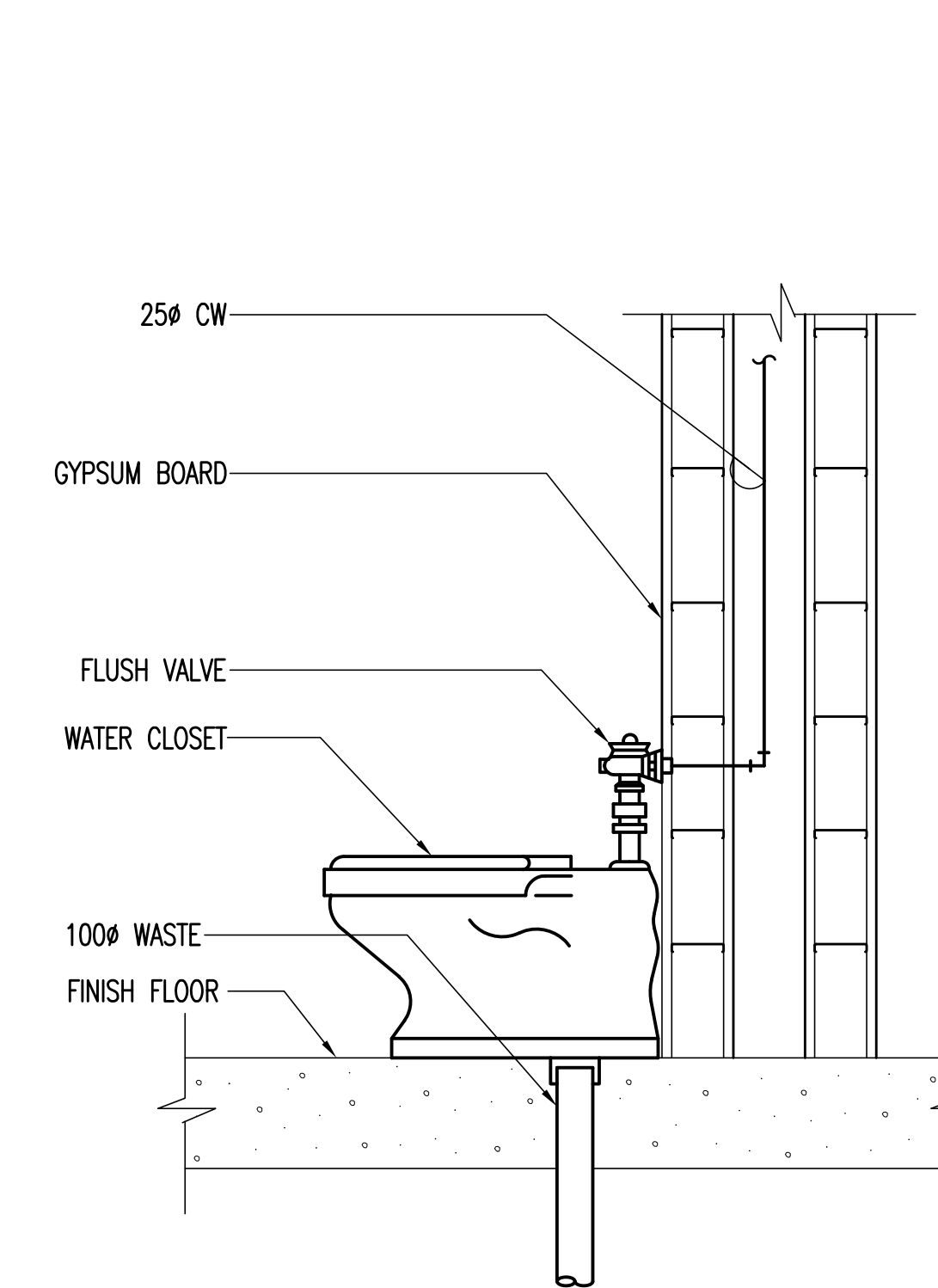
REVISIONS			
SYM	DESCRIPTION	DATE	APPROVAL
AMERICAN ENGINEERING CORP. 4100 N. 10TH AVE., SUITE 100, DENVER, CO 80202 TEL: (303) 733-1100 FAX: (303) 733-1101 WWW.AEC-SABER.COM			
U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		FORT BUCKNER OKINAWA, JAPAN	
CONT. DESIG. J. M. S. CONT. ENG. H. U.		RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA	
CONT. CIV. A. S. C. CONT. CIV. ENGR. J. RUTHERFORD		MECHANICAL	
FIRE PROTECTION ENGINEER		PARTIAL PLUMBING PLAN, PARTIAL PLUMBING CW & HW PLAN, DIAGRAMS, PLUMBING FIXTURE SCHEDULE AND KEY PLAN	
PROJECT NUMBER		CONTRACT NUMBER	SEQ. NO.
JOK34982		TO#7206(RFP#0027)	P-1
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER) 8 JUL '13		DATE: 8 JUL '13	SCALE: AS SHOWN
FIRM NUMBER (TITLE)		DATE: 8 JUL '13	SHEET: 9 of 18
BLDG. NO.		DRAWING NO.	
		FILE SEQ. NO.	



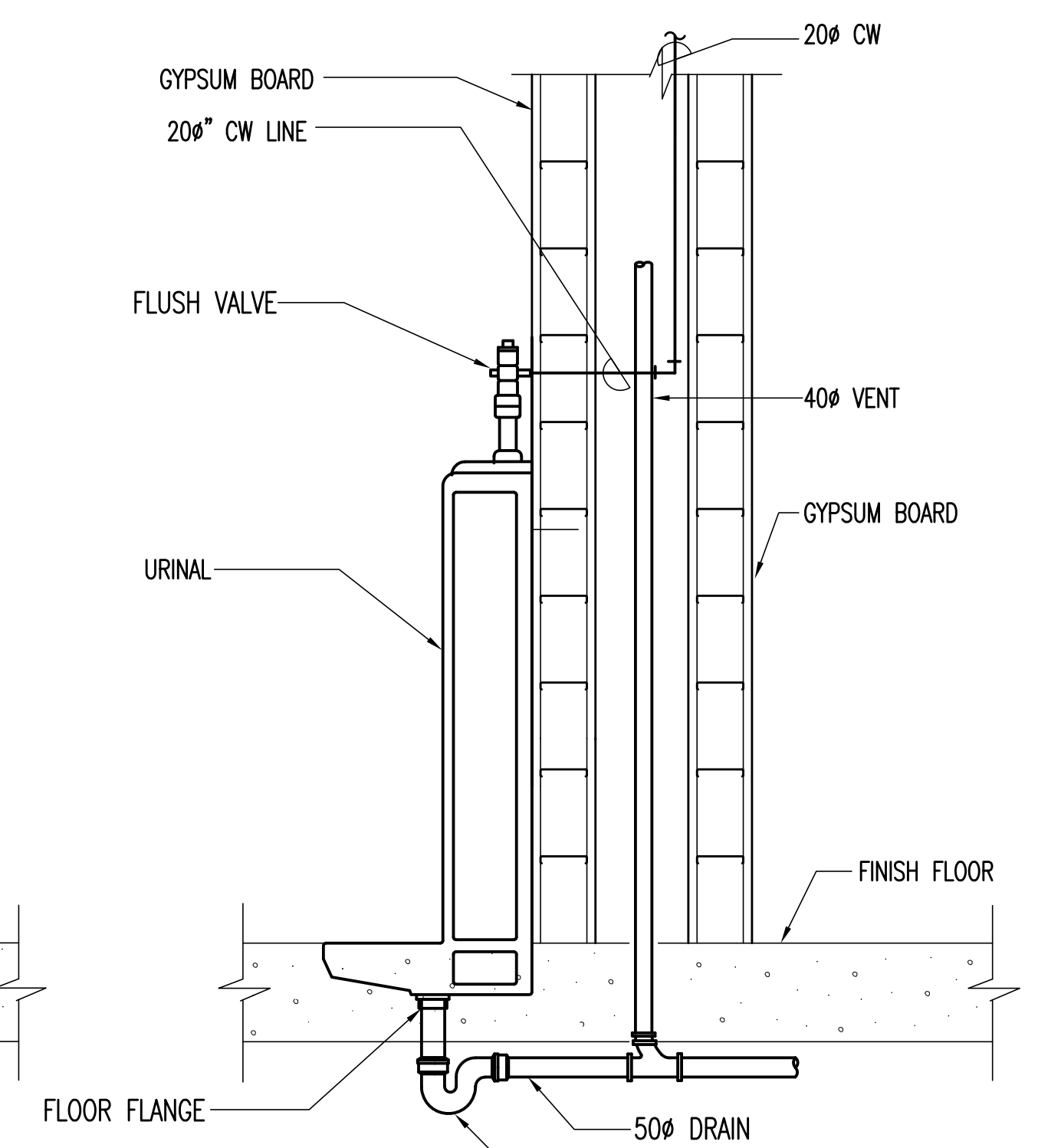
1 ENLARGED DWV PLAN (EAST ENTRANCE)
P-1 P-2 SCALE 1/50



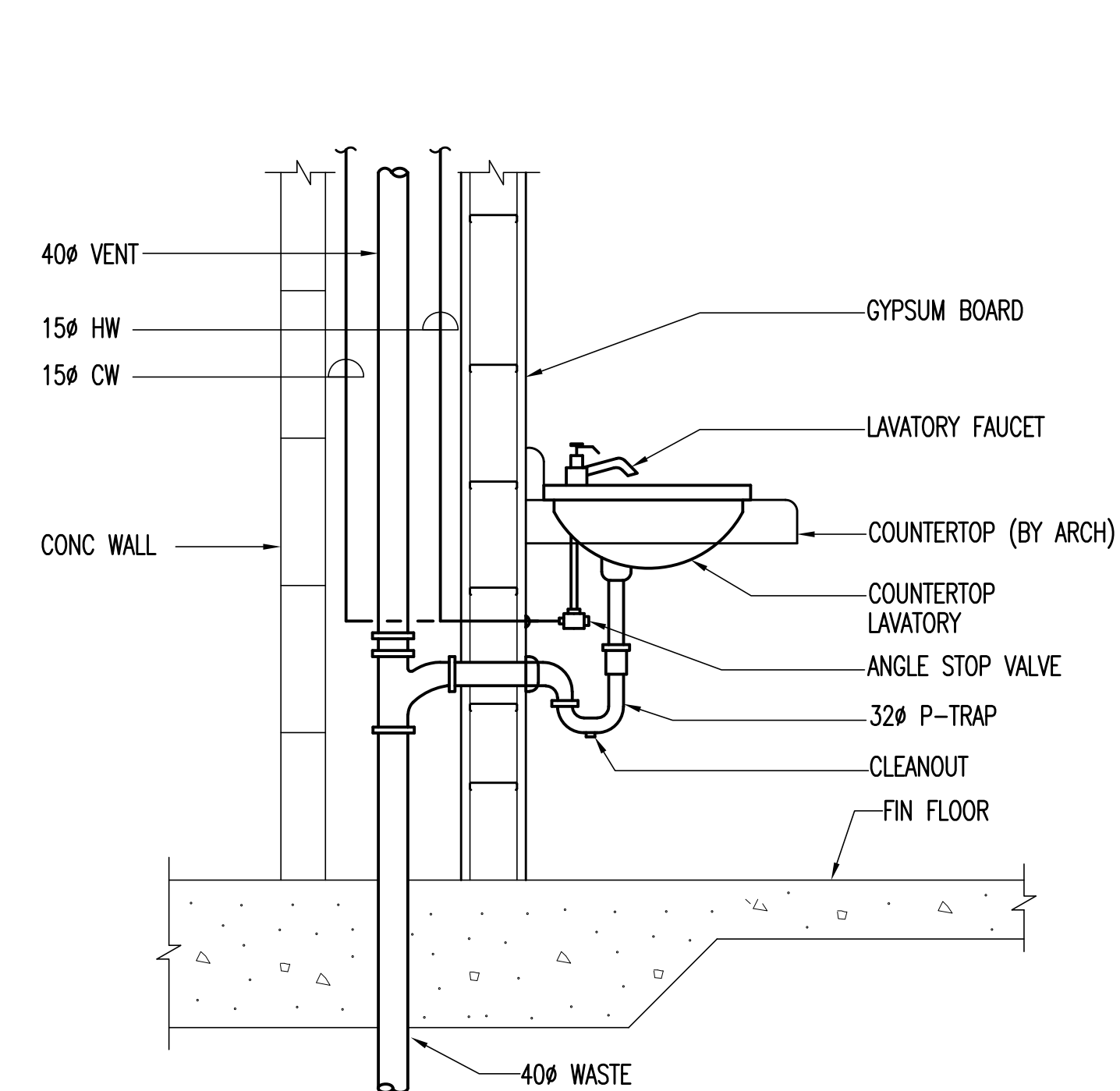
2 ENLARGED CW & HW PLAN (EAST ENTRANCE)
P-1 P-2 SCALE 1/50



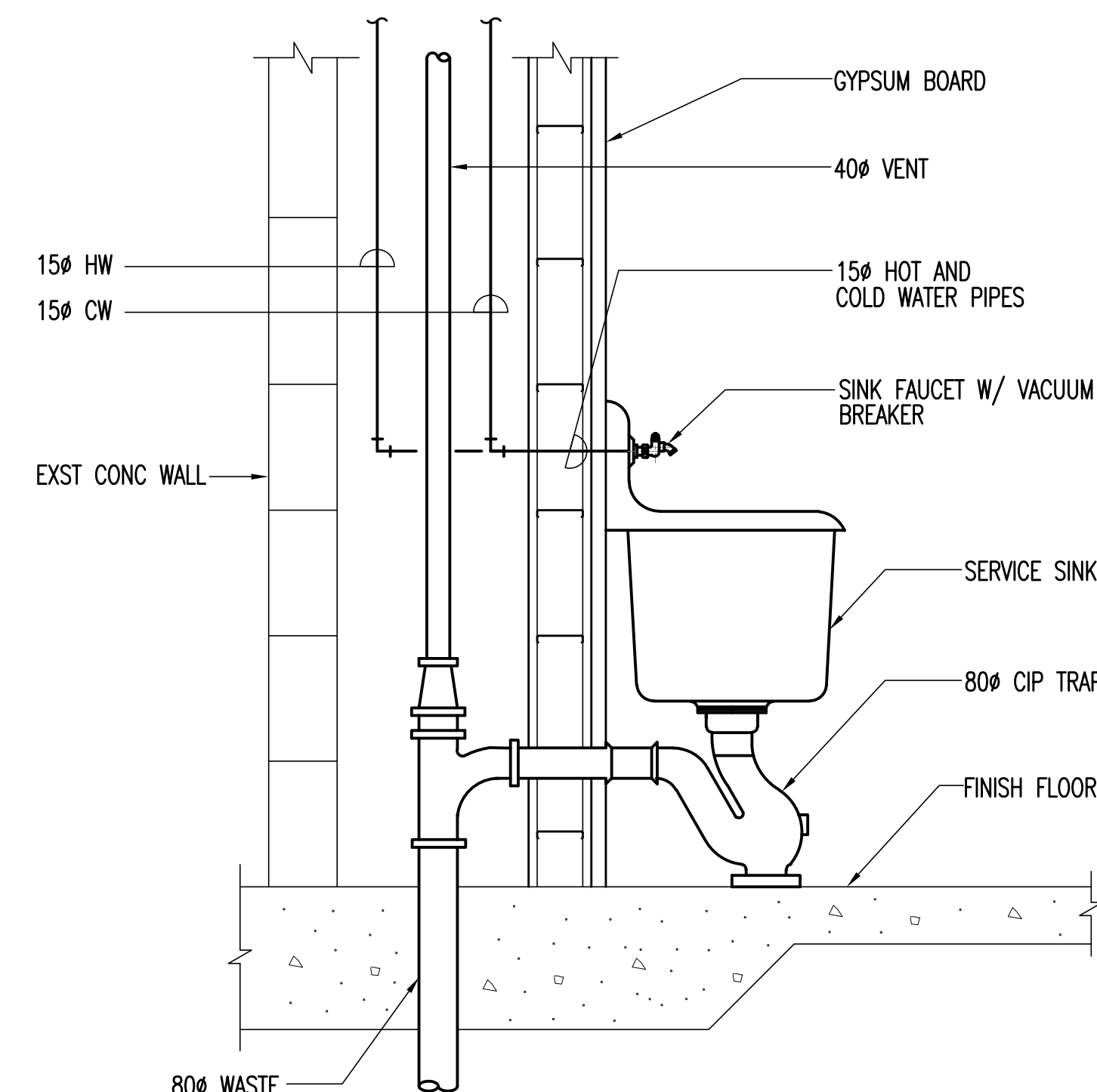
3 WATER CLOSET
P-2 P-2 NOT TO SCALE



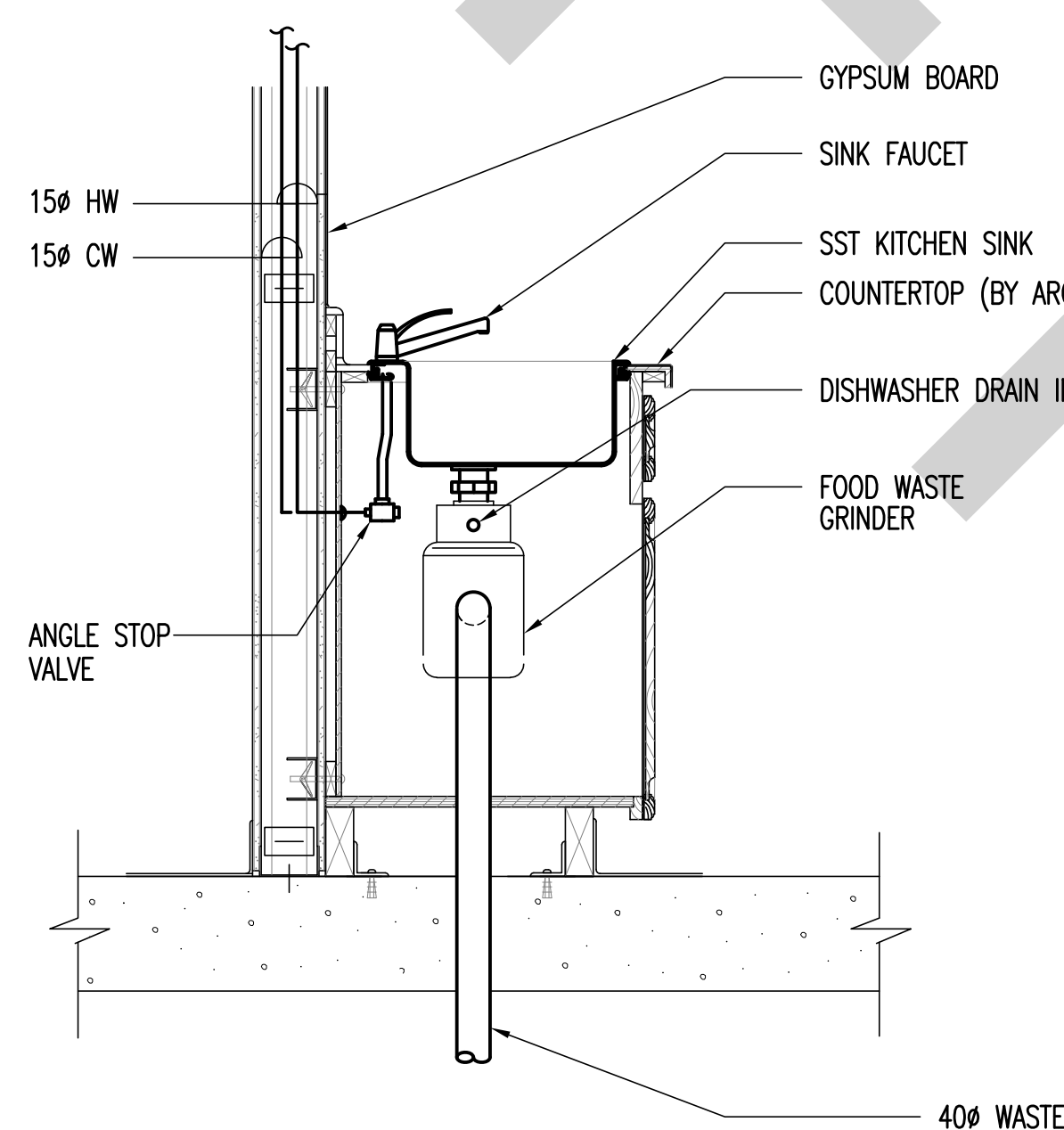
4 URINAL STALL TYPE
P-2 P-2 NOT TO SCALE



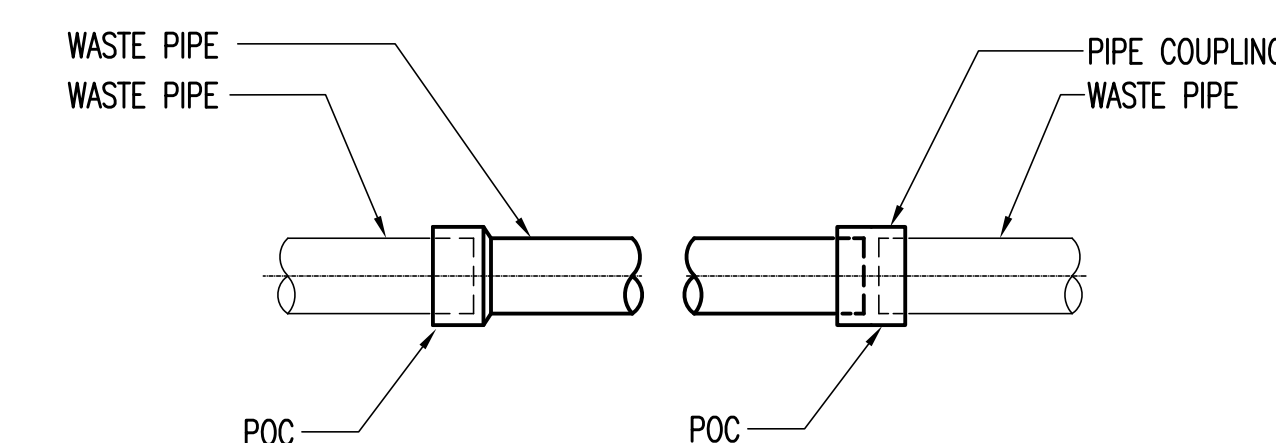
5 LAVATORY
P-2 P-2 NOT TO SCALE



6 SERVICE SINK DETAIL
P-2 P-2 NOT TO SCALE



7 KITCHEN SINK W/ FOOD
WASTE GRINDER
P-2 P-2 NOT TO SCALE



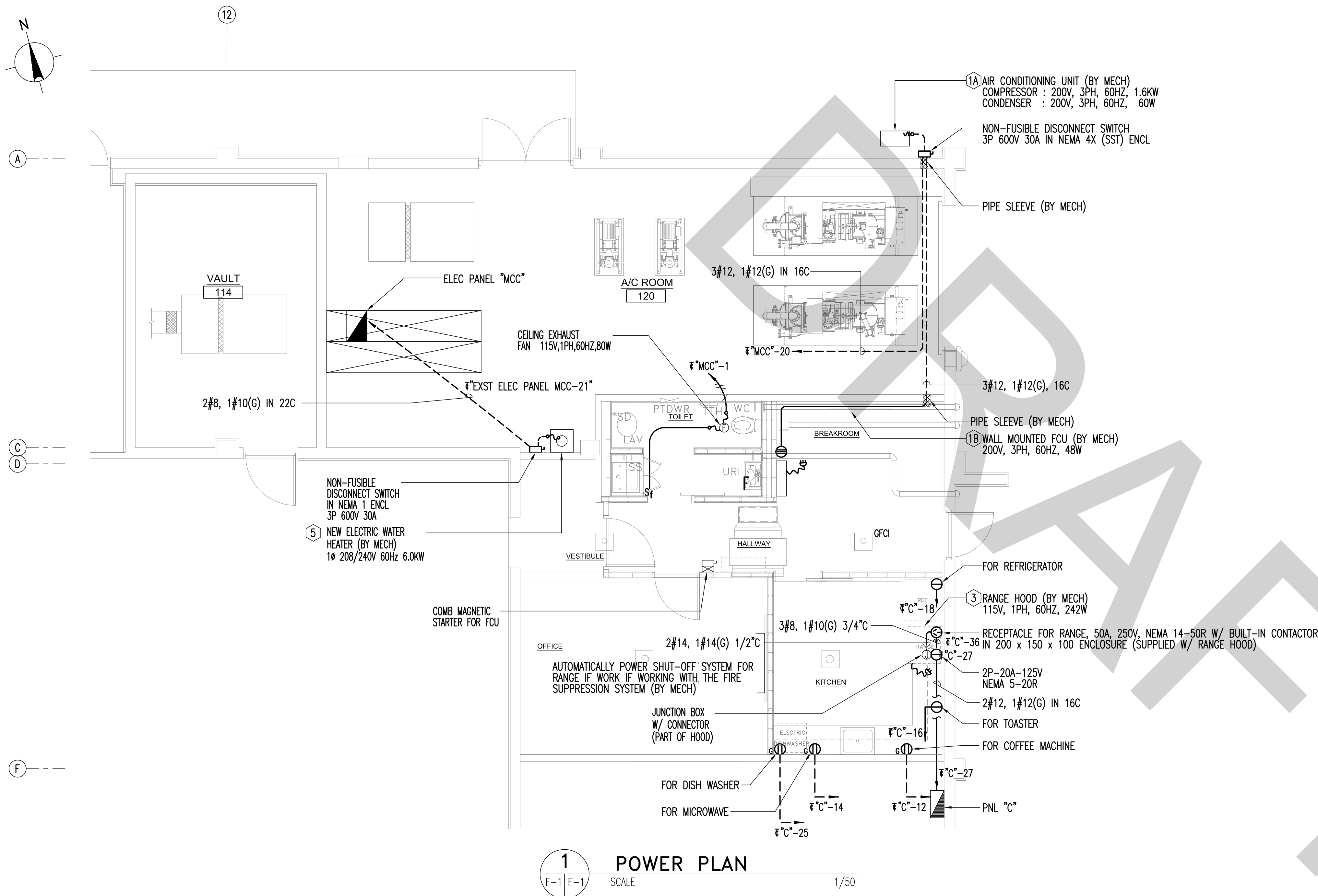
8 PIPING CONNECTION DETAIL
P-2 P-2 NOT TO SCALE

**AEC-SABER
AS-BUILT**
DATE: 5 SEP '14

GRAPHIC SCALES



SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. W. 11th Street, Suite 100, Fort Lauderdale, FL 33309 TEL: (954) 574-1111 FAX: (954) 574-1112 WWW.AEC-SABER.COM				
CONTRACTOR J. M. S.		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
CONTRACTOR A. S. C.		FORT BUCKNER, OKINAWA, JAPAN		
FIRE PROTECTION ENGINEER		RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
DATE		MECHANICAL		
DATE		ENLARGED DWV PLAN (EAST ENTRANCE), ENLARGED CW & HW PLAN (EAST ENTRANCE) AND MISCELLANEOUS DETAILS		
DATE		PROJECT NUMBER	CONTRACT NUMBER	SEQ. NO.
DATE		JOK34982	TO#7206(RFP#0027)	P-2
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER)		DATE	SCALE	SHEET
8 JUL '13		8 JUL '13	AS SHOWN	10 of 18
FILE MEMBER (TITLE)		BLDG. NO. DRAWING NO. FILE SEQ. NO.		



LEGEND :

- FLUORESCENT LIGHTING FIXTURE, RECESSED/PENDANT MOUNTED, 2' x 4'
- FLUORESCENT LIGHTING FIXTURE, RECESSED MOUNTED, 2' x 4' W/ EMERGENCY BATTERY UNIT
- FLUORESCENT LIGHTING FIXTURE, RECESSED/PENDANT MOUNTED, 1' x 4'
- FLUORESCENT LIGHTING FIXTURE, WALL MOUNTED, 1' x 4'
- COMPACT FLUORESCENT LIGHTING FIXTURE
- EXIST SIGN LIGHTING FIXTURE, WALL MOUNTED
- S WALL SWITCH, 1P, SINGLE GANG, 20A
- Sub DITTO EXCEPT TWO GANG
- OCCUPANCY SENSOR FOR LIGHTING CONTROL, CEILING MOUNTED
- POWER PACK FOR CEILING MOUNTED OCCUPANCY SENSOR
- RECEPTACLE, 3P-20A-125V/250V NEMA L-14-20R (PROVIDE MATCHING PLUG)
- SINGLE RECEPTACLE, 2P-3W-30A-125V GROUNDING TYPE,
- DUPLEX RECEPTACLE, 2P-3W-20A-125V GROUNDING TYPE,
- DUPLEX RECEPTACLE, 2P-3W-20A-125V GROUNDING TYPE, W/ GROUND FAULT CIRCUIT INTERRUPTER
- PANELBOARD
- "MCC" MOTOR CONTROL CENTER
- NON-FUSIBLE DISCONNECT SWITCH, RATING AS INDICATED
- Sr FRACTIONAL HORSEPOWER MANUAL STARTER, 1P-120V IN NEMA 1 ENCL
- Sel ON-OFF-AUTO SELECT SWITCH IN NEMA 1 ENCL
- MOTOR, RATING AS INDICATED (BY MECH)
- FIRE ALARM CONTROL PANEL (CONVENTIONAL)
- PHOTOELECTRIC SMOKE DETECTOR
- HEAT DETECTOR 190°F FIXED TEMPERATURE
- MANUAL STATION
- FA HORN
- AUDIO-VISUAL ALARM HORN
- VOLUME
- PA SPEAKER
- ACCESS CONTROL PANEL
- KEY ACCESS PAD
- GATE LOCK RELAY
- LAN OUTLET (RJ-45)
- TV OUTLET
- 2WAY-SPLITTER
- CONDUIT RUN CONCEALED IN CEILING OR WALLS
- CONDUIT RUN CONCEALED UNDERFLOOR
- CONDUIT RUN CONCEALED UNDER RAISED FLOOR
- CONDUIT RUN EXPOSED
- FLEXIBLE CONDUIT
- WP WEATHERPROOF
- CONDUIT, WIRING I. C.
- CONDUIT, WIRING F. A.
- CONDUIT, TEL WITH PULLING NYLON CORD
- SOUND CONDUIT
- JSIDS CONDUIT, GRS
- CONDUIT, WIRING TV

ELECTRICAL NOTES :

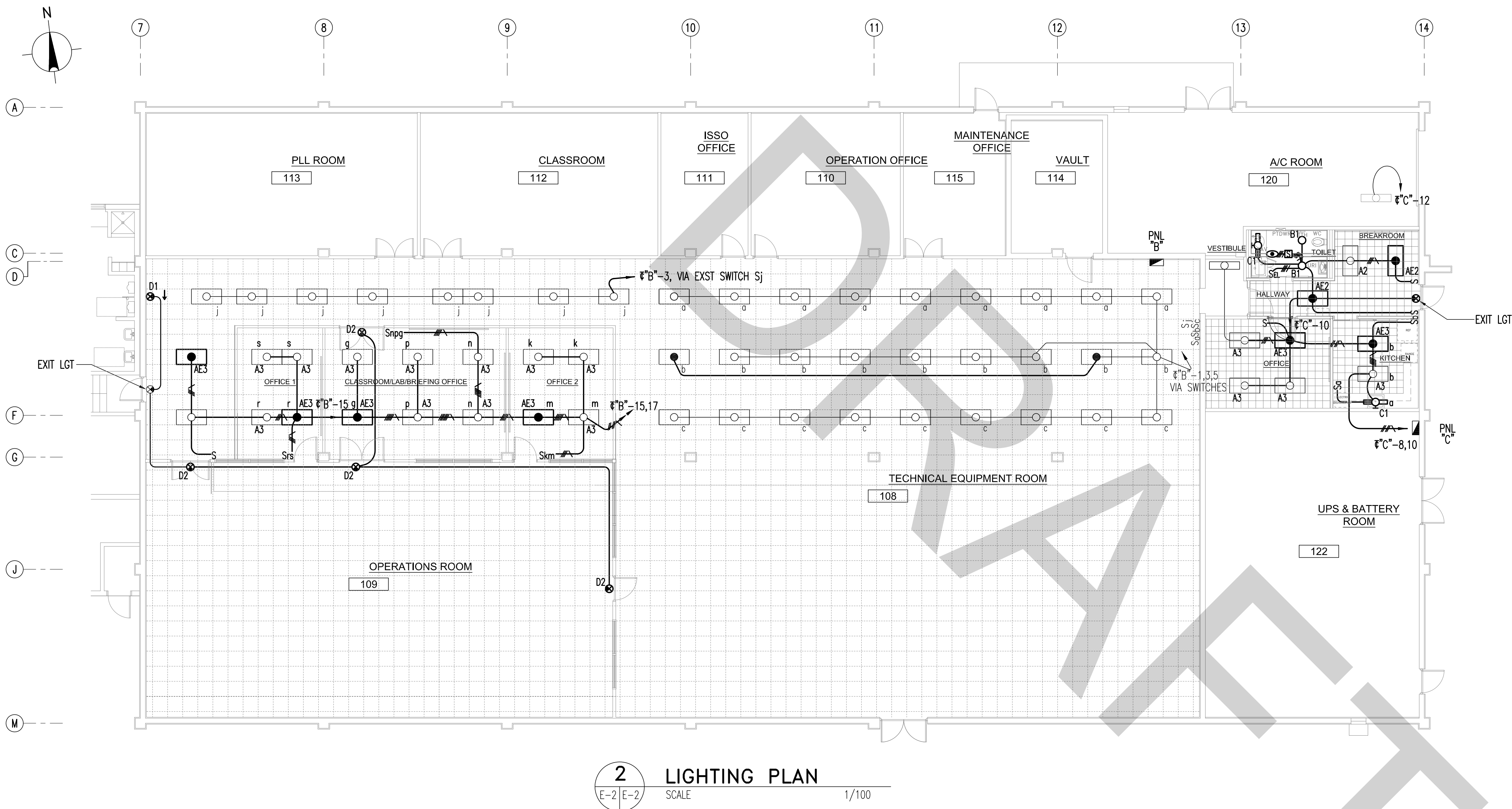
- ANY LIGHTING AND POWER CIRCUIT WITHOUT HATCH MARKS INDICATES 2#12, 1#12(G), IN 16"C. GREATER NUMBER OF WIRES ARE INDICATED AS FOLLOWS:
 - 3#12, 1#12 (G) IN 16C
 - 4#12, 1#12 (G) IN 16C
 - 5#12, 1#12 (G) IN 22C
 - 6#12, 1#12 (G) IN 22C
- UNLESS OTHERWISE INDICATED ON THE DRAWINGS, MOUNTING HEIGHTS ABOVE FINISHED FLOOR OR GROUND LEVEL SHALL BE AS FOLLOWS:
 - DISCONNECT SWITCH 1,830 mm FROM TOP OF ENCLOSURE
 - FRACTIONAL HP MANUAL STARTER 1,220 mm FROM CENTER OF DEVICE
 - WALL LIGHT SWITCH 1,220 mm FROM CENTER OF DEVICE
- ALL LIGHTING, POWER, AND CONTROL CIRCUIT CONDUIT RUNS SHALL BE PROVIDED WITH GREEN COLORED GROUNDING CONDUCTOR.
- ALL CONDUIT RUNS ABOVE DROP CEILING AND BELOW RAISED FLOOR SHALL BE ELECTRICAL METALLIC TUBING (EMT). ALL CONDUIT RUNS EXPOSED SHALL BE RIGID STEEL CONDUIT, THICK WALL.
- ALL FLEXIBLE METAL CONDUITS EXPOSED TO WEATHER SHALL BE LIQUID TIGHT.
- ALL BOXES AND ENCLOSURES FOR OUTDOOR INSTALLATIONS SHALL BE WEATHERPROOF STAINLESS STEEL.
- ALL SURFACES PREVIOUSLY HIDDEN BY ELECTRICAL ITEMS REMOVED SHALL BE PAINTED TO MATCH ADJACENT SURFACE FINISH.
- PROVIDE BLANK COVER PLATES ON ALL ABANDONED BOXES AFTER ELECTRICAL EQUIPMENT IS REMOVED.
- PANT ALL NEW EXPOSED CONDUITS TO MATCH ADJACENT SURFACES.
- EQUIP ALL DISCONNECT SWITCHES WITH GROUNDING LUG FOR EQUIPMENT GROUNDING CONNECTION.
- RESTORE OR PATCH UP ALL OPENINGS CAUSED BY THE REMOVAL OF EXISTING CONDUITS, WIRING DEVICES, OUTLET BOXES AND OTHERS WITH CONCRETE OR ANY APPROVED MATERIAL TO MATCH EXISTING AND PAINT TO MATCH THE COLOR OF EXISTING ADJACENT FINISH. ALL HOLES ON UNPAINTED WALLS SHALL BE PATCHED UP ONLY.
- FOR REMOVAL WORK:
 - PULL OUT WIRES.
 - REMOVE ALL EXPOSED CONDUIT RUNS INDICATED TO BE REMOVED INCLUDING THOSE ABOVE DROP CEILING.
 - FOR ALL THOSE CONDUITS CONCEALED IN CONCRETE CEILING, WALLS AND FLOORS, INDICATED TO BE REMOVED; CUT EXPOSED CONDUITS, PLUG BOTH ENDS AND ABANDON CONDUIT.
- HEAVY LINES INDICATES NEW WORK, LIGHT LINES INDICATES EXISTING CONDITION.
- TEL/LAN OUTLET LABELING SHALL FOLLOW EXISTING LABELING METHODS.

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DATE: 5 SEP '14

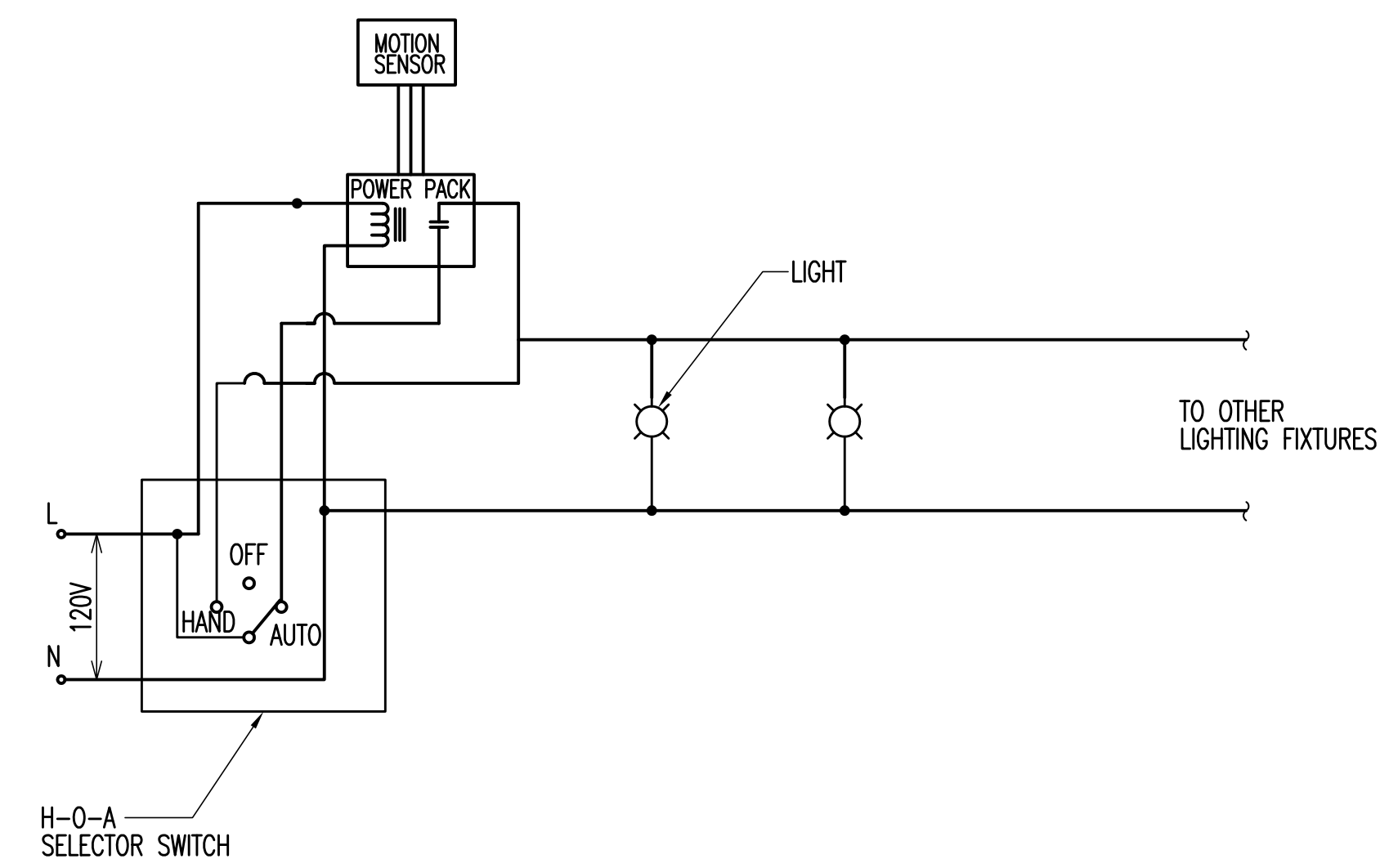
GRAPHIC SCALE



SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. 10th Street, Suite 100, Phoenix, AZ 85018 TEL: (602) 998-1234 FAX: (602) 998-1235 WWW.AEC-SABER.COM				
CONT. DESIG. T. U.	CONT. DR. H. O.	U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
CONT. CIVIL R. S. S.	CONT. CIVIL ENGR J. RUTHERFORD	FORT BUCKNER, OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
FIRE PROTECTION ENGINEER		ELECTRICAL POWER PLAN, LEGEND AND NOTES		
DATE		PROJECT NUMBER	CONTRACT NUMBER	SEQ. NO.
DATE		JOK34982	TO#7206(RFP#0027)	E-1
SUBMITTED BY K. EXETERSTEN (CONTRACT MANAGER)		DATE 8 JUL '13	SCALE AS SHOWN	SHEET 11 of 18
FIRM MEMBER (TITLE)		DATE	FILE	SEQ. NO.
BLDG. NO.		DRAWING NO.		FILE SEQ. NO.



2 LIGHTING PLAN
SCALE 1/100



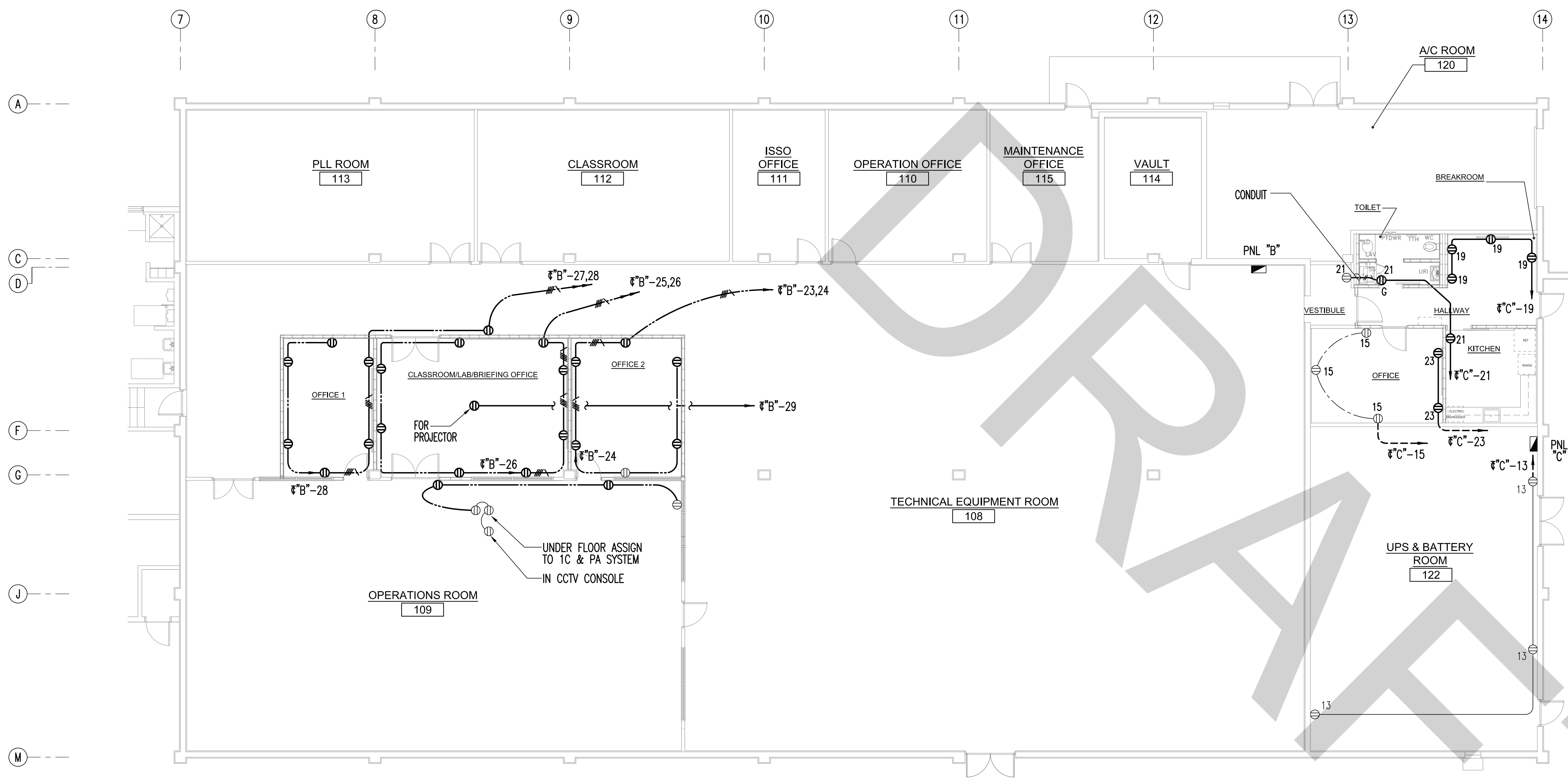
3 TYP OCCUPANCY SENSOR CONTROL DIAGRAM
(FOR TOILET)
SCALE



AEC-SABER
AS-BUILT
DATE: 5 SEP '14

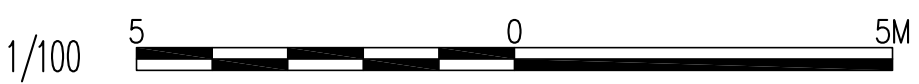
SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 1100 N. 10TH AVE., SUITE 100 DENVER, CO 80202 TEL: 303.733.1100 FAX: 303.733.1101		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
CONT. DESIG. T. U.		CONT. DR. H. O.		FORT BUCKNER OKINAWA, JAPAN
CONT. CIVIL R. S. S.		CONT. CIVIL ENGR. J. RUTHERFORD		RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA
FIRE PROTECTION ENGINEER		DATE		ELECTRICAL
DATE		DATE		LIGHTING PLAN AND TYP OCCUPANCY SENSOR CONTROL DIAGRAM (FOR TOILET)
PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.
JOK34982		TO#7206(RFP#0027)		E-2
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER)		DATE 8 JUL '13		SCALE AS SHOWN
FIRM MEMBER (TITLE)		DATE		SHEET 12 of 18

BLDG. NO. DRAWING NO. FILE SEQ. NO.



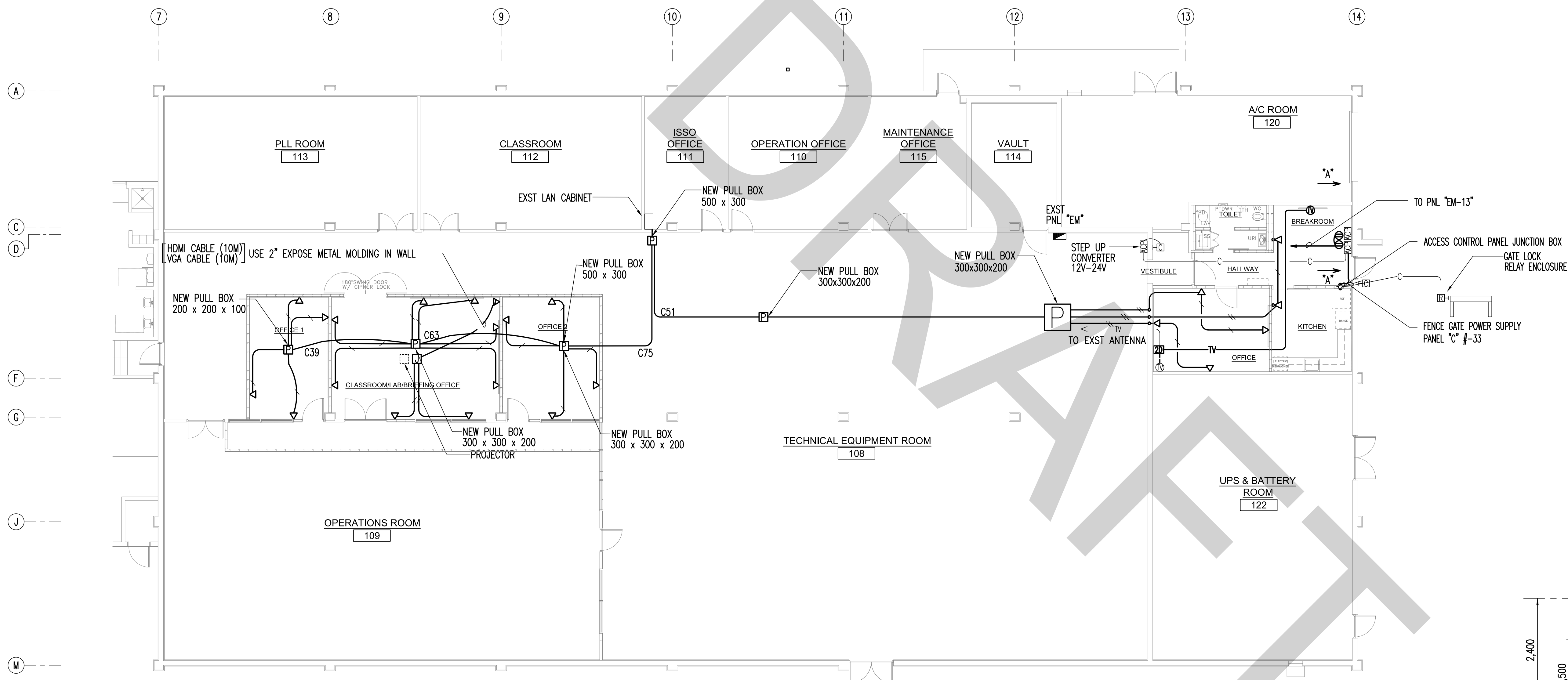
1 RECEPTACLE PLAN
E-3 E-3 SCALE 1/100

GRAPHIC SCALES



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AS-BUILT
DATE: 5 SEP '14

SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS				
FORT BUCKNER OKINAWA, JAPAN				
RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA				
ELECTRICAL RECEPTACLE PLAN				
PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.
JOK34982		TO#7206(RFP#0027)		E-3
SUBMITTED BY		DATE	SCALE	SHEET
K. EXTERSTEIN (CONTRACT MANAGER)		8 JUL '13	AS SHOWN	13 of 18
FIRM MEMBER (TITLE)		DATE	SCALE	SHEET
BLDG. NO. DRAWING NO. FILE SEQ. NO.				

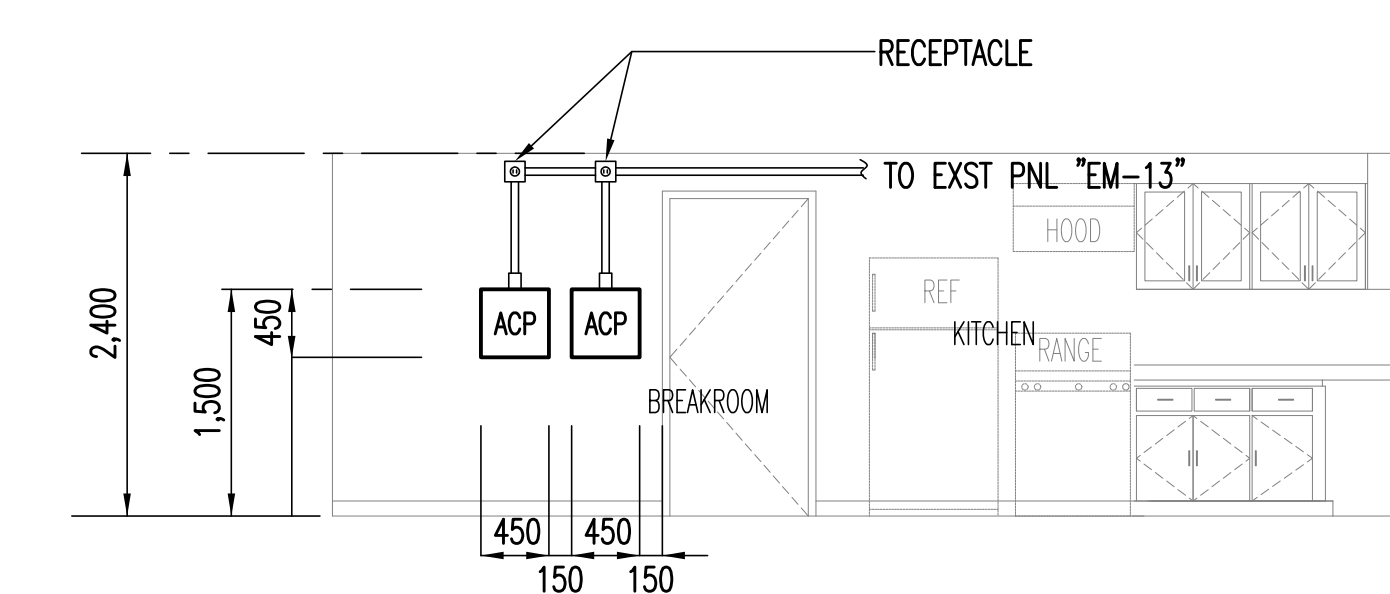


1 ACCESS CONTROL, INTERCOM, TV & TELEPHONE PLAN

E-4 E-4 SCALE 1/100

WIRING SCHEDULE

C	6/C#18,22C
T	1 CATEGORY 6 IN 28C
TV	RG11 COAXIAL CABLE IN 28C
T	2 CATEGORY 6 IN 28C



SECTION "A" - "A"

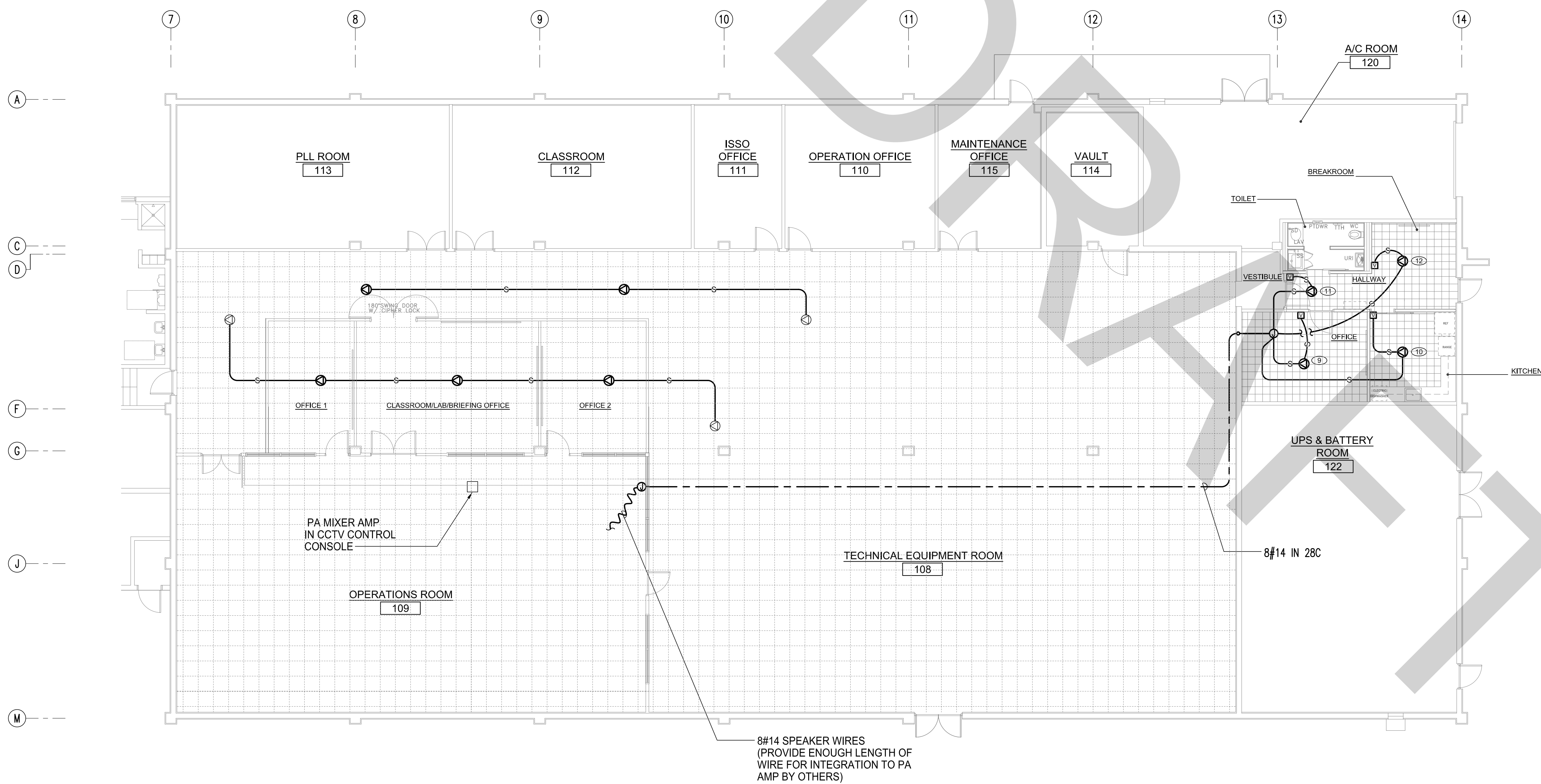
AEC-SABER
AS-BUILT
DATE: 5 SEP '14

GRAPHIC SCALES



SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. W. 11th Ave., Suite 100, Fort Lauderdale, FL 33309 TEL: (954) 584-1234 FAX: (954) 584-1235		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS		
CONT. DESIG. T. U.	CONT. DESIG. H. O.	FORT BUCKNER, OKINAWA, JAPAN		
CONT. CIVIL R. S. S.	CONT. CIVIL ENGR. J. RUTHERFORD	RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
FIRE PROTECTION ENGINEER		ELECTRICAL		
DATE		ACCESS CONTROL, INTERCOM, TV & TEL PLAN, WIRING SCHEDULE AND SECTION		
DATE		PROJECT NUMBER	CONTRACT NUMBER	SEQ. NO.
DATE		JOK34982	TO#7206(RFP#0027)	E-4
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER)		DATE	8 JUL '13	SCALE: AS SHOWN
FIRM MEMBER (TITLE)		DATE	8 JUL '13	SHEET: 14 of 18

BLDG. NO. DRAWING NO. FILE SEQ. NO.



WIRING SCHEDULE

—S— 2/C #14 IN 22C

2 PA SYSTEM PLAN
E-6 E-6 SCALE

1/100

GRAPHIC SCALES



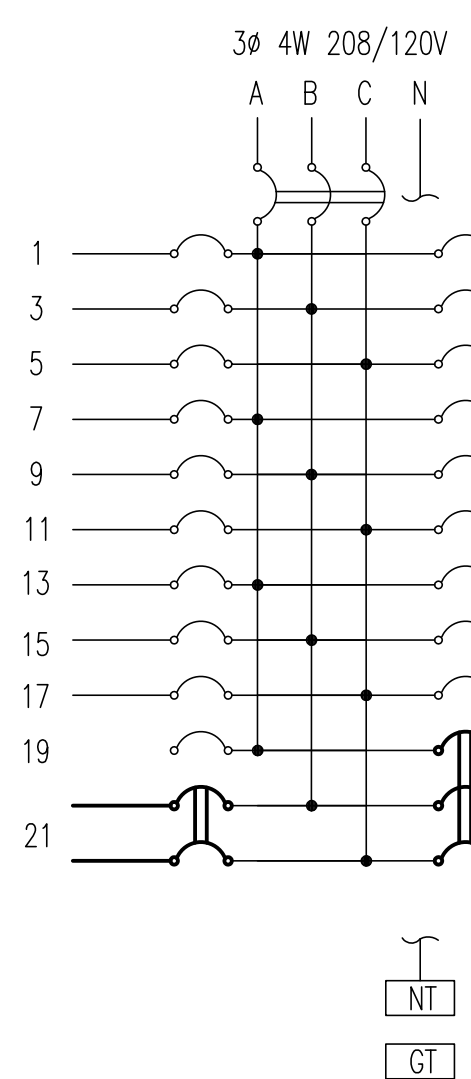
AEC-SABER
AS-BUILT
DATE: 5 SEP '14

SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. 10TH AVE. SUITE 100, DENVER, CO 80202 TEL: 303.733.1100 FAX: 303.733.1101 WWW.AEC-SABER.COM U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS FORT BUCKNER OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA ELECTRICAL PA SYSTEM PLAN AND WIRING SCHEDULE				
PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.
JOK34982		TO#7206(RFP#0027)		E-5
SUBMITTED BY		DATE	SCALE	SHEET
K. EXTERSTEIN (CONTRACT MANAGER)		8 JUL '13	AS SHOWN	15 of 18
FIRM MEMBER (TITLE)		DATE	DATE	DATE
BLDG. NO.		DRAWING NO.		FILE SEQ. NO.

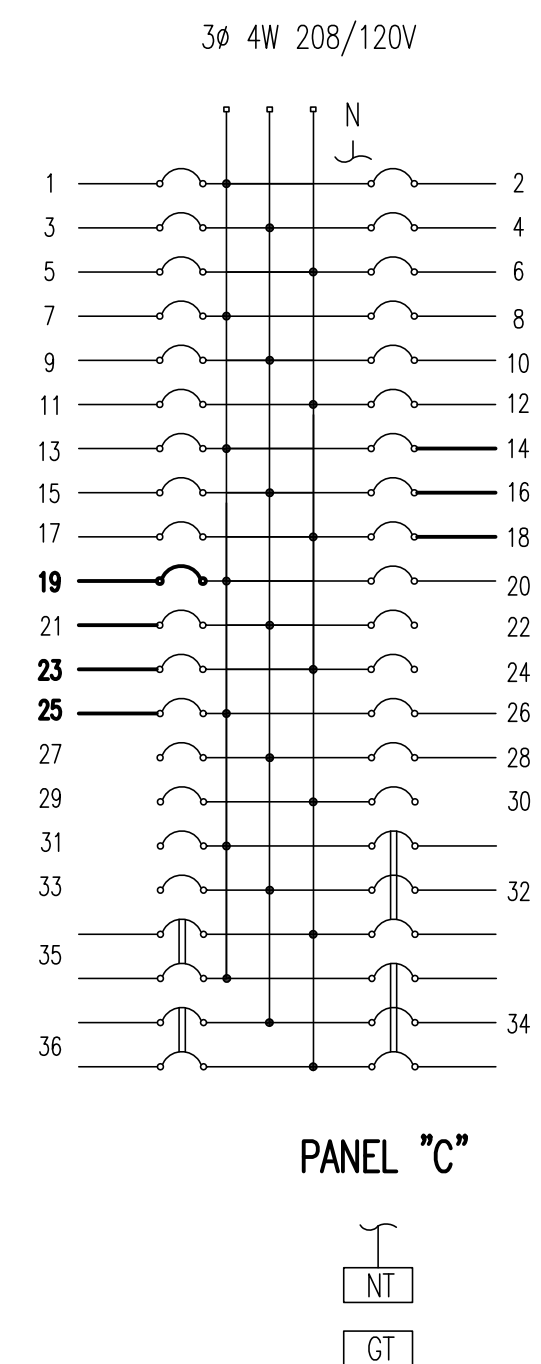
PANELBOARD (IN EXST MCC)						TYPE : FLUSH MTD, TOP FED				
SYSTEM : 208Y/120V, 3-PH, 4W						BUS SIZE : 100A ENC : INDOOR TYPE				
MIN KAIC : 10						LOC : IN EXST MCC				
CKT NO.	DESCRIPTION	TOTAL VA PER CKT	PHASE LOAD (VA)			CIRCUIT BREAKER			REMARKS	
			A	B	C	P	AF	AT		
1	EXHAUST FAN (MEN'S TOILET, 102)	290	290	—	—	—	1	100	20	
2	DESPER HEAT PUMP (EXST)	830	830	—	—	—	1	100	20	
3	SPARE	900	—	900	—	—	1	100	20	
4	HOT WATER PUMP (EXST)	830	—	830	—	—	1	100	20	
5	FAN COIL UNIT (RECEPTION, 101)	830	—	—	830	1	100	20		
6	HOT WATER HEATER SV (EXST)	200	—	—	200	1	100	20		
7	T/H RECORDER (EXST)	500	500	—	—	1	100	20		
8	TIME CLOCK (EXST)	100	100	—	—	1	100	20		
9	SPARE	900	—	900	—	1	100	20		
10	LIGHTS, MECHANICAL RM	715	—	715	—	1	100	20		
11	RECEPTACLE (EXST)	1,080	—	—	1,080	1	100	20		
12	ROLL FILTER (EXST)	500	—	—	500	1	100	20		
13	SPARE	900	900	—	—	1	100	20		
14	DDC	500	500	—	—	1	100	20		
15	VAV BOXES	300	—	300	—	1	100	20		
16	IN-LINE FAN	510	—	510	—	1	100	20		
17	EXST EXHAUST FAN (MECH RM)	450	—	—	450	1	100	20		
18	SPARE	900	—	—	900	1	100	20		
19	SPARE	900	900	—	—	1	100	20		
20	AIR CONDITIONING UNIT (NEW)	2,880	960	960	960	3	100	20		
21	ELECTRIC WATER HEATER (NEW)	6,000	—	3,000	3,000	2	100	40		
TOTAL		21,015	4,980	8,115	7,920	MAIN : 3P-100AF-100AT				

PANELBOARD (IN MCC) LOAD SCHEDULE

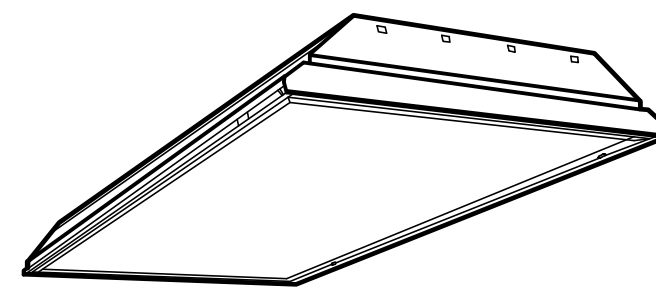
D I A G R A M



PANELBOARD ("C")			TYPE : SURFACE MTD, TOP FED						
SYSTEM : 208Y/120V, 3-PH, 4W			BUS SIZE : 225A ENC : INDOOR TYPE						
MIN KAC : 10			LOC : IN ELECTRICAL RM						
CKT NO.	DESCRIPTION	TOTAL VA PER CKT	CONNECTED LOAD (VA)			CIRCUIT BREAKER			REMARKS
			A	B	C	P	AF	AT	
1	RCPT, MAINT & STORAGE	1,200	1,200			1	100	20	
2	LTGS, BATTERY RM	1,600	1,600			1	100	20	
3	RCPTS, CONF RM/SUPERVISOR RM	1,200		1,200		1	100	20	
4	LTGS, ELECTRICAL RM	1,200		1,200		1	100	20	
5	RCPTS, SUPERVISOR/CONTR RM	1,200			1,200	1	100	20	
6	LTGS, ELECTRICAL RM	1,600			1,600	1	100	20	
7	RCPTS, CONTRACTOR RM	1,200	1,200			1	100	20	
8	LTGS, NCDC & OFFICE	1,600	1,600			1	100	20	
9	RCPTS, SUPPLY RM	1,200		1,200		1	100	20	
10	LTGS, MEN'S & WOMEN'S TOILET	1,600		1,600		1	100	20	
11	RCPT, VAULT RM	1,200			1,200	1	100	20	
12	COFFEE MACHINE	1,000			1,000	1	100	20	
13	UPS & BATTERY ROOM	1,200	1,200			1	100	20	
14	MICRO WAVE	1,000	1,000			1	100	20	
15	OFFICE	1,200		1,200		1	100	20	
16	TOASTER	1,000		1,000		1	100	20	
17	RCPTS, MECH RM	1,200			1,200	1	100	20	
18	REFRIGERATOR	1,000			1,000	1	100	20	
19	RANGE HOOD FAN	250	250			1	100	20	
20	EXHAUST FAN CONTROL	500	500			1	100	20	
21	RCPT, KITCHEN, TOILET	1,080		1,080		1	100	20	
22	SPARE	900				1	100	20	
23	RCPT, EAST OFFICE	900			900	1	100	20	
24	SPARE	900			900	1	100	20	
25	DISH WASHER	1,000	1,000			1	100	20	
26	FLOODLIGHT CONTROL	500	500			1	100	20	
27	SPARE	900		900		1	100	20	
28	PHOTOCELL SW	500		500		1	100	20	
29	SPARE	900			900	1	100	20	
30	SPARE	900			900	1	100	20	
31	SPARE	900	900			1	100	20	
32	EXHAUST FAN	1,656	552	552	552	3	100	20	
33	SPARE	900		900		1	100	20	
34	SECURITY LIGHTS	14,100	4,700	4,700	4,700	3	100	70	
35	220V KITCHEN OUTLETS	1,080	540		540	2	100	20	
36	RANGE	8,000		4,000	4,000	2	100	40	
	TOTAL	58,166	16,642	20,932	20,592				MAIN : 3P-225A MLO



LIGHTING FIXTURE DETAILS



TYPE "A2", "AE2", "A3", "AE3"
(600MM X 1200MM)

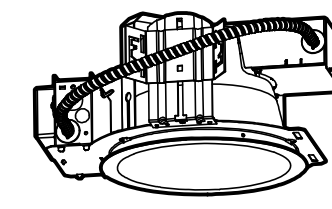
RECESSED MTD FLUORESCENT FIXTURE :

LAMP TYPE: 25W T8

CONSTRUCTION:
FRAME/SHIELDING FLUSH-MOUNTED EXTRUDED ALUMINUM
DOOR FRAME WITH MITERED CORNERS AND BAKED WHITE
ENAMEL FINISH. LENS IS CLEAR PATTERN 12 ACRYLIC.

ELECTRICAL:
STANDARD BALLAST IS ELECTRONIC, THERMALLY
PROTECTED, RESETTNG, CLASS P, HPF, NON PCB, UL
LISTED, CSA CERTIFIED BALLAST.

FIXTURES : REUSE
TYPE "AE-2" "AE-3" : NEW PROVIDE
EMERGENCY BATTERY UNIT
AS FOLLOWS



TYPE "B1"

RECESSED MTD COMPACT FLUORESCENT DOWNLIGHT :

LAMP TYPE: 18W CFL

CONSTRUCTION :
GALVANIZED STEEL FRAME WITH 1/2" PLASTER LIP.
PROVIDED WITH (2) REMODEL CLIPS TO SECURE FRAME
WHEN INSTALLED FROM BELOW THE CEILING.

REFLECTOR :
HIGHLY EFFICIENT ONE PIECE ALUMINUM REFLECTOR.
POLISHED FLANGE STANDARD WITH WHITE PAINTED
FLANGE.

ELECTRICAL:
THERMALLY PROTECTED, RESETTNG, UL LISTED AND CSA
CERTIFIED BALLAST.



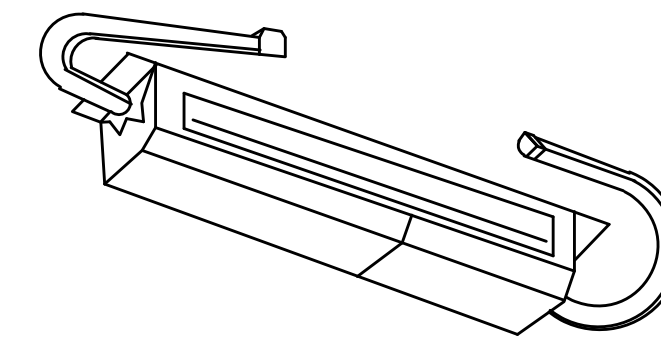
TYPE "C1"

WALL MTD FLUORESCENT LIGHTING FIXTURE :

LAMP TYPE: 17W T8

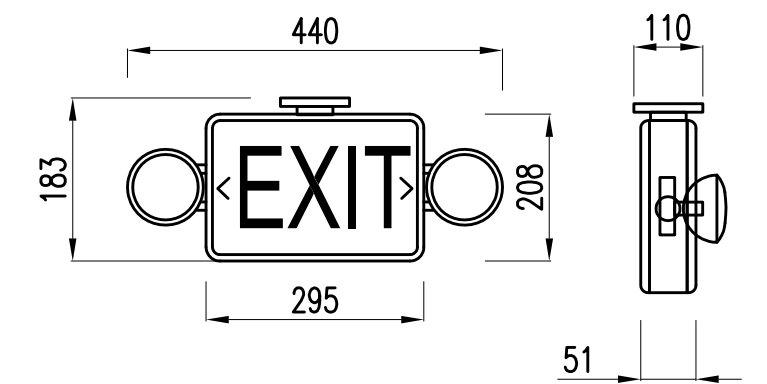
CONSTRUCTION:
EASY MAINTENANCE DESIGN FOR INDIVIDUAL OR TANDEM
MOUNTING. HEAVY-DUTY CODE GRADE STEEL. METAL
PARTS ARE DIE-FORMED.

FINISH:
FIVE-STAGE IRON-PHOSPHATE PRETREATMENT ENSURES
SUPERIOR PAINT ADHESION AND RUST RESISTANCE.



EMERGENCY BATTERY UNIT :

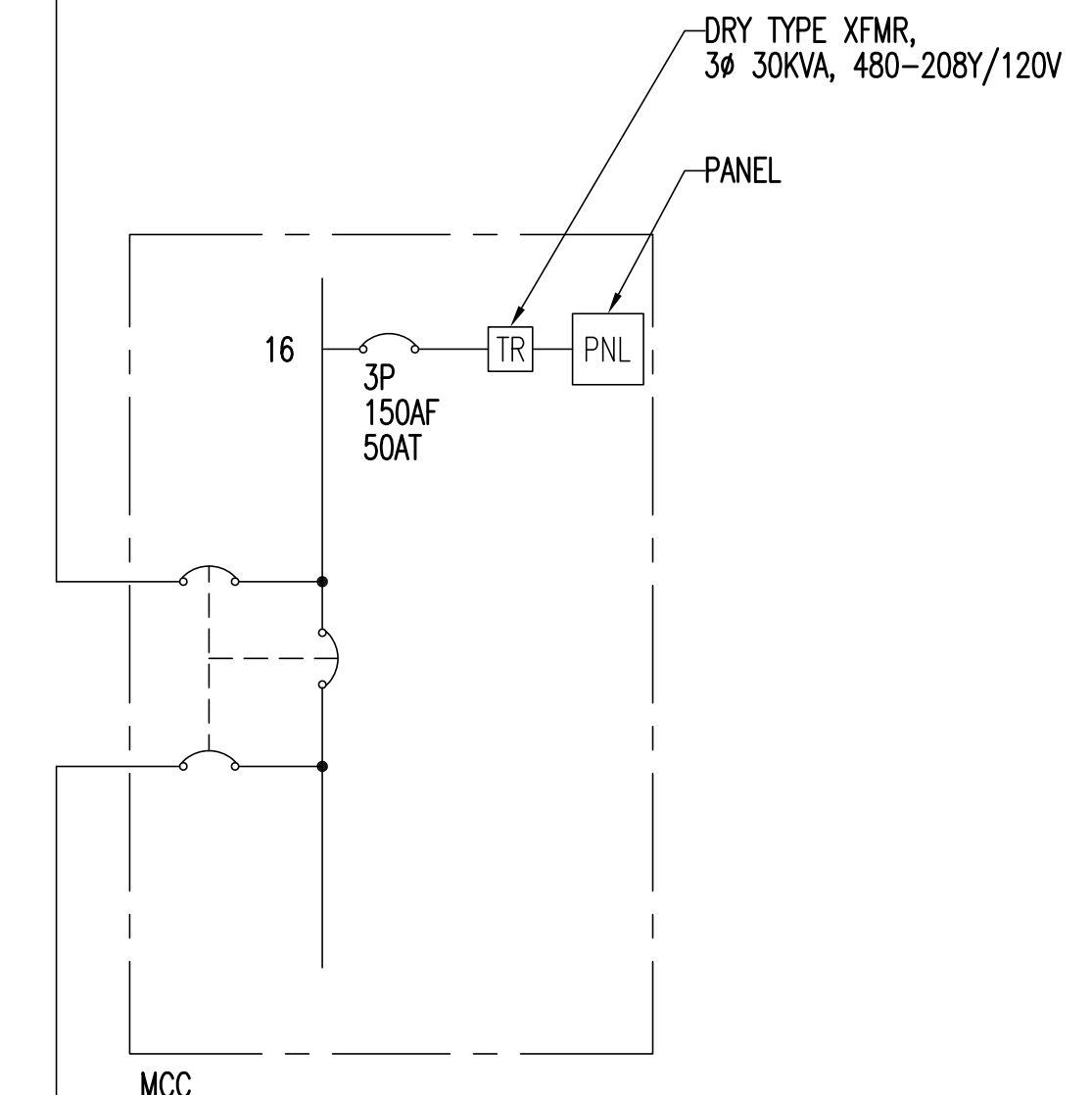
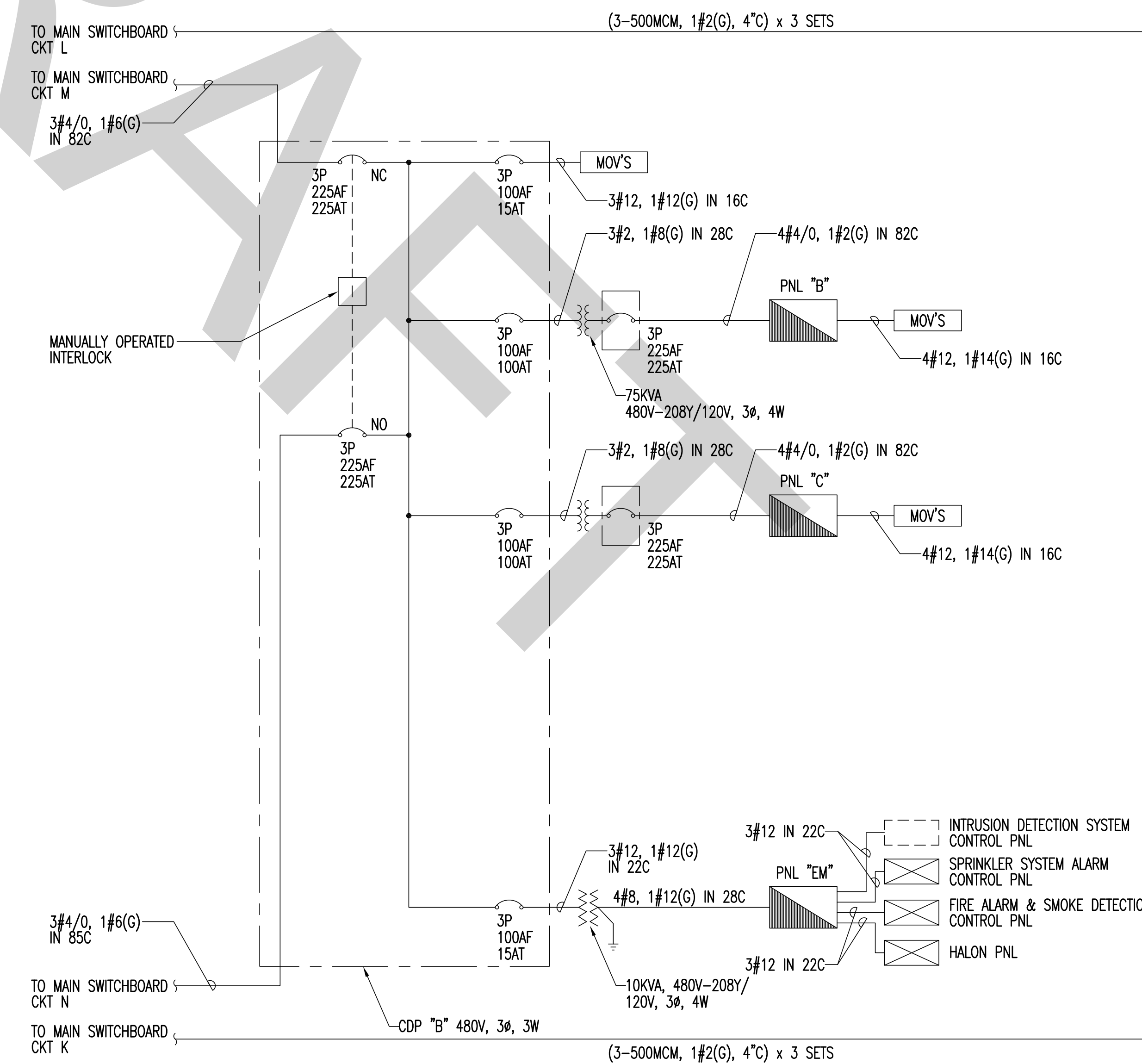
THE UNIT SHALL AUTOMATICALLY SUPPLY POWER TO ONE 25WATT T8 FLUORESCENT LAMP IN THE EVENT OF FAILURE OF THE NORMAL POWER. WHEN NORMAL POWER. WHEN NORMAL POWER RETURNS, THE UNIT SHALL AUTOMATICALLY RECHARGE FROM FULL DISCHARGE TO FULL CHARGE IN 24 HOURS.



EXIT/EMERGENCY LIGHTING WITH DEPENDABLE

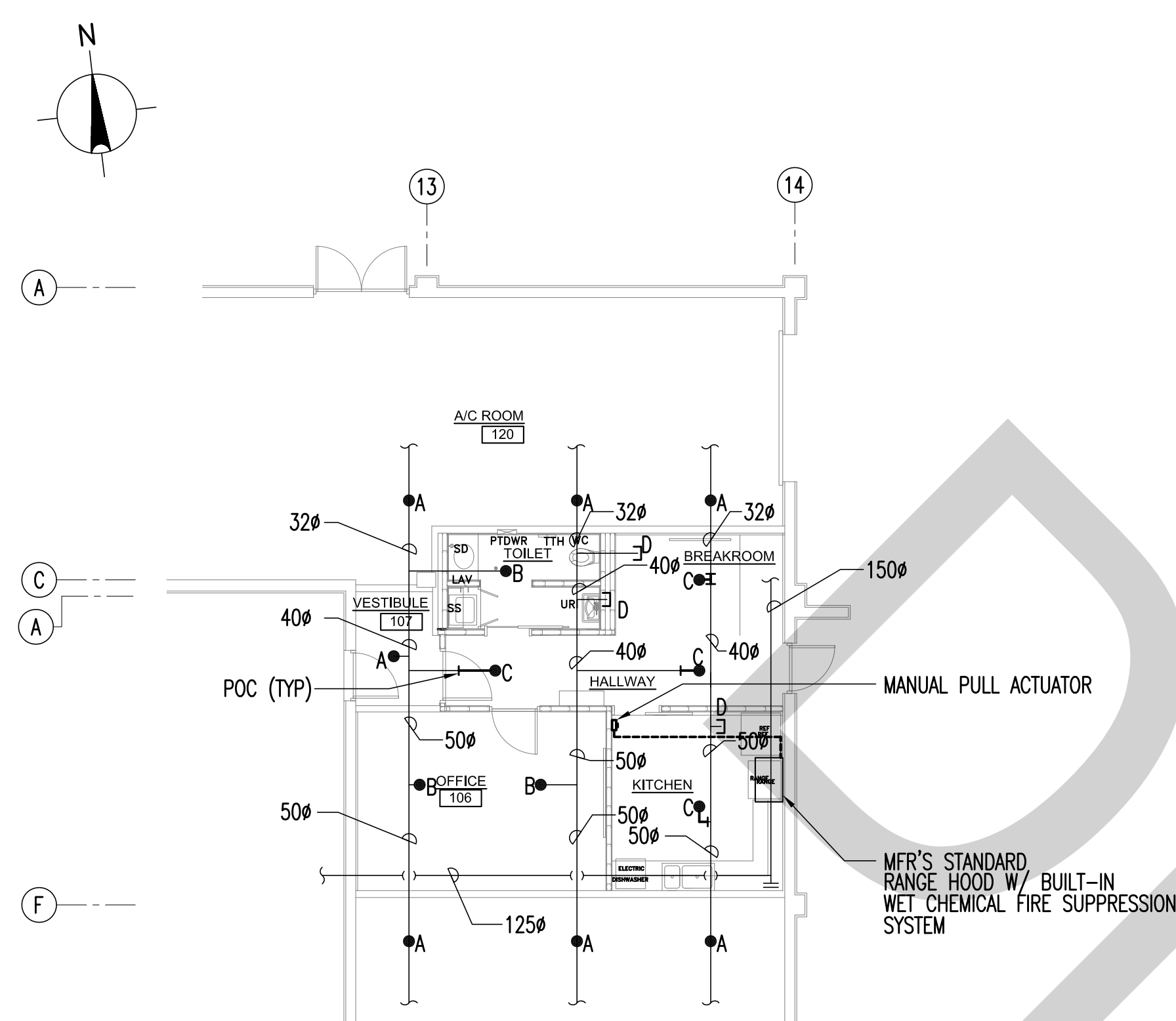
LIGHT EMITTING DIODE(LED) ENERGY EFFICIENCY :
EXIT LIGHTING FIXTURE. HOUSING SHALL BE PRECISION-
MOLDED POLYCARBONATE. UNIT SHALL BE EQUIPPED WITH
LED HAVING AN EMERGENCY RUN TIME OF 1 1/2 HOURS
MINIMUM. LETTERS SHALL BE 150mm TALL WITH 20mm WIDE
STROKES AND STENCIL PUNCH-OUT SINGLE FACE
FOR LED'S GREEN ILLUMINATION, PROVIDE
EMERGENCY BATTERY PACK WITH SOLID STATE SURGE
CHARGER. FIXTURE SHALL BE RATED 120VAC, 60Hz

TYPE	NO.OF LAMPS	VOLTS	REMARKS
"D1"	LED	120	SINGLE FACE, W/CHEVRON DIRECTION, WALL MOUNTED
"D2"	LED	120	SINGLE FACE, WALL MOUNTED

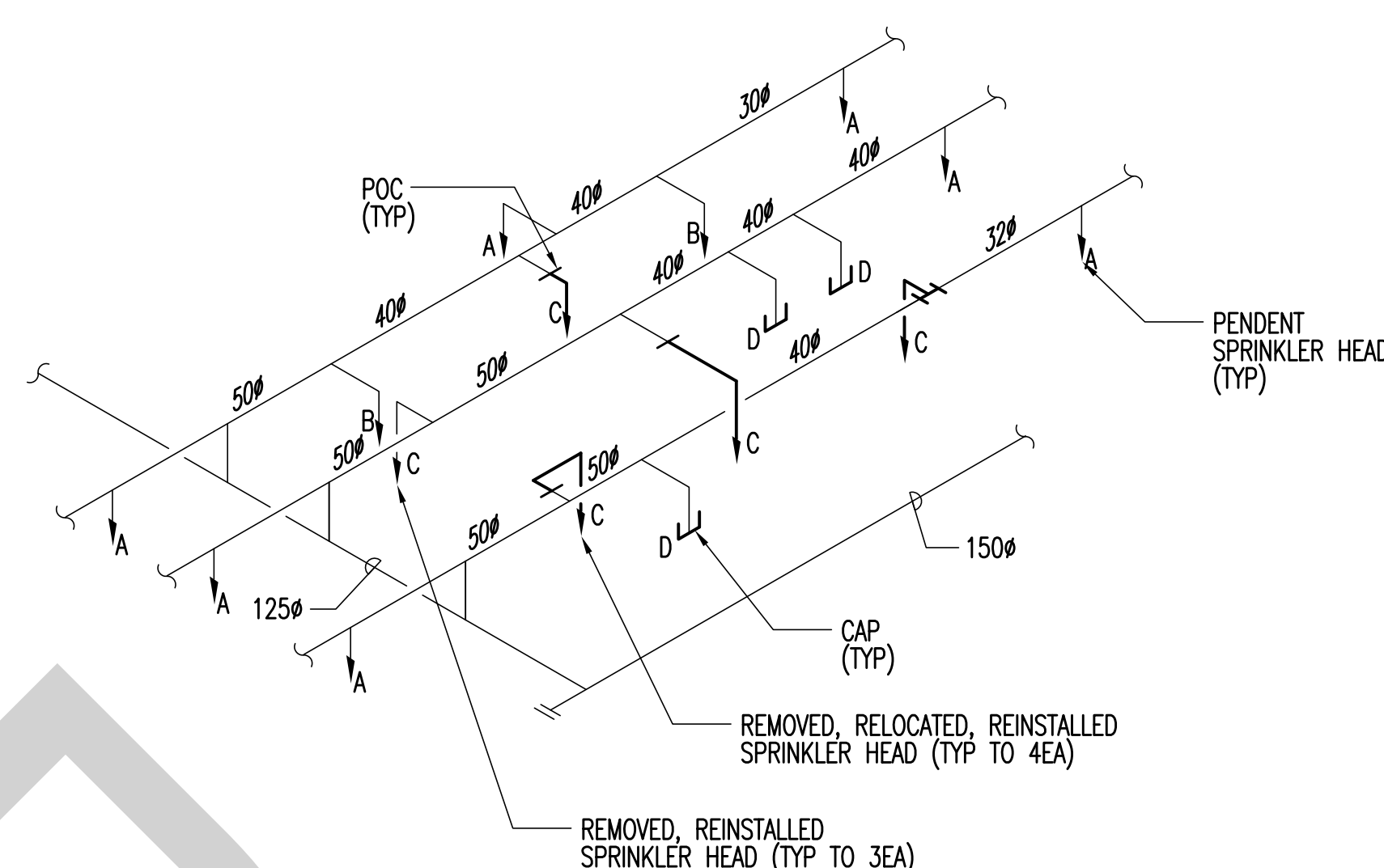


AEC-SABER
AS-BUILT
DATE: 5 SEP '14

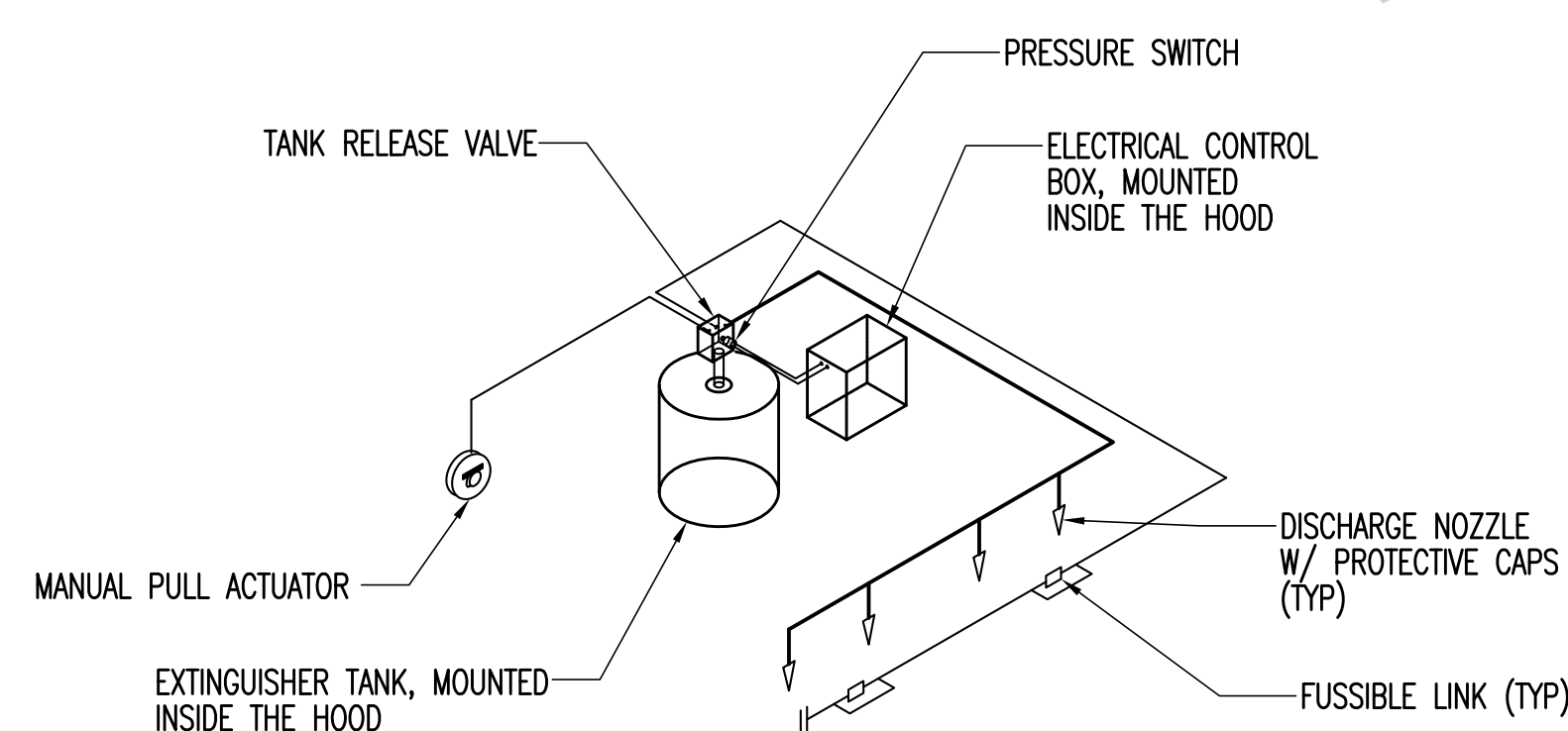
[illegible]



1 PARTIAL FIRE PROTECTION PLAN
SCALE 1/100

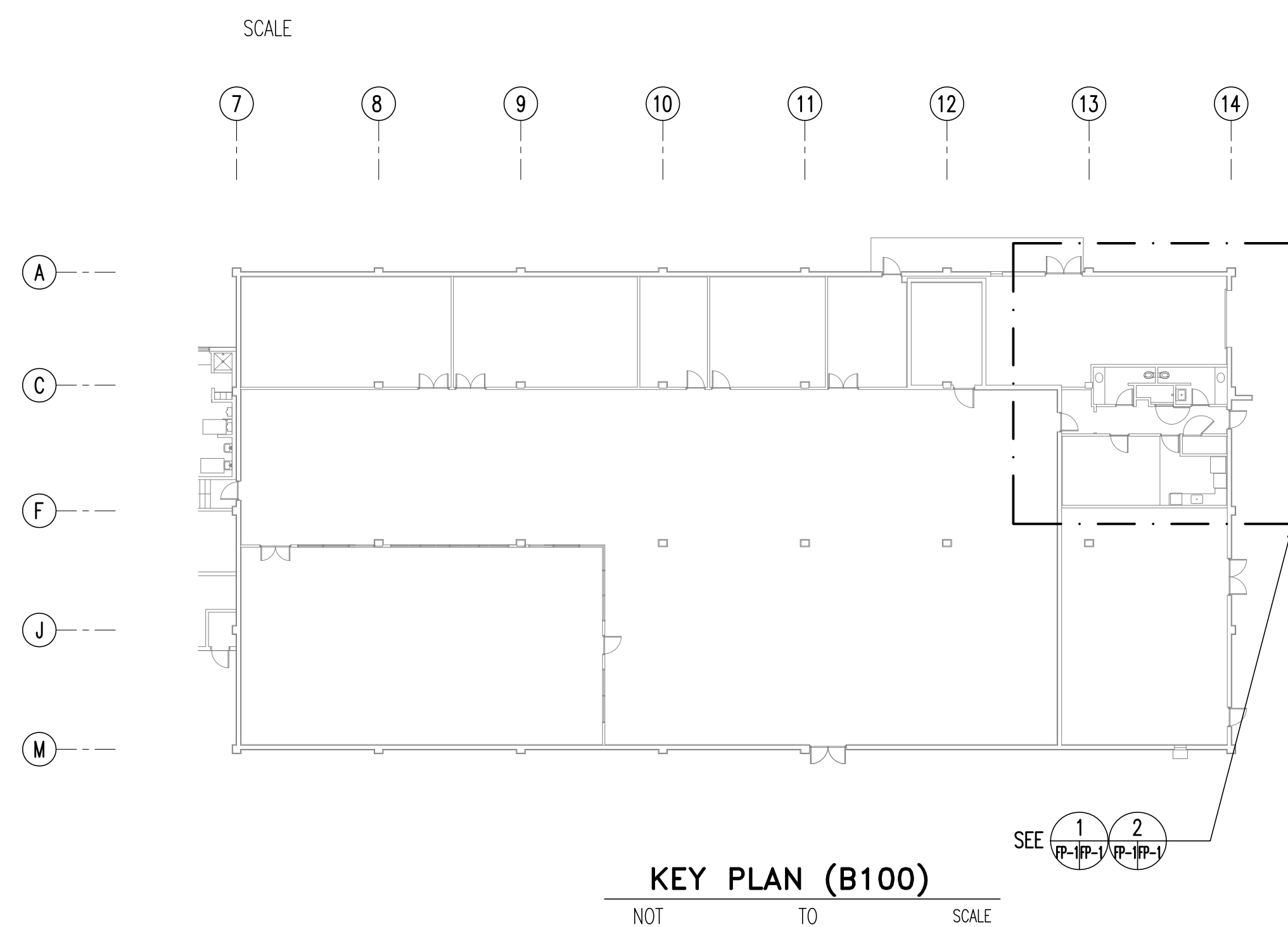


2 SPRINKLER ISOMETRIC PIPING DIAGRAM
NOT TO SCALE



3 RANGE HOOD BUILT-IN WET CHEMICAL FIRE SUPPRESSION SYSTEM ISOMETRIC PIPING DIAGRAM
NOT TO SCALE

- LEGEND :**
- SPRINKLER PIPE
 - EXST SPRINKLER HEAD TO REMAIN
 - EXST SPRINKLER HEAD TO BE REMOVE & REINSTALLED
 - EXST SPRINKLER HEAD TO BE REMOVED & RELOCATED
 - EXST SPRINKLER HEAD TO BE REMOVED & PROVIDE CAP TO EXST PIPE



KEY PLAN (B100)
NOT TO SCALE

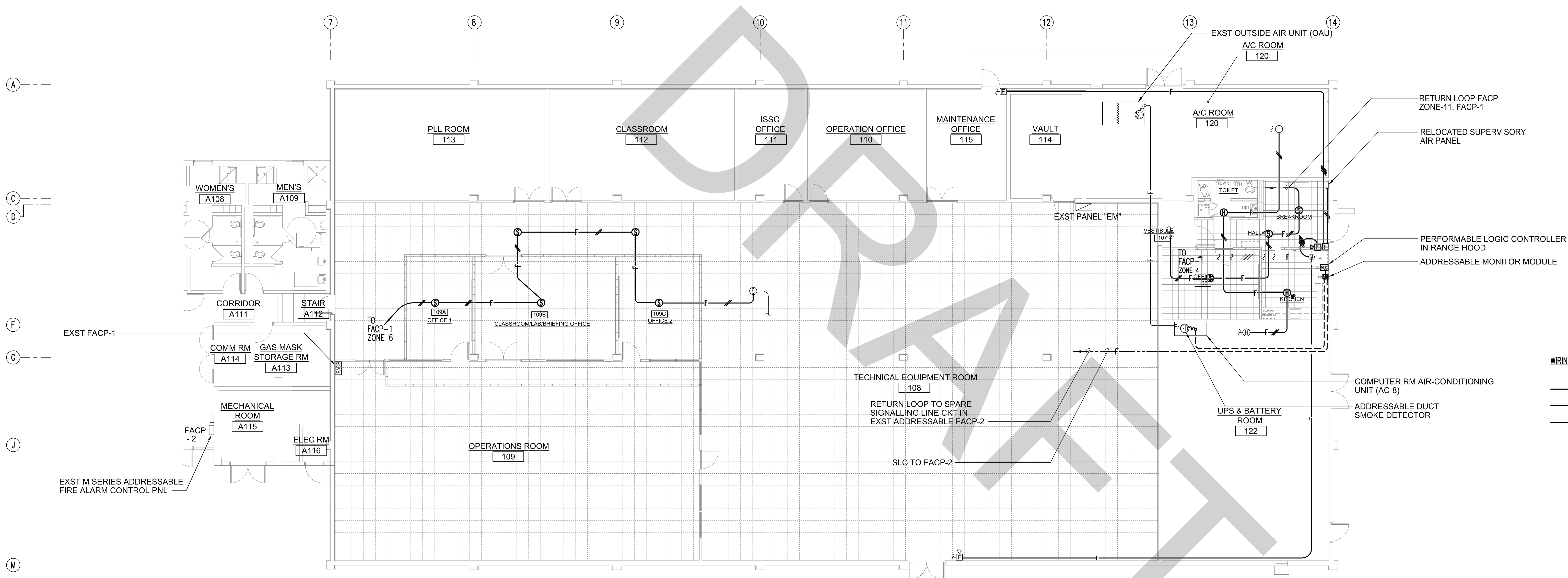
AEC-SABER
AS-BUILT
DATE: 5 SEP '14

GRAPHIC SCALES



SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
AMERICAN ENGINEERING CORP. 4101 N. W. 12th Street, Suite 100, Fort Lauderdale, FL 33304 TEL: (954) 584-1234 FAX: (954) 584-1235		U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS FORT BUCKNER OKINAWA, JAPAN RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA		
CONT. DESIG. J. M. S.	CONT. ENG. H. U.	MECHANICAL		
CONT. CIV. A. S. C.	CONT. CIV. ENGR. J. RUTHERFORD	PARTIAL FIRE PROTECTION PLAN, SPRINKLER ISOMETRIC PIPING DIAGRAM, LEGEND AND KEY PLAN		
FIRE PROTECTION ENGINEER		PROJECT NUMBER	CONTRACT NUMBER	SEQ. NO.
		JOK34982	TO#7206(RFP#0027)	FP-1
SUBMITTED BY K. EXTERSTEIN (CONTRACT MANAGER)		DATE 8 JUL '13	SCALE AS SHOWN	SHEET 17 of 18
FIRM MEMBER (TITLE)		DATE	DATE	DATE

BLDG. NO. DRAWING NO. FILE SEQ. NO.



WIRING SCHEDULE:

- 2/C #16 FOR DETECTION CKT 3/4°C
- 2/C #14 FOR NOTIFICATION 2/C #16 FOR DETECTION CKT 3/4°C
- 2/C #14 FOR NOTIFICATION 2-2/C #16 FOR DETECTION CKT 3/4°C

1 FIRE ALARM SYSTEM PLAN
FP-2FP-2 SCALE 1/100

**AEC-SABER
AS-BUILT**
DATE: 5 SEP '14

GRAPHIC SCALES

1/100 5 0 5M

SYMBOL		DESCRIPTION	DATE	APPROVAL
REVISIONS				
U.S. ARMY GARRISON, JAPAN OKINAWA PUBLIC WORKS				
FORT BUCKNER OKINAWA, JAPAN				
RENOVATE BLDG 100, EAST ENTRANCE & OFFICE AREA				
FIRE PROTECTION FIRE ALARM SYSTEM PLAN AND WIRING SCHEDULE				
PROJECT NUMBER		CONTRACT NUMBER		SEQ. NO.
JOK34982		TO#7206(RFP#0027)		FP-2
SUBMITTED BY		DATE		SCALE
K. EXTERSTEIN (CONTRACT MANAGER)		8 JUL '13		AS SHOWN
FIRM MEMBER (TITLE)		DATE		SHEET
				18 of 18

BLDG. NO. DRAWING NO. FILE SEQ. NO.

ASBESTOS AND LEAD BASED PAINT ANALYSIS REPORT

Conducted at:

PWE02431, BULK SAMPLING AND TESTING OF
ASBESTOS & LEAD BASE PAINT (LBP), VARIOUS BLDGS,
BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

Prepared for:
NIHON MECCS CO., LTD.

Prepared by:
SHOKAIHATSU CO., LTD.

SEPTEMBER 2022

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- 2.0 ASBESTOS
- 3.0 LEAD BASED PAINT
- 4.0 CONCLUSION / RECOMMENDATIONS

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- APPENDIX A: Sampled Locations Diagram, Analytical Summary, Photographic Documentation, and Laboratory Analytical Reports
- APPENDIX B: Building Inspector(s) Certification
- APPENDIX C: Laboratory Accreditations

1.0 INTRODUCTION

SHOKAIHATSU CO., LTD. was retained by NIHON MECCS CO., LTD. to collect building material samples for presence of asbestos and lead based paint. The purpose of these inspections was to identify suspect Asbestos Containing Building Materials (ACBM's) and Lead Based Painted Materials (LBPM's) and components which may be impacted by renovation or demolition activities. The following general scope of work for this project are as follows:

- The general scope of work for this project is to sample and test all suspected ACM and LBP materials at ten (10) locations on Buckner Communication Site.

From 28 February 2022 through 7 April 2022, certified Asbestos and Lead Building Inspector(s), Mr. Takafumi Noborikawa, Mr. Tomohide Makabe, and associates completed the site inspection and upon completion, building materials samples were collected from the suspected ACM's and suspected LBPM's.

2.0 ASBESTOS

SAMPLING METHODOLOGY – ASBESTOS

The method described in 40 CFR Part 763.86, Sampling (for asbestos) for collecting bulk asbestos samples protocol were used.

ASBESTOS ANALYSIS

Suspected ACM of the bulk samples collected were analyzed by HAWAII ANALYTICAL LABORATORY, LLC., who are accredited for bulk sampling analysis through successful participation in the National Voluntary Laboratory Accreditation Program (NVLAP).

The method of analysis was based on Bulk Asbestos Analysis (PLM) EPA 600/R-93/116: Method for the Determination of Asbestos in Bulk Building Materials and the provisions of the Japan Environmental Governing Standards (JEGS). Identification was made using polarized light microscopy (PLM). For “non-detect” or “ND” PLM results, the material was considered less than 0.1% asbestos. For “trace”, or “less than one (1) percent”, PLM results was further analyzed by appropriate quantification methods using the PLM 1000 point count method.

Based on the laboratory analytical report, the sampled locations diagram, analytical summary table, photographic documentation, and laboratory's analytical report can be found in Appendix A.

3.0 LEAD BASED PAINT

SAMPLING METHODOLOGY – LEAD BASED PAINT

Sampling was conducted in accordance with the EPA – U.S. HUD guidelines and 29 CFR 1926.62 and 40 CFR 745.227.

LEAD BASED PAINT CHIP ANALYSIS

The samples collected were analyzed by HAWAII ANALYTICAL LABORATORY, LLC, 3615 Harding Avenue #308, Honolulu, Hawaii. HAWAII ANALYTICAL LABORATORY, LLC is an accredited laboratory National Lead Laboratory Accreditation Program (NLLAP).

The method of analysis was based on NIOSH Method 7082m LEAD by FAAS analysis method to determine the results. All sample results were reported on a dry weight basis as mg/cm² based on the surface area of the analyzed portion of the sample. In accordance with the JEGS/EPA definition, LBP is paint or other surface coating that contains ≥ 1.0 milligram per cm², or 0.5% by weight (5,000 mg/kg) or 5,000 ppm by weight.

Based on the laboratory analytical report, the sampled locations diagram, analytical summary table, photographic documentation, and laboratory's analytical report can be found in Appendix A.

4.0 CONCLUSION / RECOMMENDATIONS

Based on the analytical results of the samples, asbestos and lead containing materials were found. All accessible building interiors/exterior were visually inspected to identify suspect materials. The sampling methodology was non-destructive and conducted in accessible areas only and did not identify materials hidden inside walls, above hard ceilings, etc., as the buildings are currently occupied/used and pursuant to the SOW.

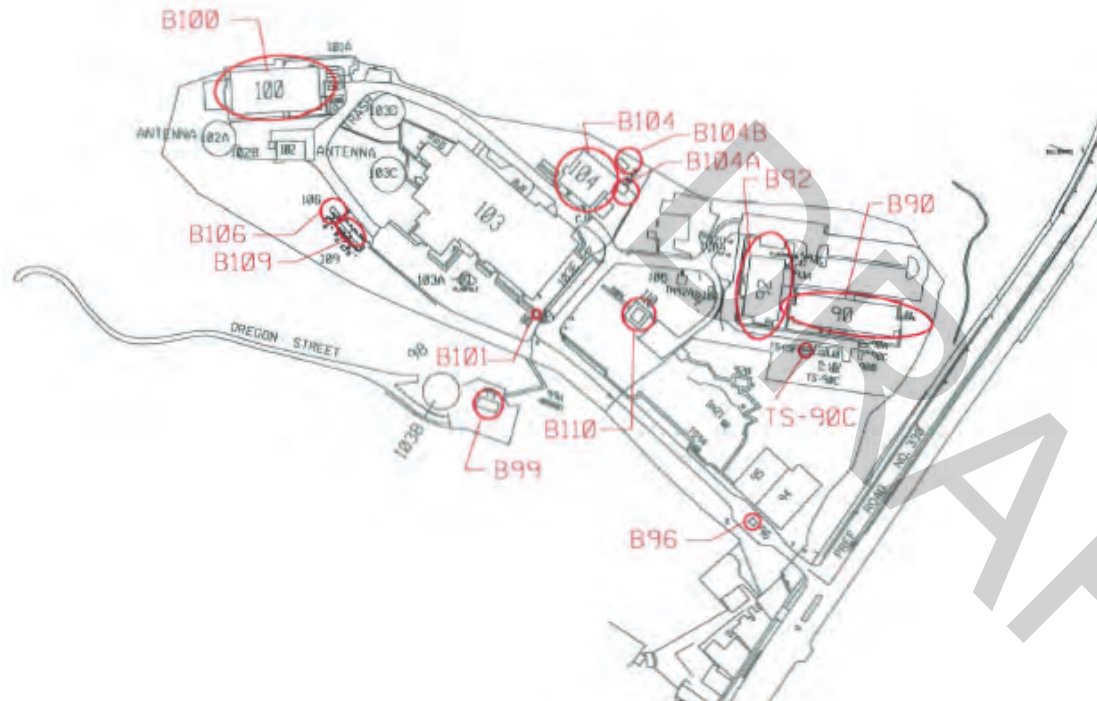
While reasonable effort was made to locate and sample all suspect materials for lead and asbestos, there are limitations inherent to physical inspections, especially for materials located behind walls, under multiple layers of flooring, above hard ceilings, inaccessible plenum spaces, or concealed pipe/mechanical chase, etc. it is possible that areas or materials may exist in the surveyed buildings that have not been identified or evaluated.

Should any potentially hazardous building materials be found during future construction activities not addressed in this report should be assumed to be asbestos containing materials or lead-based paint, and not be disturbed prior to evaluated by a licensed and competent inspector.

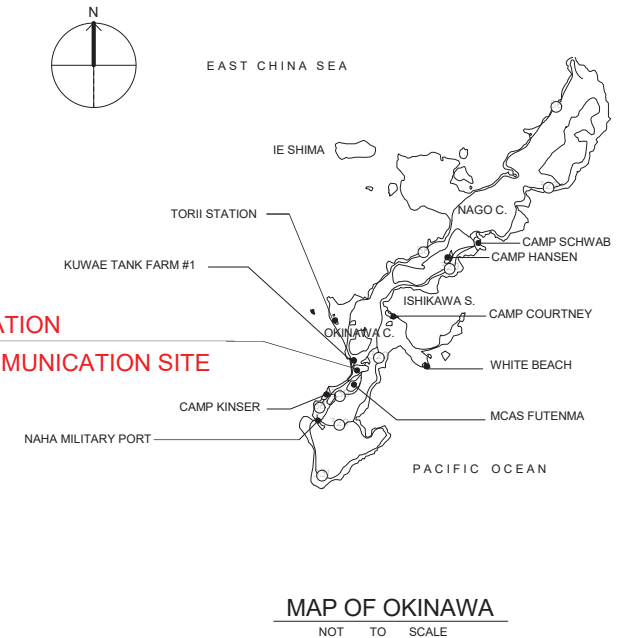
APPENDIX A:

Sampled Locations Diagram, Analytical Summary, Photographic
Documentation, and Laboratory Analytical Reports

DRAFT



BUCKNER COMMUNICATION SITE



BLDG. 100

DRAFT

SACM Analyzed Material Description	Result
B100-A-01 Exterior Concrete Wall Surface	ND*
B100-A-02 Exterior Concrete Wall Surface	ND*
B100-A-03 Exterior Concrete Wall Surface	ND*
B100-A-04 Exterior Concrete Wall Surface	ND*
B100-A-05 Exterior Concrete Wall Surface	ND*
B100-A-06 Exterior Concrete Wall Surface	ND*
B100-A-07 Exterior Concrete Wall Surface	ND*
B100-A-08 Neo Seal (Elect Box, Exterior)	ND*
B100-A-09 Neo Seal (Elect Box, Exterior)	-
B100-A-10 Caulking Material (Exterior)	ND*
B100-A-11 Concrete Floor Slab (Exterior)	ND*
B100-A-12 Concrete Floor Slab (Exterior)	ND*
B100-A-13 Concrete Floor Slab (Exterior)	ND*
B100-A-14 Floor Mortar Repair Seal (Exterior)	ND*
B100-A-15 Expansion Joint (Exterior)	ND*
B100-A-16 Concrete Pad (Condenser, Exterior)	ND*
B100-A-17 Concrete Pad (Condenser, Exterior)	ND*
B100-A-18 Concrete Pad (Condenser, Exterior)	ND*
B100-A-19 Pipe Insulation (Straight, Exterior)	-
B100-A-20 Pipe Insulation (Elbow, Exterior)	-
B100-A-21 Pipe Flange Gasket (Exterior)	ND*
B100-A-22 Concrete Pad (AC, Exterior)	ND*
B100-A-23 Concrete Pad (AC, Exterior)	ND*
B100-A-24 Concrete Pad (AC, Exterior)	ND*
B100-A-25 Pipe Insulation (Straight, Exterior)	ND*
B100-A-26 Pipe Insulation (Elbow, Exterior)	-
B100-A-27 Ceramic Floor Tile w/Grout (Front Entrance, Exterior)	ND*
B100-A-28 Ceramic Floor Tile w/Grout (Front Entrance, Exterior)	ND*
B100-A-29 Ceramic Floor Tile w/Grout (Front Entrance, Exterior)	ND*
B100-A-30 Mortar (Under Tile, Front Entrance, Exterior)	ND*
B100-A-31 Mortar (Under Tile, Front Entrance, Exterior)	ND*
B100-A-32 Mortar (Under Tile, Front Entrance, Exterior)	ND*
B100-A-33 Concrete Floor Slab (Front Entrance, Exterior)	ND*
B100-A-34 Concrete Floor Slab (Front Entrance, Exterior)	ND*
B100-A-35 Concrete Floor Slab (Front Entrance, Exterior)	ND*
B100-A-36 Expansion Joint (Wall, Exterior)	ND*
B100-A-37 Interior Concrete Wall (Rm 101)	ND*
B100-A-38 Interior Concrete Wall (Rm 101)	ND*
B100-A-39 Interior Concrete Wall (Rm 101)	ND*
B100-A-40 Ceiling Board (Rm 101)	ND*
B100-A-41 Ceiling Board (Rm 101)	ND*
B100-A-42 Ceiling Board (Rm 101)	ND*
B100-A-43 Ceiling Insulation (Rm 101)	-
B100-A-44 Ceiling Insulation (Rm 101)	-
B100-A-45 Ceiling Insulation (Rm 101)	-
B100-A-46 Ceiling Slab (Toilet)	ND*
B100-A-47 Ceiling Slab (Toilet)	ND*
B100-A-48 Ceiling Slab (Toilet)	ND*
B100-A-49 Vinyl Cove Base w/Mastic (Rm 101)	ND*
B100-A-50 Vinyl Cove Base w/Mastic (Rm 101)	Chrysotile (2%)

SACM Analyzed Material Description	Result
B100-A-51 Vinyl Cove Base w/Mastic (Rm 101)	Chrysotile (2%)
B100-A-52 Wallboard (Rm 101)	ND*
B100-A-53 Wallboard (Rm 101)	ND*
B100-A-54 Wallboard (Rm 101)	ND*
B100-A-55 Vinyl Floor Tile w/Mastic (Rm 101)	ND*
B100-A-56 Vinyl Floor Tile w/Mastic (Rm 101)	ND*
B100-A-57 Vinyl Floor Tile w/Mastic (Rm 101)	ND*
B100-A-58 Concrete Floor Slab (Rm 101)	ND*
B100-A-59 Concrete Floor Slab (Rm 101)	ND*
B100-A-60 Concrete Floor Slab (Rm 101)	ND*
B100-A-61 Neo Seal (Rm 101)	-
B100-A-62 Duct Insulation (Rm 101)	ND*
B100-A-63 Duct Insulation (Rm 101)	ND*
B100-A-64 Duct Insulation (Rm 101)	ND*
B100-A-65 Floor Carpet w/Mastic (Rm 102)	ND*
B100-A-66 Floor Carpet w/Mastic (Rm 102)	ND*
B100-A-67 Floor Carpet w/Mastic (Rm 102)	ND*
B100-A-68 Interior Concrete Wall (Rm 105)	-
B100-A-69 Interior Concrete Wall (Rm 105)	-
B100-A-70 Interior Concrete Wall (Rm 105)	-
B100-A-71 Ceiling Board (Rm 105)	ND*
B100-A-72 Ceiling Board (Rm 105)	ND*
B100-A-73 Ceiling Board (Rm 105)	ND*
B100-A-74 Ceiling Insulation (Rm 105)	-
B100-A-75 Ceiling Insulation (Rm 105)	ND*
B100-A-76 Ceiling Insulation (Rm 105)	-
B100-A-77 Ceiling Slab (Rm 105)	-
B100-A-78 Ceiling Slab (Rm 105)	ND*
B100-A-79 Ceiling Slab (Rm 105)	-
B100-A-80 Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-81 Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-82 Vinyl Cove Base w/Mastic (Rm 105)	Chrysotile (2%)
B100-A-83 Wallboard (Rm 105)	ND*
B100-A-84 Wallboard (Rm 105)	ND*
B100-A-85 Wallboard (Rm 105)	ND*
B100-A-86 Wall Insulation (Rm 105)	ND*
B100-A-87 Wall Insulation (Rm 105)	ND*
B100-A-88 Wall Insulation (Rm 105)	ND*
B100-A-89 Dampproofing Material (Rm 105)	ND*
B100-A-90 Vinyl Floor Tile w/Mastic (Rm 105)	ND*
B100-A-91 Vinyl Floor Tile w/Mastic (Rm 105)	ND*
B100-A-92 Vinyl Floor Tile w/Mastic (Rm 105)	ND*
B100-A-93 Concrete Floor Slab (Rm 105)	ND*
B100-A-94 Concrete Floor Slab (Rm 105)	ND*
B100-A-95 Concrete Floor Slab (Rm 105)	ND*
B100-A-96 Ceramic Wall Tile w/Grout (Toilet)	ND*
B100-A-97 Ceramic Wall Tile w/Grout (Toilet)	ND*
B100-A-98 Ceramic Wall Tile w/Grout (Toilet)	ND*
B100-A-99 Ceramic Floor Tile w/Grout (Toilet)	ND*
B100-A-100 Ceramic Floor Tile w/Grout (Toilet)	ND*

SACM Analyzed Material Description	Result
B100-A-101 Ceramic Floor Tile w/Grout (Toilet)	ND*
B100-A-102 Mortar (Under Tile) (Toilet)	-
B100-A-103 Mortar (Under Tile) (Toilet)	-
B100-A-104 Mortar (Under Tile) (Toilet)	-
B100-A-105 Concrete Floor Slab (Toilet)	ND*
B100-A-106 Concrete Floor Slab (Toilet)	ND*
B100-A-107 Concrete Floor Slab (Toilet)	ND*
B100-A-108 Ceiling Board (Toilet)	ND*
B100-A-109 Ceiling Board (Toilet)	ND*
B100-A-110 Ceiling Board (Toilet)	ND*
B100-A-111 Ceiling Insulation (Toilet)	ND*
B100-A-112 Ceiling Insulation (Toilet)	ND*
B100-A-113 Ceiling Insulation (Toilet)	ND*
B100-A-114 Concrete Floor Slab (Toilet)	ND*
B100-A-115 Concrete Floor Slab (Toilet)	ND*
B100-A-116 Concrete Floor Slab (Toilet)	ND*
B100-A-117 Floor Mortar Repair Seal (Rm 110)	ND*
B100-A-118 Floor Mortar Repair Seal (Rm 110)	ND*
B100-A-119 Floor Mortar Repair Seal (Rm 110)	ND*
B100-A-120 Interior Concrete Wall (Elec Rm)	Chrysotile (3%)
B100-A-121 Interior Concrete Wall (Elec Rm)	ND*
B100-A-122 Interior Concrete Wall (Elec Rm)	Chrysotile (1%)
B100-A-123 Ceiling Slab (Elec Rm)	ND*
B100-A-124 Ceiling Slab (Elec Rm)	ND*
B100-A-125 Ceiling Slab (Elec Rm)	-
B100-A-126 Neo Seal (Elec Rm)	ND*
B100-A-127 Neo Seal (Elec Rm)	ND*
B100-A-128 Neo Seal (Elec Rm)	ND*
B100-A-129 Wallboard (Elec Rm)	ND*
B100-A-130 Wallboard (Elec Rm)	ND*
B100-A-131 Wallboard (Elec Rm)	ND*
B100-A-132 Floor Slab (Elec Rm)	ND*
B100-A-133 Floor Slab (Elec Rm)	ND*
B100-A-134 Floor Slab (Elec Rm)	ND*
B100-A-135 Wall Mortar Repair Seal (Elec Rm)	-
B100-A-136 Floor Mortar Repair Seal (Elec Rm)	-
B100-A-137 Duct Flange Gasket (Elec Rm)	ND*
B100-A-138 Flexible Duct Joint (Elec Rm)	-
B100-A-139 Under Sink Coating (Elec Rm)	-
B100-A-140 Interior Concrete Wall (Mech)	ND*
B100-A-141 Interior Concrete Wall (Mech)	ND*
B100-A-142 Interior Concrete Wall (Mech)	Chrysotile (1%)
B100-A-143 Ceiling Slab (Mech)	ND*
B100-A-144 Ceiling Slab (Mech)	ND*
B100-A-145 Ceiling Slab (Mech)	ND*
B100-A-146 Neo Seal (Mech)	-
B100-A-147 Neo Seal (Mech)	-
B100-A-148 Neo Seal (Mech)	-
B100-A-149 Wallboard (Mech)	ND*
B100-A-150 Wallboard (Mech)	ND*

SACM Analyzed Material Description	Result
B100-A-151 Wallboard (Mech)	ND*
B100-A-152 Floor Slab (Mech)	Chrysotile (1%)
B100-A-153 Floor Slab (Mech)	Chrysotile (1%)
B100-A-154 Floor Slab (Mech)	ND*
B100-A-155 Wall Mortar Repair Seal (Mech)	ND*
B100-A-156 Floor Mortar Repair Seal (Mech)	Chrysotile (2%)
B100-A-157 Duct Insulation (Mech)	ND*
B100-A-158 Flexible Duct Joint (Mech)	ND*
B100-A-159 Concrete Pad (AHU, Mech)	ND*
B100-A-160 Concrete Pad (AHU, Mech)	ND*
B100-A-161 Concrete Pad (AHU, Mech)	ND*
B100-A-162 Concrete Pad (Compressor, Mech)	ND*
B100-A-163 Concrete Pad (Compressor, Mech)	ND*
B100-A-164 Concrete Pad (Compressor, Mech)	ND*
B100-A-165 Concrete Pad (Pump, Mech)	ND*
B100-A-166 Concrete Pad (Pump, Mech)	ND*
B100-A-167 Concrete Pad (Pump, Mech)	ND*
B100-A-168 Concrete Pad (Generator, Mech)	ND*
B100-A-169 Concrete Pad (Generator, Mech)	ND*
B100-A-170 Concrete Pad (Generator, Mech)	ND*
B100-A-171 Pipe Flange Gasket (Mech)	ND*
B100-A-172 Interior Concrete Wall (Rm 118)	ND*
B100-A-173 Interior Concrete Wall (Rm 118)	ND*
B100-A-174 Interior Concrete Wall (Rm 118)	ND*
B100-A-175 Ceiling Slab (Rm 118)	ND*
B100-A-176 Ceiling Slab (Rm 118)	ND*
B100-A-177 Ceiling Slab (Rm 118)	ND*
B100-A-178 Neo Seal (Elec Rm)	ND*
B100-A-179 Neo Seal (Elec Rm)	ND*
B100-A-180 Neo Seal (Elec Rm)	ND*
B100-A-181 Concrete Floor Slab (Rm 118)	ND*
B100-A-182 Concrete Floor Slab (Rm 118)	ND*
B100-A-183 Concrete Floor Slab (Rm 118)	ND*
B100-A-184 Wall Mortar Repair Seal (Rm 118)	ND*
B100-A-185 Floor Mortar Repair Seal (Rm 118)	ND*
B100-A-186 Duct Flange Gasket (Rm 118)	ND*
B100-A-187 Flexible Duct Joint (Rm 118)	-
B100-A-188 Concrete Pad (Rm 118)	ND*
B100-A-189 Concrete Pad (Rm 118)	ND*
B100-A-190 Concrete Pad (Rm 118)	ND*
B100-A-191 Wall Insulation (Rm 118)	ND*
B100-A-192 Wall Insulation (Rm 118)	ND*
B100-A-193 Wall Insulation (Rm 118)	ND*
B100-A-194 Pipe Insulation (Straight, Rm 118)	ND*
B100-A-195 Pipe Insulation (Elbow, Rm 118)	ND*
B100-A-196 Roofing System (Roof)	ND*
B100-A-197 Roofing System (Roof)	ND*
B100-A-198 Roofing System (Roof)	ND*
B100-A-199 Roofing System (Roof)	ND*
B100-A-200 Roofing System (Roof)	ND*

BLDG-100, ACM SAMPLED MATERIAL LIST

SACM Analyzed Material Description	Result
B100-A-201	Roofing System (Roof)
B100-A-202	Roofing System (Roof)
B100-A-203	Roof Concrete Surface (Roof)
B100-A-204	Roof Concrete Surface (Roof)
B100-A-205	Roof Concrete Surface (Roof)
B100-A-206	Roof Concrete Surface (Roof)
B100-A-207	Roof Concrete Surface (Roof)
B100-A-208	Roof Concrete Surface (Roof)
B100-A-209	Roof Concrete Surface (Roof)
B100-A-210	Vent Pipe Flashing Sheet (Roof)
B100-A-211	Vent Pipe Flashing Sheet (Roof)
B100-A-212	Vent Pipe Flashing Sheet (Roof)
B100-A-213	Coating Material of Ventilator (Roof)
B100-A-214	Coating Material of Ventilator (Roof)
B100-A-215	Coating Material of Ventilator (Roof)
B100-A-216	Ventilator Foundation Concrete (Roof)
B100-A-217	Ventilator Foundation Concrete (Roof)
B100-A-218	Ventilator Foundation Concrete (Roof)
B100-A-219	Siding Board (Exterior)
B100-A-220	Siding Board (Exterior)
B100-A-222	Flange Gasket (Exterior)
B100-A-223	Flange Gasket (Exterior)
B100-A-226	Waterproofing (Exterior)
B100-A-227	Concrete Wall (Exterior)
B100-A-300	Floor Caulking (Rm 118)
B100-A-301	Wall Caulking (Rm 118)
B100-A-302	Concrete Wall (Rm 119)
B100-A-303	Concrete Wall (Rm 119)
B100-A-304	Concrete Wall (Rm 119)
B100-A-305	Ceiling Slab (Rm 119)
B100-A-306	Ceiling Slab (Rm 119)
B100-A-307	Ceiling Slab (Rm 119)
B100-A-308	Wallboard (Rm 119)
B100-A-309	Wallboard (Rm 119)
B100-A-310	Wallboard (Rm 119)
B100-A-311	Floor Slab (Rm 119)
B100-A-312	Floor Slab (Rm 119)
B100-A-313	Floor Slab (Rm 119)
B100-A-314	Wall Caulking (Rm 119)
B100-A-315	Floor Caulking (Rm 119)
B100-A-316	Wallboard (Rm 118)
B100-A-317	Wallboard (Rm 118)
B100-A-318	Wallboard (Rm 118)
B100-A-319	Duct Flange Gasket (Rm 118)
B100-A-400	Ceiling Insulation (Kitchen)
B100-A-401	Ceiling Insulation (Kitchen)
B100-A-402	Ceiling Insulation (Kitchen)
B100-A-403	Ceiling Board (Kitchen)
B100-A-404	Ceiling Board (Kitchen)
B100-A-405	Ceiling Board (Kitchen)

SACM Analyzed Material Description	Result
B100-A-406	Wallboard (Kitchen)
B100-A-407	Wallboard (Kitchen)
B100-A-408	Wallboard (Kitchen)
B100-A-409	Vinyl Cove Base w/Mastic (Kitchen)
B100-A-410	Vinyl Cove Base w/Mastic (Kitchen)
B100-A-411	Vinyl Cove Base w/Mastic (Kitchen)
B100-A-412	Wall Tile w/Grout (Kitchen)
B100-A-415	Floor Tile w/Mastic (Kitchen)
B100-A-416	Floor Tile w/Mastic (Kitchen)
B100-A-417	Floor Tile w/Mastic (Kitchen)
B100-A-418	Floor Slab (Kitchen)
B100-A-419	Floor Slab (Kitchen)
B100-A-420	Floor Slab (Kitchen)
B100-A-421	Caulking (Gray) (Kitchen)
B100-A-422	Caulking (White) (Kitchen)
B100-A-425	Ceiling Board (Toilet)
B100-A-426	Ceiling Board (Toilet)
B100-A-427	Ceiling Board (Toilet)
B100-A-428	Wallboard (Toilet)
B100-A-429	Wallboard (Toilet)
B100-A-430	Wallboard (Toilet)
B100-A-431	Wall Ceramic Tile w/Grout (Toilet)
B100-A-432	Wall Ceramic Tile w/Grout (Toilet)
B100-A-433	Wall Ceramic Tile w/Grout (Toilet)
B100-A-434	Floor Tile w/Grout (Toilet)
B100-A-435	Floor Tile w/Grout (Toilet)
B100-A-436	Floor Tile w/Grout (Toilet)
B100-A-437	Floor Slab (Toilet)
B100-A-438	Floor Slab (Toilet)
B100-A-439	Floor Slab (Toilet)
B100-A-440	Ceiling Insulation (Kitchen)
B100-A-441	Pipe Insulation, Straight (Kitchen)
B100-A-442	Pipe Insulation, Elbow (Kitchen)
B100-A-500	Ceiling Board (Server Rm)
B100-A-501	Ceiling Board (Server Rm)
B100-A-502	Ceiling Board (Server Rm)
B100-A-503	Ceiling Board (Server Rm)
B100-A-504	Ceiling Board (Server Rm)
B100-A-505	Ceiling Slab (Server Rm)
B100-A-506	Ceiling Slab (Server Rm)
B100-A-507	Ceiling Slab (Server Rm)
B100-A-508	Ceiling Slab (Server Rm)
B100-A-510	Wall Board (Server Rm)
B100-A-511	Wall Board (Server Rm)
B100-A-512	Wall Board (Server Rm)
B100-A-513	Wall Board (Server Rm)
B100-A-514	Wall Board (Server Rm)
B100-A-515	Vinyl Cove Base w/Mastic (Server Rm)
B100-A-516	Vinyl Cove Base w/Mastic (Server Rm)
B100-A-517	Vinyl Cove Base w/Mastic (Server Rm)

SACM Analyzed Material Description	Result
B100-A-518	Vinyl Cove Base w/Mastic (Server Rm)
B100-A-519	Vinyl Cove Base w/Mastic (Server Rm)
B100-A-520	Floor Slab (Server Rm)
B100-A-521	Floor Slab (Server Rm)
B100-A-522	Floor Slab (Server Rm)
B100-A-523	Floor Slab (Server Rm)
B100-A-524	Floor Slab (Server Rm)
B100-A-530	Acoustic Board (Server Rm)
B100-A-531	Acoustic Board (Server Rm)
B100-A-532	Acoustic Board (Server Rm)
B100-A-533	Wainscot Board (Server Rm)
B100-A-534	Wainscot Board (Server Rm)
B100-A-535	Wainscot Board (Server Rm)
B100-A-539	Floor Board (Server Rm)
B100-A-540	Floor Board (Server Rm)
B100-A-541	Floor Board (Server Rm)
B100-A-543	Wall Tile w/Grout (Kitchen)
B100-A-544	Ceiling Board (Janitor)
B100-A-545	Wallboard (Janitor)
B100-A-546	Floor Slab (Janitor)
B100-A-547	Dampproofing Material (Rm 110)
B100-A-548	Wall Insulation (Rm 110)

*ND=None Detected (Less than 0.1% based on the provision of JEGS)

BLDG-100, ACM SAMPLED MATERIAL LIST

SLBP Analyzed Material Description	Result
B100-L-01 Exterior Wall Paint (Ivory) (Exterior)	< 40 mg/kg
B100-L-02 Exterior Wall Trim Paint (Brown) (Exterior)	< 40 mg/kg
B100-L-03 Pipe Paint (Ivory) (Exterior)	< 40 mg/kg
B100-L-04 Pipe Paint (Red) (Exterior)	13000 mg/kg*
B100-L-05 Pipe Paint (Blue) (Exterior)	< 40 mg/kg
B100-L-06 Pipe Paint (Silver) (Exterior)	< 48 mg/kg
B100-L-07 Conduit Paint (Ivory) (Exterior)	230 mg/kg
B100-L-08 Elect Box Paint (Gray) (Exterior)	< 61 mg/kg
B100-L-09 Steel Shutter Door/Frame Paint (Brown) (Exterior)	1900 mg/kg
B100-L-10 Support Angle Paint (Ivory) (Exterior)	1800 mg/kg
B100-L-11 Bollard Paint (Ivory) (Exterior)	150 mg/kg
B100-L-12 Condenser Paint (Gray) (Exterior)	< 40 mg/kg
B100-L-13 Condenser Paint (Light Gray) (Exterior)	2000 mg/kg
B100-L-14 Safety Bumper Pole Paint (Yellow) (Exterior)	98 mg/kg
B100-L-15 Chiller Paint (Gray) (Exterior)	< 100 mg/kg
B100-L-16 Conduit Paint (Gray) (Exterior)	5100 mg/kg*
B100-L-17 Interior Wall Paint (Ivory) (Rm 101)	< 48 mg/kg
B100-L-18 Conduit Paint (Ivory) (Rm 101)	1200 mg/kg
B100-L-19 Elec Box Paint (Red) (Rm 101)	< 100 mg/kg
B100-L-20 Steel Door/Frame Paint (Tan) (Rm 102)	15000 mg/kg*
B100-L-21 Wood Door/Frame Paint (Ivory) (Rm 102)	12000 mg/kg*
B100-L-22 Interior Wall Paint (Ivory) (Rm 102)	< 40 mg/kg
B100-L-23 Wood Door/Frame Paint (Ivory) (Rm 105)	22000 mg/kg*
B100-L-24 Interior Wall Paint (Ivory) (Rm 105)	< 49 mg/kg
B100-L-25 Ceiling Paint (Ivory) (Toilet)	< 78 mg/kg
B100-L-26 Interior Wall Paint (White) (Elect Rm)	< 40 mg/kg
B100-L-27 Interior Wall Paint (Ivory) (Elect Rm)	< 40 mg/kg
B100-L-28 Floor Paint (Gray) (Elect Rm)	52 mg/kg
B100-L-29 Conduit Paint (Gray) (Elect Rm)	7800 mg/kg*
B100-L-30 Conduit Paint (Dark Gray) (Elect Rm)	2400 mg/kg
B100-L-31 Pipe Paint (Red) (Elect Rm)	25000 mg/kg*
B100-L-32 Pipe Paint (Ivory) (Elect Rm)	210 mg/kg
B100-L-33 Elect Box Paint (Gray) (Elect Rm)	77 mg/kg
B100-L-34 Elect Box Paint (Light Gray) (Elect Rm)	1300 mg/kg
B100-L-35 Duct Paint (Ivory) (Elect Rm)	2200 mg/kg
B100-L-36 Duct Paint (Blue) (Elect Rm)	< 49 mg/kg
B100-L-37 Duct Paint (Gray) (Elect Rm)	11000 mg/kg*
B100-L-38 Support Angle Paint (Gray) (Elect Rm)	2400 mg/kg
B100-L-39 Interior Wall Paint (Ivory) (Mech)	< 40 mg/kg
B100-L-40 Conduit Paint (Gray) (Mech)	5700 mg/kg*
B100-L-41 Pipe Paint (Red) (Mech)	2800 mg/kg
B100-L-42 Pipe Paint (Gray) (Mech)	120 mg/kg
B100-L-43 Elect Box Paint (Ivory) (Mech)	< 64 mg/kg
B100-L-44 Elect Box Paint (Gray) (Mech)	< 53 mg/kg
B100-L-45 Floor Paint (Gray) (Mech)	280 mg/kg
B100-L-46 Pipe Support Angle Paint (Gray) (Mech)	4100 mg/kg
B100-L-47 Pump Paint (Green) (Mech)	< 65 mg/kg
B100-L-48 Generator Paint (Gray) (Mech)	< 53 mg/kg
B100-L-49 Boiler Paint (Gray) (Mech)	< 78 mg/kg
B100-L-50 Pipe Paint (Silver) (Mech)	< 160 mg/kg

SLBP Analyzed Material Description	Result
B100-L-51 Conduit Paint (Ivory) (Rm 118)	3800 mg/kg
B100-L-52 Pipe Paint (Red) (Rm 118)	< 46 mg/kg
B100-L-53 Pipe Paint (White) (Rm 118)	33000 mg/kg*
B100-L-54 Elect Box Paint (Tan) (Rm 118)	< 110 mg/kg
B100-L-55 Support Angle Paint (Ivory) (Rm118)	730 mg/kg
B100-L-100 Elect Box Paint (Gray) (Rm 119)	< 85 mg/kg
B100-L-101 Wall Paint (Ivory) (Rm119)	< 40 mg/kg
B100-L-102 Elect Box Paint (Ivory) (Rm119)	< 40 mg/kg
B100-L-103 Wall Paint (Ivory) (Rm118)	< 68 mg/kg
B100-L-400 Elect Box Paint (Gray) (Interior)(Rm 100)	< 140 mg/kg
B100-L-401 Wall Paint (Ivory) (Interior)	< 40 mg/kg
B100-L-402 Slide Door Paint (Ivory) (Interior)	< 40 mg/kg
B100-L-403 Conduit Paint (Ivory) (Interior)	< 47 mg/kg
B100-L-500 Elect Box Paint (Gray) (Rm108)	10000 mg/kg*
B100-L-501 Elect Box Paint (Tan) (Rm 108)	< 64 mg/kg
B100-L-502 Wall Paint (Ivory) (Rm108)	45 mg/kg
B100-L-503 Frame Paint (Gray) (Rm108)	7900 mg/kg*
B100-L-504 Pipe Paint (Red) (Rm 108)	610 mg/kg
B100-L-505 Conduit Paint (Ivory) (Rm108)	3800 mg/kg
B100-L-506 Door/ Frame Paint (Orange) (Conference Room)	65 mg/kg
B100-L-507 Wall Paint (Orange) (Conference Room)	< 40 mg/kg
B100-L-600 Steel Door/ Frame Paint (Ivory) (Janitor)	8000 mg/kg*

* The definition of LBP with EPA is that it is greater than or equal to 0.5% (5,000 mg/kg) by weight.

BLDG-100, LBP SAMPLED MATERIAL LIST

25SF0004



ASBESTOS

Analytical Summary, Photographic Documentation, and
Laboratory Analytical Reports

(Bldg. 100)

DRAFT

Asbestos Bulk Sampling Results Summary - Bldg. 100, Buckner Comm Site

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-01	202236023	Exterior	Exterior Concrete Wall Surface	Ivory paint / gray concrete	ND*
B100-A-02	202236024	Exterior	Exterior Concrete Wall Surface	Ivory paint / gray concrete	ND*
B100-A-03	202236025	Exterior	Exterior Concrete Wall Surface	1. Gray concrete 2. Ivory textured paint	ND*
B100-A-04	202236026	Exterior	Exterior Concrete Wall Surface	1. Gray concrete 2. Ivory textured paint	ND*
B100-A-05	202236027	Exterior	Exterior Concrete Wall Surface	1. Gray concrete 2. Ivory textured paint	ND*
B100-A-06	202236028	Exterior	Exterior Concrete Wall Surface	Gray concrete	ND*
B100-A-07	202236029	Exterior	Exterior Concrete Wall Surface	Gray concrete	ND*
B100-A-08	202236030	Exterior	Neo Seal (White) · Elect Box	White neo seal	ND*
B100-A-10	202236031	Exterior	Caulking Material	White caulk / paint	ND*
B100-A-11	202236032	Exterior	Concrete Floor Slab	Gray concrete	ND*
B100-A-12	202236033	Exterior	Concrete Slab	Gray concrete	ND*
B100-A-13	202236034	Exterior	Concrete Slab	Gray concrete	ND*
B100-A-14	202236035	Exterior	Floor Mortar Repair Seal	Ivory paint / gray concrete	ND*
B100-A-15	202236036	Exterior	Expansion Joint	1. Off-white expansion joint / paint 2. Orange adhesive	ND*
B100-A-16	202236037	Exterior	Concrete Pad · Condenser	Gray concrete	ND*
B100-A-17	202236038	Exterior	Concrete Pad · Condenser	Gray concrete	ND*
B100-A-18	202236039	Exterior	Concrete Pad · Condenser	Gray concrete	ND*
B100-A-21	202236040	Exterior	Pipe Flange Gasket	Blue gasket / red paint	ND*
B100-A-22	202236041	Exterior	Concrete Pad · AC	Gray concrete	ND*
B100-A-23	202236042	Exterior	Concrete Pad · AC	Gray concrete	ND*
B100-A-24	202236043	Exterior	Concrete Pad · AC	Gray concrete	ND*
B100-A-25	202236044	Exterior	Pipe Insulation · Straight	White pipe insulation / paint	ND*
B100-A-27	202236045	Front Entrance, Exterior	Ceramic Floor Tile w/Grout	1. Brown ceramic tile 2. Gray grout 3. Gray mortar	ND*
B100-A-28	202236046	Front Entrance, Exterior	Ceramic Floor Tile w/Grout	1. Brown ceramic tile 2. Gray grout 3. Gray mortar	ND*
B100-A-29	202236047	Front Entrance, Exterior	Ceramic Floor Tile w/Grout	1. Brown ceramic tile 2. Gray grout 3. Gray mortar	ND*
B100-A-30	202236048	Front Entrance, Exterior	Mortar · Under Tile	Brown / gray mortar	ND*
B100-A-31	202236049	Front Entrance, Exterior	Mortar · Under Tile	Brown / gray mortar	ND*
B100-A-32	202236050	Front Entrance, Exterior	Mortar · Under Tile	Gray mortar	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-33	202236051	Front Entrance, Exterior	Concrete Floor Slab	Gray concrete	ND*
B100-A-34	202236052	Front Entrance, Exterior	Concrete Floor Slab	Gray concrete	ND*
B100-A-35	202236053	Front Entrance, Exterior	Concrete Floor Slab	Gray concrete	ND*
B100-A-36	202236054	Exterior	Expansion Joint · Wall	White expansion joint / paint	ND*
B100-A-37	202236055	Rm 101	Interior Concrete Wall	Ivory paint / gray concrete	ND*
B100-A-37	202236055	Rm 101	Interior Concrete Wall	Ivory paint / gray concrete	ND*
B100-A-38	202236056	Rm 101	Interior Concrete Wall	Ivory paint / gray concrete	ND*
B100-A-39	202236057	Rm 101	Ceiling Board	White ceiling board	ND*
B100-A-40	202236058	Rm 101	Ceiling Board	White ceiling board	ND*
B100-A-41	202236059	Rm 101	Ceiling Board	White ceiling board	ND*
B100-A-42	202236060	Rm 101	Ceiling Board	White ceiling board	ND*
B100-A-46	202236061	Toilet	Ceiling Slab	Ivory paint / gray concrete	ND*
B100-A-47	202236062	Toilet	Ceiling Slab	Ivory paint / gray concrete	ND*
B100-A-48	202236063	Toilet	Ceiling Slab	Ivory paint / gray concrete	ND*
B100-A-49	202236064	Rm 101	Vinyl Cove Base w/Mastic	1. Beige vinyl cove base 2. Gray cementitious material 3. Yellow mastic (limited)	ND*
B100-A-50	202236065	Rm 101	Vinyl Cove Base w/Mastic	1. Beige mastic / white paint 2. Beige vinyl cove base 3. Gray cementitious material	1. Chrysotile (2%) 2. ND* 3. ND*
B100-A-51	202236066	Rm 101	Vinyl Cove Base w/Mastic	1. Beige mastic (limited) / white paint 2. Beige vinyl cove base 3. Gray cementitious material	1. Chrysotile (2%) 2. ND* 3. ND*
B100-A-52	202236067	Rm 101	Wallboard	White paint / gypsum board	ND*
B100-A-53	202236068	Rm 101	Wallboard	White paint / gypsum board	ND*
B100-A-54	202236069	Rm 101	Wallboard	White paint / gypsum board	ND*
B100-A-55	202236070	Rm 101	Vinyl Floor Tile w/Mastic	1. Beige mastic 2. Beige vinyl floor tile 3. Gray cementitious material	ND*
B100-A-56	202236071	Rm 101	Vinyl Floor Tile w/Mastic	1. Beige mastic 2. Beige vinyl floor tile	ND*
B100-A-57	202236072	Rm 101	Vinyl Floor Tile w/Mastic	1. Beige mastic / gray cementitious material 2. Beige vinyl floor tile	ND*
B100-A-58	202236073	Rm 101	Concrete Floor Slab	Gray concrete	ND*
B100-A-59	202236074	Rm 101	Concrete Floor Slab	Gray concrete	ND*
B100-A-60	202236075	Rm 101	Concrete Floor Slab	Gray concrete	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-62	202236076	Rm 101	Duct Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-63	202236077	Rm 101	Duct Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-64	202236078	Rm 101	Duct Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-65	202236079	Rm 102	Floor Carpet w/Mastic	1. Gray/black carpet 2. Yellow mastic 3. Yellow/gray carpet pad	ND*
B100-A-66	202236080	Rm 102	Floor Carpet w/Mastic	1. Gray/black carpet 2. Yellow mastic 3. Yellow/gray carpet pad	ND*
B100-A-67	202236081	Rm 102	Floor Carpet w/Mastic	1. Gray/black carpet 2. Yellow mastic 3. Yellow/gray carpet pad	ND*
B100-A-71	202236082	Rm 105	Ceiling Board	1. White gypsum board 2. White joint compound / white paint	ND*
B100-A-72	202236083	Rm 105	Ceiling Board	White gypsum board / white paint	ND*
B100-A-73	202236084	Rm 105	Ceiling Board	White/gray ceiling tile	ND*
B100-A-75	202236085	Rm 105	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-78	202236086	Rm 105	Ceiling Slab	Gray concrete	ND*
B100-A-80	202236087	Rm 105	Vinyl Cove Base w/Mastic	1. Beige mastic 2. Beige vinyl covebase	1. Chrysotile (2%) 2. ND*
B100-A-81	202236088	Rm 105	Vinyl Cove Base w/Mastic	1. Beige mastic 2. Beige vinyl covebase	1. Chrysotile (2%) 2. ND*
B100-A-82	202236089	Rm 105	Vinyl Cove Base w/Mastic	1. Beige mastic 2. Beige vinyl covebase	1. Chrysotile (2%) 2. ND*
B100-A-83	202236090	Rm 105	Wallboard	1. White gypsum board 2. White joint compound / white paint / mesh	ND*
B100-A-84	202236091	Rm 105	Wallboard	White gypsum board / white paint	ND*
B100-A-85	202236092	Rm 105	Wallboard	1. White gypsum board (1) / white paint 2. White gypsum board (2)	ND*
B100-A-86	202236093	Rm 105	Wall Insulation	Yellow insulation	ND*
B100-A-87	202236094	Rm 105	Wall Insulation	Yellow insulation	ND*
B100-A-88	202236095	Rm 105	Wall Insulation	Yellow insulation	ND*
B100-A-89	202236096	Rm 105	Dampproofing Material	Black dampproofing material	ND*
B100-A-90	202236097	Rm 105	Vinyl Floor Tile w/Mastic	1. Tan vinyl floor tile 2. Yellow mastic	ND*
B100-A-91	202236098	Rm 105	Vinyl Floor Tile w/Mastic	1. Tan vinyl floor tile 2. Yellow mastic	ND*
B100-A-92	202236099	Rm 105	Vinyl Floor Tile w/Mastic	1. Tan vinyl floor tile 2. Yellow mastic	ND*
B100-A-93	202236100	Rm 105	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-94	202236101	Rm 105	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-95	202236102	Rm 105	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-96	202236103	Toilet	Ceramic Wall Tile w/Grout	1. Gray mortar 2. White ceramic tile 3. White grout	ND*
B100-A-97	202236104	Toilet	Ceramic Wall Tile w/Grout	1. Gray mortar 2. White ceramic tile 3. White grout	ND*
B100-A-98	202236105	Toilet	Ceramic Wall Tile w/Grout	1. Gray mortar 2. White ceramic tile 3. White grout	ND*
B100-A-99	202236106	Toilet	Ceramic Floor Tile w/Grout	1. Beige ceramic tile 2. Gray grout 3. Gray mortar	ND*
B100-A-100	202236107	Toilet	Ceramic Floor Tile w/Grout	1. Beige ceramic tile 2. Gray cementitious material 3. Gray grout 4. Gray mortar	ND*
B100-A-101	202236108	Toilet	Ceramic Floor Tile w/Grout	1. Beige ceramic tile 2. Gray cementitious material 3. Gray grout 4. Gray mortar	ND*
B100-A-105	202236109	Toilet	Concrete Floor Slab	1. Gray cementitious material 2. Gray mortar	ND*
B100-A-106	202236110	Toilet	Concrete Floor Slab	1. Gray cementitious material 2. Gray mortar	ND*
B100-A-107	202236111	Toilet	Concrete Floor Slab	1. Gray cementitious material 2. Gray mortar	ND*
B100-A-108	202236112	Toilet	Ceiling Board	White ceiling board / off-white paint	ND*
B100-A-109	202236113	Toilet	Ceiling Board	White ceiling board / off-white paint	ND*
B100-A-110	202236114	Toilet	Ceiling Board	White ceiling board / off-white paint	ND*
B100-A-111	202236115	Toilet	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-112	202236116	Toilet	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-113	202236117	Toilet	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-114	202236118	Toilet	Concrete Slab	Gray cementitious material	ND*
B100-A-115	202236119	Toilet	Concrete Slab	Gray cementitious material	ND*
B100-A-116	202236120	Toilet	Concrete Slab	Gray cementitious material	ND*
B100-A-117	202236121	Rm 110	Floor Mortar Repair Seal	Gray cementitious material	ND*
B100-A-118	202236122	Rm 110	Floor Mortar Repair Seal	Gray cementitious material	ND*
B100-A-119	202236123	Rm 110	Floor Mortar Repair Seal	Gray cementitious material	ND*
B100-A-120	202236124	Elec Room	Interior Concrete Wall	1. Gray cementitious material 2. Gray plaster / white paint	1. ND* 2. Chrysotile (3%)
B100-A-121	202236125	Elec Room	Interior Concrete Wall	Gray cementitious material / white paint	ND*
B100-A-122	202236126	Elec Room	Interior Concrete Wall	Gray plaster/cementitious material / white paint	Chrysotile (1%)
B100-A-123	202236127	Elec Room	Ceiling Slab	1. Gray cementitious material 2. Gray plaster material	ND*
B100-A-124	202236128	Elec Room	Ceiling Slab	Gray cementitious material / white paint	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-126	202236129	Elec Room	Neo Seal (Black)	Gray neo seal	ND*
B100-A-127	202236130	Elec Room	Neo Seal (Black)	Gray neo seal	ND*
B100-A-128	202236131	Elec Room	Neo Seal (Black)	Gray neo seal	ND*
B100-A-129	202236132	Elec Room	Wallboard	White drywall / off-white paint	ND*
B100-A-130	202236133	Elec Room	Wallboard	White drywall / off-white paint	ND*
B100-A-131	202236134	Elec Room	Wallboard	White drywall / off-white paint	ND*
B100-A-132	202236135	Elec Room	Floor Slab	Gray cementitious material	ND*
B100-A-133	202236136	Elec Room	Floor Slab	Gray cementitious material	ND*
B100-A-134	202236137	Elec Room	Floor Slab	Gray cementitious material / paint	ND*
B100-A-137	202236138	Elec Room	Duct Flange Gasket	Black gasket / white paint	ND*
B100-A-140	202236139	Mech Room	Interior Concrete Wall	1. Gray cementitious material (1) / off-white paint 2. Gray cementitious material (2)	ND*
B100-A-141	202236140	Mech Room	Interior Concrete Wall	Gray cementitious material / off-white paint	ND*
B100-A-142	202236141	Mech Room	Interior Concrete Wall	1. Gray cementitious material (1) 2. Gray cementitious material (2) 3. White joint compound (limited) / off-white paint	1. ND* 2. ND* 3. Chrysotile (1%)
B100-A-143	202236142	Mech Room	Ceiling Slab	Gray cementitious material	ND*
B100-A-144	202236143	Mech Room	Ceiling Slab	1. Gray cementitious material (1) 2. Gray cementitious material (2)	ND*
B100-A-145	202236144	Mech Room	Ceiling Slab	Gray cementitious material	ND*
B100-A-149	202236145	Mech Room	Wallboard	White drywall / off-white paint	ND*
B100-A-150	202236146	Mech Room	Wallboard	1. Off-white joint compound (limited) / paint 2. White drywall	ND*
B100-A-151	202236147	Mech Room	Wallboard	1. Off-white joint compound (limited) / paint 2. White drywall	ND*
B100-A-152	202236148	Mech Room	Floor Slab	Gray cementitious material	Chrysotile (1%)
B100-A-153	202236149	Mech Room	Floor Slab	1. Gray cementitious material (1) / gray paint 2. Gray cementitious material (2)	1. Chrysotile (1%) 2. ND*
B100-A-154	202236150	Mech Room	Floor Slab	Gray cementitious material / gray paint	ND*
B100-A-155	202236151	Mech Room	Wall Mortar Repair Seal	Gray cementitious material / white paint	ND*
B100-A-156	202236152	Mech Room	Floor Mortar Repair Seal	1. Gray cementitious material (1) / gray paint 2. Gray cementitious material (2)	1. Chrysotile (2%) 2. ND*
B100-A-157	202236153	Mech Room	Duct Insulation	1. Silver foil wrap 2. Yellow insulation	ND*
B100-A-158	202236154	Interior	Flexible Duct Joint	Black flex joint	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-159	202236155	Mech Room	Concrete Pad · AHU	Gray cementitious material	ND*
B100-A-160	202236156	Mech Room	Concrete Pad · AHU	Gray cementitious material	ND*
B100-A-161	202236157	Mech Room	Concrete Pad · AHU	1. Gray cementitious material 2. Off-white cementitious material	ND*
B100-A-162	202236158	Mech Room	Concrete Pad · Compressor	Gray cementitious material	ND*
B100-A-163	202236159	Mech Room	Concrete Pad · Compressor	Gray cementitious material	ND*
B100-A-164	202236160	Mech Room	Concrete Pad · Compressor	Gray cementitious material	ND*
B100-A-165	202236161	Mech Room	Concrete Pad · Pump	Gray cementitious material	ND*
B100-A-166	202236162	Mech Room	Concrete Pad · Pump	Gray cementitious material	ND*
B100-A-167	202236163	Mech Room	Concrete Pad · Pump	Gray cementitious material	ND*
B100-A-168	202236164	Mech Room	Concrete Pad · Generator	Gray cementitious material	ND*
B100-A-169	202236165	Mech Room	Concrete Pad · Generator	Gray cementitious material	ND*
B100-A-170	202236166	Mech Room	Concrete Pad · Generator	Gray cementitious material	ND*
B100-A-171	202236167	Mech Room	Pipe Flange Gasket	Black gasket	ND*
B100-A-172	202236168	Rm 118	Interior Concrete Wall	1. Gray cementitious material (1) 2. Gray cementitious material (2) 3. Gray cementitious material (3)	ND*
B100-A-173	202236169	Rm 118	Interior Concrete Wall	1. Gray cementitious material (1) 2. Gray cementitious material (2) 3. Gray cementitious material (3)	ND*
B100-A-174	202236170	Rm 118	Interior Concrete Wall	1. Gray cementitious material (1) 2. Gray cementitious material (2)	ND*
B100-A-175	202236171	Rm 118	Ceiling Slab	Gray cementitious material	ND*
B100-A-176	202236172	Rm 118	Ceiling Slab	Gray cementitious material	ND*
B100-A-177	202236173	Rm 118	Ceiling Slab	Gray cementitious material	ND*
B100-A-178	202236174	Elec Room	Neo Seal (Black)	Black neo seal	ND*
B100-A-179	202236175	Elec Room	Neo Seal (Black)	Black neo seal	ND*
B100-A-180	202236176	Elec Room	Neo Seal (White)	White neo seal	ND*
B100-A-181	202236177	Rm 118	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-182	202236178	Rm 118	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-183	202236179	Rm 118	Concrete Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-184	202236180	Rm 118	Wall Mortar Repair Seal	Gray concrete / yellow adhesive	ND*
B100-A-185	202236181	Rm 118	Floor Mortar Repair Seal	Gray concrete / yellow adhesive	ND*
B100-A-186	202236182	Rm 118	Duct Flange Gasket	Black gasket	ND*
B100-A-188	202236183	Rm 118	Concrete Pad	Gray concrete	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-189	202236184	Rm 118	Concrete Pad	Gray concrete	ND*
B100-A-190	202236185	Rm 118	Concrete Pad	Gray concrete	ND*
B100-A-191	202236186	Rm 118	Wall Insulation	Yellow insulation	ND*
B100-A-192	202236187	Rm 118	Wall Insulation	Yellow insulation	ND*
B100-A-193	202236188	Rm 118	Wall Insulation	Yellow insulation	ND*
B100-A-194	202236189	Rm 118	Pipe Insulation · Straight	1. Silver wrap 2. Yellow insulation	ND*
B100-A-195	202236190	Rm 118	Pipe Insulation · Elbow	1. Silver wrap 2. Yellow insulation	ND*
B100-A-196	202236191	Roof	Roofing System	1. Off-white patch / white paint 2. Yellow mastic	ND*
B100-A-197	202236192	Roof	Roofing System	1. Off-white patch / white paint 2. Yellow mastic / gray cementitious material	ND*
B100-A-198	202236193	Roof	Roofing System	1. Off-white patch / white paint 2. Yellow mastic / gray cementitious material	ND*
B100-A-199	202236194	Roof	Roofing System	1. Black felt / tar (1) 2. Black felt / tar (2) 3. Black felt / tar (3) 4. Black felt / tar (4) 5. Black roofing tar (1) 6. Black roofing tar (2) 7. Black shingle (1) 8. Black shingle (2) 9. Tan acoustic tile 10. Yellow foam insulation	ND*
B100-A-200	202236195	Roof	Roofing System	1. Black felt / tar (1) 2. Black felt / tar (2) 3. Black felt / tar (3) 4. Black felt / tar (4) 5. Black roofing tar (1) 6. Black roofing tar (2) 7. Black shingle (1) 8. Black shingle (2) 9. Tan acoustic tile 10. Yellow foam insulation	ND*
B100-A-201	202236196	Roof	Roofing System	1. Black felt / tar (1) 2. Black felt / tar (2) 3. Black felt / tar (3) 4. Black felt / tar (4) 5. Black roofing tar (1) 6. Black roofing tar (2) 7. Black shingle (1) 8. Black shingle (2) 9. Tan acoustic tile 10. Yellow foam insulation	ND*
B100-A-202	202236197	Roof	Roofing System	1. Black felt / tar (1) 2. Black felt / tar (2) 3. Black felt / tar (3) 4. Black felt / tar (4) 5. Black roofing tar (1) 6. Black roofing tar (2) 7. Black shingle (1) 8. Black shingle (2) 9. Tan acoustic tile 10. Yellow foam insulation	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-203	202236198	Roof	Roof Concrete Surface	1. Gray cementitious material 2. Yellow mastic	ND*
B100-A-204	202236199	Roof	Roof Concrete Surface	1. Gray cementitious material 2. Yellow mastic	ND*
B100-A-205	202236200	Roof	Roof Concrete Surface	1. Gray cementitious material 2. Yellow mastic	ND*
B100-A-206	202236201	Roof	Roof Concrete Surface	1. Black tar 2. Gray cementitious material	ND*
B100-A-207	202236202	Roof	Roof Concrete Surface	1. Black tar 2. Gray cementitious material	ND*
B100-A-208	202236203	Roof	Roof Concrete Surface	1. Black tar 2. Gray cementitious material	ND*
B100-A-209	202236204	Roof	Roof Concrete Surface	Gray cementitious material	ND*
B100-A-213	202236205	Roof	Coating Material Of Ventilator	1. Off-white patch / white paint 2. Yellow mastic / gray cementitious material	ND*
B100-A-214	202236206	Roof	Coating Material Of Ventilator	Off-white patch / white paint	ND*
B100-A-215	202236207	Roof	Coating Material Of Ventilator	1. Off-white patch / white paint 2. Yellow mastic / gray cementitious material	ND*
B100-A-216	202236208	Roof	Ventilator Foundation Concrete	Gray concrete	ND*
B100-A-217	202236209	Roof	Ventilator Foundation Concrete	1. Gray concrete 2. Yellow mastic	ND*
B100-A-218	202236210	Roof	Ventilator Foundation Concrete	1. Gray concrete 2. Yellow mastic	ND*
B100-A-219	202236211	Exterior	Siding Board (Brown)	1. Brown painted siding board 2. White foam insulation	ND*
B100-A-220	202236212	Exterior	Siding Board (Ivory)	1. Ivory painted siding board 2. White foam insulation	ND*
B100-A-222	202236213	Exterior	Flange Gasket	Beige gasket	ND*
B100-A-223	202236214	Exterior	Flange Gasket	Black gasket / red paint	ND*
B100-A-226	202236215	Exterior	Waterproofing	Black waterproofing	ND*
B100-A-227	202236216	Exterior	Concrete Wall	1. Gray cementitious material 2. Ivory textured paint	1. Chrysotile (5%) 2. ND*
B100-A-300	202236217	Rm 118	Floor Caulking	Gray caulk	ND*
B100-A-301	202236218	Rm 118	Wall Caulking	Gray caulk	ND*
B100-A-302	202236219	Rm 119	Concrete Wall	Gray concrete	ND*
B100-A-303	202236220	Rm 119	Concrete Wall	Gray concrete	ND*
B100-A-304	202236221	Rm 119	Concrete Wall	Gray concrete	ND*
B100-A-305	202236222	Rm 119	Ceiling Slab	Gray concrete	ND*
B100-A-306	202236223	Rm 119	Ceiling Slab	Gray concrete	ND*
B100-A-307	202236224	Rm 119	Ceiling Slab	Gray concrete	ND*
B100-A-308	202236225	Rm 119	Wallboard	White gypsum board / white paint	ND*
B100-A-309	202236226	Rm 119	Wallboard	White gypsum board / white paint	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-310	202236227	Rm 119	Wallboard	White gypsum board / white paint	ND*
B100-A-311	202236228	Rm 119	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-312	202236229	Rm 119	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-313	202236230	Rm 119	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-314	202236231	Rm 119	Wall Caulking	1. Gray caulk 2. White foam	ND*
B100-A-315	202236232	Rm 119	Floor Caulking	Gray caulk	ND*
B100-A-316	202236233	Rm 118	Wallboard	White gypsum board / white paint	ND*
B100-A-317	202236234	Rm 118	Wallboard	White gypsum board / white paint	ND*
B100-A-318	202236235	Rm 118	Wallboard	White gypsum board / white paint	ND*
B100-A-319	202236236	Rm 118	Duct Flange Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-400	202236237	Kitchen	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-401	202236238	Kitchen	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-402	202236239	Kitchen	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-403	202236240	Kitchen	Ceiling Board	White/gray ceiling board	ND*
B100-A-404	202236241	Kitchen	Ceiling Board	White/gray ceiling board	ND*
B100-A-405	202236242	Kitchen	Ceiling Board	White/gray ceiling board	ND*
B100-A-406	202236243	Kitchen	Wallboard	White gypsum board / white paint	ND*
B100-A-407	202236244	Kitchen	Wallboard	White gypsum board	ND*
B100-A-408	202236245	Kitchen	Wallboard	1. Off-white adhesive 2. White gypsum board	ND*
B100-A-409	202236246	Kitchen	Vinly Cove Base w/Mastic	1. Beige vinyl covebase 2. Off-white mastic	ND*
B100-A-410	202236247	Kitchen	Vinly Cove Base w/Mastic	1. Beige vinyl covebase 2. Off-white mastic	ND*
B100-A-411	202236248	Kitchen	Vinly Cove Base w/Mastic	1. Beige vinyl covebase 2. Off-white mastic	ND*
B100-A-412	202236249	Kitchen	Wall Tile w/Grout	1. Off-white adhesive 2. Pale pink ceramic tile 3. White grout	ND*
B100-A-415	202236250	Kitchen	Floor Tile w/Mastic	1. Gray concrete 2. Pale yellow floor tile 3. Yellow mastic	ND*
B100-A-416	202236251	Kitchen	Floor Tile w/Mastic	1. Gray concrete 2. Pale yellow floor tile 3. Yellow mastic	ND*
B100-A-417	202236252	Kitchen	Floor Tile w/Mastic	1. Gray concrete 2. Pale yellow floor tile 3. Yellow mastic	ND*
B100-A-418	202236253	Kitchen	Floor Slab	Gray concrete	ND*
B100-A-419	202236254	Kitchen	Floor Slab	Gray concrete	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-420	202236255	Kitchen	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-421	202236256	Kitchen	Caulking (Gray)	Gray caulk	ND*
B100-A-422	202236257	Kitchen	Caulking (White)	White caulk	ND*
B100-A-425	202236258	Toilet	Ceiling Board	White gypsum board / white paint	ND*
B100-A-426	202236259	Toilet	Ceiling Board	White gypsum board / white paint	ND*
B100-A-427	202236260	Toilet	Ceiling Board	White gypsum board / white paint	ND*
B100-A-428	202236261	Toilet	Wallboard	White gypsum board	ND*
B100-A-429	202236262	Toilet	Wallboard	White gypsum board	ND*
B100-A-430	202236263	Toilet	Wallboard	White gypsum board	ND*
B100-A-431	202236264	Toilet	Wall Ceramic Tile w/Grout	1. Off-white adhesive 2. Pale pink ceramic tile 3. White grout	ND*
B100-A-432	202236265	Toilet	Wall Ceramic Tile w/Grout	1. Gray concrete 2. Pale pink ceramic tile 3. White grout	ND*
B100-A-433	202236266	Toilet	Wall Ceramic Tile w/Grout	1. Gray concrete 2. Pale pink ceramic tile 3. White grout	ND*
B100-A-434	202236267	Toilet	Floor Tile w/Grout	Tan ceramic tile	ND*
B100-A-435	202236268	Toilet	Floor Tile w/Grout	1. Gray concrete 2. Tan ceramic tile 3. White grout	ND*
B100-A-436	202236269	Toilet	Floor Tile w/Grout	1. Gray concrete 2. Tan ceramic tile	ND*
B100-A-437	202236270	Toilet	Floor Slab	Gray concrete	ND*
B100-A-438	202236271	Toilet	Floor Slab	Gray concrete	ND*
B100-A-439	202236272	Toilet	Floor Slab	Gray concrete	ND*
B100-A-440	202236273	Kitchen	Ceiling Insulation	1. Silver wrap 2. Yellow insulation	ND*
B100-A-441	202236274	Kitchen	Pipe Insulation · Straight	1. Silver wrap 2. Yellow insulation	ND*
B100-A-442	202236275	Kitchen	Pipe Insulation · Elbow	1. Silver wrap 2. Yellow insulation	ND*
B100-A-500	202236276	Server Rm	Ceiling Board	White/gray ceiling tile	ND*
B100-A-501	202236277	Server Rm	Ceiling Board	White/gray ceiling tile	ND*
B100-A-502	202236278	Server Rm	Ceiling Board	White/gray ceiling tile	ND*
B100-A-503	202236279	Server Rm	Ceiling Board	White/gray ceiling tile	ND*
B100-A-504	202236280	Server Rm	Ceiling Board	White/gray ceiling tile	ND*
B100-A-505	202236281	Server Rm	Ceiling Slab	Gray concrete	ND*
B100-A-506	202236282	Server Rm	Ceiling Slab	Gray concrete	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-507	202236283	Server Rm	Ceiling Slab	Gray concrete	ND*
B100-A-508	202236284	Server Rm	Ceiling Slab	Gray concrete	Chrysotile (2%)
B100-A-510	202236285	Server Rm	Wallboard	White gypsum board / white paint	ND*
B100-A-511	202236286	Server Rm	Wallboard	1. Blue joint compound 2. White gypsum board	1. Chrysotile (1%) 2. ND*
B100-A-512	202236287	Server Rm	Wallboard	White gypsum board	ND*
B100-A-513	202236288	Server Rm	Wallboard	White gypsum board	ND*
B100-A-514	202236289	Server Rm	Wallboard	1. White gypsum board 2. White joint compound / white paint	1. ND* 2. Chrysotile (1%)
B100-A-515	202236290	Server Rm	Vinyl Cove Base w/Mastic	1. Black vinyl covebase 2. Yellow mastic	ND*
B100-A-516	202236291	Server Rm	Vinyl Cove Base w/Mastic	1. Black vinyl covebase 2. Cream caulk 3. Yellow mastic	ND*
B100-A-517	202236292	Server Rm	Vinyl Cove Base w/Mastic	1. Black vinyl covebase 2. Yellow mastic	ND*
B100-A-518	202236293	Server Rm	Vinyl Cove Base w/Mastic	1. Black vinyl covebase 2. Yellow mastic	ND*
B100-A-519	202236294	Server Rm	Vinyl Cove Base w/Mastic	1. Black vinyl covebase 2. Yellow mastic	ND*
B100-A-520	202236295	Server Rm	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-521	202236296	Server Rm	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-522	202236297	Server Rm	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-523	202236298	Server Rm	Floor Slab	Gray concrete / yellow adhesive	Chrysotile (2%)
B100-A-524	202236299	Server Rm	Floor Slab	Gray concrete / yellow adhesive	ND*
B100-A-530	202236300	Server Rm	Acoustic Board	White/gray acoustic board	ND*
B100-A-531	202236301	Server Rm	Acoustic Board	White/gray acoustic board	ND*
B100-A-532	202236302	Server Rm	Acoustic Board	White/gray acoustic board	ND*
B100-A-533	202236303	Server Rm	Wainscot Board	White gypsum board / white paint	ND*
B100-A-534	202236304	Server Rm	Wainscot Board	White gypsum board / silver paint	ND*
B100-A-535	202236305	Server Rm	Wainscot Board	White gypsum board / silver paint	ND*
B100-A-539	202236306	Server Rm	Floor Board	Off-white floor board	ND*
B100-A-540	202236307	Server Rm	Floor Board	Off-white floor board	ND*
B100-A-541	202236308	Server Rm	Floor Board	Off-white floor board	ND*
B100-A-543	202236309	Kitchen	Wall Tile w/Grout	1. Off-white ceramic tile 2. White grout	ND*
B100-A-544	202236310	Janitor	Ceiling Board	White gypsum board	ND*
B100-A-545	202236311	Janitor	Wallboard	Gray concrete / ivory paint	ND*

Sample ID No.	LAB ID No.	Location	Material Description	Layer(s)	Asbestos Content
B100-A-546	202236312	Janitor	Floor Slab	Gray concrete	ND*
B100-A-547	202236313	Rm 110	Dampproofing Material	1. Black dampproofing material 2. Gray concrete	ND*
B100-A-548	202236314	Rm 110	Wall Insulation	Yellow insulation	ND*

*ND=None Detected (less than 0.1% based on the provisions of JEGS)

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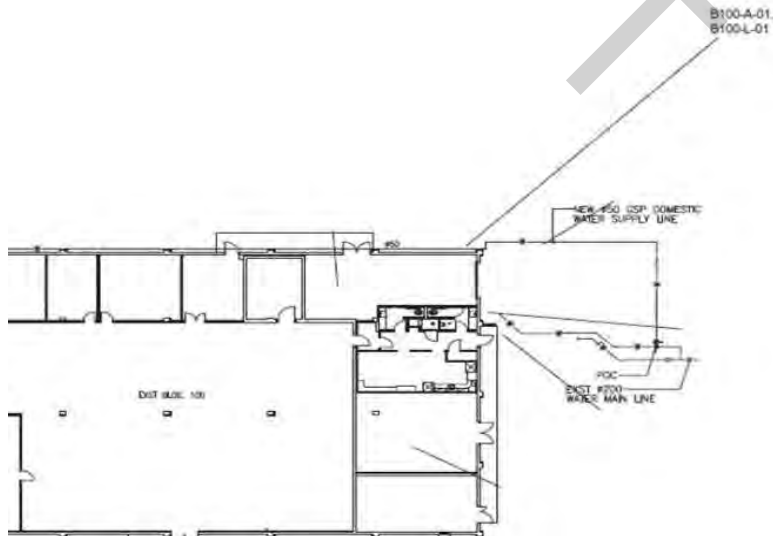
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-01	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



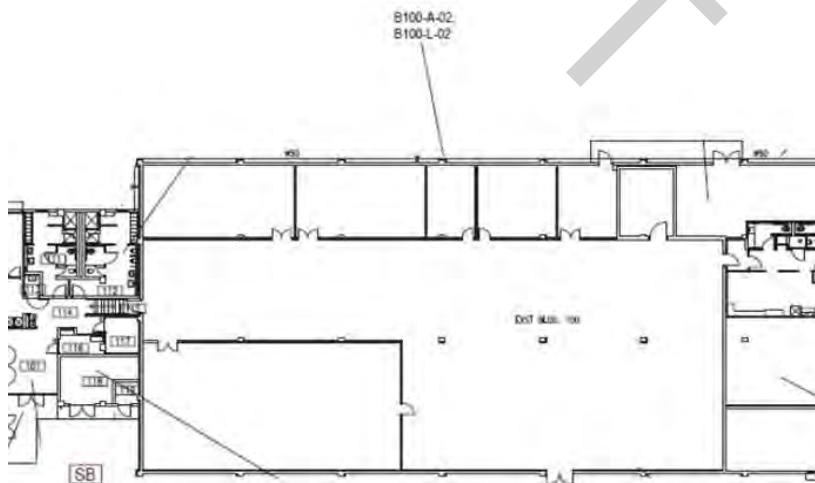
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-02	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



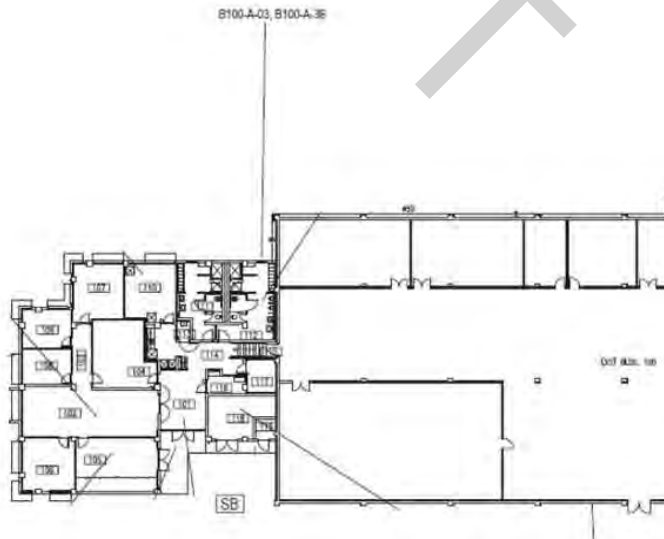
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-03	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



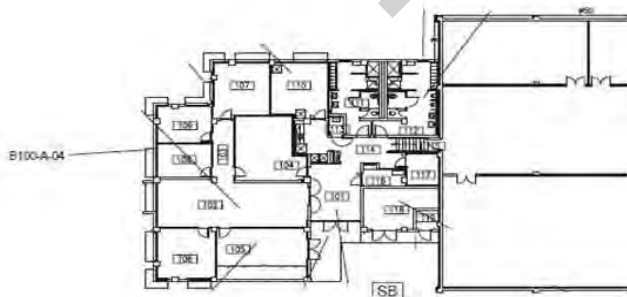
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-04	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



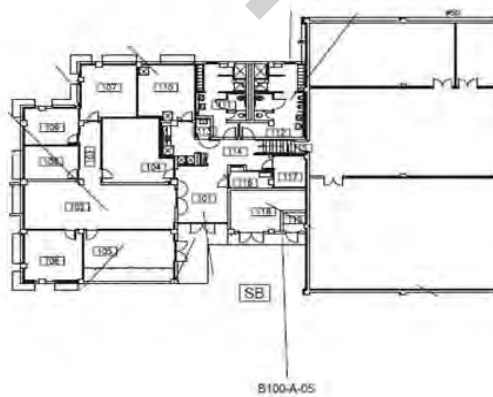
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-05	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



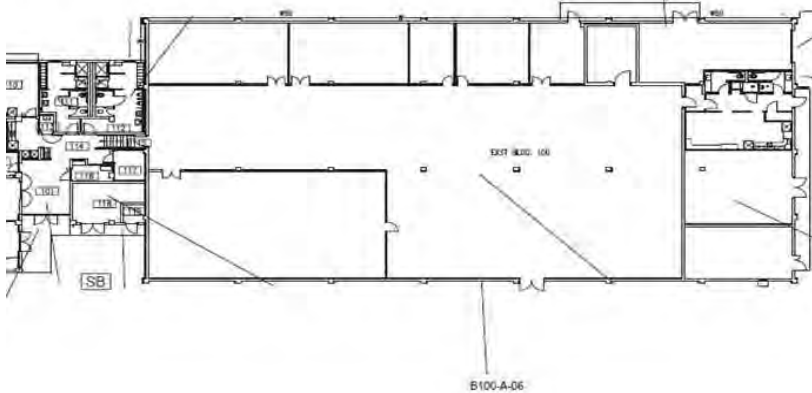
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-06	TYPE OF MATERIAL	: Exterior Concrete Wall Surface
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



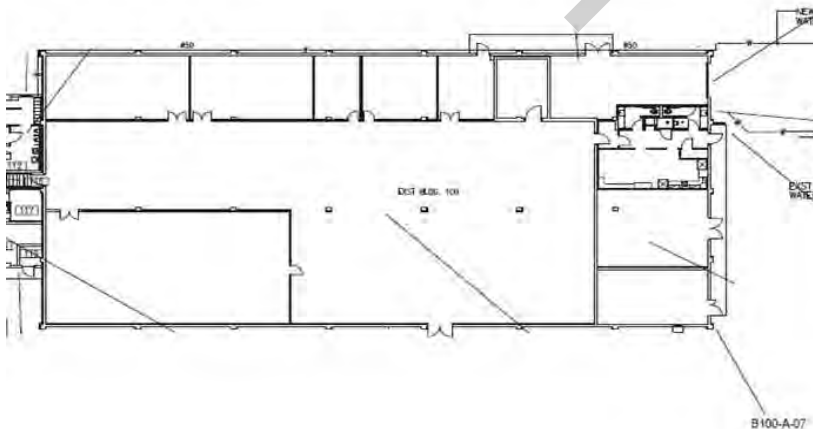
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-07	TYPE OF MATERIAL :	Exterior Concrete Wall Surface
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



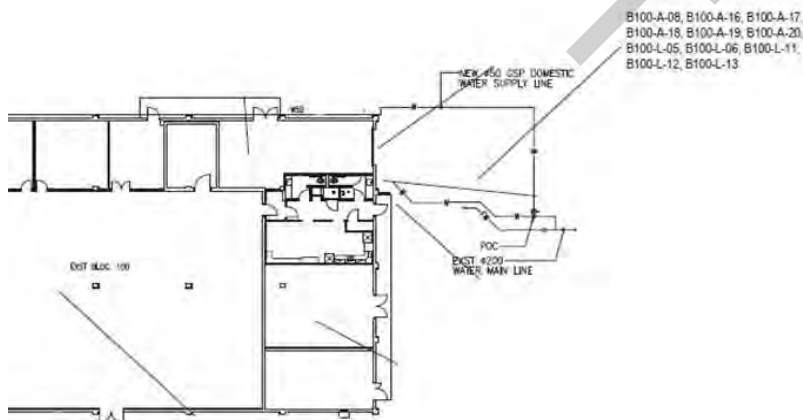
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-08	TYPE OF MATERIAL	: Neo Seal
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Elect Box, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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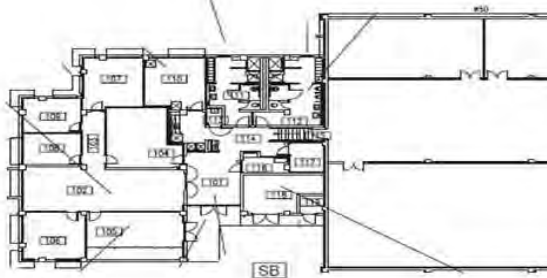
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-10	TYPE OF MATERIAL	: Caulking Material
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26,
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B100-L-16



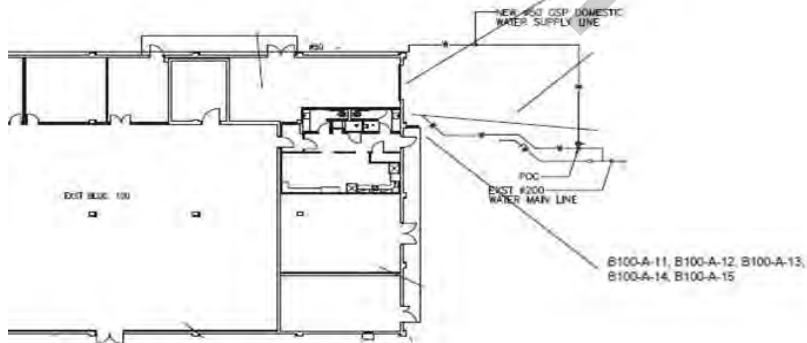
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-11	TYPE OF MATERIAL :	Concrete Floor Slab
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



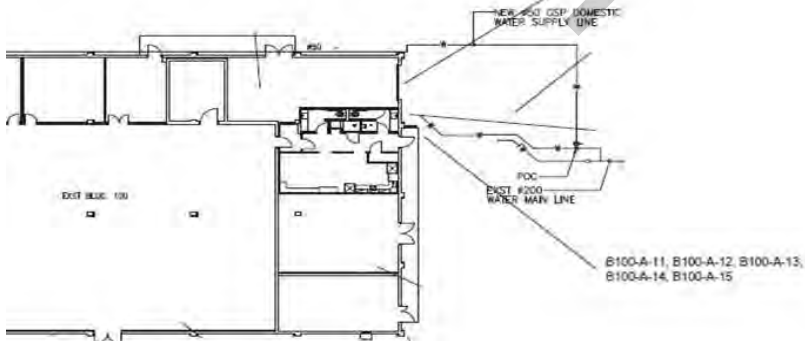
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-12	TYPE OF MATERIAL	: Concrete Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



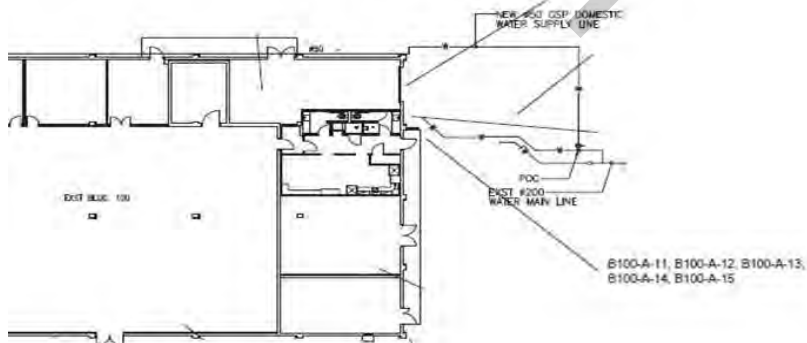
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-13	TYPE OF MATERIAL :	Concrete Slab
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



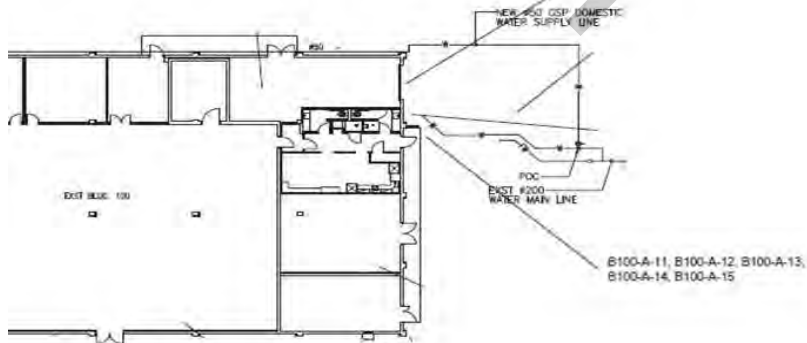
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-14	TYPE OF MATERIAL	: Floor Mortar Repair Seal
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



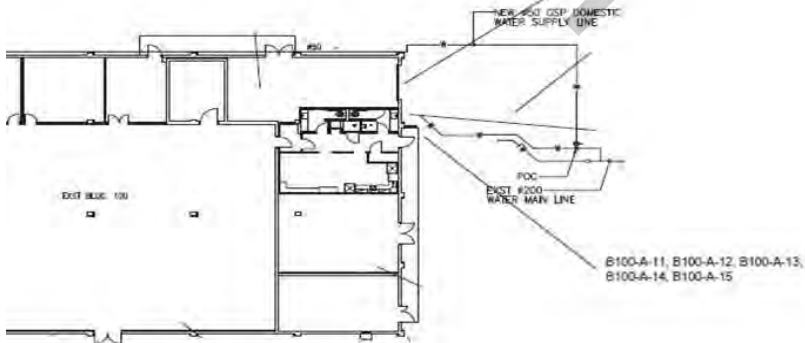
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-15	TYPE OF MATERIAL	: Expansion Joint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



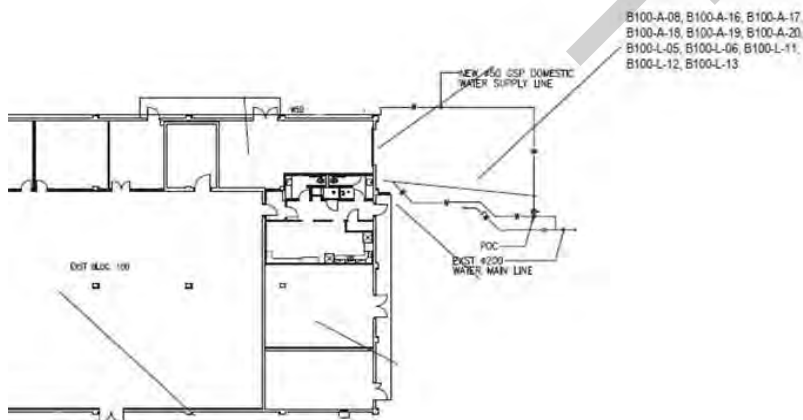
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-16	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Condenser, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



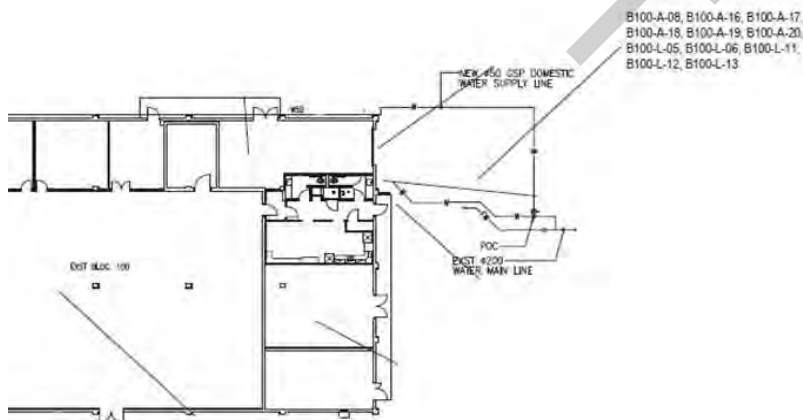
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-17	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Condenser, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



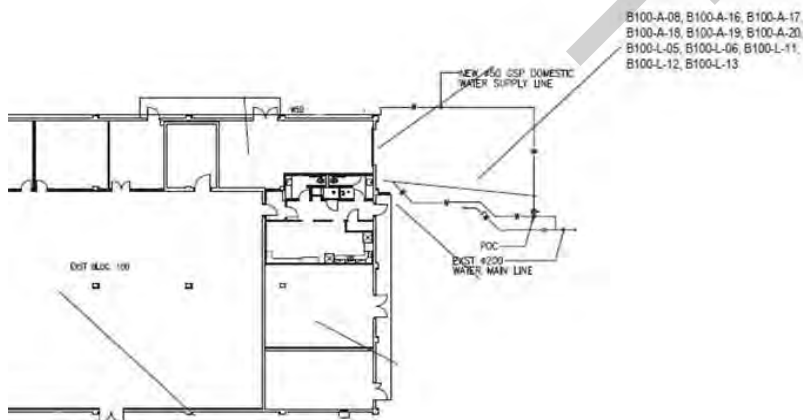
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

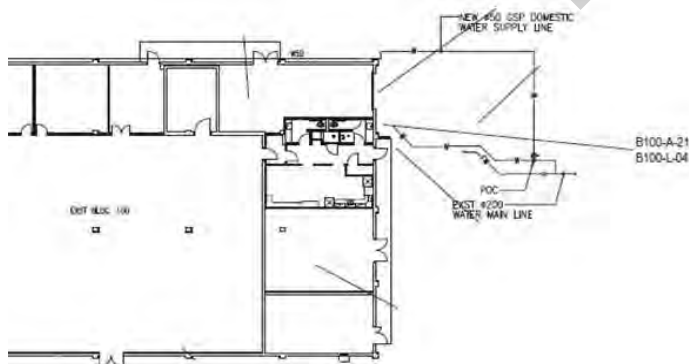
SAMPLE ID NO.	: B100-A-18	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Condenser, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



(Bldg. 100, BUCKNER COMMUNICATION SITE)

PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-21	TYPE OF MATERIAL	:	Pipe Flange Gasket
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
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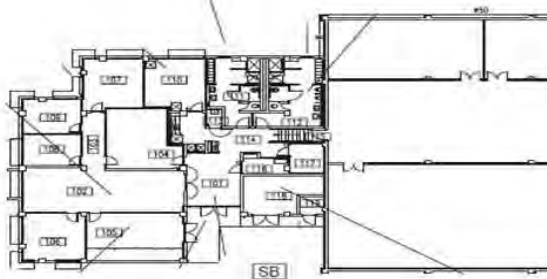
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-22	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: AC, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26,
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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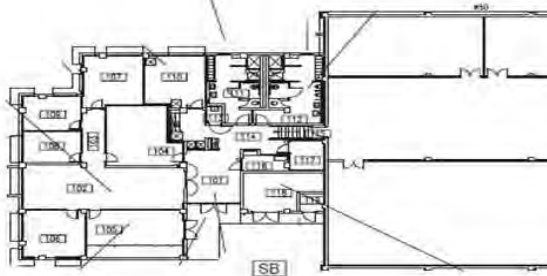
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-23	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: AC, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26,
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

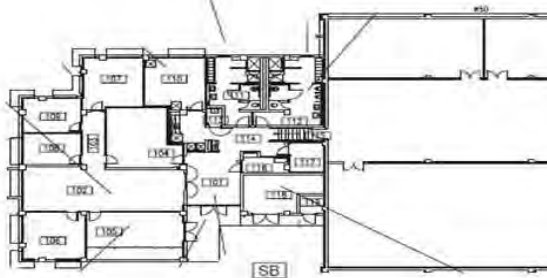
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PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-24	TYPE OF MATERIAL	: Concrete Pad
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: AC, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-09, B100-A-10, B100-A-22,
B100-A-23, B100-A-24, B100-A-25,
B100-A-26,
B100-L-07, B100-L-08, B100-L-15,
B100-L-16



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

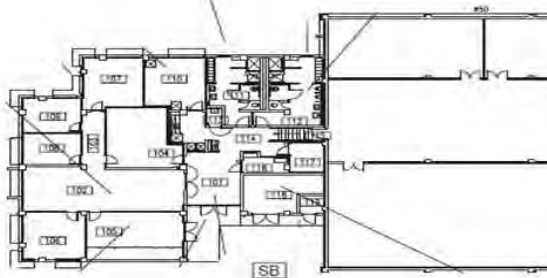
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-25	TYPE OF MATERIAL	:	Pipe Insulation
SAMPLE DATE	:	1-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Straight, Exterior	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-09 B100-A-10 B100-A-22
B100-A-23 B100-A-24 B100-A-25
B100-A-26
B100-L-07 B100-L-08 B100-L-15
B100-L-16



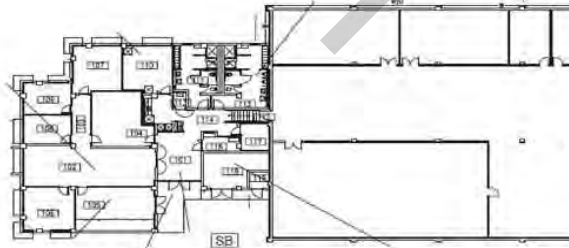
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-27	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

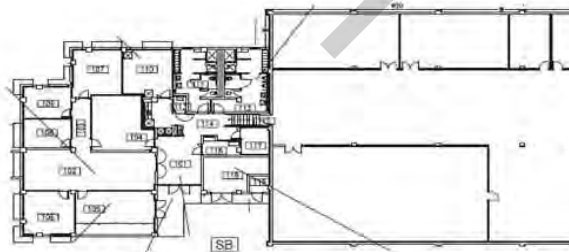
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-28	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

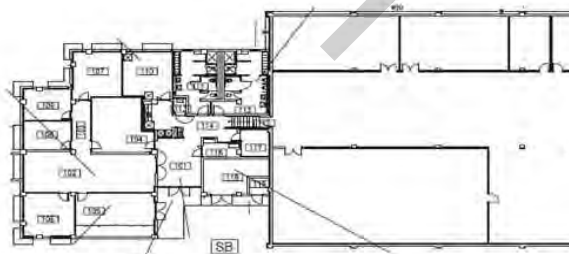
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-29	TYPE OF MATERIAL	: Ceramic Floor Tile w/Grout
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

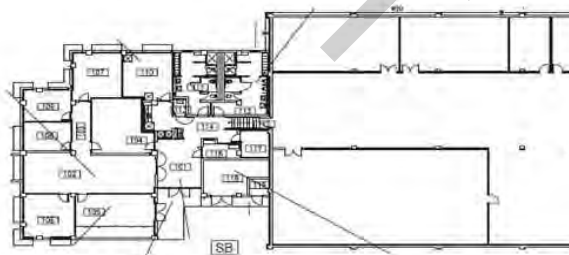
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-30	TYPE OF MATERIAL	: Mortar
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Under Tile, Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

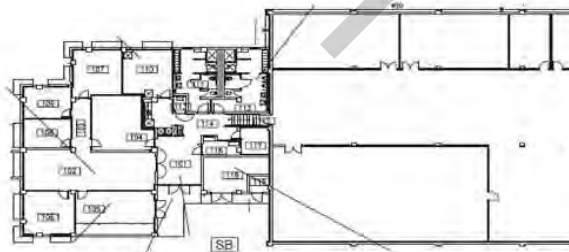
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-31	TYPE OF MATERIAL	: Mortar
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Under Tile, Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

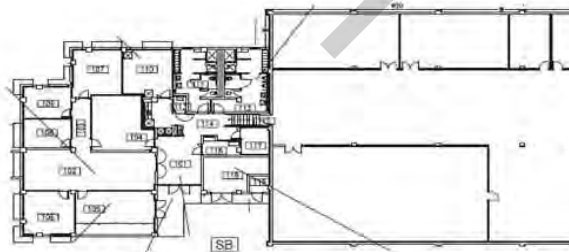
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-32	TYPE OF MATERIAL	: Mortar
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Under Tile, Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

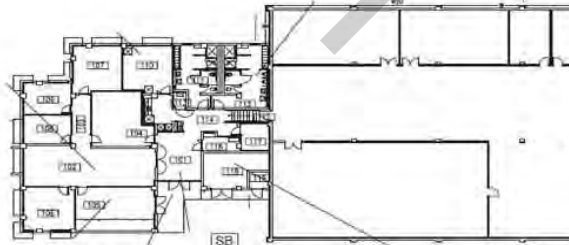
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-33	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

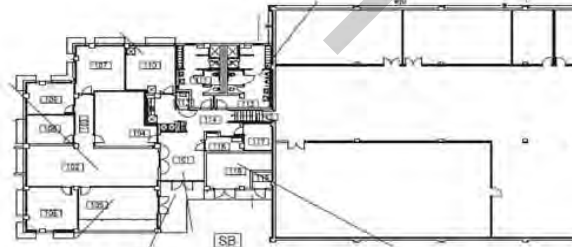
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-34	TYPE OF MATERIAL :	Concrete Floor Slab
SAMPLE DATE :	1-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Front Entrance, Exterior	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

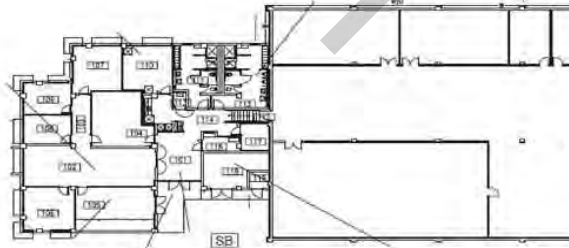
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-35	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Front Entrance, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-27, B100-A-28, B100-A-29,
B100-A-30, B100-A-31, B100-A-32,
B100-A-33, B100-A-34, B100-A-35

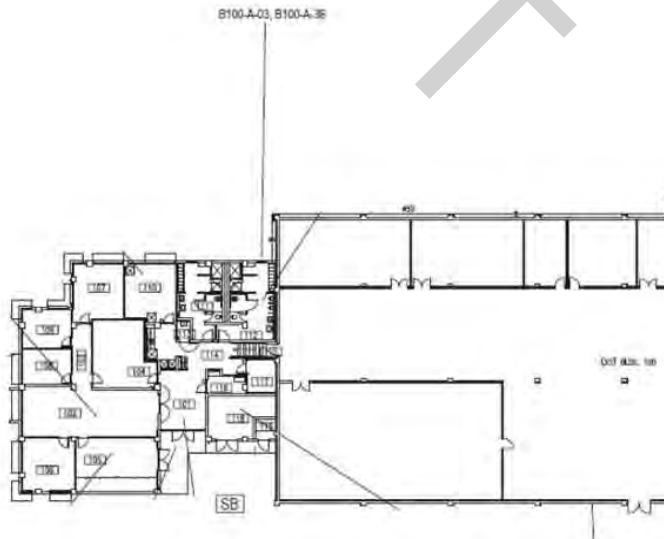
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-36	TYPE OF MATERIAL	: Expansion Joint
SAMPLE DATE	: 1-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Wall, Exterior	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



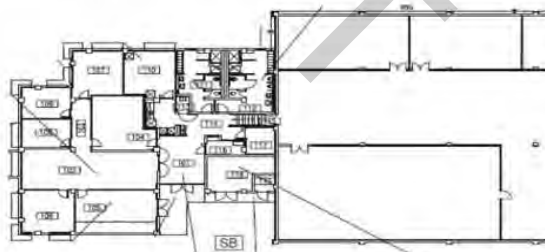
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

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(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-37	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
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B100-A-64,
B100-L-17, B100-L-18, B100-L-19

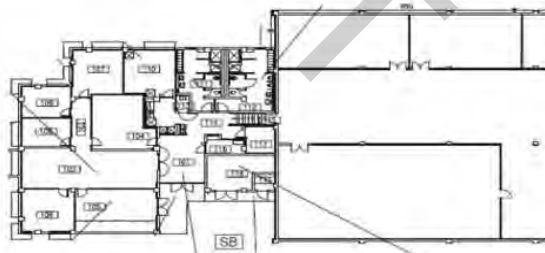
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-38	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
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B100-A-64,
B100-L-17, B100-L-18, B100-L-19

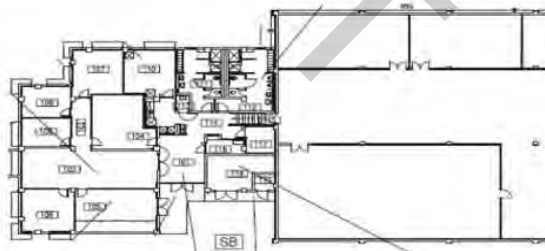
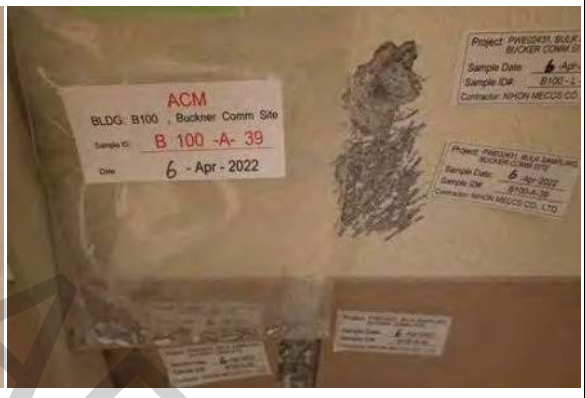
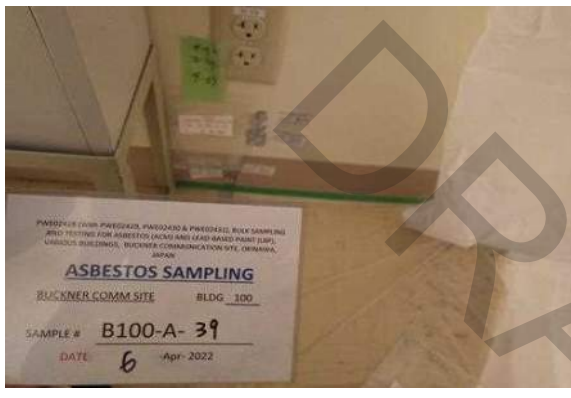
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-39	TYPE OF MATERIAL	: Interior Concrete Wall
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
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B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
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B100-L-17, B100-L-18, B100-L-19

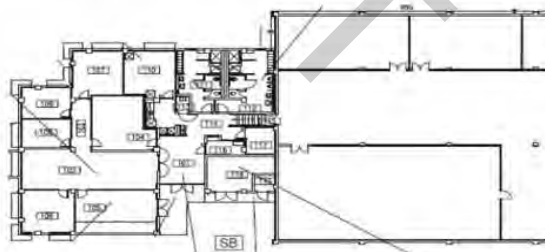
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ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-40	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
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B100-A-64,
B100-L-17, B100-L-18, B100-L-19

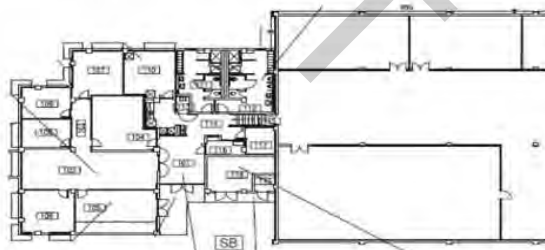
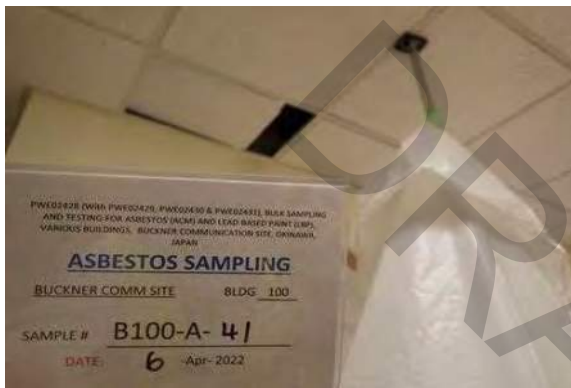
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CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-41	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

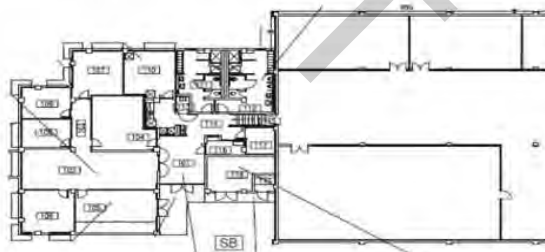
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-42	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

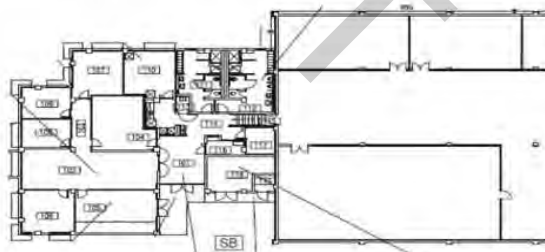
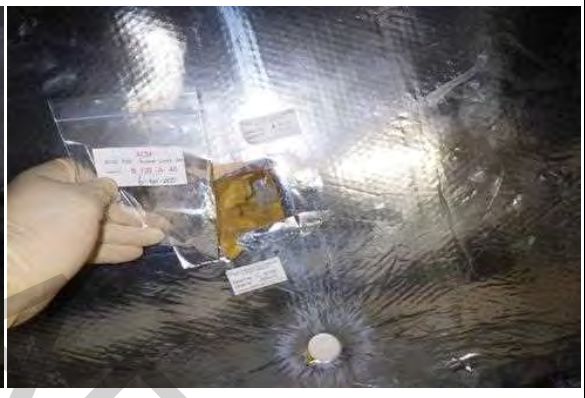
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-46	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

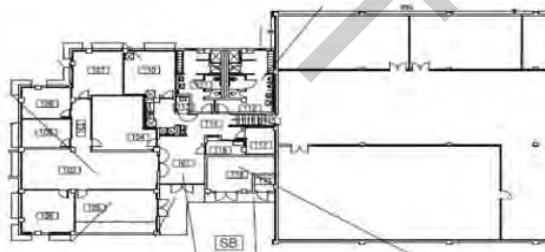
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-47	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

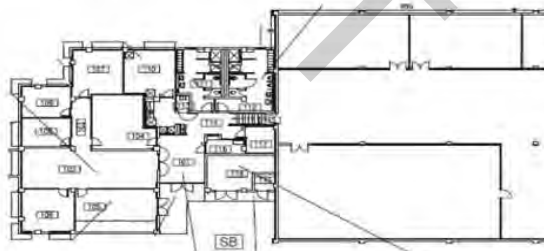
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-48	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Toilet	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

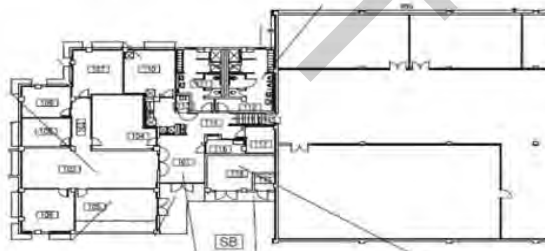
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-49	TYPE OF MATERIAL :	Vinyl Cove Base w/Mastic
SAMPLE DATE :	6-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Tomohide Makabe
ROOM NO./NAME :	Rm 101	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

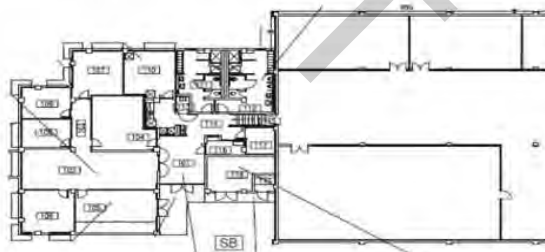
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-50	TYPE OF MATERIAL	:	Vinyl Cove Base w/Mastic
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

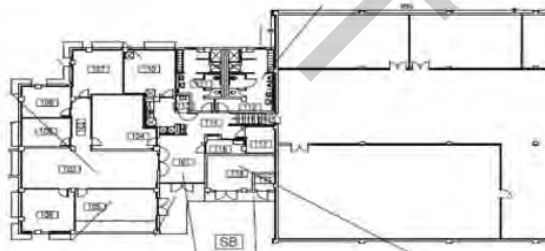
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-51	TYPE OF MATERIAL	:	Vinyl Cove Base w/Mastic
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

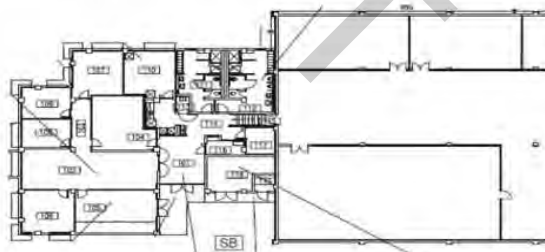
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-52	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

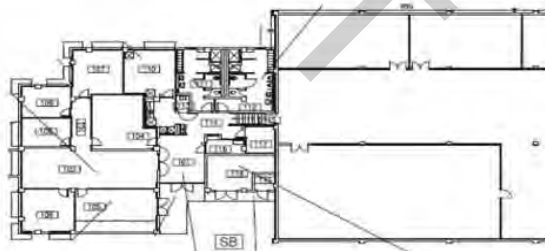
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-53	TYPE OF MATERIAL	:	Wallboard
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
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B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

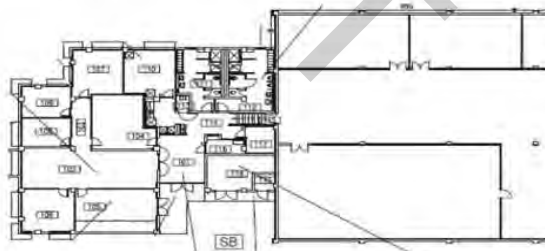
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-54	TYPE OF MATERIAL	: Wallboard
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
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B100-A-58, B100-A-59, B100-A-60,
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B100-A-64,
B100-L-17, B100-L-18, B100-L-19

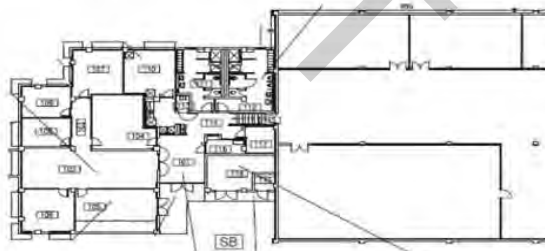
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-55	TYPE OF MATERIAL	:	Vinyl Floor Tile w/Mastic
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

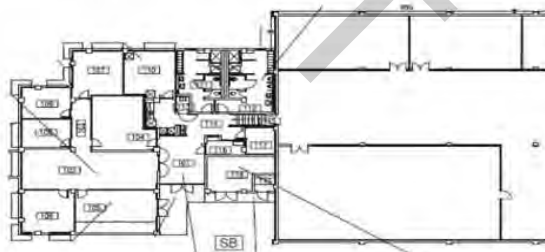
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-56	TYPE OF MATERIAL	: Vinyl Floor Tile w/Mastic
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
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B100-A-55, B100-A-56, B100-A-57,
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B100-L-17, B100-L-18, B100-L-19

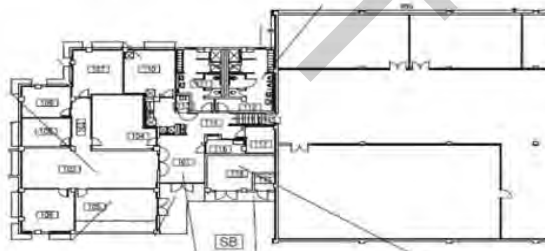
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-57	TYPE OF MATERIAL	: Vinyl Floor Tile w/Mastic
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-37, B100-A-38, B100-A-39,
B100-A-40, B100-A-41, B100-A-42,
B100-A-43, B100-A-44, B100-A-45,
B100-A-46, B100-A-47, B100-A-48,
B100-A-49, B100-A-50, B100-A-51,
B100-A-52, B100-A-53, B100-A-54,
B100-A-55, B100-A-56, B100-A-57,
B100-A-58, B100-A-59, B100-A-60,
B100-A-61, B100-A-62, B100-A-63,
B100-A-64,
B100-L-17, B100-L-18, B100-L-19

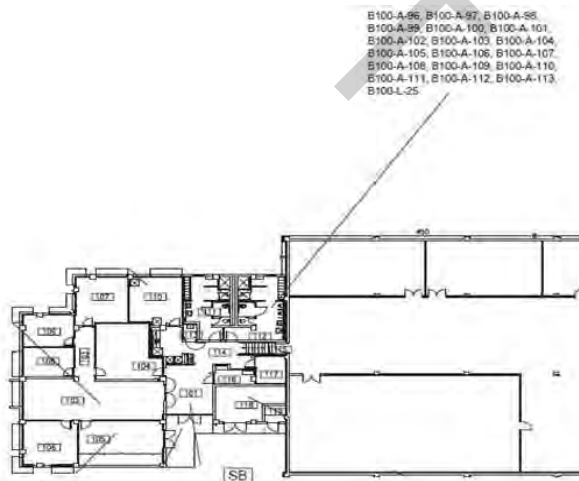
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-58	TYPE OF MATERIAL	:	Concrete Floor Slab
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



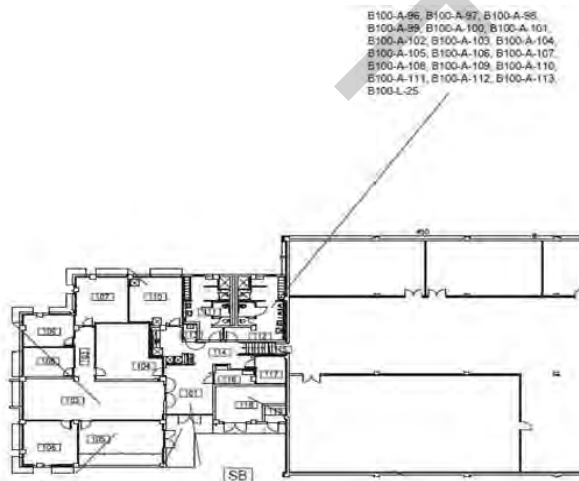
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-59	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



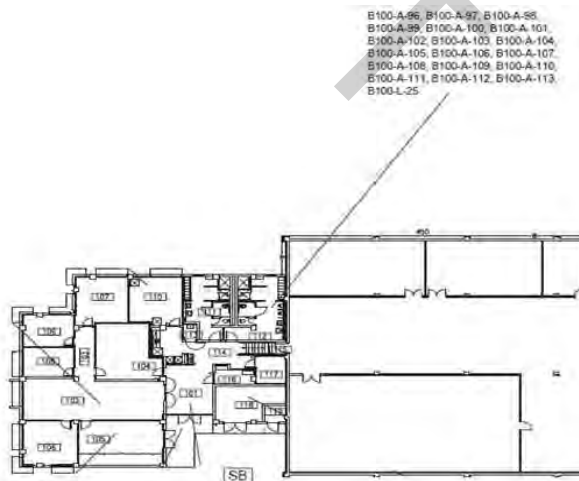
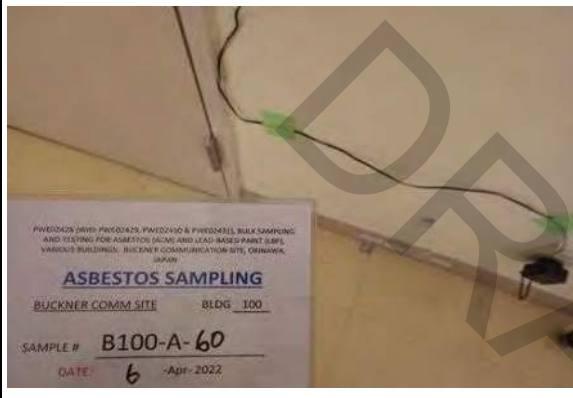
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-60	TYPE OF MATERIAL	: Concrete Floor Slab
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



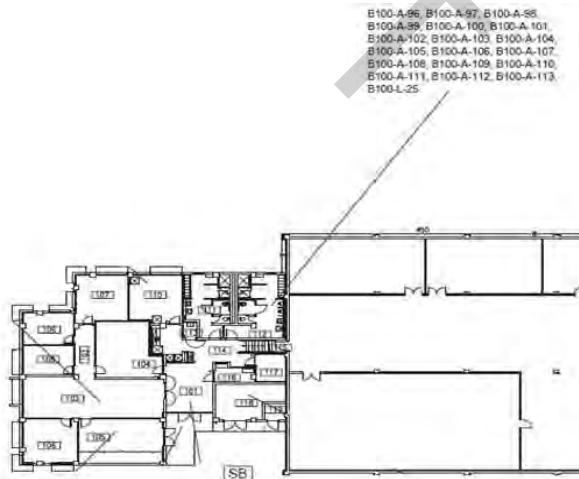
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP), VARIOUS BLDGS, TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-62	TYPE OF MATERIAL	: Duct Insulation
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



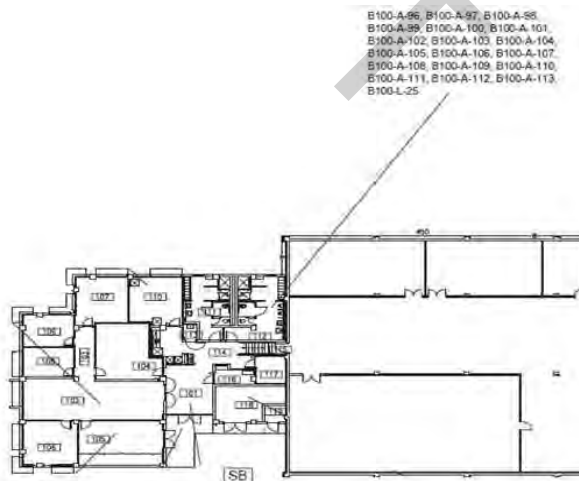
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-63	TYPE OF MATERIAL	: Duct Insulation
SAMPLE DATE	: 6-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 101	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



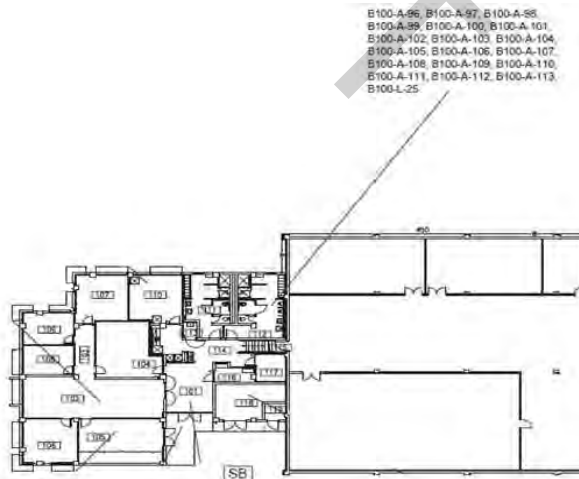
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-64	TYPE OF MATERIAL	:	Duct Insulation
SAMPLE DATE	:	6-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	None Detected
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 101	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



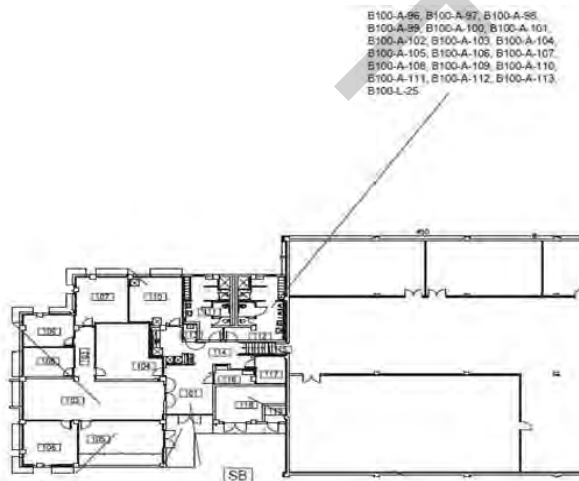
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-65	TYPE OF MATERIAL	: Floor Carpet w/Mastic
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 102	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



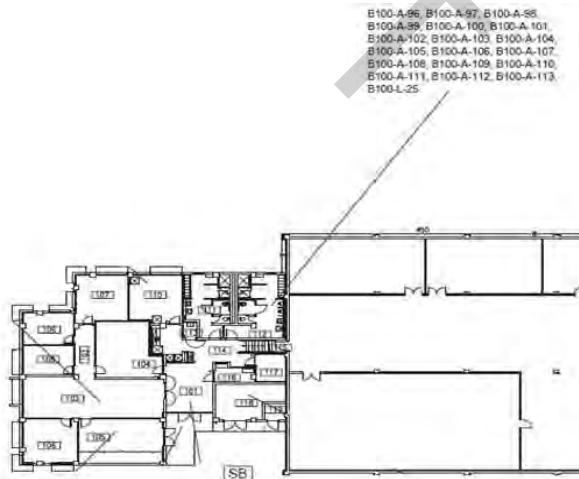
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO. :	B100-A-66	TYPE OF MATERIAL :	Floor Carpet w/Mastic
SAMPLE DATE :	7-Apr-22	CONDITION :	Good
SAMPLE LOCATION :	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS :	None Detected
BUILDING NO./LOCATION :	Bldg. 100	ASBESTOS BUILDING INSPECTOR :	Ryota Tomoyose
ROOM NO./NAME :	Rm 102	TESTING LABORATORY :	Hawaii Analytical Laboratory, LLC



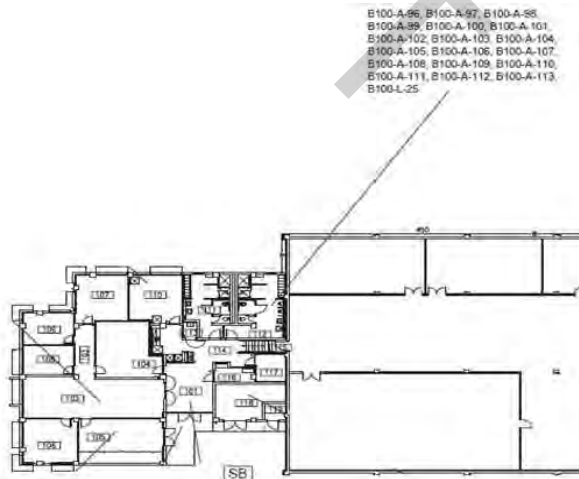
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-67	TYPE OF MATERIAL	: Floor Carpet w/Mastic
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 102	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



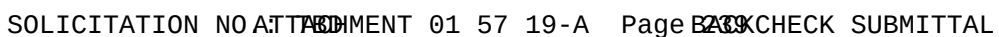
PROJECT TITLE: PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1, NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

The left photograph shows a white label attached to a piece of equipment. The label contains the following text:

POTENTIAL FIBER RELEASE FROM
POTENTIAL FIBER SAMPLES AND TESTING
FOR ASBESTOS IN AIR AND SURFACES
AND/OR FIBER RELEASE FROM SURFACES
CONTAINING ASBESTOS (ASBESTOS)
ASBESTOS SAMPLING
Building Construction Site - BLDG. 100
SAMPLE # **B100-A-71**
DATE **7-Apr-2022**

The right photograph shows a clear plastic bag containing a sample. A small white label is attached to the bag with the following information:

MATERIAL: **ACRA**
Location: **H 100-A-71**
Date: **7 Apr. 2022**



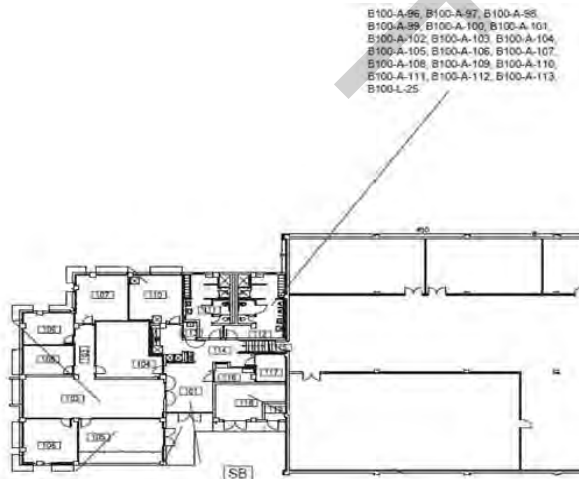
REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-72	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

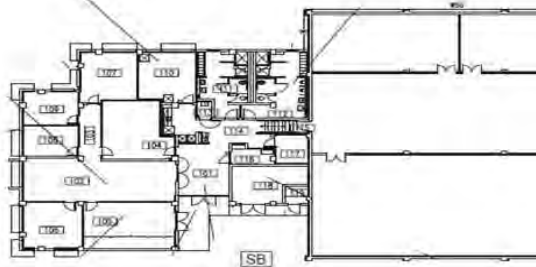
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-73	TYPE OF MATERIAL	: Ceiling Board
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Tomohide Makabe
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

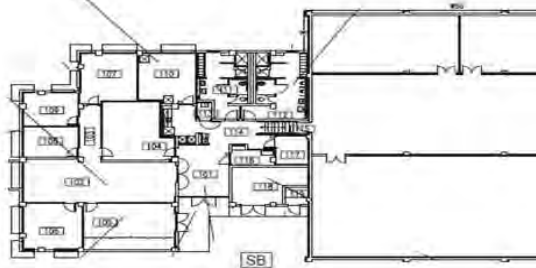
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-75	TYPE OF MATERIAL	: Ceiling Insulation
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

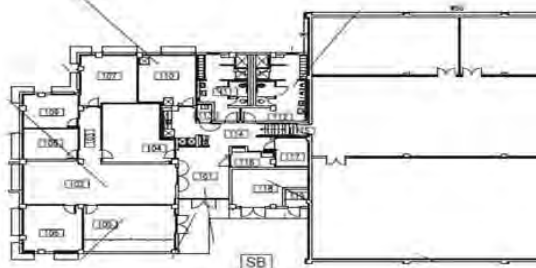
(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	: B100-A-78	TYPE OF MATERIAL	: Ceiling Slab
SAMPLE DATE	: 7-Apr-22	CONDITION	: Good
SAMPLE LOCATION	: BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	: None Detected
BUILDING NO./LOCATION	: Bldg. 100	ASBESTOS BUILDING INSPECTOR	: Ryota Tomoyose
ROOM NO./NAME	: Rm 105	TESTING LABORATORY	: Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119



REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX, OKINAWA, JAPAN
ASBESTOS ANALYSIS REPORT

25SF0004

(Bldg. 100, BUCKNER COMMUNICATION SITE)

PWE02429 (With PWE02429, PWE02430&PWE02431), BULK SAMPLING AND TESTING FOR
PROJECT TITLE: ASBESTOS (ACM) & LEAD BASE PAINT (LBP),VARIOUS BLDGS,TORII STATION, KUWAE TANK FARM #1,
NAHA PORT AND BUCKNER COMMUNICATION SITE, OKINAWA, JAPAN

SAMPLE ID NO.	:	B100-A-80	TYPE OF MATERIAL	:	Vinyl Cove Base w/Mastic
SAMPLE DATE	:	7-Apr-22	CONDITION	:	Good
SAMPLE LOCATION	:	BUCKNER COMMUNICATION SITE	QUALITATIVE ANALYSIS	:	Chrysotile (2%)
BUILDING NO./LOCATION	:	Bldg. 100	ASBESTOS BUILDING INSPECTOR	:	Tomohide Makabe
ROOM NO./NAME	:	Rm 105	TESTING LABORATORY	:	Hawaii Analytical Laboratory, LLC



B100-A-114, B100-A-115, B100-A-116,
B100-A-117, B100-A-118, B100-A-119

